ISPE®

GUIDE:

Oligonucleotide and Peptide Manufacturing Facilities



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GUIDE:

Oligonucleotide and Peptide Manufacturing Facilities

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ISBN 978-1-969543-07-4

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Preface

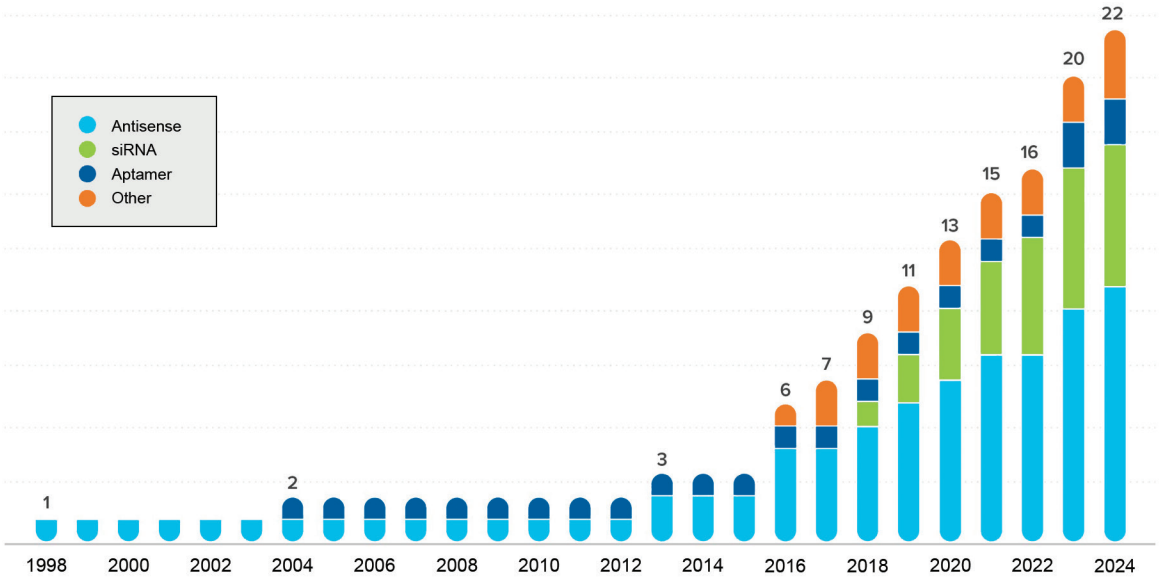
The pharmaceutical industry has long faced the challenge of balancing speed to market, cost, and regulatory compliance. As seen with new modalities such as ATMPs and others, striking this balance becomes even more challenging when designing facilities intended to manufacture new products where facility templates have yet to be developed. As new modalities continue to develop, an emphasis on establishing a compliant and cost-effective baseline approach is a necessary part of the pharmaceutical lifecycle. This foundation supports the reliable production of therapeutics for patients while establishing an industry knowledge base to support future development within these modalities or adjacent modalities.

This First Edition of the ISPE Guide for Oligonucleotide and Peptide Manufacturing Facilities defines how manufacturing processes for oligonucleotides and peptides can be aligned with the facilities that manufacture them in a manner that protects the process, product and ultimately the patient, while ensuring operations remain cost-effective and sustainable throughout the facility lifecycle. This Guide was assembled by a team of global subject matter experts with vast experience in oligonucleotide and peptide manufacturing using Good Manufacturing Practices (GMPs).

Although peptide manufacturing has been established for several decades, with some sources citing 100 commercial approvals by the year 2024, this Guide is being published at a time when commercial oligonucleotide therapeutics are rapidly emerging.

Therapeutic approvals have increased sharply: as of 2024, the US Food and Drug Administration (FDA) had approved 22 oligonucleotide products, up from only three therapies a decade earlier (see Figure 1). These therapeutics are administered parenterally, with some including intrathecal administration routes. Most oligonucleotide drugs have a unique mode of action, moving the therapeutic effect up the biological cascade to largely work on RNA rather than proteins.

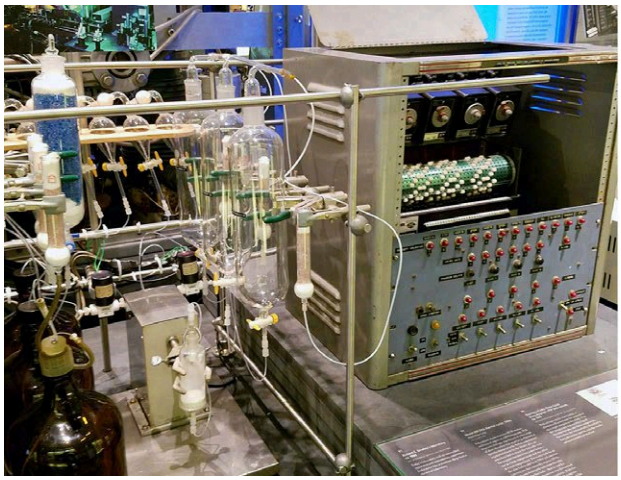
Figure 1: History of US FDA Oligonucleotide Approvals

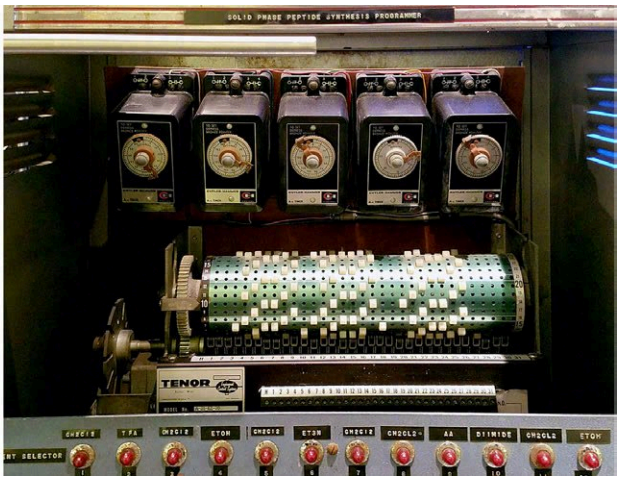


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A more established technology, peptides production was initially founded on the work of R. B. Merrifield whose publication in 1963 [1], on solid-phase peptide synthesis methods, coincided with his design and development of the first automatic peptide synthesizer. This was a significant milestone on the route to modern chemical manufacturing of both peptides and oligonucleotides. His continued work eventually culminated in the total synthesis of bovine pancreatic Ribonuclease A in 1971 and earned him the Nobel Prize in Chemistry in 1984.

Figure 2: Original Merrifield Peptide Synthesizer and Sequencer [2]
Images courtesy of Science History Institute, www.sciencehistory.org.





Today, peptides are seeing growth and future potential that is arguably unprecedented in pharmaceutical industry history, as the success and commercial momentum of Incretin drugs is driving massive investment in the technology globally, with implications for production scale, facility needs, and supply chain that have yet to be fully appreciated.

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Acknowledgements

The effort to create a new ISPE facility guide for oligonucleotides and peptides was a significant undertaking that lasted more than two years from initial inception of the idea to final publication. Here we do our best to acknowledge the tremendous industry involvement and support that made it happen.

The group of Co-Leads formed a critical steering committee that was with the project for the duration, developing the initial strategy of what to write and how to organize the document. They were indispensable in the effort to harmonize content and level of detail across the guide and provided unflagging support at every stage in the process.

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Key Industry Support

The Team wishes to extend sincere thanks to the following additional individuals for their unique contributions to the success of the guide. Many reviewed content, helped with networking and promotion, and generally provided intangible support, without which we could not have achieved our mission.

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Special Thanks

The Team would like to thank the many individuals and companies from around the world who supported the effort of producing this industry guide, in particular by providing valuable industry feedback; although they are too numerous to list here, their input is greatly appreciated.

The Leads would also like to thank ISPE for technical writing and editing support by Gabriela Kantor and Nina Wang (ISPE Guidance Documents Technical Editors), production support by Lynda Goldbach (ISPE Senior Publications Manager), and to Madi Scribner, of CRB Group, who provided indispensable graphic design support for the guide.

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1 Introduction

1.1 Background

This *ISPE Guide: Oligonucleotide and Peptide Manufacturing Facilities* addresses the unique position these modalities occupy at the intersection of biologically and chemically derived APIs, and their respective requirements. Although they rely on solvent-based processes aligned with current best practices for traditional small molecule manufacturing, their specialized chemistries and work-up differ from traditional small molecules or biologics manufacturing. As the modalities have expanded, these unique elements have contributed to varied interpretations of facility requirements.

The design, construction, commissioning and qualification of pharmaceutical facilities continue to pose significant challenges for manufacturers, engineering and construction professionals, and equipment suppliers. Achieving success requires an aligned understanding of both the guiding philosophy and its practical implementation in the context of these facility modalities.

This Guide is modeled after previous ISPE Baseline® Guides, including *Volume 1 – Active Pharmaceutical Ingredients (Second Edition)* [3], *Volume 3 – Sterile Product Manufacturing Facilities, (Third Edition)* [4], and *Volume 6 – Biopharmaceutical Manufacturing Facilities (Third Edition)* [5] – and seeks to address the requirements of oligonucleotide and peptide production facilities.

The group of authors and contributors to this Guide likewise reflect the unique position of oligonucleotides and peptides within the pharma landscape. They include experts whose organizations supported and designed some of the earliest facilities and conducted early oligonucleotide clinical Good Manufacturing Practice (GMP) batches in the mid 1990s; leaders who drove the rapid growth of the peptide generics sector in in the early 2000s, a decade that saw more new peptide drug approvals than the previous two decades combined; and innovators who leveraged biologics experience from the continued acceleration of monoclonal Antibodies (mAbs) over the last 25 years to develop novel strategies for oligonucleotide purification and final processing.

The 2007 *ISPE Baseline® Guide: Active Pharmaceutical Ingredients (Second Edition)* [3], established foundational approaches for APIs produced via chemical synthesis. This document introduced early concepts of cleanliness zones of a facility by defining levels of protection, helping the industry develop clearer facility and process requirements at a time when discretionary capital costs were escalating. Although influential, Volume 1 has not been updated since its second edition was published nearly two decades ago.

Since then, the biopharmaceutical industry has undergone significant maturation, particularly in its understanding and approach to process and facility design. Risk assessments and contamination control strategies are now recognized as core elements of GMP compliance. The recently published *ISPE Baseline® Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5], reflects this evolution and served as an important reference for this document, especially where analogous aqueous-based bioprocessing unit operations are used in oligonucleotide and peptide manufacturing and facility design philosophies can be applied.

Today, metric-ton scale peptide manufacturing is emerging and has potentially far-reaching implications for sustainability considerations and the global organic solvent supply chain. These trends suggest possible changes in facility design approaches that may soon reach commercial feasibility. In parallel, oligonucleotide solid-phase synthesis technologies are poised for disruption by a significant anticipated expansion of the patient populations that these therapeutics can serve.

This Guide aims to provide a robust, contemporary baseline from which the industry can navigate these evolving challenges and continue advancing its mission of delivering high-quality medicines to patients.

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1.2 Scope

This Guide focuses on compliance with US and EU regulatory guidelines, as well as requirements of additional PIC/S members throughout the world. Project specific considerations and regulations must be evaluated for the applicable geography.

The Guide addresses drug substance facilities for oligonucleotides and peptides within the GMP framework described in ICH Q7 for Active Pharmaceutical Ingredients (APIs) [6].

Nof1 oligonucleotides and guide RNA are out of scope due to potential differences in regulatory requirements (see Chapter 2), though the principles described herein generally provide a good basis for facilities using similar production technologies.

Conjugation of synthetic oligonucleotides to biologics (e.g., monoclonal antibodies) is an emerging field. The GMP requirements for synthetic oligonucleotides and for the resulting conjugates, often regulated as biologics, may differ. This topic is not addressed directly in this Guide.

Additionally, mRNA production, cell and gene therapy, plasmid production, or other facility types that potentially use oligonucleotide or peptides as raw materials or intermediates are out of scope. An upcoming ISPE guide focused on mRNA drug substance facilities is expected for publication in 2026.

This Guide is intended primarily for large-scale GMP production facilities for oligonucleotides and peptides. It provides specific considerations for scale-up, such as boundaries for management of flammable solvents and waste, and its philosophies and interpretations are expected to apply to clinical-scale facilities for human trials and other smaller facilities. Even so, the risk assessments and philosophical framework for facilities dedicated to those scales may be viewed differently by regulators and operators.

While written for new facilities, the concepts herein can and should be leveraged when considering modifications and expansions to legacy facilities.

1.3 Key Concepts

Biotechnology processes where molecules are large and complex have been traditionally defined by the notion that the process is the product, whereas small molecule API production focused on well characterized molecules. These facility approaches are fundamentally different, and an approach rooted in either will miss the mark for oligonucleotide and peptide production facilities. This Guide defines facility requirements in the specific context of oligonucleotide and peptide processes.

Environmental contamination considerations are likewise distinct from both aqueous facilities where the processes are designed for cell growth and small molecule chemical processing facilities. The nature of oligonucleotide and peptide facilities, including synthetic chemical processes, organic solvent backgrounds used in the operations, and transition to aqueous environments in downstream processes require unique strategies.

Scale directly affects facility decisions, for both operational approaches and code compliance for flammable solvents. Important inflection points in facility configuration are introduced during scale-up of these facilities.

Multiproduct versus single-product – oligonucleotide and peptide facilities are commonly multiproduct. Coupled with the unique aspects of the processes, this places heightened emphasis on cross-contamination controls.

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Future trends – These discussions of emerging technology are included to acknowledge the rapid change afoot within these industries. For oligonucleotides, the pace of therapeutic development and the inclement scale horizon are poised to change manufacturing. For peptides, current scale needs and implications are being made a reality by the burgeoning GLP-1 market. Facilities will need to adapt beyond footprint alone to handle these changes and the discussion sets the stage for future editions of this Guide.

1.4 Use of this Guide

Although oligonucleotide and peptide manufacturing facilities share enough similarities to merit their inclusion together in a single guide, there are unique elements to both that require special focus within this Guide. The initial publication of this Guide provides a comprehensive oligonucleotide and peptide facility guide, with appropriate specific content included for each modality where needed.

1.5 Structure of this Guide

Due to substantial differences between core oligonucleotide and peptide processes, the following chapters have dedicated sections and provide specific guidance for each technology.

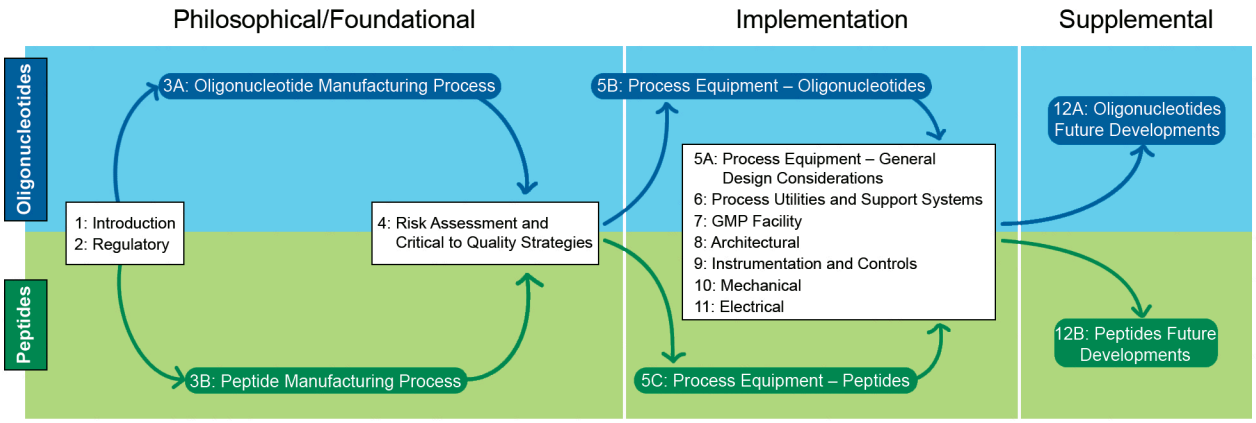
- Chapter 3: Manufacturing Process
- Chapter 5: Process Equipment
- Chapter 12: Future Developments Impacting Production Facilities

Due to substantial similarities in the philosophies, buildings and systems that support the core processes for the two technologies, the following chapters will be largely agnostic and provide guidance for both oligonucleotides and peptides, with specific discussions included as appropriate.

- Chapter 2: Regulatory
- Chapter 4: Risk Assessments and Critical to Quality Strategies
- Chapter 6: Process Utilities and Support Systems
- Chapter 7: GMP Facility Approaches
- Chapter 8: Architecture
- Chapter 9: Instrumentation and Controls
- Chapter 10: Mechanical
- Chapter 11: Electrical

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Figure 1.1: Oligonucleotide and Peptide Guide Outline



1.5.1 Philosophical and Foundational Chapters Synopsis

Philosophical chapters form the basis from a regulatory, product quality, and process perspective and provide the "why" for oligonucleotide and peptide facilities, giving a foundation to the later implementation chapters.

- Chapter 2: Regulatory – Establishes the regulatory basis for drug substance production of oligonucleotides and peptides.

- Chapter 3: Manufacturing Process – In two modality-specific subsections, outlines the core technology basis, chemistry, and typical unit operations for each oligonucleotide and peptide process, and sets the stage for the equipment and facility design. This chapter provides a thorough treatment of current process methods, and in the case of oligonucleotides, references a robust Appendix on process chemistry, providing valuable background for the continued emergence of the industry.

- Chapter 4: Risk Assessments and Critical to Quality Strategies – Provides a framework and examples for unique risk assessment and critical to quality considerations and strategies for these molecule types over the course of their production processes. This chapter includes specific process assessment discussions for both oligonucleotide and peptide processes.

1.5.2 Implementation Chapters Synopsis

Implementation chapters respond to the philosophical and foundational chapters, interpreting the "how" and outlining baseline facility considerations and approaches.

- Chapter 5: Process Equipment contains three subsections:

  - General design considerations

  - Oligonucleotide specific design requirements

  - Peptide specific design requirements

The general design considerations subsection addresses the unique process conditions typical to these modalities and their impact on equipment design. The technology specific sections describe core process equipment, instrumentation needs and selection and considers the support of critical process parameters established in Chapter 3.

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- Chapter 6: Process Utilities and Support Systems – Provides the basis for utility design and identifies unique systems required to support the process and facility needs.

- Chapter 7: GMP Facility Approaches – Addresses scale, throughput, facility layout approaches, room classification examples, and functional adjacency for core process and support areas, including the management of solvent and waste aspects typical of these facilities.

- Chapter 8: Architecture – Ranges from site to building, describing code approaches necessary for proper planning and incorporation of remaining support functions such as warehouse, laboratories, and offices. Design considerations to integrate manufacturing elements with people and culture elements as well as approaches for sustainable design are included.

- Chapter 9: Instrumentation and Controls – Provides automation considerations that respond to unique code aspects and describe where automation is employed in the facility to support the needs of the processes.

- Chapter 10: Mechanical – Provides mechanical equipment and design considerations that respond to unique code aspects of these facilities and describes cost, code, sustainability, and maintenance elements.

- Chapter 11: Electrical – Provides electrical equipment and design considerations that respond to unique code aspects of these facilities and describes cost, code, sustainability, and maintenance elements.

1.5.3 Supplemental Chapter Synopsis

The supplemental chapter addresses potential facility implications and future of changes that are currently underway in the industry.

- Chapter 12: Future Developments Impacting Production Facilities – In two subsections, provides a look at anticipated changes to process and operations responding to the rapid growth of oligonucleotide and peptide modalities and the challenges they must overcome to continue thriving.

1.5.4 Place within Previous Guides

This Guide aims to provide a holistic view of GMP production facilities for oligonucleotides and peptides. This naturally includes cursory treatment of well established industry concepts from adjacent biotechnology and API industries as discussed previously in this chapter. This approach is meant to provide value to readers as a comprehensive, readable guide that stands alone without requiring significant cross referencing.

Within the context of prior facility guides, this Guide highlights the areas of discussion where thinking has evolved, where earlier treatments provided a foundation for further relevant discussion, and where baseline facility concepts remain to be established.

Above all, this Guide focuses on unique aspects of oligonucleotide and peptide facilities that may diverge from traditional approaches taken in adjacent industries. This Guide represents a tool to challenge ambient assumptions and provides a baseline for a compliant oligonucleotide and peptide facility design now and in the future.

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2 Regulatory

2.1 Introduction

Globally, regulatory agencies such as the Food and Drug Administration (FDA) [7] in the US and European Medicines Agency (EMA) [8] in the European Union, have the responsibility of ensuring safety, efficacy, and quality of medicines available to the public by overseeing their development, manufacturing, and marketing. The key concepts behind the current regulatory framework for oligonucleotides and peptides are highlighted in Section 2.2 along with references relevant to the manufacture of oligonucleotides, peptides, or both.

Oligonucleotide and peptide manufacturing requires the use of large quantities of flammable and hazardous chemicals. In addition to drug regulatory considerations, several safety and environmental regulations are applicable to oligonucleotide and peptide production facilities. References to those regulations are included in Section 2.3 to support further sections of this Guide focused on facility design considerations affected by these elements.

2.2 Product Regulatory Requirements and Guidance

2.2.1 General Oligonucleotide and Peptide Regulatory Considerations

This Guide is focused specifically on the manufacture of drug substance, also referred to as Active Pharmaceutical Ingredient (API), and therefore does not address regulatory requirements associated with sterile drug product manufacturing. This topic is already discussed in the *ISPE Baseline® Guide: Sterile Product Manufacturing Facilities (Third Edition)* [4]. The focus of this Guide is to clarify which requirements must be satisfied for drug substance manufacturing.

From a regulatory perspective, both oligonucleotides and peptides are at the interface of small molecule and biopharmaceutical products, with considerations that reflect this. For example, oligonucleotides, peptides and their impurities such as fragment sequences or residual solvents are more easily characterized than large-molecule therapeutics, where the cell culture process defines the product. Nevertheless, endotoxins can affect oligonucleotide and peptide products destined for parenteral administration routes. In the case of oligonucleotides, many function by modulating gene expression through different interactions with nucleic acids, a distinct mechanism compared to traditional small molecules, which generally target protein receptors.

Regardless of their mode of action, (modulation of gene expression or other mechanisms) the manufacture of oligonucleotide and peptide drug substances is generally considered a low bioburden process designed to protect the product from contamination and to produce the required quality of material while also protecting operators from any potential safety hazards. Specific requirements may vary depending on the type of manufacturing process and the quality attributes desired for the drug substance.

The need for operator protection in oligonucleotide and peptide manufacturing facilities often results in closed systems that also protect the process and final product from environmental contaminants and from neighboring concurrent operations in a multiproduct facility.

Additionally, the use of bacteriostatic and bactericidal organic solvents is typical of the oligonucleotide and peptide synthesis processes that precede purification. These conditions provide additional protection of the product and process from environmental contaminants. These elements provide a basis for facility design that can be markedly different from that of a biopharmaceutical manufacturing facility.

Prior to designing a facility, it is essential to have a clear understanding of the manufacturing process, contamination risks, and specific product requirements for the oligonucleotides and/or peptides to be produced. Further discussion on these topics is presented in Chapter 4.

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2.2.2 Current Oligonucleotide and Peptide Regulatory Landscape

In the US, the majority of oligonucleotides and peptides have been regulated under the FDA's Center for Drug Evaluation and Research (CDER) [9], which traditionally has dealt with small molecule products. For peptides, the revision in 2020 of the Biologics Price Competition and Innovation Act of 2009 (BPCI) [10] removed the exception for the synthetic production pathway of peptides greater than 40 amino acids in length. With the updated act, these peptides may be regulated as biological products, which are regulated under the FDA's Center for Biologics Evaluation and Research (CBER) [11] instead of CDER. Furthermore, the FDA may determine CBER jurisdiction for other reasons besides just the length of the peptide being produced. For US filings, this implies that PPQ studies be submitted and manufacturing facilities be ready for inspection at the time of submission.

Synthetic oligonucleotides also appear as components in other drug substance processes that are regulated by CBER (e.g., plasmid templates for mRNA processes or sgRNA for CRISPR gene editing therapies, Antibody-Oligonucleotide Conjugates (AOCs), etc.). Synthetic oligonucleotides and peptides are at a crossroads between small molecule APIs and biological products. For example, ICH Q3A [12] notably excludes both modalities from its requirements on drug substance impurities, leaving a gap in the regulatory guidance.

To address the unique attributes of these modalities that extend beyond the guidance available in ICH Q3A [12], the EMA [8] has issued specific guidance documents for both modalities that discuss "specific aspects regarding the manufacturing process, characterization, specifications and analytical control" and build on foundational ICH Q3A elements. In particular, they elaborate on impurity profiling strategies using orthogonal analytical methods to address specific impurity considerations for oligonucleotide and peptide modalities. This includes unique elements arising from the synthetic processes such as truncation and insertion sequences, mis-incorporation of bases, as well as aggregation and deletion impurities in peptides.

- EMA Guideline on the Development and Manufacture of Synthetic Peptides – Effective June 2026 [13]
- Draft Guideline on the Development and Manufacture of Oligonucleotides – June 2024 [14]

While the EMA oligonucleotide guidance document remains in draft form, the "Legal Basis and Relevant Guidelines" sections of these documents directly reference multiple sections in the ICH guidelines as well as EU GMP Guide Part II [15]. This underscores the applicability of these documents and implies that development and facility strategies should take into account the body of guidance available, including the principles contained in these documents, while focusing on applicability for specific product and patient needs. In the subsequent section, examples of other applicable sections and the context are listed.

For peptides, alignment has been further addressed by USP <1503> and <1504> [16], providing recommendations for the minimum quality attributes of raw materials:

- <1503> Quality Attributes of Synthetic Peptide Drug Substances
- <1504> Quality Attributes of Starting Materials for the Chemical Synthesis of Therapeutic Peptides

In addition, the FDA has recently published two oligonucleotide-specific guidance documents focused on nonclinical and clinical considerations, but at this time there is still no specific US FDA guidance regarding CMC or quality expectations for oligonucleotides. The two existing (as of the time of writing this guide) FDA guidance documents are:

- Draft Nonclinical Safety Assessment of Oligonucleotide-Based Therapeutics – Nov 2024 [17]
- Clinical Pharmacology Considerations for the Development of Oligonucleotide Therapeutics – Final Jun 2024 [18]

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In 2021, the FDA also published additional guidance documents on oligonucleotides, specifically for the development of “individualized” or “personalized” oligonucleotide therapies (e.g., Draft Guidance: IND Submissions for Individualized Antisense Oligonucleotide Drug Products for Severely Debilitating or Life-Threatening Diseases: Clinical Recommendations – Dec 2021 [19]).

As discussed in Chapter 1, these therapies have a unique regulatory context and are outside of the regulatory scope of this Guide.

2.2.3 Broader Regulatory Reference Landscape

While specific oligonucleotide and peptide guidance continues to develop, there are numerous additional regulatory resources that can be used to determine the appropriate requirements for the manufacture of these modalities. As discussed above, cGMP regulations and guidance from around the world contain considerations that should be assessed for applicability in the manufacturing of oligonucleotide drug substance and drug product, such as:

- FDA Guidance for Industry: cGMP for Phase 1 Investigational Drugs, issued July 2008 [20]
- 21 CFR 210 and 211: Current Good Manufacturing Practice in Manufacturing, Processing, Packing, or Holding of Drugs [21, 22]
- ICH Q3C/ICH Q3D and ICH M7 [23, 24, 25]: Regarding reagent and solvent impurities used in the manufacture of peptides and oligonucleotides
- ICH Q7: Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients [6]. Oligonucleotide and peptides are largely produced synthetically, and within the USA, filed with CDER [9]. These attributes should also be considered for applicability within the production of oligonucleotides and peptides through similar manufacturing methods where destined as API intermediates of other processes (e.g., sgRNA for CRISPR)
- ICH Q6B [26]: for evaluation of synthetic peptides and oligonucleotides with regard to biological activity, purity, and stability
- ICH Q8, Q9, Q10, Q11 [27, 28, 29, 30]: For pharmaceutical development, quality risk management, pharmaceutical quality systems, and development and manufacture of drug substances
- Eudralex–Volume 3: Guideline on the Sterilization of the Medicinal Product, Active Substance, Excipient and Primary Container [31]. As peptides and oligonucleotides are typically administered as parenteral products, manufacturing process guidelines should be considered
- EudraLex–Volume 4: Good Manufacturing Practice (GMP) [32] guidelines including multiple Annexes which should be assessed for applicability
- PIC/S Pharmaceutical Inspection Convention, Pharmaceutical Inspection Cooperation Scheme Guide to Good Manufacturing Practice for Medicinal Products [33]
- Pharmacopeia specifications and validated analytical methods: For characterization and quality control of therapeutic biological products

In addition to guidance from regulatory agencies, there are also industry organizations that are focused on helping to advance oligonucleotide therapeutics.

- The Oligonucleotide Safety Working Group (OSWG) [34] has a mission to facilitate the exchange of information between industry, academia, and regulatory authorities and establish consensus scientific recommendations. OSWG subcommittees have published multiple whitepapers (e.g., toxicology assessment guidance for oligonucleotides)

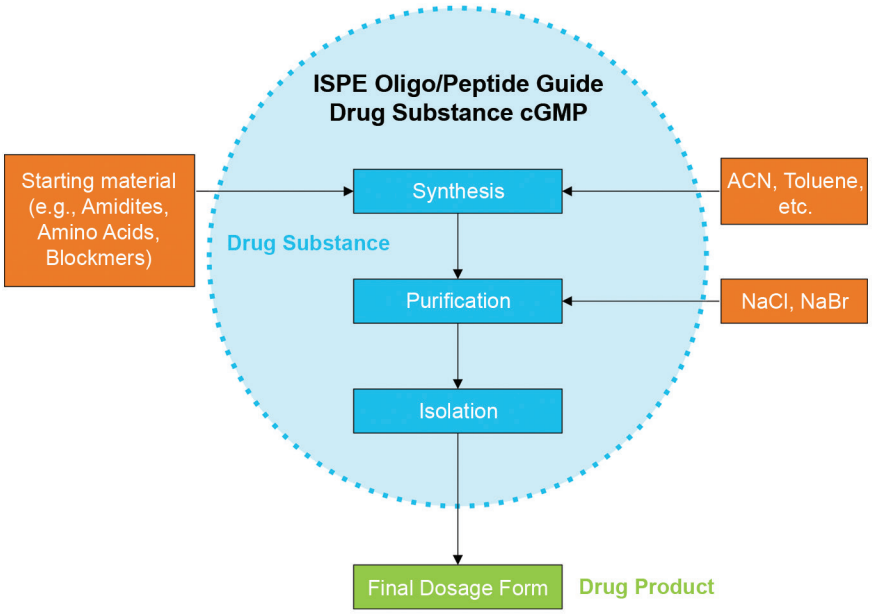
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- The Unites States Pharmacopeia (USP) [16] is also working on developing standards associated with oligonucleotides

- The European Pharma Oligonucleotide Consortium (EPOC) [35] is a group of pharmaceutical companies that works to build consensus around the development and commercialization of oligonucleotides. EPOC has published technical white papers that provide recommendations (e.g., the use of solution APIs in oligonucleotide drugs)

Figure 2.1 shows the intended Guide scope as aligned with ICH Q7 [6] regarding GMP facility boundaries.

Figure 2.1: Oligonucleotide and Peptide Facility Guide Scope Boundary



2.2.4 Global Regulatory Agencies

Regulatory expectations continue to evolve with time and technology even though the regulations themselves may not change for decades. As a result, regulatory agencies use more frequently issued guidance documents to represent current thinking.

While not an exhaustive list, the following are some of the global regulatory authorities and harmonization organizations. A listing of global agency links is provided by ISPE at www.ispe.org/regulatory-affairs.

- Regulatory Bodies
  - US Food and Drug Administration (FDA) [7]
  - European Medicines Agency (EMA) [8]
  - Medicines and Healthcare products Regulatory Agency (MHRA) [36]
  - China National Medical Products Administration (NMPA) [37]
  - Japan Pharmaceuticals and Medical Devices Agency (PMDA) [38]
  - India Central Drugs Standard Control Organization (CDSCO) [39]

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- Harmonization Organizations
  - International Conference on Harmonization (ICH) [40]
  - Pharmaceutical Inspection Convention and Pharmaceutical Inspection Cooperation Scheme (jointly referred to as PIC/S) [41]
- Advisory Organizations
  - European Commission (EC), Public Health [42]
  - World Health Organization (WHO) [43]
  - US Centers for Disease Control and Prevention (CDC) and the US National Institutes of Health (NIH) (in partnership to publish biosafety guidelines) [44, 45]
  - International Federation of Biosafety Associations (IFBA) [46]
  - Association of South-East Asian Nations (ASEAN) [47]

2.3 Safety and Environmental Regulations

As stated in the introduction to this chapter, the manufacture of oligonucleotides and peptides requires the use of large quantities of flammable and hazardous chemicals. This section provides an overview of applicable regulations. Additional detail is provided throughout this Guide to help the reader understand the regulatory and safety framework applicable to the various aspects of facility design, construction, and operation.

Applicability of safety and environmental requirements varies at the local, state, and national level. Architectural and Engineering teams should work closely with local permitting authorities (AHJ) to understand the regulatory code framework that applies to a given project and site.

2.3.1 Flammable Storage and Processing

Flammable storage, flammable processing requirements, and associated area electrical classification requirements are typically enforced through a combination of local and state fire codes, as well as local and state building codes. Local and state fire code often references applicable NFPA or EU standards. Local Environmental Health Authorities or CUPA (Certified Unified Program Authority) may also represent the governing regulatory body.

Building codes often reference International Building Code, which similarly draws from other standards including International Mechanical Code, NFPA NEC 70 [48], EN 13501-2 [49], ATEX [50], and IEC [51].

2.3.2 Occupational Health and Safety Regulations

Federal and state occupational health and safety regulations applicable to general industrial operation also apply to oligonucleotide and peptide manufacturing facilities. These are set forth by the US OSHA [52], EU-OSHA [53] or other similar health and safety agencies applicable to a specific jurisdiction.

These regulations may not directly mandate or directly dictate the design or construction of a facility, but their applicability and compliance strategies should be considered early in the process as they may significantly influence use, operational efficiency, and capacity of facilities and equipment. Examples include material handling and storage, transportation, confined spaces, fall protection, chemical exposure, and potent materials handling.

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In some cases, these federal laws (e.g., in the US, OSHA Process Safety Management (21 CFR 1910.119) [54], in Europe Seveso-III Directive (Directive 2012/18/EU) [55], COMAH regulations [56] supported by ISO 45001 [57], IEC 61511 [58]). The EPA Risk Management Program (40 CFR 68) [59]) mandate additional risk assessments be performed for the program and facility.

2.3.3 Environmental Regulations

The US EPA [60], EEA [61], and other similar nationally recognized environmental standards typically establish environmental requirements that may guide the design, operation, and maintenance of facilities and systems.

Additional environmental protection regulations apply depending on the location, size, design, and purpose of the site, including, but not limited to:

- Greenhouse gas and hazardous air pollutants
- Emissions regulations such as IED [62] in Europe; Title V [63] in the US, depending on emissions quantities, as one consideration. US state regulations can require additional reporting and stringency above and beyond the federal requirements. In addition, the use of certain chemicals can trigger jurisdictional requirements (e.g., use of DCM)
- Waste discharge and stormwater discharge regulations (IED in Europe; CWA in the US)

In the EU, the Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH) [64] is a binding EU regulation that collects and assesses hazard information of industrial materials and categorizes them according to Carcinogenic, Mutagenic, Reprotoxic (CMR) considerations. The Industrial Emission Licensing directive has begun using the CMR as a means of identifying potentially problematic solvents. Some key peptide production solvents have been classified as CMR materials, including NMP, DMF and DMAC. This has implications for emission control requirements, among other considerations for these facilities and can be expected to drive the evolution of alternative solvents and chemistry. For existing filings, changes to chemistry and process sequences require post approval change management and represent a barrier to implementation. See Chapter 12 for further discussion on these topics.

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3 Manufacturing Process

3.1 Introduction

This chapter serves as an introduction to the terminology, chemistry, and considerations behind synthetic oligonucleotide production. Process descriptions are provided to assist in the process understanding. Reaction diagrams are provided in Appendices 2 and 3.

The information contained within this chapter is intended to:

- Introduce readers to the common oligonucleotide and peptide synthesis reaction cycles, core terminology, reagents, and potential Critical Process Parameters (CPPs)
- Discuss specific oligonucleotide and peptide considerations and potential CPPs associated with common post-synthesis unit operations
- Inform the facility design team of identity testing considerations, typical operational bottlenecks, and relevant supply chain considerations

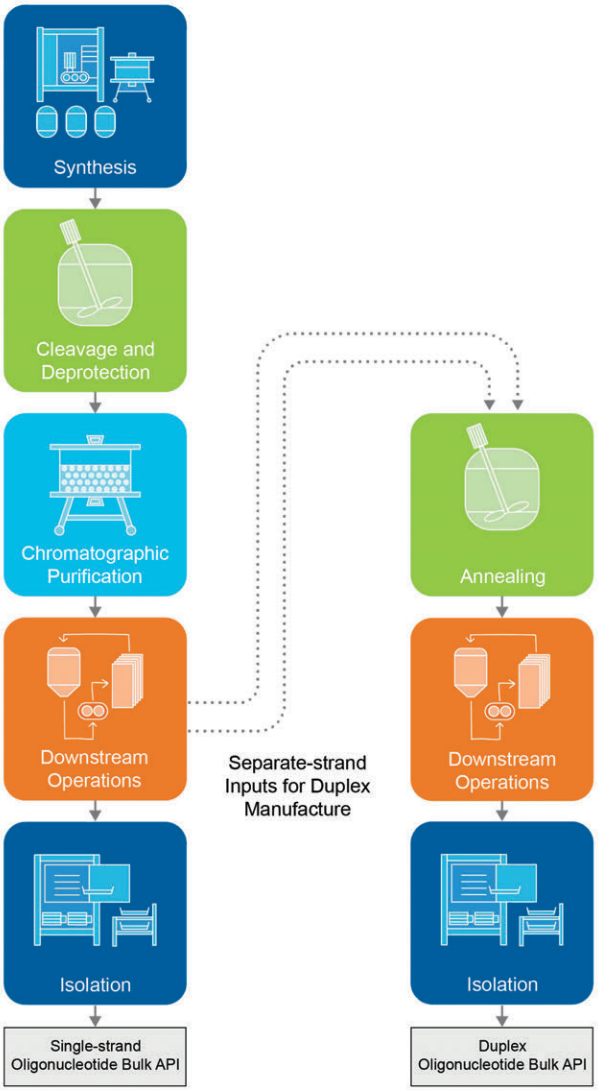
3.2 Common Oligonucleotide Unit Operations

This section provides an in-depth description of each oligonucleotide manufacturing process, outlining the purpose of each stage and the common operations used to execute it. Different synthetic oligonucleotides may require different combinations of process stages, and certain operations may be more appropriate for specific oligonucleotide types. This section highlights examples of these distinctions, but in general, it is assumed that the appropriate process will be selected and refined during pilot scale development work for each oligonucleotide.

Figure 3.1 identifies the common sequence of unit operations used to produce bulk oligonucleotide API. Each of these operations is described within this section.

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Figure 3.1: Block Flow Diagram of Common Oligonucleotide Manufacturing Unit Operations



For the manufacture of duplex molecules, a sense strand and an antisense strand are manufactured and purified separately, then annealed into a duplex.

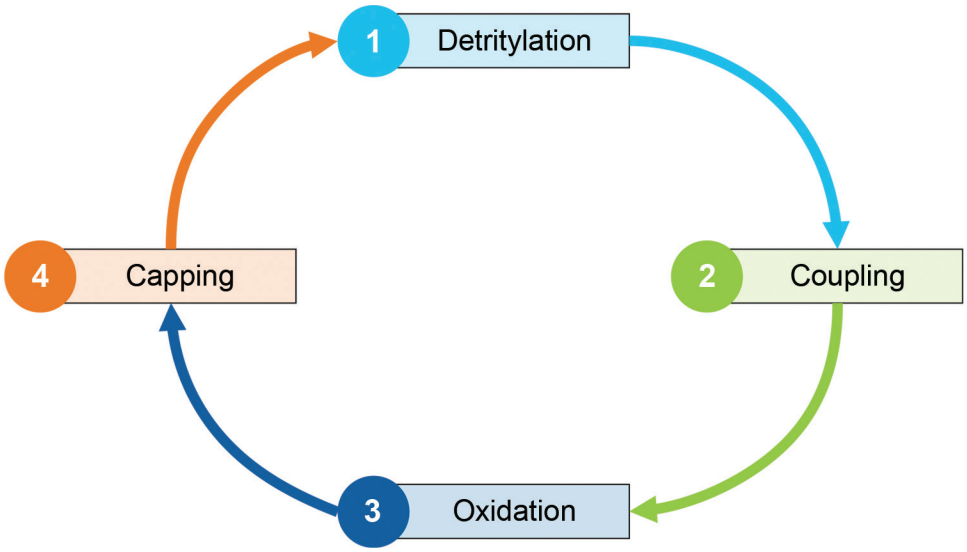
3.2.1 Solid-Phase Oligonucleotide Synthesis (SPOS)

Although other chemistries exist for the synthesis of oligonucleotides, this discussion is limited to the (currently prevalent approach) solid support bed in a static compression column phosphoramidite chemistry, commonly referred to as SPOS. Forward-looking approaches such as enzymatic and hybrid enzymatic-chemical synthesis, liquid phase synthesis, stirred bed, and P(V) chemistry are discussed in Chapter 12.

SPOS consists of stepwise elongation of a nucleotide immobilized on a solid support through the sequential attachment of activated phosphoramidites to form the desired sequence. Each phosphoramidite is incorporated through a four-step chemical reaction cycle as shown in Figure 3.2.

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Figure 3.2: Standard SPOS Reaction Cycle



This reaction cycle is repeated until an oligonucleotide of the desired sequence and length is fully synthesized. During synthesis, reactants are pumped through a synthesis column (typically from top to bottom) where the oligonucleotide is attached to a solid support. Each reaction step is followed by solvent rinses to wash reactants and reaction by-products from the column.

This phosphoramidite process enables highly efficient reactions with minimal by-product formation. The phosphoramidites themselves, described in Appendix 1, are broadly analogous to RNA or DNA nucleotides. They are commonly defined by their nucleobase (adenine, cytosine, guanine, thymine, or uracil) and by the functional group of the 2' carbon of the ribose sugar. For example, “2'-O-Methyl A” refers to an adenine nucleoside with a 2'-O-methyl group.

Synthesis Solid Support

The solid support serves as the foundation upon which the oligonucleotide is synthesized. Solid support is filled into a synthesis column in a manner similar to chromatography column packing and is commonly described in those terms.

The solid support is prepared for synthesis with protected reaction sites attached via a cleavable linkage. The most commonly used protected reaction sites are the first nucleoside and UnyLinker™, a commercially available universal solid support linkage that gets removed from the oligonucletide molecule during the base deprotection step. There are two main classes of solid supports commonly used:

- Controlled Pore Glass (CPG) supports
- Polymeric supports, predominantly polystyrene-based

Polymeric supports exhibit swelling and contraction when switching between solvents (for example, more hydrophobic solvents typically induce greater swelling). Polymeric supports will also gradually swell over the course of the synthesis due to the bulk of the elongating oligonucleotide chains. Consequently, special consideration must be taken to set piston heights to accommodate this expansion and contraction. CPG supports, by contrast, do not swell or contract but have lower reaction site density (i.e., loading) than polymeric supports. Spent solid support is discarded after the crude synthesized oligonucleotide has been cleaved from the support, and new support must be used for subsequent syntheses. Typical CPPs for the column packing step should include scale (i.e., the number of available reaction sites) and the piston height. Control of the scale is based on both the analysis of the loading of the solid support (typically expressed in units of µmols of reaction sites per gram of solid support) and the quantity of solid support added to the column during packing.

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Detritylation

The DMT protecting group is removed from the end of the elongating support-bound oligonucleotide chain or the initial solid support reaction site via a detritylation reaction prior to coupling the next phosphoramidite (see Appendix 2 for more information). This reaction is achieved via treatment with a strong acid solution, most commonly Dichloroacetic Acid (DCA) in toluene, which is commonly referred to as deblock solution. Potential CPPs for this step typically include acid concentration, molar equivalents, solution quantity, contact time, linear velocity, and/or temperature.

Because of the DMT group's hydrophobicity, there are times when the final DMT group is retained on the fully assembled molecule to facilitate chromatographic separations. In such cases, the detritylation of the final nucleoside in the oligonucleotide may be deferred until a later stage in the manufacturing process.

Deblock solution is commonly sourced from specialty chemical suppliers in stainless steel pressure-rated totes, but may also be prepared on-site. However, DCA is highly corrosive and may be incompatible with stainless steel under long term storage conditions. It can be blended in-line with toluene, prepared as a batched solution, and may even be mixed in-line directly on the synthesizer. Deblock solution has demonstrated a stable shelf life of greater than one year when stored at ambient temperatures.

Deblock treatment of the support-bound oligonucleotide is typically followed by multiple column volumes of rinse with solvent (commonly acetonitrile) to clear the synthesis column of excess acid and remove DMT cation.

Coupling

The coupling reaction proceeds with the addition of the desired phosphoramidite in the sequence, typically dissolved in acetonitrile or other co-solvents such as an amidite solution along with an activator solution (see Appendix 2 for more information). Potential CPPs for the coupling step typically include solution water content, amidite/activator solution ratio, molar equivalents, total solution quantity, contact time, linear velocity, and/or temperature.

Amidites are typically sourced as a dry solid packed in an inert atmosphere and exhibit long-term stability when stored at 2–8°C or lower temperatures (shelf stability may be more limited if the parent container is opened/closed multiple times). Amidites in solution; however, have variable expiry of several days up to weeks in some cases. Because the coupling reaction is sensitive to moisture, amidites are commonly dissolved in solvent then stored and/or mixed with molecular sieves for several hours to reduce residual water content. For these reasons, amidite solutions are often prepared on-site immediately prior to synthesis.

Activator solutions are generally sourced from specialty chemical manufacturers in inert bottles or stainless steel pressure-rated totes, and have demonstrated stability of more than one year when stored at ambient temperatures. However, some activator formulations may precipitate azole solids if stored or transported below ambient temperature; therefore, mild heating capability for activator storage may be necessary. Activator solution can also be prepared on-site to maintain control over formulation and temperature.

Coupling is typically followed by an acetonitrile rinse to remove excess reactant and coupling activation by-products.

Backbone Oxidation

The newly coupled phosphite group undergoes oxidation or sulfurization (also commonly referred to as thiolation) to produce phosphate or phosphorothioate oligonucleotides, respectively, as directed by the design of the therapeutic molecule. Potential CPPs for this step typically include molar equivalents, solution quantity, contact time, solution age, rinse volume (for oxidation steps), linear velocity, and/or temperature.

Oxidation is most commonly performed using an iodine reagent which contains water and a weak base, such as pyridine. Sulfurization is performed using a sulfurizing agent, most often xanthane hydride or phenylacetyldisulfide (PADS) in the presence of a weak base, such as pyridine, lutidine, or picoline. See Appendix 2 for more information on the reactions.

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Oxidation solutions are typically sourced from specialty chemical suppliers in stainless steel pressure-rated totes but may be mixed on-site. Oxidation solution has demonstrated a stable shelf life over several months when stored at ambient temperatures. Many oxidation solution mixtures require aging when used for manufacturing oligonucleotides containing a mixture of phosphate and phosphorothioate backbones.

The PADS formulation of sulfurization solutions have a narrow window of effective use, typically 12 to 72 hours after preparation, after which its efficacy declines sharply. Consequently, PADS solutions must be prepared on-site immediately before use. Xanthane hydride-based sulfurization solutions display far improved stability and may even be sourced from specialty chemical suppliers in stainless steel pressure-rated totes and stored at ambient temperatures for several months without loss in reactivity.

Both oxidation and sulfurization reactions steps are followed by an acetonitrile rinse to remove excess reagents and reaction by-products from the synthesis column. For oxidation reactions, the rinse step must also remove residual water.

Capping

Capping solutions are employed to protect any unreacted sites after the coupling reaction. Commonly, a mixture of two solutions, one of which contains an azole catalyst and the other which contains acetic anhydride, are pumped through the synthesis column to block the unreacted sites (see Appendix 2 for more reaction information). Potential CPPs for this step typically include capping solution ratio, molar equivalents, total solution quantity, contact time, linear velocity, and/or temperature.

Capping solutions are commonly sourced from specialty chemical suppliers in stainless steel pressure-rated totes but may be mixed on-site. Capping solutions have generally demonstrated a stable shelf life of greater than one year when stored at ambient temperatures.

Capping reactions are followed with an acetonitrile rinse to remove excess reagents and reaction by-products from the synthesis column.

3.2.2 Deprotection

Following synthesis, the crude oligonucleotide is bound to the solid support and has numerous remaining base-labile protecting groups that were necessary during the chemical synthesis process. At this point, the oligonucleotide must be deprotected and cleaved from the solid support.

Backbone Deprotection

A treatment with a base, usually Diethylamine (DEA) or Triethylamine (TEA) in acetonitrile, drives off the cyanoethyl protecting groups on the phosphate and phosphorothioate linkages, resulting in P-deprotection (see Appendix 3 for more reaction information). Potential CPPs for this reaction typically include molar equivalents, solution quantity, contact time, linear velocity, and/or temperature.

P-deprotection reagent may be prepared by blending neat DEA or TEA with acetonitrile on the synthesizer skid, prepared in batch solutions, or may be supplied pre-mixed from a specialty chemical manufacturer. These solutions have demonstrated a stable shelf life of greater than one year when stored at ambient temperatures.

Following the P-deprotection treatment, the synthesis column is typically rinsed with solvent and blown dry with an inert gas.

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Base Deprotection/Cleavage from the Solid Support

Base deprotection and cleavage from the solid support are achieved via treatment with a deprotection reagent, commonly ammonium hydroxide or a mixture of ammonium hydroxide and methylamine in water, at elevated temperature (see Appendix 3 for more reaction information). Potential CPPs for this reaction typically include molar equivalents, solution quantity, concentration, pH, reaction time, temperature, and/or agitation.

The solid support may be physically removed from the synthesis column and placed in a filter tank for the cleavage and deprotection operations, or cleavage may be performed on-column by recirculating cleavage reagent through the column.

Following base deprotection and cleavage operations, the resulting solution is highly basic and many oligonucleotides are unstable in such conditions. Consequently, most processes incorporate a treatment step to reduce the basicity and overall hazardous nature of the solution, increase the allowable storage time, and/or prepare the solution for subsequent chromatographic purification. Common practices include dilution with water, ultra-filtration, and buffer exchange to remove reagents and reaction by-products and replace the ammonium ions with sodium ions (typically with a sodium chloride buffer), vacuum distillation to remove ammonium hydroxide and residual synthesis solvents, and freezing.

In the case of RNA oligonucleotides, an additional desilylation step is required to remove the specialized TBDMS protecting groups from the 2'-hydroxyl positions. This step can be carried out with triethylamine trihydrofluoric acid (TEA-3HF) after the oligonucleotide has been cleaved from the solid support. Treatment with TEA-3HF is relatively fast and exhibits minimal sensitivity to moisture.

3.2.3 Chromatographic Purification

After the completion of the synthesis, deprotection operations, and any subsequent operations used to dilute or remove ammonia, the resulting “crude” solution contains the desired product mixed with three core types, as depicted in Figure 3.3, of product-related impurities:

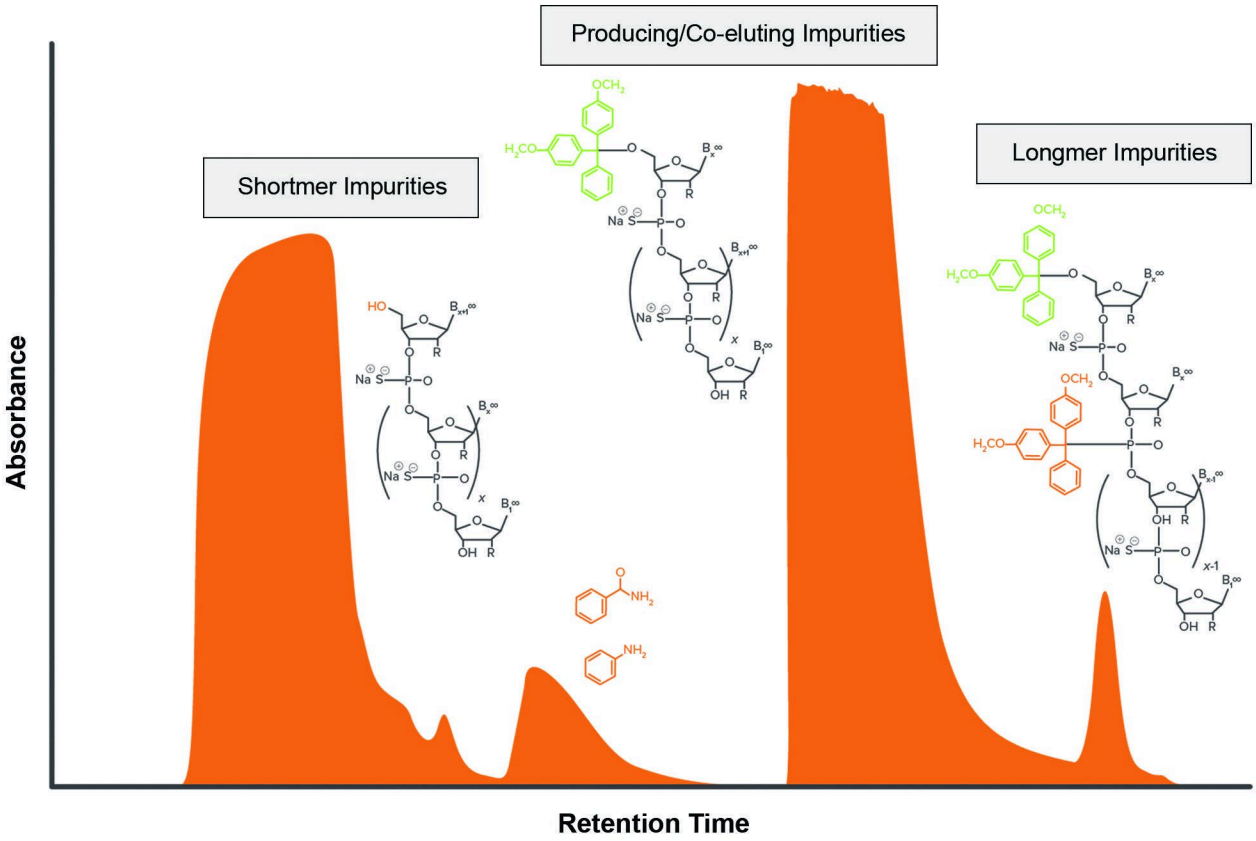
- Shortmer Impurities: Impurities that did not reach the desired full length and were capped during the synthesis process. These impurities are shorter than the desired product.

- Longmer Impurities: Impurities that arise when two strands combined to form a single chain or when a single strand “branches” to continue as two chains. In either case, the resulting molecule contains more bases than the intended product.

- Co-eluting Impurities: Impurities that possess approximately the desired number of bases (typically within ±1 nucleobase), but are structurally modified due to side reactions or the incorporation of an impure starting material. Therefore, the resulting molecule, although similar in length, is not the intended product.

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Figure 3.3: Impurities Overview



At this stage, column chromatography is used to separate these highly similar molecules from the desired product. This operation typically also significantly reduces small molecule impurities and remaining reagents and reaction by-products from the deprotection operations, although the chromatography operations are typically not optimized for this purpose.

There are two potential separation mechanisms at this stage and both mechanisms are commonly used in oligonucleotide facilities depending on process requirements and product characteristics.

Hydrophobicity-based Chromatography

Hydrophobicity-based chromatography strategies rely on Reversed-Phase High Performance Liquid Chromatography (RP-HPLC), Hydrophobic Interaction Chromatography (HIC), or Ion-Paired Reversed-Phase High Performance Liquid Chromatography (IP-RP-HPLC) for achieving separation. This section focuses on RP-HPLC; brief overviews of HIC and IP-RP-HPLC are provided in Chapter 12.

This approach is only applicable when the final nucleobase retains the large, hydrophobic DMT protecting group (or, in rare cases, if a 3' conjugate can serve as an alternative hydrophobic chromatographic handle). In this case, the core separation strategy relies on the different number of DMT protecting groups that will typically be found on the various oligonucleotide species in the crude solution. The desired product will only contain a single DMT protecting group, while the shortmer and longer impurities will commonly have zero or multiple DMT protecting groups, respectively. This purification strategy can often completely remove oligonucleotide species lacking the desired single DMT protecting group, but generally affords limited resolution of the co-eluting impurity species that also contain a single DMT protecting group.

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RP-HPLC is performed using a mixture of organic solvent (typically acetonitrile or methanol) and water and an acetate salt (e.g., sodium acetate or ammonium acetate). Typical RP-HPLC resins are non-functionalized polymeric resins or C-18 silica resins. RP-HPLC purifications are typically run at ambient temperature or slightly elevated temperatures (22–35°C) and moderate pressures (20–50 bar). The product elutes as a relatively concentrated fraction under high organic, low salt conditions and requires further downstream processing to remove the final DMT protecting group (see Section 3.2.4) and exchange the buffer/solvent make-up to water.

Potential CPPs for the purification typically include column packing quality, column bed height, column loading, linear velocity, shortmer elution conditions, baseline separation between shortmer peak and product peak, product elution conditions, product collection parameters, and/or temperature.

Charge-based Chromatography

Charge-based chromatography strategies rely on Anion Exchange Chromatography (AEX), specifically Strong Anion Exchange Chromatography (SAX), for achieving separation. This strategy is effective regardless of whether the final hydrophobic DMT protecting group remains on the molecule. In this case, the core separation is driven primarily by oligonucleotide length, which corresponds to the number of negatively charged backbone sites.

Shortmer impurities are of a shorter length and have fewer charged sites than the target product, while longmer impurities have more. Additionally, many co-eluting impurities indirectly affect the number or behavior of the charged sites and can also elute separately from the desired product. While this purification strategy has the potential to separate all manner of impurities from the desired product, nearly all species elute within a relatively narrow chromatographic window and most purification is achieved by using shallow gradients and pooling numerous fractions from a broad, dilute product peak.

AEX is generally performed using water, sodium chloride or sodium bromide, and oftentimes low levels of acetonitrile (5–15%). Commonly, low levels of sodium hydroxide (25–50 mM) are used as well. Typical AEX resins are quaternary ammonium-functionalized polymeric resins. AEX purifications may be run at a wide variety of temperatures (22–60°C) and a wide range of operating pressures are commonly used, primarily based on the pressure/flow limitations of the selected resin and equipment. The product elutes as a dilute fraction under high salt conditions and requires further downstream processing to concentrate the product and remove the salts. Depending on the oligonucleotide and conjugate, if present, stability at high pH, the collected fractions from both the AEX process may need to be neutralized shortly after collection.

Potential CPPs for the purification typically include column packing quality, column bed height, column loading, elution conditions, linear velocity, pH, and/or temperature.

When performing the AEX process with DMT-on material, a solid-phase detritylation reaction can be performed. This process is described in Section 3.2.4.

3.2.4 Final Detritylation

In some oligonucleotide processes the final DMT protecting group is not removed during the synthesis process (see Section 3.2.1) and is left on to aid in the chromatographic operations (see Section 3.2.3). It is possible to remove the protecting group either as part of the AEX chromatography process or in a solution-phase reaction. Potential CPPs for this reaction typically include acid concentration, oligonucleotide concentration, pH, time, agitation, and/or temperature.

Solid-Phase Detritylation

Because most AEX resins are manufactured from the same polystyrene base beads used in the reverse-phase resins, they exhibit low levels of hydrophobic interaction effects that can be utilized to elute the DMT-off shortmers from the AEX column before eluting the desired product. Once the DMT-off shortmers are eluted, the packed column may be rinsed with water and then stored under acidic conditions (typically 80% glacial acetic acid) and ambient temperature until the detritylation is completed.

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Once completed, the acid and liberated DMT group may be rinsed from the column and then the purification operation proceeds as normal.

Solution-Phase Detritylation

The solution-phase reaction operation is very straightforward: reducing the solution pH allows for the detritylation reaction to proceed and increasing the solution pH stops both the reaction and the depurination reaction, if deoxy (2'-H) purine nucleobases are present.

After the reaction is completed, the detritylated oligonucleotide compounds remains in solution with the freed DMT species, which forms a poorly water-soluble DMT-OH in the solution-phase conditions. When performing the reaction at very low oligonucleotide concentrations, the DMT-OH will typically stay in solution, but at reaction conditions above approximately 0.5 wt% oligonucleotide, DMT-OH aggregates will begin to form. This aggregated material may be separated from the oligonucleotide via precipitation in ethanol (see Section 3.2.5) or solubilized by adding an organic co-solvent.

3.2.5 Concentration Operations

A variety of concentration operations are often performed as part of oligonucleotide manufacturing. In some cases, the concentration operations also serve to purify the material as well. In general, only parameters associated with oligonucleotide purification are considered CPPs in these operations. All of these are common processes and only the oligonucleotide-specific considerations will be discussed.

Precipitation

Oligonucleotides are, broadly speaking, insoluble in alcohols that have longer alkane chains than methanol. When solutions of oligonucleotides with sodium acetate at approximately 0.2M or above are combined with an excess of alcohol (typically 4–8 volumetric equivalents), the oligonucleotide generally precipitates and forms aggregates. Depending on the base chemical modifications (DNA-rich sequences are most amenable), the aggregates will grow into floccules and settling can be achieved. Potential CPPs typically include incoming oligonucleotide concentration, alcohol quantity, salt content, pH, agitation, and/or temperature.

After settling is completed, the supernatant consists primarily of alcohol-soluble small molecules, water, and sodium acetate which can be manually decanted from the precipitated solid. Given effective decantation of the supernatant, a single precipitation typically results in a ~100-fold reduction in soluble small molecules and non-ethanol solvents. Product concentrations of ~20 wt% are readily achievable via the precipitation operation, while the maximum practical oligonucleotide concentration that can be reached is ~30 wt%.

Tangential Flow Filtration

Tangential Flow Filtration (TFF), also known as Cross Flow Filtration or colloquially as Ultra-filtration, is commonly used in oligonucleotide processes to remove small molecule impurities, exchange backbone ions, exchange solvents, and concentrate dilute oligonucleotide solutions.

Membranes that can effectively retain oligonucleotides typically have Molecular Weight Cut-Off (MWCO) values of 1–10 kDa. The most common membrane materials of construction within this range are regenerated cellulose and polyethersulfone (PES) both of which are suitable for oligonucleotide processing. The MWCO selection will be determined by the molecular weights of the oligonucleotide and other dissolved species, and the solvent matrix in use. In some conjugated oligonucleotide processes, larger MWCOs may be used to allow unconjugated oligonucleotide to exit through the permeate.

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Potential CPPs for the TFF process typically include the number of diavolumes exchanged, pH, oligonucleotide concentration, and/or permeate conductivity. As concentration of the oligonucleotide solution in the retentate increases, permeate flux decreases, to the point where flux can, for all intents and purposes, cease to occur (less than 0.01 L/m^2/min). For most oligonucleotides, this flux limit occurs at concentrations of 4–8 wt% without a salt present. With salt present, this limit is approximately doubled.

Additional factors that can impact permeate flux are the number and type of functional group modifications on the oligonucleotide chain, the solvent matrix, and the lack of counter-ions in the retentate solution, all of which can negatively affect flux. Finally, based on the characteristics of the specific oligonucleotide being processed, the solvent matrix, and the filter pore size, there is typically a correlation, oftentimes exponential, between oligonucleotide concentration in the retentate and breakthrough of the product into the permeate stream that may be used to inform maximum operating concentrations.

Thin Film Evaporation and Vacuum Distillation

Thin Film Evaporation (TFE) and vacuum distillation are very similar processes in which a vacuum and heat are applied to a chamber/vessel/apparatus (referred to as “vessel” from here on out) containing a portion of the product solution in order to drive off the solvent (generally water) without impacting or degrading the product. For both processes, the vacuum pressure is set to allow the solvent or reagent to boil off of the product solution at a temperature which has been demonstrated to not impact the oligonucleotide product. During this process, the formation of bubbles or foam may need to be managed as the oligonucleotide product can act as a surfactant. Potential CPPs for these processes typically include temperature, pressure, oligonucleotide concentration, and/or residence time in the vessel.

The appropriate set points may be sequence dependent, both due to degradation rates and viscosity properties. The maximum achievable concentration for these operations are typically approximately 20–30 wt%.

3.2.6 Annealing/Hybridization

After completion of all synthesis and purification steps for the individual single strands, the annealing, or hybridization, process is performed to make duplex oligonucleotides from complementary single strands. The annealing step is a heat mediated process utilized to form an oligonucleotide duplex from the constituent complementary single strands. The process begins by mixing the individual strand solutions in a defined ratio, determined either from pilot studies or a fixed molar ratio. The mixture is then heated up to a target temperature (process ranges vary from room temperature to above 80°C, depending on the oligonucleotide sequences), held at the temperature for a specified duration, and is finally cooled down at a target ramp rate to ensure proper duplex formation.

Potential CPPs for this process typically include the defined ratio of the oligonucleotide sequences, target temperature, hold time, and/or cooling ramp rate. Target temperature and hold time are of particular concern as oligonucleotides tend to degrade at elevated temperature; minimizing the target temperature and time held at temperature will help to minimize the degradation of the product.

3.2.7 Final Isolation

The final isolation step transitions the oligonucleotide to its final drug substance state, typically either a liquid or powder state. Currently, lyophilization is the most common method for powder isolation used in industry and is therefore the only method discussed in this chapter. Spray drying is also performed and is discussed in Chapter 12.

Lyophilization

Lyophilization, which is also known as freeze-drying, is a low-pressure process which is utilized to remove solvent (primarily water when producing an oligonucleotide) from the product.

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There are three phases in a lyophilization operation:

- Freezing
- Primary Drying/Sublimation
- Secondary Drying/Desorption

The freezing phase is utilized to transform the product solution from a liquid state to a solid state. Most oligonucleotide solutions in water freeze at temperatures of approximately –30 °C. As the majority of oligonucleotides do not have secondary structures which must be preserved, freezing phase parameter optimization is purely focused on optimal drying rates and completely freezing the solution. Potential CPPs, if any, for this phase are the temperature, cooling rate, and time. The quantity of material in the freeze-drying vessel (either a tray or a vial), in particular the fill depth, is also a potential CPP for the freeze-drying operation.

The primary drying phase is utilized to remove the majority of the solvent via sublimation. Much like the freezing phase, optimization of the primary dry step parameters, typically chamber pressure and shelf temperature, can be focused primarily on optimal drying rates and completeness of drying without inducing meltback. Potential CPPs for this phase are the temperature, pressure, and phase time.

The secondary drying step reduces the residual moisture content of the product as far as possible by removing the solvent molecules physically adsorbed to the product matrix. The appropriate secondary drying settings are based on the conditions necessary to induce desorption for the specific product, which can vary based on the chemical modifications and the conjugate present (if any). In many oligonucleotide processes, this additional moisture removal is not considered critical and no CPPs are identified for this step (although it is still typically performed). Potential CPPs for this phase still include temperature, pressure, and phase time.

Product bearing containers can then be removed from the lyophilizer, inspected for predefined lyophilized cake attributes, and packaged. The powder properties are highly dependent on the incoming solution, particularly the oligonucleotide concentration and solvent content. Incoming solutions below 10 wt% can be very difficult to handle (due to their high static, high dispersibility properties), but there are few handling benefits from incoming solutions above 20 wt%, which can require mechanical milling to break up (discussed further in Chapter 12).

Liquid Drug Substance

There are some limited examples of the use of a liquid bulk drug substance in commercial manufacturing, though this approach is not covered here. EPOC, the European Pharma Oligonucleotide Consortium has published a position paper on the topic in 2020 [124].

3.2.8 Summary of Potential CPPs

Many potential CPPs were listed in the above discussion. The final CPPs for a process will vary depending on a number of factors, determined via a risk assessment and justified based on experiments and platform knowledge.

While most CPPs for the synthesis unit operation are common regardless of the target end molecule or sequence, strategies for achieving these results may differ between equipment and end users. The main goal of the synthesis process is to achieve accurate and efficient coupling of the nucleotide bases, which requires reaction completion for each consecutive reaction associated with the assembly of the oligonucleotide chain. Like all chemical reactions, potential CPPs include effective mixing, equivalents, reaction time, and temperature. While not all of these parameters will be critical for every synthesis reaction, process parameters representing these terms must be assessed with appropriate controls in place.

A summary of the potential CPPs listed in this chapter is provided in Table 3.1 for reference. For the Synthesis potential CPPs, the reaction terms that these parameters represent are listed as well.

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Table 3.1: Potential CPPs per Reaction/Operation

Process Step	Reaction/Operation	Potential CPPs
Synthesis	Column Packing/Filling	Scale (based on reaction sites) [Equivalents] Piston height [Mixing]
	Detritylation	Acid Concentration [Equivalents] Quantity [Equivalents] Contact Time [Time] Linear Velocity [Mixing/Time/Equivalents] Temperature
	Coupling	Water Content [Equivalents] Amidite/Activator Ratio [Equivalents] Quantity [Equivalents] Contact Time [Time] Linear Velocity [Mixing/Time/Equivalents] Temperature
	Oxidation/Thiolation	Quantity [Equivalents] Solution Age [Equivalents/Side Reactions] Contact Time [Time] Linear Velocity [Mixing/Time/Equivalents] Temperature Rinse Volume (Oxidation only) [Side Reactions]
	Capping	Capping Solution Ratio [Equivalents] Quantity [Equivalents] Contact Time [Time] Linear Velocity [Mixing/Time/Equivalents] Temperature
	Backbone Deprotection	Quantity [Equivalents] Contact Time [Time] Linear Velocity [Mixing/Time/Equivalents] Temperature
	Base Deprotection and Cleavage	Quantity Concentration pH Reaction Time Temperature Agitation
Chromatography	RP-HPLC	Column Packing (HETP) Bed Height Loading Linear Velocity Temperature Shortmer Elution Conditions Baseline Separation Product Elution Conditions Product Collection Conditions
	AEX	Column Packing (HETP) Bed Height Loading Linear Velocity Temperature Elution Conditions

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Table 3.1: Potential CPPs per Reaction/Operation (continued)

Process Step	Reaction/Operation	Potential CPPs
Final Detritylation	Detritylation	Acid Concentration Oligonucleotide Concentration pH Reaction Time Temperature Agitation
Concentration	Precipitation	Oligonucleotide Concentration Alcohol Quantity Salt Content pH Temperature Agitation
	TFF	Number of Diavolumes (Permeate Volume) pH Oligonucleotide Concentration Permeate Conductivity
	TFE/Distillation	Temperature Pressure Oligonucleotide Concentration Residence Time
Anneal-ing/Hybridization	Anneal-ing/Hybridization	Strand Ratio Maximum Temperature Hold Time Cooling Ramp Rate
Freeze Drying	Vessel Filling	Fill Depth
	Freeze	Temperature Cooling Rate Time
	Primary Drying	Temperature Pressure Time
	Secondary Drying	Temperature Pressure Time

3.3 Additional Oligonucleotide Considerations

This section discusses a variety of oligonucleotide-specific facility considerations that inform the design process.

3.3.1 Identity Test Considerations

Oligonucleotides are commonly produced in multiproduct facilities, making identity testing an important part of the overall control strategy. Analytical verification that the appropriate product is present in the bulk API, and is not mixed with, nor mislabeled as, another product, is essential. For an in-depth discussion on identity testing, see *Strategies for Identity Testing of Therapeutic Oligonucleotide Drug Substance and Drug Products* [65].

Following sequence confirmation as part of the synthesis testing, most of the oligonucleotide identity tests rely on the intact oligonucleotide mass. However, for a multiproduct facility with a limited number of phosphoramidites and recurring sequence chemistry patterns, it is conceivable that intact mass alone may not be a sufficiently robust identity test.

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In such cases, the likelihood that two different sequences made from phosphoramidites with the same chemical modifications share the same intact mass may be around 1 in 20.

This concern does not necessarily apply to the single strands from oligonucleotide duplexes, which can utilize the duplex formation as a secondary identity test.

3.3.2 Capacity and Scheduling

Capacity represents the maximum sustainable output of an oligonucleotide facility in the production of the final product. There are many factors which will drive the maximum output including the allowable scale of the equipment and the duration of the unit operations and activities supporting the unit operations. Utilizing the number of cycles, along with the duration of the unit operation (including all support operations such as cleaning and regeneration) will determine where the operational bottlenecks are in the process.

Two common bottlenecks for oligonucleotide manufacturing facilities include:

- Complex analytical testing requirements, particularly the synthesis and SAX chromatography testing
- Lyophilization operation, which will generally define the maximum batch size

Analytical testing timelines can be mitigated by budgeting additional storage space, usually refrigerated, and appropriate storage vessels while testing occurs. Sizing large lyophilization equipment, that can process multiple combined syntheses, can increase the maximum batch size and reduce the final analytical testing demands. For multiproduct facilities, efficient equipment cleaning and release practices can also significantly debottleneck operations as well.

Effective scheduling for oligonucleotide processes should consider allowable points in the operation where the in-process material can be held, what condition it is required to be held in (considering both stability and microbial growth potential), and the maximum acceptable duration. Oftentimes, these parameters can be determined and applied on a platform basis for oligonucleotides, based on the sequence chemistry and conjugate (if present).

3.3.3 Supply Chain

Manufacturers should maintain sufficient inventory levels to ensure raw material disruptions do not impact the supply of therapeutic products to patients. Due to the platform-based nature of solid-phase synthesis, a limited number of distinct raw materials are required as inputs.

The growing interest in oligonucleotides as a therapeutic modality, has led to the development of a healthy global market for the key raw materials. As a result, the primary synthesis solutions, solvents, and common amidites are nearly all readily available globally from multiple specialty chemical suppliers.

Synthesis solid supports and specialty amidites may continue to prove difficult to procure, as well as key starting materials like ligands for conjugation. Shifts from fossil fuels as primary energy sources in the global market may result in the reduction of global production capability for key solvent raw materials like low water content acetonitrile. It is conceivable that acetonitrile supply may not be able to meet the rising demand from the oligonucleotide market. Evaluating solvent recycling strategies, as well as alternative solvents, is advisable both to mitigate risk and to advance sustainability initiatives to reduce the amount of incinerated waste produced by these facilities (see Chapter 12 for further discussion).

Cold chain availability is another key consideration for manufacturers. Solid bulk oligonucleotide API is routinely stored at –20°C to ensure stability of the molecule. Due to moderate bulk density of lyophilized oligonucleotide and modest quantities in demand in the market, a healthy global infrastructure exists for the transport of clinical and commercial quantities of bulk API to drug product manufacturers. However, a significant shift towards the production of liquid oligonucleotide API would require reassessment of cold-chain capabilities and capacity.

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3.4 Typical Peptide Unit Operations

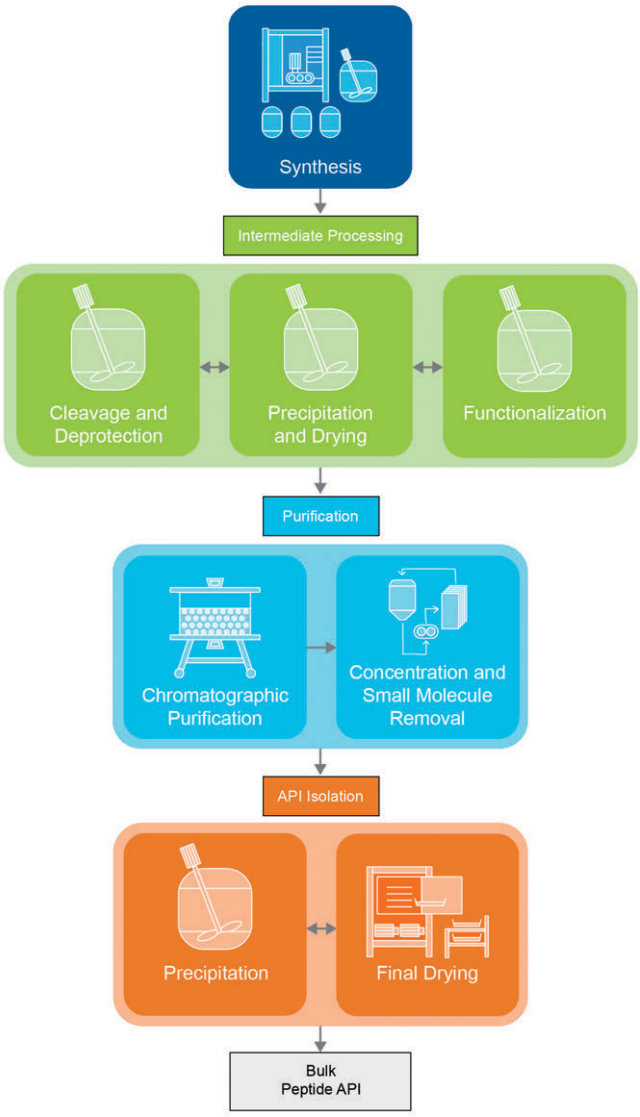
This section will transition to a focus on the peptide manufacturing process, providing further detail on the specific unit operations and introduce critical chemistry and design concepts.

There are two general routes for peptide manufacturing:

- **Biological:** which relies on microbial fermentation or recombinant DNA expression
- **Synthetic chemistry:** which utilizes sequential synthesis reactions, in either liquid or solid phases, to conjoin single amino acids or multimer fragments via ligation/condensation reactions

This Guide focuses on the commercially prevalent synthetic routes and associated downstream operations. It is important to note that the complexity of the chemistry and the number of processing steps can vary greatly from peptide to peptide, including synthesis route selection. Figure 3.4 identifies the general sequence of unit operations for production of bulk peptide API, each of which will be described in this section.

Figure 3.4: Sequence of Unit Operations for Peptide Manufacturing



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- **Synthesis:** Unit operations in which the peptide is built out from base amino acids, or peptide fragments through ligation/condensation reactions
- **Intermediate Processing:** Unit operations that modify, functionalize, or (most commonly) deprotect the growing peptide chain
- **Purification:** Unit operations that remove impurities and reaction byproducts, concentrate the peptide, and/or perform salt exchange
- **Isolation:** Unit operations that isolate the peptide intermediates or final products in either solid or solution form

3.4.1 Synthesis

All peptide manufacturing begins with the synthesis of the desired peptides. Selection of the synthesis methodology strongly influences the necessary downstream operations. There are three categories of synthetic route peptide synthesis:

- Solid Phase Peptide Synthesis (SPPS)
- Liquid Phase Peptide Synthesis (LPPS)
- Fragment Condensation (Hybrid)

Though all are commercially used, SPPS has been the dominant approach due to its flexibility and scalability. However, hybrid approaches are quickly emerging as more efficient for large scale synthesis where chemistry can support fragment coupling as opposed to purely sequential synthesis. Additional detail on LPPS and hybrid approaches is provided in Chapter 12.

Nevertheless, the complexity of peptide synthesis can influence process choices that will dictate the equipment train. There are several factors to consider, such as length of the peptide chain, branching, cyclization, location of unnatural amino acids in the sequences and coupling difficulty of particular amino acids, as well as stability of these starting materials and intermediates.

3.4.1.1 Peptide Synthesis Approaches

SPPS is based on sequential condensation of a growing amino acid chain utilizing a solid support, typically a resin. This sequential addition of amino acid units is generally referred to as “elongation”. The use of a solid support adds a few advantages. The main benefit is that the sequential elongation steps can be repeated in batch operations in a stirred tank fritted/filter reactor. Since the support bed can be retained, it can be washed to remove reactants and reaction byproducts, allowing for sequential and different reaction steps without requiring complex isolation between additions. This allows for an efficient platform to synthesize complex peptides. This approach differs from the column-based solid support approach used in oligonucleotide synthesis due to the comparatively slower reaction kinetics of the peptide synthesis chemistry.

LPPS can also be applied to stepwise peptide elongation. The preference for commercial use of LPPS is based on several advantages, including the use of a simplified equipment train leveraging traditional small-molecule stirred tank reactors, lower raw material costs compared to solid-support based synthesis, better stoichiometric ratios that generally require lower reagent quantities, and the ability to more easily produce certain difficult sequences.

Despite these benefits, commercial LPPS is often limited to these specialized chemistry cases or short chain peptides primarily because elongation requires isolation and purification after each stepwise addition, whereas SPPS can be isolated once after the entire chain is produced on the solid support. As a result, SPPS is better suited for automation and large-scale production sequences.

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Hybrid strategies combine the strengths of both approaches, coupling peptide fragments produced via either SPPS or LPPS within LPPS-based stirred tank systems. The reduction in sequential coupling reactions improves yield, and purification can be simplified due to size differences between fragments and coupled sequences. Recent commercial successes have also included purification via TFF in lieu of isolation, allowing the process to be telescoped into subsequent fragment couplings within a single reactor.

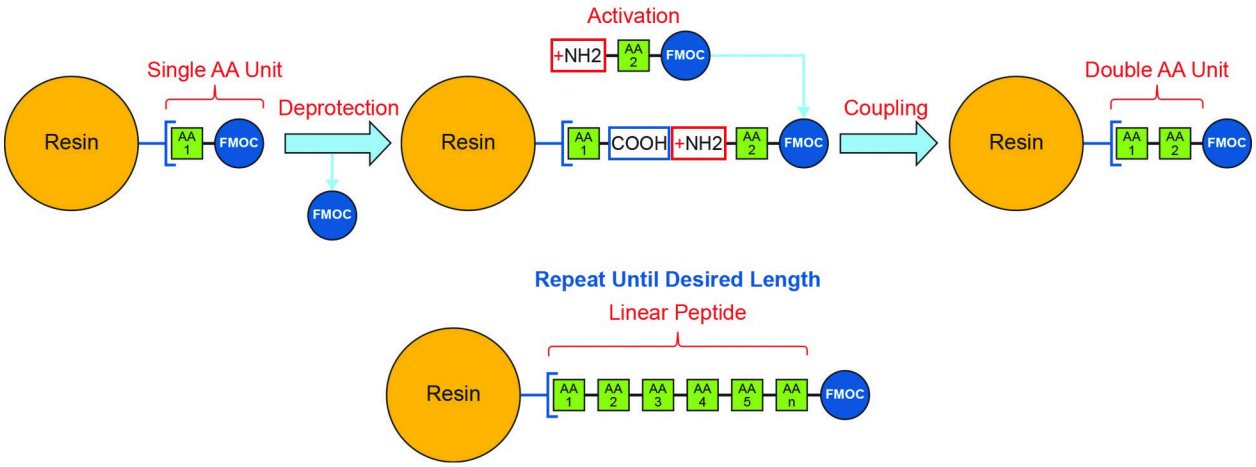
Chapter 12 discusses emerging innovations aimed at further advancing these approaches. The remainder of this chapter will focus primarily on commercially prevalent SPPS technology and subsequent unit operations, and will capture specific aspects of LPPS and hybrid approaches that meaningfully impact equipment and facilities.

3.4.1.2 Peptide Synthesis Chemistry (SPPS) and Operational Sequence

Currently the predominant chemistry for synthesis of peptides via SPPS uses Fmoc, (9-Fluorenylmethyloxycarbonyl) as the amine protecting group. Fmoc is generally considered to be more versatile than other protecting groups, providing better containment of side reactions and gentler reaction conditions. In select cases, an older BoC chemistry (tert-butyloxycarbonyl) is used where Fmoc based methods cannot achieve the desired synthesis outcome or in the case of processes that have been previously developed and filed.

Regardless of the protecting group selection, all SPPS cycles follow the same general sequence detailed in the following sections.

Figure 3.5: SPPS Elongation Sequence



SPPS Solid Support and Loading

Most synthesis solid supports used for industrial peptide manufacturing are based on swellable, cross-linked poly(styrene-divinylbenzene) co-polymers. Typical synthesis solid supports are composed of spherical beads formed of a polystyrene core resin crosslinked by 1%–2% of divinylbenzene. In organic solvents, these resins can swell considerably, by as much as a factor of 5, providing ample reaction sites for peptide chain elongation. As a result, the shape, size, and swelling of the solid support in organics are critical determiners of reaction efficiency. However, this underscores an inherent limitation for SPPS at elongation reaction sites due to steric hindrance.

The synthetic process in SPPS begins by covalently anchoring the C-terminal amino acid to the solid support. To enable the reaction, the solid support needs to be functionalized with corresponding linkers or handles. The choice of linker will depend on the chemical nature of the target peptide and on the selected synthesis strategy.

The solid support is typically discarded after it is cleaved from the peptide chain.

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SPPS Deprotection

Generally, the first chemical reaction of the iterative SPPS cycle is the deprotection of the Fmoc (or alternate) protecting group on the solid support-linked peptide. The selection of the protecting group will determine the type of deprotection reagents and can vary in specialized chemistry.

Following this step, the deprotection reagent and adducts are washed away from the solid support bed. This is often the most solvent intensive step of the entire SPPS process and most commonly conducted in Dimethylformamide (DMF). DMF is a hazardous chemical solvent and industry efforts are underway to replace its use in commercial processes as indicated in Chapter 2. See further discussion about alternative approaches and challenges that commercial manufacturers face in implementing process changes to support usage of alternative solvents in Chapter 12.

Deprotection enables the subsequent coupling step, which begins the elongation of the peptide chain. Residual deprotection reagents have the potential to deprotect incoming amino acids, which then results in loss of control of the reaction selectivity and subsequent impurity formation. Verification of deprotection completion (using colorimetric testing such as Kaiser testing) and deprotection reagent removal (pH, UV-Vis, NIR, Raman, etc.) are typical control parameters.

Activation and Coupling

The coupling reaction is preceded by the activation of amino acid carboxyl groups to facilitate coupling with the next amino acid in the sequence. Activation may be performed either as a discrete step prior to addition to the reactor (pre-activation) or directly within the reactor (in situ). Pre-activated solution offers the benefits of parallel processing within the SPPS reactor, minimizing competing functional groups on the coupling partner; however, it relies on the stability of the activated solution and requires additional equipment. Additional equipment may also be required if pre-dissolution of in-situ activator is required. Common solvents used in the activation and coupling steps include DMF, NMP, DCM, and DMSO. Nitrogen headspace in the reactors is often used because both activation and coupling steps can be sensitive to moisture.

Capping

Capping solutions are used to prevent deletion impurities that arise from incomplete coupling. Particularly for longer peptides, these impurities pose a significant challenge to downstream purification. The capping step addresses this by acetylating unreacted amine coupling sites that create different physicochemical properties than deletion impurities, which is often sufficient to achieve separation of product peptide and impurity by HPLC purification.

While capping is a useful tool to reduce deletion impurities, the step lengthens the cycle time and can affect yield. A solution of acetic anhydride is typically used in the presence of a base (e.g., DIPEA).

Key factors influencing the synthesis process include:

- Reaction time and temperature
- Solid support size and fining characteristics (SPPS)
- Solid support swelling characteristics in target solvents (SPPS)
- Reaction stoichiometry and relation to reagent delivery and excesses required
- Wash time parameters and solvent volumes to achieve required conditions for next reaction
- Deprotection reaction time and temperature
- Agitation speed and mixing

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3.4.2 Intermediate Processing

This process phase discussion covers all post-synthesis wet chemistry occurring before purification, including cleavage, deprotection, and any solution-phase reactions after solid support removal, as well as precipitation and drying. For LPPS, this will include routing the peptide through purification and isolation phases to prepare it for the next sequential coupling. Typical unit operations include stirred tanks and precipitation and drying reactors as discussed further within this section.

3.4.2.1 Cleavage and Deprotection

After Solid Phase Peptide Synthesis (SPPS), the peptide remains attached to the solid support with protecting groups intact.

Cleavage detaches the peptide from the solid support, while deprotection removes side-chain protecting groups for purification. These steps can occur together (**hard cleavage**) or by removing only the solid support (**soft cleavage**) in a vessel equipped for filtration and peptide precipitation.

Hard Cleavage

The most common method combines solid support cleavage and full deprotection in one step using a strong acid, typically 80%+ Trifluoroacetic acid (TFA) solutions with acid-labile linkers. Side reactions from reactive byproducts (e.g., tert-butylation) are minimized by adding scavengers like triisopropylsilane (TIPS), water, and thiols (e.g., 1,2-ethanedithiol (EDT), 3,6-dioxa-1,8-octanedithiol (DODT), thioanisole).

Hard cleavage is the harshest condition the peptide faces, so stability and impurity risks must be considered during process design.

Soft Cleavage

Soft cleavage involves removing the solid support linker while leaving protecting groups intact. This approach is often chosen to minimize impurity formation under harsh conditions or to allow further solution-phase chemistry. Common solvents include 20% HFIP in DCM or < 5% TFA in organic solutions. After completing the necessary steps, full deprotection is carried out under appropriate labile conditions.

Factors influencing the process:

- Exothermic Reaction: Cleavage reactions are highly exothermic and require temperature control to prevent thermal runaway and minimize impurity formation
- Order of Addition: Adding solid support to the cleavage solution provides better thermal control than adding the solution to solid support, though the latter is often simpler from an equipment standpoint
- Impurity Control: Designing the cleavage step to minimize hard-to-remove impurities is often more important than maximizing yield, as downstream purification (e.g., chromatography) can result in significant losses and added cost. Early control of impurity formation improves overall process efficiency and robustness
- Scalability:
    - Optimize solution-to-solid support ratio and concentration
    - Extra volume can help manage heat but increases anti-solvent needs (see Section 3.4.2.3)
    - Avoid large-scale evaporation of organic acids

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- Pre-cooling can improve thermal control, but consider plant utility limits

- Material Handling and Waste:
  - Many scavengers are sensitizers, requiring strict containment and PPE
  - Cleavage reagents are often corrosive
  - Common carrier solvents are highly flammable
  - Waste streams are often halogenated due to cleavage acids or halide reagents and require special handling
  - Even mild functionalization steps may also require neutralization. Reagent selection should consider plant environmental systems, and neutralization should occur either in-line or at the facility level

3.4.2.2 Functionalization

After cleavage, additional liquid-phase chemistry may include cyclization, oxidation, pegylation, ligandization, or fragment condensation. If crude purity is sufficient, the peptide can proceed directly to these steps; otherwise, further purification is necessary. Complex peptides may require multiple intermediate cycles, each ending in isolation before final chromatographic purification. While intermediate wet chemistry adds operational complexity, it is often necessary for challenging products.

3.4.2.3 Precipitation

Precipitation may occur as part of LPPS elongation cycles and is also typically the final processing/isolation step after intermediate processing has been completed, preparing the peptide for further downstream processing. Solid peptide from the precipitation is filtered, washed of residual cleavage reagents and often dried to remove organic compounds and salts not compatible with purification.

Precipitation is an isolation technique in which dissolved product (in an aqueous or organic solvent), is forced out of solution by adding an anti-solvent in which the product has low solubility. This reduces the solubility until the target molecule precipitates as a solid.

The morphology of the precipitate is critical to process performance; poorly controlled conditions may result in “oiling out”, where the product forms a gel phase, which later solidifies into a sticky material that is difficult to process.

A well-designed precipitation can provide both isolation and partial purification. Achieving ordered crystals (advantageous morphology) becomes increasingly difficult with larger peptides, often resulting in amorphous agglomerates that complicate isolation and purification and negatively impact processing times.

In peptide processes, organic anti-solvents such as Methyl Tert-Butyl Ether (MTBE) or ethanol are commonly used in Fmoc-based processes due to their availability, low peptide solubility, and, in the case of intermediate isolation, their ability to quench cleavage reagents. An effective precipitation step should achieve:

- High product recovery
- Formation of a filterable, non-sticky cake
- Efficient washing to remove residual cleavage agents, solvents and process impurities

Factors influencing the process:

- Peptide length: Longer peptides precipitate more easily

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- Concentration: Higher peptide concentration favors precipitation
- Temperature: Lower temperatures reduce solubility and colder anti-solvent promotes smaller particles
- Temperature change and aging: Increased agglomeration/particle size
- Anti-solvent type and volume: Poor peptide solubility and higher anti-solvent volumes promote precipitation
- Anti-solvent addition mode: Forward (anti-solvent into peptide solution), reverse (peptide solution into anti-solvent), or co-addition of solution and anti-solvent in controlled ratios
- Additional anti-solvent additions: Increase agglomeration/particle size
- Addition rate: Slower addition favors smaller particles
- Agitation speed: Higher agitation speed promotes smaller particles
- pH and common ion effect: Can influence solubility but are less commonly used in crude API precipitation
- Solubility curve: Concentration versus solvent/anti-solvent ratio
- Seeding: Introducing seed particles may induce precipitation
- Water content: Depending on the peptide both too high or too low water content can result in failed, partial, or gel out of the peptide

3.4.2.4 Drying

Once precipitated, the product must be separated from the process liquors and dried. These unit operations are typically performed using either a filter dryer or a combination of a centrifuge followed by a separate drying step (e.g., vacuum dryer). Effective washing and drying of the solid peptide crude are critical, as residual solvents can adversely impact stability and downstream purification.

Filter dryers are often preferred because they combine two unit operations, solid-liquid separation and drying, within a single piece of equipment. This reduces capital cost, facility footprint, and operational complexity. However, if either filtration or drying is slow, the equipment can become a production bottleneck. Filtration delays are commonly encountered when the precipitate consists of very fine particles or dense agglomerates, which significantly reduce filtration rates.

In cases where filter cake resistance is high, increasing filtration area can help reduce cake thickness and shorten filtration time. For filter dryers, however, expanding filtration area significantly increases capital cost without a proportional reduction in overall cycle time. In such cases, a centrifuge may be an advantageous alternative. The main drawback of the centrifuge is the requirement for additional material handling required to transfer wet solids to a separate drying unit operation.

Lyophilization may also be used for intermediate phase drying, although it is more prevalent for final API isolation. Refer to Section 3.4.4 for further discussion on lyophilization.

Factors influencing the process (Filter Drying):

- Filter cake resistance and integrity: Filter cake thickness influences washing and drying characteristics.
- Filter cake formation and integrity: This can also affect washing and drying and will be influenced by morphology as discussed for the precipitation operation.

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- Temperature/vacuum: A balance is struck between processing time and thermal exposure and stability of the peptide.

- Agitation type: Conical agitation versus plough agitation can influence shear on the product. This is more typically associated with isolation phase drying than intermediate phase drying.

3.4.3 Purification

Chromatographic Purification

Crude peptide from intermediate operations typically requires multiple purification steps to meet quality requirements. Common impurities include diastereomers, hydrolysis products, truncated or deletion sequences (SPPS), insertion peptides, and by-products from deprotection. Raw materials may also introduce impurities such as isomers or β-alanine. Generally, as peptide length increases, so does the number of undesired sequences, making purification more complex.

Because these impurities vary in their physicochemical properties, no single purification step is sufficient. Instead, multiple techniques, or multiple versions of the same technique, are often required. Chromatography is the primary method, with reversed-phase HPLC (RP-HPLC) being the most common. Ion Exchange Chromatography (IEX), usually anion exchange, is also widely used, often as an orthogonal step. Normal-phase HPLC is less frequently used due to its heavy organic solvent requirements.

Reversed-Phase Chromatography

RP-HPLC separates molecules based on hydrophobicity, making it ideal for peptide purification where impurities closely resemble the target product. Columns typically use silica-based resins with long-chain hydrocarbons or polymer-based stationary phases for greater chemical stability and broader pH tolerance.

Factors influencing the process:

- Stationary phase properties: Particle shape and size affect surface area and diffusion rates. Spherical particles are preferred for durability under dynamic axial compression

- Chromatographic conditions: Flow rate, gradient slope, pH, peptide loading, pressure, buffer modifier and conductivity

- Temperature: The temperature can affect retention and elution properties of the peptide among other factors such as selectivity and efficiency

The process begins by loading the peptide onto the column with a binding buffer and ion-pairing agent (e.g., dilute TFA) to retain the peptide while hydrophilic impurities wash through. Elution then uses an organic solvent gradient (such as acetonitrile or alcohol), which releases peptides based on hydrophobicity. Fraction collection and pooling decisions can be guided by PAT tools (e.g., UV, NIR) or by offline analytical testing.

Ion Exchange Chromatography

IEX separates peptides based on ionic interactions between the stationary phase and peptide charge groups. Both cation and anion exchangers are used, with stationary phases typically composed of agarose-based resins or synthetic polymers.

Factors influencing the process:

- pH: Strongly influences elution order and selectivity

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- Isoelectric point: Determines peptide charge state and interaction with the stationary phase
- Stationary phase properties: Ionic form, porosity, and particle size and impact on binding capacity and mass transfer

Elution is achieved by gradually increasing the ionic strength of the mobile phase, which weakens peptide-resin interactions. Longer, more highly charged peptides elute later. Elution fractions are collected and analyzed to provide batch acceptance cutoffs and establish pooling decisions for the purified material to be processed downstream.

Concentration of the purified peptide can be achieved either through chromatography or by UF/DF. Reloading columns with minimal cycles improves concentration. For peptides prone to aggregation, intermediate lyophilization (see Section 3.4.4) may be used, followed by redissolution at lower volumes to achieve higher concentration.

Multicolumn Chromatography

At commercial scale, single-column purification often results in low overall yields because achieving high purity requires discarding material contaminated with weakly or strongly adsorbing impurities at the start and end of the separation. These product losses become increasingly costly at larger scales. Furthermore, impacts are more notable for analytical testing workflows and process throughput constraints.

To address these limitations, multicolumn chromatography techniques have been developed. Instead of collecting fewer lower purity shoulder fractions for offline reprocessing, these systems enable automated, continuous transfer of fractions directly to another column for further purification.

One increasingly adopted approach at large scale is Multicolumn Countercurrent Solvent Gradient Purification (MCSGP), which uses two or more columns operating in a coordinated manner. This configuration enables continuous or semi-continuous operation. This technique is particularly effective when the selectivity difference between the desired product and related impurities, such as deletion or truncation sequences, is low, a common scenario in peptide purification. See further discussion in Chapter 12.

It is important to note that if this technique is intended for large-scale implementation, development work must begin early in the process lifecycle. Retrofitting a conventional single-column process to a multicolumn system later in the lifecycle is technically challenging and often impractical.

Concentration and Small Molecule Removal Operations

Tangential Flow Filtration (TFF)

Tangential Flow Filtration (TFF) is a versatile unit operation in which a membrane filtration, driven by pressure and concentration gradients, leads to separation of components in a mixed solution based on molecular size. In typical peptide processes, TFF operations can include Ultrafiltration (UF), Diafiltration (DF), and Nanofiltration (NF).

Ultrafiltration and Diafiltration

Ultrafiltration (UF) is widely used in the biopharmaceutical and peptide industries, particularly after chromatographic steps that produce dilute solutions containing buffer salts. UF serves two main purposes: removing small molecules such as buffer components and concentrating the product solution.

Further purification is typically achieved through diafiltration. In this process, Water for Injection (WFI) or Purified Water (PW) replaces the solution that permeates the membrane, gradually diluting and removing salts and other impurities to reach acceptable levels. Solvents other than water may be used if aqueous solubility is limited.

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Evaporative Concentration

When concentrations beyond the limits of tangential flow filtration are required, evaporation can be employed. To protect temperature-sensitive peptides, the process is typically performed under vacuum, which lowers the boiling point and minimizes thermal degradation.

For further temperature control and reduced exposure time, technology such as thin film evaporators can be used. In these systems, the solution flows down the inner wall of a heated cylindrical column under low pressure, while rotating blades spread it into a thin film. This design maximizes surface area and promotes rapid solvent removal through flash evaporation, significantly reducing the risk of product degradation. Thin film evaporation can achieve concentrations of up to 150 mg/mL and is relatively unaffected by solution viscosity, making it a flexible option for concentrating peptides.

3.4.4 API Isolation

3.4.4.1 Precipitation

For intermediate processing isolation, precipitation is also commonly used to facilitate final API isolation in peptide processes. Specific GMP considerations that may apply at this stage of the process are discussed in Chapter 4. Otherwise, the precipitation step before isolation should generally follow the same process considerations as discussed section 3.4.2.3 following cleavage and deprotection.

3.4.4.2 Final Drying

Chapter 4 should be referenced for discussion on particular requirements for this unit operation at this stage of the drug substance process. Additionally, the need for longer stable storage periods between production of the bulk API and eventual DP filling can influence the choice of unit operation.

3.4.4.3 Lyophilization

Peptides are not fully stable in aqueous solutions and will gradually be subject to degradation. While freezing can slow or halt degradation, repeated freeze–thaw cycles may compromise molecular integrity. Bulk lyophilization, which avoids solubility considerations, is very effective at preserving product stability. It typically produces a solid with high specific surface area, which facilitates downstream formulation. As such, lyophilization remains one of the most widely used methods for isolating stable peptide solids from aqueous peptide solutions.

Advantages:

- Enhanced stability of the dried product
- Water removal without exposure to excessive heat
- Rapid and efficient reconstitution
- Reduced cold storage requirements

Disadvantages:

- Increased handling and processing time
- Risk of contamination
- High equipment cost and complexity

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- Time required to develop and qualify the lyophilization cycle

Factors influencing the process:

- Freezing temperature
- Rate of temperature drop
- Vacuum pressure
- Rate of temperature increase
- Primary drying time and temperature
- Secondary drying time and temperature

3.4.4.4 Spray Drying

When precipitation is not feasible, spray drying offers an alternative method for isolating a dry solid from solution. This method, widely used across industries, has recently gained increased attention in biopharmaceutical manufacturing as a potential alternative to lyophilization.

This method provides advantages such as better control of particle size distribution, greater batch consistency, and significant reductions in energy consumption, processing time, and resource requirements, particularly at a large scale. Additionally, it can also operate as a continuous process, which may benefit certain production strategies.

Spray drying peptides, however, can cause thermal and shear stress, leading to degradation or loss of biological activity. Exposure to higher oxygen environments can increase the risk of oxidation and hydrolysis. Formulation challenges include excipient selection and particle stability. Additionally, amorphous powders produced by spray drying may agglomerate and process scale-up complexity may increase costs compared to gentler methods such as lyophilization.

Critical factors include:

- Temperature: This achieves a balance of drying while minimizing thermal degradation
- Peptide concentration: Concentration of peptides in the feed solution can influence particle size
- Colloidal matrix properties: The type of carrier solution will have a large impact on morphology and thermal stability

3.4.4.5 Filter Dryers, Centrifuges and Cone Dryers

As discussed in Section 3.4.2, filter dryers have the advantage of combining solid-liquid separation and drying into a single piece of equipment. Additionally, they are totally enclosed operations that can be beneficial from both product and operator exposure standpoints.

In processes using centrifuges, cone dryers are often employed downstream. Cone dryers can provide the additional advantages of gentle agitation compared to the plough-style agitator of filter dryers and less hold-up volume and are more often chosen at the API isolation phase than in the intermediate drying phase.

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Potential Additional Steps

Precipitated solids often form agglomerates that can trap solvents, making drying to below ICH residual solvent limits difficult and time-consuming. Humidification, applied either prior to or during drying, can plasticize the solid, releasing occluded solvent and reducing drying time. Specific GMP considerations that may apply at this stage of the process are discussed in Chapter 4.

Packaging

After drying, the bulk drug substance is packaged into final product containers. The approach to this operation depends on several factors, including manufacturing scale, product quality requirements, and operator exposure requirements.

Key considerations include:

- Potency and operator exposure requirements
- Final product bioburden specification
- Hygroscopic effects and moisture control
- Product stability

3.5 Additional Peptide Considerations

3.5.1 Capacity and Scheduling

Capacity is the maximum output a peptide facility can sustain in the production of the final product. This output is influenced by multiple factors, including equipment scale limitations and the duration of both unit operations and supporting activities.

Scheduling processes should consider allowable points in the operation where the in-process material can be held, including allowed storage conditions and durations. These are influenced by both product stability and the potential for microbial growth.

3.5.2 Supply Chain

Peptide synthesis and purification rely on a robust supply chain for high-purity amino acids, specialized reagents, and chromatography media. Timely access to controlled temperature shipping and GMP-compliant materials is critical to maintain product quality.

Common challenges include long lead times for rare amino acids, variability in solid support performance, and disruptions in cold-chain logistics. Additional complexity arises from regulatory compliance and batch traceability. Furthermore, global shortages of solvents or specialty chemicals can halt production. To mitigate these risks, strategic vendor diversification and proactive inventory planning are essential.

3.5.3 Cost, Sustainability, and Available Capacity for Commercialization

As discussed in Chapter 12, fundamental challenges with the core SPPS technology are driving innovation across the product lifecycle. Sustainability, cost of commercial production, and capital investment in manufacturing facilities are all factors receiving increased attention as the industry grapples with the rapid growth of incretin mimetic therapies and their large production demands, to say nothing of other novel approaches in the development of therapeutic uses for peptides.

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Chapter 12 discusses the current and future state in more detail and provides context to the challenges for commercial adoption. Peptide development and scale-up strategies must adapt accordingly with a strong focus on continuity and robustness of scalability.

Peptide Development and Scale-Up Considerations

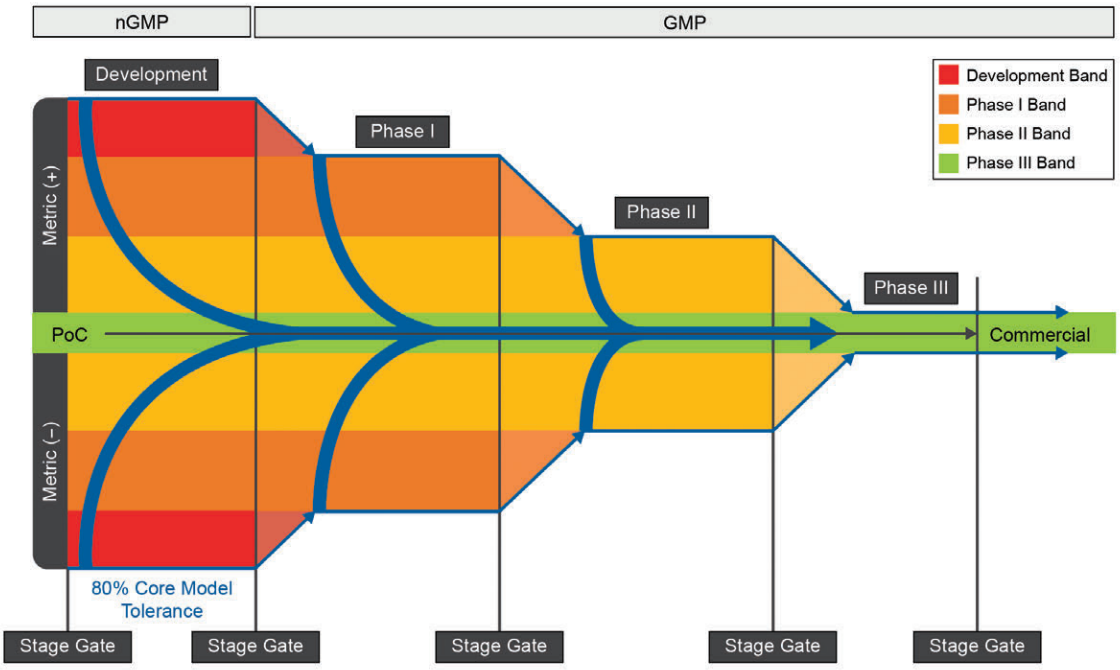
Selecting an appropriate process design is critical to long term manufacturing success. The following are some key areas where a forward-looking development approach can significantly reduce the risk of costly changes later in the lifecycle.

- Late phase changes can affect the validity of clinical trials, particularly with respect to impurity profiles and isolation characteristics (e.g., morphology).
- Selection of reagents and materials should anticipate eventual commercial scale requirements. In particular, reagents that pose particular challenges for sourcing, handling, or disposal can become problematic at larger scales.
- Differences in equipment considerations and methodology for handling, dispensing, metering and tempering reagents must be understood for the full anticipated scale horizon.
- Process unit operation selection must also be guided by anticipated future-scale objectives, including facility capabilities and equipment constraints. For example, while lyophilization is commonly used during development, if the commercial scale process will require isolation involving precipitation, this approach should be incorporated and evaluated in the earlier phases and supported by the process development approach.

Peptide APIs generally follow the same development lifecycle as other pharmaceutical products as they progress from discovery through commercialization. Figure 3.6 illustrates this progression, highlighting the tolerance for process changes as the development lifecycle progresses and moves towards a robust, repeatable commercial process that can be validated.

These tolerances naturally must be aligned with appropriate equipment and facilities to produce material at each phase.

Figure 3.6: Peptide Phasing Diagram



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Metrics and Metrics Range: Can include CQA, CPP, etc. Metric (+) and Metric (−) represent the tolerance or applicable range, for example: Yield (Target 30 ± 5% as a CQA).

Phase Bands: Indicates where a metric falls relative to its development phase. As a process progresses from development to commercialization, there is less tolerance for variance and each phase is expected to demonstrate better control and refinement than the previous one. For example: Purity: 90% Development, 95% Phase I – II, 98% Phase III and Commercial.

Stage Gates: Before moving through phases, control strategies, design parameters, and scale-up expectations should be formally reviewed and documented, often via a Quality by Design (QbD) system. In the case of technology transfer, all incoming data and supporting information should meet the requirements of the preceding phase's stage gate before progressing further.

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4 Risk Assessments and Critical to Quality Strategies

4.1 Introduction

In this chapter, facility Risk Assessments (RAs) and Critical to Quality (CTQ) strategy elements relevant to oligonucleotide and peptide manufacturing are reviewed.

Risk assessment practices are discussed briefly to provide a framework specific to oligonucleotide and peptide facilities, leveraging existing industry practice and knowledge. In addition, specific considerations related to hazardous environmental risk assessments are outlined.

A CTQ strategy is central to the evaluation of any GMP process and should be used to ensure that the desired product quality profile is consistently achieved. The appropriate CPPs, Critical Material Attributes (CMAs), and Contamination Control Strategy (CCS) are determined based on risk assessments and the Critical Quality Attributes (CQAs) of the product and process. While potential CPPs are discussed in Chapters 3 and 5, product-specific CPPs, CMAs, and CQAs are purposefully excluded from this Guide. Instead, this chapter provides a high-level approach focused on establishing the CCS for both oligonucleotide and peptide manufacturing facilities.

Many of the elements of a CCS have already been described in regulatory and ISPE guidance documents. This chapter will reference those established resources to provide a framework, and place primary emphasis on CCS elements that are specific to oligonucleotide and peptide manufacturing.

Given the unique properties of many of the steps in synthesizing oligonucleotides and peptides, with extensive use of bactericidal solutions, the typical CCS for these modalities differs significantly from aqueous-based biopharmaceutical manufacturing schemes. To help further clarify this context, we present a process assessment with example CCS approaches and considerations for typical oligonucleotide and peptide manufacturing processes.

4.2 Oligonucleotide and Peptide Facility Risk Assessment Considerations

As described in *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5], risk assessment and risk management have become the cornerstone of the quality control of biopharmaceutical manufacturing. Elements of a robust risk assessment program are described in detail in other ISPE Baseline Guides and are listed here for reference:

- Risk Management Principles and Practices, including integration of a risk management master plan within a broader project master plan, initiation and execution of the risk assessment, control of risks via recommendations, and communication of risks, using QRM techniques to appropriately conduct assessments and evaluate risks. ISO 31000:2018 — *Risk Management — Guidelines* [66], *ISPE Guide: Quality Risk Management* [67].

- Risk Management Tools, Techniques and Platforms, including HAZOP, HACCP, FMEA, FTA [ICH Q9(R1) [28]. *Quality Risk Management,* European Medicines Agency/US FDA, 2023 revision.

Building on these established industry techniques and approaches, this Guide focuses on unique risk assessment considerations for oligonucleotide and peptide facilities that will help readers design and implement appropriate assessments for such facilities.

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Oligonucleotide and peptide facilities typically handle hazardous materials and solutions that may be toxic, flammable, reactive, and/or explosive. For these reasons, oligonucleotide and peptide manufacturing programs require additional, more specific risk assessments that also include safety considerations aimed at preventing the release of these hazardous agents and minimizing exposure to personnel and the environment. These considerations are discussed in the following sections.

4.2.1 Process Safety Management (PSM)

As discussed in Chapter 2, OSHA PSM regulations require that additional risk assessments be performed in certain situations. These laws come into effect when handling specific chemicals, or when a threshold quantity of hazardous material is met. For example, in the US, if a site utilizes 10,000 pounds or more of flammable liquid(s) in one location/process, (equivalent to roughly 1,000 gallons), PSM regulations may apply. Many oligonucleotide and peptide manufacturing facilities will meet this threshold. While OSHA PSM is US-specific, equivalent process safety requirements and risk management frameworks need to be similarly considered outside of the USA, as enumerated in Chapter 2.

There is notable overlap between cGMP compliance and PSM compliance, including the need for detailed characterization of facility processes and chemicals, appropriate engineering, maintenance, and operational documentation, employee training, routine audits and inspection, and robust recordkeeping.

Safety-related chemical information, including SDS documentation, should be used in conjunction with test data to identify and thoroughly characterize hazards that may be encountered in the process. Common hazards encountered include toxicity, flammability, and reactivity. Maintaining accurate inventories for on-site chemicals and records of their usage is also essential.

Automated system controls and logic are a key component of the risk assessment. In some high-risk scenarios, a Safety Instrumented System (SIS) may be recommended to address the risk. Refer to Chapter 9 for further discussion.

Typically, PSM regulations require the execution of a Process Hazard Analysis (PHA). A PHA is a risk assessment focused specifically on protecting the safety of personnel, facilities, and the environment.

4.2.2 Process Hazard Analysis

Regardless of whether it is driven by PSM regulations, a PHA should be performed independently from other product quality and safety risk assessments to promote a focus on process and system safety. For example, a vessel containing acetonitrile may be low risk when evaluated in a product contamination risk assessment but represents a significant flammability and explosion hazard to operators and surroundings when evaluated in a PHA. Additionally, PSM regulations require revalidation of the initial PHA at least every five years to ensure risks are consistent with the current process.

4.2.3 Change Management and Hazard Analysis

Change management is a critical element in oligonucleotide and peptide facilities from GMP, safety, and regulatory compliance standpoints. The modification of process conditions, operations, materials, or equipment without properly assessing potential effects can cause troublesome consequences. All changes should be formally evaluated, documented, and incorporated to ensure ongoing compliance and accurate procedures and drawings. When changes are substantial, a PHA should be performed as part of the change management process.

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4.3 Oligonucleotide and Peptide Facility Contamination Control Strategies (CCS)

An effective CCS needs to be developed for both oligonucleotide and peptide manufacturing facilities and programs. The CCS serves to establish a program that protects the process and product from contaminants that could adversely impact product quality or patient safety.

4.3.1 Classes of Contamination

ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition) [5], identifies three classes of contamination, based on the potential impact they may have on the process and product quality:

- Biological (viable): Viruses, yeasts, molds, bacteria, including endotoxins. Risk is posed primarily under aqueous process conditions where proliferation is possible. They represent lower risk under bactericidal or bacteriostatic process conditions.

- Particulate (non-viable-inert): Typically of low risk as these are generally non-reactive and removed naturally in most purification steps.

- Chemical (non-viable): May react or co-purify with the product and therefore can be of concern if not properly controlled. They often originate from raw materials and are a particular concern for multiproduct facilities.

In oligonucleotide and peptide manufacturing, biological contamination is primarily a concern when transitioning from bacteriostatic to “growth-supporting” (i.e., non-bacteriostatic) conditions, when latent viable contaminants may proliferate. Even when subsequently removed, the lysis of a microbial culture may generate endotoxins that still pose downstream risks.

4.3.2 Sources of Contamination

Having established the types of contaminants that can potentially affect oligonucleotide and peptide manufacturing processes, an understanding of the potential sources of these contaminants is essential to developing a program that mitigates the risk of ingress and impact of these contaminants and forms the basis of the CCS.

ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition) [5], defines the process zone to include all the elements that can affect a process step.

Figure 4.1 depicts examples of the process zone for three distinct operations:

- Figure A: Open synthesis operation performed in bactericidal process conditions

- Figure B: Closed system housing an aqueous “growth supporting” operation

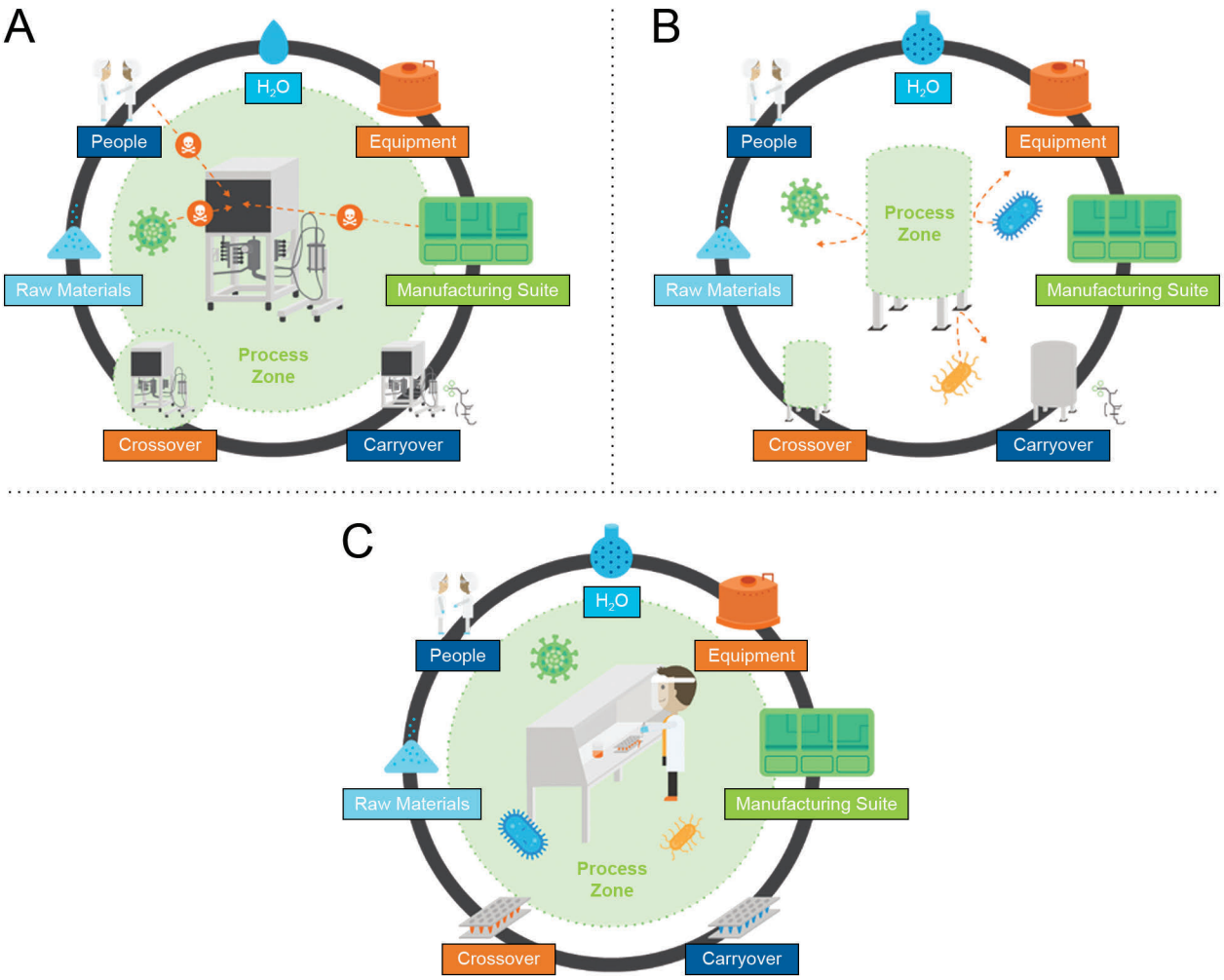
- Figure C: Open system “growth supporting” operation

Although all three operation types are used in oligonucleotide and peptide manufacturing schemes, Figure 4.1 Sections B and C represent operations also performed routinely in biopharmaceutical manufacturing that are thoroughly addressed in *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5]. This Guide focuses primarily on the type of operations represented by Figure 4.1 Section A.

Notably, there are no biological risks under bactericidal conditions, as any ingress is inherently non-viable and thus inert. Instead, the primary contamination risks arise from chemical contamination, such as water ingress or carryover/cross-contamination between reagents and products in shared equipment.

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Figure 4.1: Sources of Contamination



Understanding the sources of biological, chemical and inert contaminants and their potential impact on the process and product is a prerequisite for developing the CCS.

4.3.2.1 Operations Performed in Bactericidal Solvents

Oligonucleotide and peptide synthesis steps are performed in organic solvents. Few (if any) viable environmental contaminants can affect these synthesis steps. For this reason, manufacturing areas used for synthesis typically do not require classified cleanrooms for process and product protection. However, emerging oligonucleotide manufacturing programs that use alternate synthesis methods performed in “growth-supporting” conditions may require additional controls, including closed systems or classified cleanrooms as part of the CCS to protect the processes appropriately. See Chapter 12 for further discussion of these program approaches.

In multiproduct facilities, protection of one process stream from another to prevent cross-contamination may be required, particularly during concurrent manufacturing operations. In all cases, proper control of the ingredients, raw materials, consumables, solvents and buffers is required.

A robust materials management program, with thorough documentation, is a critical element of the CCS.

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4.3.2.2 Operations Performed in Bacteriostatic Conditions

Special attention is required for operations that transition from organic-solvent-based to aqueous-based. This is particularly important when the organic solvent used is bacteriostatic rather than bactericidal, as transitioning from a bacteriostatic solvent to an aqueous environment may enable the proliferation of latent bioburden. Solvents, buffers, and intermediate storage solutions should be identified or qualified early as either non-bacteriostatic, bacteriostatic, or bactericidal via Antimicrobial Effectiveness Testing (AET).

Since latent viable contaminants such as endotoxins cannot be removed by sterile filtration, specification limits should be defined for drug substances that will not undergo further purification. This is true of other contaminants that may interact with the resulting drug product and affect overall quality and performance.

4.3.2.3 Operations Performed in Aqueous Solutions.

The CCS for low bioburden downstream processes should follow the approaches described in *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5], as the risks of contamination are essentially the same. As with biological Drug Substance (DS) manufacturing, closed systems are recommended wherever feasible. Classified cleanrooms should be employed for open, non-bacteriostatic, low bioburden operations.

Even within classified cleanrooms, an understanding of the impact of all controlled conditions (e.g., duration, temperature, pH, and initial microbial load) is critical. For an oligonucleotide or peptide used as an ingredient in subsequent DS manufacturing (e.g., antibody oligonucleotide conjugates), or a DS that will be sterile-filtered into a closed container and stored in liquid form or frozen (or other bacteriostatic state), the CCS may not require a classified cleanroom. In this context, a bacteriostatic state is defined as a condition in which the risk of bioburden increase is negligible throughout the controlled period.

4.3.2.4 Drying Operations

Most oligonucleotide and peptide manufacturing operations include a final drug substance drying step. The CCS applied during drying must be commensurate with the final drug substance specifications.

If the DS requirements include very low bioburden limits, the CCS may need to include system and environmental controls similar to those used in drug product manufacturing. Isolation technology is an ideal choice to appropriately mitigate the risk of contamination from the environment.

For an oligonucleotide or peptide intended for subsequent drug substance manufacturing, or a final drug substance that will be subsequently sterile-filtered into a closed container and stored in liquid or frozen form, the CCS may not require a classified cleanroom.

4.3.3 Elements for Developing a Robust Contamination Control Strategy

As in biologics manufacturing, the GMP manufacture of low bioburden oligonucleotides and peptides requires a holistic approach to developing a robust CCS. A clear understanding of the types and sources of contamination and communication of these elements of scientific knowledge to the overall quality organization builds operator awareness and fosters a quality culture throughout the organization, yielding an effective quality risk management program.

Contamination risks include environmental contaminants, carryover/cross-contamination from previous operations in shared systems, and cross-contamination from neighboring operations. This should serve as the goal and motivation and the foundation of a robust CCS.

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4.3.4 Elements of the Contamination Control Strategy

A well-developed CCS evaluates the risks of contamination at each process step and establishes appropriate controls to mitigate these risks. A well-defined approach to CCS development considers the following elements of the CCS individually:

- Materials: Specifications are documented and potential impact to product quality is evaluated
- Process: Unit operations and conditions are assessed
- People: Gowning, hygiene, and operational practices are defined
- Equipment: Closure of process (e.g., CLARA principles) and appropriate design (e.g., ASME BPE standards) should be considered
- Environment: Cleanrooms are controlled per EMA classifications (A, B, C, D or equivalent) via facility design and appropriate operational controls
- Testing and detection: Appropriate analytical methods should be used to confirm adherence to the CCS

These elements have been previously defined (*ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5]). The following oligonucleotide and peptide process assessments describe specific considerations for CCS development within the context of typical unit operations and manufacturing areas.

4.4 Oligonucleotide Process CCS Risk Assessment

Chapter 3 provides an overview of the SPOS manufacturing process. In the following sections, a typical assessment of the CCS risks will be performed for each phase of the oligonucleotide manufacturing process considering the above CCS elements.

4.4.1 Synthesis Process Assessment

Synthesis operations are performed using solvents such as acetonitrile, toluene, and lutidine/pyridine, or highly caustic ammonium hydroxide and methylamine solutions. This results in bactericidal process conditions that inherently protect the product from biological contaminants. The synthesis operation is at minimal risk of endotoxin contamination because all reagents and solutions used in the synthesis process step are bactericidal, thereby preventing the opportunity for any significant microbial population to be present. Therefore, the risk of bioburden or endotoxin is extremely low.

The synthesis process also provides significant protection from inert and chemical contaminants: hundreds of column volumes of reagents and wash solvents flow over the column during synthesis, effectively removing potential contaminants while the product remains chemically bound to the solid support in the filled column. The process inherently involves a filtration step to remove the solid support from the oligonucleotide after synthesis is complete, which will also serve to remove foreign inert particulates.

Although the chemical synthesis process is rugged for most chemical contaminants, it remains susceptible to water contamination (or other hydroxyl-containing compounds) and reactive raw and starting materials and impurities. Protection against water contamination, either from the environment or the materials, may necessitate dedicated process step controls such as molecular sieves to limit water content in prepared solutions.

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Preventing moisture as a chemical contaminant is a significant challenge. Bulk Acetonitrile (ACN) delivered in tanker trucks and stored in large volume tanks is a potential source of contamination for the process. Recycling of ACN from the process may also contain traces of water and reagents that could lead to issues with further processing. The phosphoramidites and activators themselves are also common sources of moisture and should be tested and controlled for water content.

Typical analytical testing of the synthesis crude intermediate includes ion-paired chromatography, paired with high resolution mass spectrophotometry. This analytical regime provides a high degree of assurance of detecting product impurities arising from chemical contamination prior to further processing.

One of the highest chemical contamination risks in oligonucleotide manufacturing is associated with the phosphoramidite solution preparation. These solutions, often 8–12 unique solutions, are commonly prepared in shared spaces with an open powder dispensing and dissolution operation, often in non-dedicated and visually similar vessels. This creates the risk of cross-contamination during the preparations as well as physical errors such as mix up of solutions/vessels, misconnection of lines, or insufficient rinsing/cleaning. Because phosphoramidite solution cross-contamination directly alters the oligonucleotide sequence and cannot be removed downstream, a focused risk assessment should be performed to define appropriate procedural controls, and potentially in-process analytical controls, to prevent contamination of this nature.

The synthesis solid support is a single-use consumable with very low leachable/extractables risk based on the large volume of solvents that are delivered during the synthesis process while the product is chemically bound to the synthesis support.

Lastly, in this paradigm, no environmental room particulate classification controls are necessary for product protection against contamination. However, based on a risk assessment for water contamination, humidity control may be considered and room temperature control is recommended.

4.4.2 Chromatography Process Assessment

Chromatography operations may be performed in solutions ranging from fully aqueous conditions to primarily solvent-based (typically ACN or methanol). Despite this variety of solutions, most oligonucleotide chromatography buffers and conditions are inherently bacteriostatic or bactericidal, even when fully aqueous, due to their high concentration of dissolved salts and typically caustic nature. Each chromatography process, including the prepared buffers and intermediate storage solution/conditions, should be assessed via AET.

For the purposes of this assessment, the oligonucleotide chromatography process conditions will be assumed to be bacteriostatic; however, product fraction hold times and storage conditions, particularly if aqueous fractions are stored at neutral pH, may need to be defined with the intention of providing protection from additional biological contamination.

The chromatographic separation is effective at removing synthesis impurities and reaction by-products from the drug substance. This step is also effective at removing most chemical contaminants, based on the bind-elute mechanism. Filtration of the chromatography feeds prevents suspended solids from reaching the column and ultimately the product.

Chromatography resins are typically dedicated to a single product; however, resin re-use, either across lots of the same product or across different products may be enabled via resin cleaning if a robust risk assessment and validation confirms the practice does not constitute a risk. Resin leachables and extractables represent a potential risk and some purification resins may require conditioning prior to use. At a minimum, a risk assessment of resin leachables and extractables is required.

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Chromatography column packing is an open operation. Depending on the antimicrobial effectiveness of the chromatography operation, procedural controls to close, clean, and secure the system after packing may be required. After performing a robust risk assessment for the facility specific chromatography process, environmental room classification controls may not be necessary for product protection against contamination.

4.4.3 Downstream Process Assessment

Downstream operations comprise a wide variety of aqueous and non-aqueous steps, but eventually the process transitions to fully aqueous conditions. For non-aqueous steps, a thorough risk assessment should be performed to evaluate the risk of biological contamination for that specific step, while considering the growth-inhibiting properties of the process solutions. For purely aqueous steps, such as thin film evaporation, *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5], provides specific guidance on risk assessments and appropriate contamination control measures.

Other steps, such as TFF and precipitation, involve a transition from bacteriostatic/bactericidal conditions to purely aqueous conditions. These steps (for example, TFF, including buffer preparation, TFF system operation, filter element installation, and diafiltration) require a risk assessment that defines appropriate layers of protection to ensure a low bioburden drug substance, while taking into account the microbial control benefits of the organic solvents and/or base content present at the start of the process. Importantly, transitions from bacteriostatic conditions may require additional controls to prevent proliferation of latent bioburden. Most chemical and particulate contamination risks for these steps are mitigated by the process itself, although additional particulate filtration steps may be desirable.

4.4.4 Isolation Process Assessment

Lyophilization (freeze-drying) is a typical isolation step in oligonucleotide manufacturing. The CCS should include an evaluation of the loading of the final solution in freeze-drying trays or vials, lyophilization of the material, unloading of the API, sampling, and storage. Although lyophilization of the API results in a moisture content of < 15% for the product and is therefore in a bacteriostatic state, any latent bioburden constitutes a risk for proliferation once the product is rehydrated.

Loading and unloading the freeze-drying trays or vials is an inherently open operation. It is common to perform sterile filtration of the process solution prior to filling. Given that single-use trays are typically used, the lyophilizer itself is considered non-product-contacting equipment. However, cleaning and drying of the surfaces (potentially including a sanitizing cycle) between lots is recommended as an element of the process microbial control strategy. Release testing of the lyophilizer is not necessarily required for this cleaning, due to the minimal risk of cross-contamination. With vials and re-usable trays, the lyophilizer may be considered product-contacting and a more robust and validated cleaning cycle may be required.

During unloading, single-use trays remain closed to the environment until they are physically opened, often via cutting. Documentation of the risk of tray material shedding during the operation (typically controlled via downstream particulate filtration steps as part of the drug product process), as well as procedures defining appropriate and safe techniques, is required. Stoppered vials allow for the product to be unloaded in an already closed state.

For the unloading of disposable trays, powder handling and cleaning concerns arise, as the material is highly static and has a low bulk density. Disposable isolators reduce the cleaning burden and potential testing requirements for the unloading equipment.

Both filling and emptying of the single-use freeze-drying trays require a classified space consistent with the drug substance bioburden specification and a risk assessment of the effectiveness of any isolation equipment used. Oligonucleotide powder is highly hygroscopic; therefore, humidity control of the environment is important. At a minimum, humidity should be maintained at levels that ensure consistent water content in the drug substance, even if water content at this stage of the process is not always considered a CQA.

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4.5 Oligonucleotide Carryover/Cross-Contamination Assessment

Oligonucleotide facilities are often multiproduct, and preventing carryover/cross-contamination of raw materials, intermediates, and products is a critical focus. Because oligonucleotide processes commonly rely on shared, multiuse equipment, contamination control depends heavily on temporal segregation of materials and effective cleaning between batches. This section will discuss commonly justifiable cleaning thresholds and considerations for shared reagent and product-contacting equipment.

4.5.1 Synthesis Reagent Equipment

Where possible, facilities should dedicate synthesis reagent preparation and delivery systems to the type of reagent (for example, oxidizer solution), unrelated to which product it will be used to manufacture. If the same formulation is being used in the equipment, depending on the justified reagent expiry, no cleaning is necessarily required. If different formulations of the reagent are required for different products (for example, different acid concentrations for the Deblock solution), chemical reactivity arguments can generally justify maximum reagent formulation cross-contamination levels of ~1% (as the previous formulation is also capable of affecting the desired chemical reaction). Given such a high threshold, regular testing of the rinse material can generally be justified as not required.

If different types of synthesis reagents share equipment (for example, a tank that may be used to prepare either sulfurizer or oxidizer solution or the phosphoramidite tanks), then the maximum reagent cross-contamination level should be determined by a risk assessment of the impact of this reagent cross-contamination on the desired synthesis reaction and the product or reaction risk that the alternative reaction could pose.

In particular, carryover of the previous phosphoramidite solution to the next phosphoramidite solution will directly result in an equal length, incorrect sequence that cannot be removed by the chromatography operation. Furthermore, some phosphoramidites have the same mass when incorporated into the oligonucleotide sequence, making the cross-contamination undetectable to the standard mass spectrophotometry tests commonly used in oligonucleotide analysis.

Maximum reagent cross-contamination levels of 10–100 ppm can typically be justified (1–2 orders of magnitude below the limit of detection by mass spectrophotometry). Rinsing and any associated testing of the reagent equipment must also include the synthesizer inlet and valve via cycling.

4.5.2 Synthesizer and Column

The synthesizer is generally not product-contacting and therefore does not require cleaning or analytical testing for product carryover. However, synthesizers typically share delivery system components, including pumps, valves, and piping, across all the synthesis reagents. Therefore, it is imperative to determine the minimum rinse volume for these systems to achieve the justified reagent carryover limits determined in the previous section.

While the synthesis column may be considered product-contacting, it is unique in that, during the synthesis process, the oligonucleotide remains chemically bound to the solid support and does not enter the liquid phase or contact the column surface directly. In this setting, the risk of product cross-contamination is nearly negligible, as any product remaining will certainly be washed away over the course of synthesis. A visual inspection of the synthesis column can generally be justified, instead of a quantitatively defined cleaning limit.

4.5.3 Downstream Unit Operations

For downstream operations, carryover must be more carefully considered, as the product is solubilized, contact with the process fluid can leave residues on the equipment that can be resolubilized during later batches. Product contacting stainless steel equipment will need to follow an Maximum Allowable Carryover (MAC) framework as part of the risk assessment.

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In addition, non-stainless steel product-contact components, such as chromatography resins, must be assessed for carryover in multiproduct or multibatch scenarios where this can impact future products passing through shared equipment. In some cases, resins may be dedicated to individual products.

Any additional downstream reactions (such as conjugations) would also require specific focus within the risk assessment to consider alternate reactions and removal of reagents, byproducts, and potential degradants from product contact surfaces.

In the case of lyophilization, many processes are able to both seal the product in trays or vials, often employing single-use technology. As discussed previously, the lyophilizer itself can often be considered non-product-contacting. In the case where reusable trays are used, the risk assessment needs to consider the potential exposure of the lyophilizer itself as product contacting.

Further discussion in this section on MAC should be captured in the context of these unit operations and specific product considerations to form the CCS.

4.5.4 Product-Contacting Equipment

Each facility must perform its own risk assessment to evaluate the potential for cross-contamination across the portfolio of products. A generalized assessment, including an illustrative rinse-threshold calculation, is provided for consideration.

Oligonucleotides are typically highly water-soluble and can be readily removed from polished stainless steel surfaces through contact with water. Therefore, post-cleaning rinse concentrations are representative of the remaining product in a piece of equipment.

Most oligonucleotides produced on shared equipment will have broadly similar molecular weights and potencies within two orders of magnitude of each other. Applying a commonly accepted safety factor of 1,000 corresponds to a maximum allowable carryover concentration of 10 ppm (0.01/1,000). Oligonucleotide facility cleaning programs must ensure that the total carryover contribution from all shared product-contacting equipment in a process does not exceed this approximate concentration limit when calculated as carryover mass/product mass.

For example, if the oligonucleotide process includes ten pieces of shared product-contacting equipment, then the cleaning threshold should ensure that each piece of equipment contributes no more than 1 ppm of carryover. As a point of reference, a tank that has been determined to retain up to 0.1% of its operating volume would require a final rinse concentration of at least three orders of magnitude below the operating concentration to target carryover below 1 ppm.

4.6 Peptide Process CCS Risk Assessment

Chapter 3 provides an overview of the peptide manufacturing process. As with oligonucleotide manufacturing, potential carryover/cross-contamination in multiproduct facilities, as well as batch to batch contamination in a campaign scenario, should be carefully considered. CCS considerations for each phase of the peptide manufacturing process are discussed in the following sections, considering the general CCS elements introduced earlier in this chapter.

4.6.1 SPPS Process Assessment

Similarly to the SPOS process, SPPS operations are performed using organic solvents. In the case of SPPS, these include chemicals such as dimethylformamide, piperidine, and other organic chemicals used for washes and/or resin swelling. Such conditions create a bactericidal environment throughout the operation, providing effective protection of the product from biological contaminants.

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The SPPS synthesis process also provides significant protection from inert and chemical contaminants, as large amounts of reagent and wash solvents flow through the reactor, washing away potential contaminants while the product remains chemically bound to the solid support within the reactor. Care should be taken during solids additions and when removing the resin to observe any potential presence of inert contaminants, prior to proceeding to cleavage operations.

For most peptide compounds, the peptide on resin intermediate is generally stable. Prior to process performance qualification (PPQ), and process validation (PV), stability studies should be conducted to define appropriate storage conditions and allowable hold times before cleavage.

The synthesis chemical process is developed to be quite rugged to most chemical contaminants, but there is potential for excess moisture to inhibit activation or coupling reactions.

Qualitative testing of synthesis intermediate coupling reactions (e.g., Kaiser/Chloranil tests) can help detect incomplete coupling reactions that may result from chemical contaminants. Additionally, intermediate microcleavage with UPLC or HPLC analysis can provide data to show if any contaminants have led to significant formation of impurities. Together, these analytical approaches offer a high degree of confidence in detecting chemical contamination product impurities prior to further processing. However, there is a tradeoff with extending processing time to incorporate these methods that must be balanced against the use of other more time-efficient production controls to reduce the presence of contaminants.

A relatively higher risk of chemical contamination in peptide manufacturing is associated with the amino acid solutions. Multiple different amino acids are frequently dispensed in open or partially protected environments in the vicinity of each other, which introduces the possibility for cross-contamination. A risk assessment should be performed to define appropriate procedural and/or engineering controls to mitigate such risks.

The synthesis resin is a single use consumable with very low leachable/extractables risk to the product. In particular, large volumes of solvents are delivered during the synthesis process and serve to flush the reactor while product remains chemically bound to the synthesis resin.

Lastly, at this stage, no environmental room classification controls for particulates are typically required due to the protected API being bound to the resin within a bactericidal environment. These product risk mitigating factors should be captured in the form of a risk assessment to justify the CCS around the entire operation, including for example activities such as open addition of amino acids to the activation vessel.

4.6.2 LPPS and Cleavage Process Assessment

LPPS and cleavage operations still maintain a high degree of bactericidal process conditions based on the solvents in use. The primary differences requiring increased controls and validation of cleaning methods for these processes arise from the fact that these processes have the peptide product in solution and as a solid precipitate.

In these operations, inert and chemical contaminants become significant considerations. Chemical contaminants may react or bind directly to the peptide product while inert contaminants may be retained by the filter with the precipitated peptide. A comprehensive risk assessment should consider a holistic view of the operations, including procedural and operational controls that influence the likelihood of contamination. This would also include removal and detection capabilities of purification processes downstream that generally reduce contaminant propagation.

At this stage, the peptide crude is isolated and dried for further processing downstream, where those processes can eventually transition to being entirely aqueous. As discussed in Section 4.3.1, these transition points must be evaluated carefully within the contamination control strategy as propagation of latent bioburden can present product impact risk. As a result, this is the first process stage where environmental room classification may be considered for evaluation.

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4.6.3 Chromatography Process Assessment

Chromatography operations are generally analogous to those used for oligonucleotides; however, the content is repeated here for continuity of the section. These operations may be performed in a range of organic solvent systems, from fully aqueous conditions to primarily solvent based (typically ACN). Despite this range of solvents, most peptide chromatography buffers and process conditions are either bacteriostatic or bactericidal. This is often the case even under aqueous conditions due to higher concentration of dissolved salts or highly acidic materials. While individual peptide processes, including the prepared buffers, should each be assessed, most peptide chromatography process conditions sufficiently protect the product from biological contamination. Nevertheless, fraction hold times and storage conditions, particularly if aqueous fractions are stored at neutral pH, should be clearly defined with the intention of providing additional protection against microbial proliferation.

Chromatographic separation is effective at removing synthesis related impurities and reaction by-products from the API. This step also likely removes the majority of chemical contaminants, based on the bind-elute mechanism. In addition, pre-filtration of the chromatography feeds prevents suspended solids from entering the column and ultimately contaminating the product. Depending on the process design, salt exchange may also be performed within the bind-elute mechanism.

Chromatography resins are typically dedicated to a specific product. However, controlled resin cleaning may be used if robust risk assessment and validation confirms the practice does not constitute a risk. Leachables and extractables from resins represent a potential concern, and as such, should be evaluated as part of the overall control strategy.

Chromatography column packing is, by nature, an operation that cannot be closed by filtration from a contamination control perspective. Although closed connection transfers can be made from resin slurry vessels to columns, the contamination risk around the entire operation must still be assessed to determine the level of appropriate environmental controls. For the chromatography operation itself, outside of column packing, environmental room classification controls may not be necessary for product protection, provided that a risk-based assessment supports this conclusion.

4.6.4 Downstream Process Assessment

Downstream operations comprise a wide range of steps. As the process transitions to fully aqueous conditions, risk assessments are more likely to identify the need for enhanced environmental contamination controls. For non-aqueous steps, a thorough risk assessment should still be performed considering the potential for biological contamination, while considering the inherent growth inhibiting properties of the process solutions.

For example, in aqueous operations such as ultrafiltration/diafiltration via TFF, particular attention should be given to contamination control. Key elements to be evaluated include buffer feed systems and preparation, TFF system operation, and filter element installation. These aspects should be assessed within the context of the overall control strategy to ensure appropriate layers of protection are in place to deliver a low bioburden drug substance. Most chemical and particulate contamination risks for these steps are mitigated by the process itself although particulate filtration steps may be desirable.

4.6.5 Isolation Process Assessment

Lyophilization is a common isolation step in peptide manufacturing and is considered here in the context of CCS development. The CCS should include evaluation of the loading of the final solution in single-use freeze-drying trays, lyophilization of the material, unloading of the API, sampling, and storage.

Although lyophilization of the API results in a final moisture content consistent with finished product storage Aw (water activity) levels for the product and a bacteriostatic state, the presence of latent bioburden constitutes a risk of bioburden proliferation once the product is rehydrated. Accordingly, downstream bioburden control remains critical.

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Loading and unloading the freeze-drying trays is an inherently open operation, but it may be performed within isolator systems or local HEPA zone. It is common to perform sterile filtration of the process solution prior to lyophilization. Given that single use trays are typically used at commercial scales (vials are also possible), the lyophilizer itself is generally considered non-product contacting equipment. However, cleaning and adequate drying of the surfaces between lots is recommended, even if release testing of the equipment may not be required.

During unloading, single-use trays are often designed to be cut open. Documentation of the risk of tray material shedding during the operation, as well as procedures defining appropriate and safe techniques, is required.

For unloading disposable trays, powder handling and cleaning concerns arise, as the material is a high static, low bulk density material. Disposable isolators reduce the cleaning and potential testing requirements for unloading equipment.

Both filling and emptying of the single-use freeze-drying trays requires a classified space consistent with the drug substance bioburden specification and a risk assessment of the effectiveness of any isolation equipment used. In addition, the peptide powders are often highly hygroscopic; therefore, humidity control of the product-contacting environment is important, at a minimum to ensure consistent levels in the drug substance, even if water content at this stage in the process is not defined as a CQA.

For precipitation and drying in a filter dryer, the process is able to be fully enclosed which decreases the possibility of ingress of contaminants; however, this equipment is inherently multiuse equipment and thus requires validation of proper cleaning between lots to prevent bacterial growth in the equipment or carry over of cleaning agents into the process. Additionally, the lack of ability to directly filter certain precipitate slurries, coupled with the potential for the filter dryer frits to retain particulates may potentially modify the contamination risk profile. Mitigation strategies may include filtration after downstream purification operations and more stringent controls on process closure and environmental considerations.

For processes including humidification of the drying step (e.g., to preserve hydrates), there is potential bioburden ingress that should be assessed for the humidification water, as well as ensuring appropriate filtration and quality of process gases (e.g., nitrogen).

For unloading, isolators provide concomitant product and personnel protection, which is especially important in the case of potent APIs. Disposable (single-use) isolators may be used to remove cleaning requirements, whereas fixed isolator or gloveboxes offer advantages in direct connection to the isolator and reducing likelihood of losing containment on the isolator. The risks and benefits of single use versus fixed isolators should be based on a balanced assessment in the design of the equipment to be used.

4.7 Peptide Facility Carryover/Cross-Contamination Assessment

Peptide manufacturing facilities are often designed for multiproduct operations, and there is critical focus given to prevent carryover/cross-contamination of raw materials, intermediates, and finished products. The use of shared, multiuse equipment is common and must therefore rely heavily on temporal segregation of materials and the implementation of effective, validated cleaning between batches.

This section outlines commonly justifiable cleaning thresholds and key considerations for shared reagent and product-contacting equipment. Establishing and maintaining a robust cross-contamination control strategy is a general expectation and should be supported by risk-based assessments, validated cleaning processes, and appropriate procedural and engineering controls.

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4.7.1 Synthesis Reagent Equipment

Where possible, facilities should dedicate synthesis reagent preparation and delivery systems to the type of reagent (e.g., Fmoc-deprotection solutions), unrelated to which product it will be used to manufacture. If the same formulation is being used in the equipment, depending on the justified reagent expiry, cleaning requirements can likely be reduced or omitted within the justified expiry period per a risk-based assessment. This may involve a regular rinsing strategy with periodic more extensive cleaning cycle justified by carryover and contamination risks established in the CCS. If different formulations of the reagent are required for different products (for example, different piperidine concentrations or an alternative primary solvent to DMF for the Fmoc-deprotection solution), release testing of the solution can be used to establish that the appropriate levels of the reagents/solvent are present.

Such approaches should consider a robust MAC strategy as discussed later in this chapter for each product that takes into account the ability to detect carryover risk (e.g., solution release testing may not be appropriate to detect carryover associated with nitrosamine formation, which would negatively affect subsequent peptide synthesis).

4.7.2 SPPS Reactor and Activation Vessel

The activation vessel is generally not considered a product-contacting piece of equipment and therefore, does not need to be tested for product carryover. However, carryover of the previous amino acid solution to the next amino solution can directly lead to incorporation of an incorrect sequence impurity of identical peptide length. Depending on the nature of the amnio acids used in the peptide sequence, such impurities may be difficult to remove during downstream purification. Therefore, it is imperative to define and validate the minimum rinse volume for these systems to achieve the justified reagent carryover levels. Residual amino acid level reduction of ~5000x compared to the starting solution concentration can typically be justified, based on risk assessment and process capability. Cleaning strategies and any associated testing of the reagent equipment must also include the transfer lines to the synthesis reactor.

While the synthesis reactor may be classified as product-contacting equipment, it is unique in that, during the synthesis process, the peptide remains chemically bound to the solid support and not actually in the liquid phase, contacting the reactor surface. Under these conditions, the risk of product cross-contamination is nearly negligible, as any product remaining is expected to be removed over the course of the synthesis process. A visual inspection of the synthesis reactor can generally be used to demonstrate an acceptable state of control, instead of a specific residue limit.

4.7.3 Downstream Unit Operations

For downstream operations, carryover risks require more rigorous evaluation, as the product is solubilized and in direct contact with the process fluid and can leave residues on the equipment. Under these conditions, residues can resolubilize during later batches, posing a risk of cross-contamination. Accordingly, product contacting stainless steel equipment should be evaluated using a MAC framework as part of a comprehensive risk assessment.

In addition, non-stainless steel product-contact components, such as chromatography resins, must also be assessed for potential carryover in multiproduct or multibatch scenarios where this can impact future products passing through shared equipment. In some cases, resins may be dedicated to individual products.

Any additional downstream reactions (e.g., functionalization steps) require specific focus within the risk assessment to consider alternate reactions and removal of reagents, by-products, and potential degradants from product contact surfaces to prevent unintended carryover into subsequent batches. In the case of lyophilization, many processes are able to both seal the product in trays or vials, often employing single-use technology. In such cases, the lyophilizer itself can often be considered non-product-contacting. However, Iin the case where reusable trays are utilized, the risk assessment needs to consider the potential exposure of the lyophilizer internal surfaces as product contacting and require corresponding cleaning and control strategies.

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The application of MAC principles should be clearly defined within the context of the specific unit operations and product characteristics to form an integrated CCS.

For multiproduct facilities as well as campaign manufacturing, specific attention should be given to the potential carryover of equal length, incorrect sequence peptides from a previous batch.

4.7.4 Product Contacting Equipment

Each facility should perform a product-specific risk assessment to evaluate the potential for cross-contamination across their portfolio of products.

Peptide products have varying degrees of water solubility. While some can be readily removed from polished stainless steel surfaces using aqueous cleaning solutions, others (such as polymer matrices) may require varying levels of organic solvents (e.g., ACN) or more aggressive cleaning approaches to remove the peptide. Various destructive methods of degrading the API should also be evaluated. Therefore, post-cleaning rinse compositions must be evaluated with the peptide to confirm that concentrations are representative of the remaining product in a piece of equipment. For this reason, it is also recommended that worst-case (based on solubility) products be established and used as the basis for multiproduct cleaning validation studies.

Given the high degree of variability in peptide potency and toxicity, the establishment of appropriate MAC limits is essential. A worst-case scenario should be applied, typically evaluating the most highly potent peptide against the maximum daily dosage of the subsequent product manufactured using the same equipment.

Overall, peptide facility cleaning programs should integrate risk-based principles, robust analytical methods, and scientifically justified acceptance criteria to ensure that the total carryover contribution from all shared product-contacting equipment in a process does not exceed this target concentration (carryover/product).

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5 Process Equipment

5.1 Introduction

This chapter focuses on key design considerations for process equipment commonly used in the manufacture of oligonucleotides and peptides. It covers general design considerations for both oligonucleotide and peptide process equipment and design considerations specific to each. Process utilities and support systems are discussed in Chapter 6.

5.2 General Equipment Design Considerations

5.2.1 Materials of Construction

The materials of construction for oligonucleotide and peptide manufacturing processes and equipment vary depending on the specific application. In general, materials of construction are required to be compatible with the solvents and reagents utilized in the process, and must meet standards for equipment used in pharmaceutical manufacturing.

It is ultimately the responsibility of the end user to evaluate material of construction requirements based on material compatibility, chemical concentrations, operating temperatures, hold times, the process application, and to ensure both product and personnel safety.

Metals/Alloys

The most common metal/alloy used in process equipment and piping is 316L-type stainless steel (UNS S31603 or 1.4404). This material is compatible with the majority of chemicals and reagents utilized in oligonucleotide and peptide manufacturing. However, 316L is generally unsuitable for applications involving:

- Corrosive acids: Including dichloroacetic, trifluoroacetic, hydrofluoric, and hydrochloric acid due to risk of corrosion
- Halide-induced pit corrosion: 316L is susceptible to halide pitting corrosion and unsuitable for most applications involving higher concentration halide salts
- Peptide coupling activators: Certain coupling activators can promote pitting corrosion of 316L SS above the liquid/gas interface

Figure 5.1: Examples of Chloride Pit Corrosion with 316L-type Stainless Steel



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When working with highly corrosive solutions, specialty metal alloys with a higher Pitting Resistance Equivalent (PRE) value are often required. AL-6XN (UNS N08367 or 1.4529) is a type of super austenitic stainless steel with improved corrosion resistance against halides used in NaCl and NaBr salt buffer applications. Hastelloy C-22 (UNS N06022 or 2.4602) or Hastelloy C-276 (UNS N10276 or 2.4819) are often used in oligonucleotide process applications involving hydrofluoric acid and dichloroacetic acid (DCA) applications.

For peptide related SPPS and LPPS processes, Hastelloy C-22 and C-276 are also used due to their superior corrosion resistance. However, their use comes with drawbacks, including increased cost and longer procurement lead times.

Depending on the application and qualification requirements, documentation for metal/alloy Material Test Records (MTRs) and surface finish specifications may be required.

Glass-lined and Polymer-lined

Glass-lined and polymer-lined equipment and piping is an alternative solution used in applications requiring increased chemical and corrosion resistance. Glass-lined reactors are often used in peptide LPPS. Glass-lined vessels are also used for DCA solutions used in oligonucleotide synthesis.

Halar (ECTFE) coating of steel is another option that may be a more cost-effective alternative to specialty alloys. However, when using Halar coated steel, additional considerations include avoiding vacuum conditions that could result in delamination. Use of polymer-lined equipment also requires routine maintenance inspections and testing to verify coating integrity, along with periodic recoating as needed.

Polytetrafluoroethylene (PTFE)-lined piping is an alternative to the usage of corrosion resistant metal alloys. In general, PTFE-lined piping may not be suitable for high concentrations of trifluoroacetic acid or non-conductive solvents like MTBE or toluene. The use of non-conductive solvents in PTFE-lined piping may result in static charge build up and unintended electrical discharge within the piping, resulting in safety issues and damage to the pipe lining. When using PTFE-lined piping, special grounding methods are required for static dissipation. Fluid velocity and static dissipation should be evaluated to prolong the life of the piping and to prevent safety issues related to static build up. Usage of PTFE-lined piping includes additional maintenance requirements to regularly inspect lining integrity and monitoring or leak detection systems for mechanical joints.

Elastomers and Polymers

Elastomers and polymers used in the oligonucleotide and peptide manufacturing equipment are required to be compatible with the variety of solvents and reagents used throughout the process. Common materials include:

- PTFE
- FFKM (Kalrez™)
- PFA (Perfluroalkoxy alkane)
- FEP (Fluorinated ethylene propylene), or similar fluoropolymers

Due to its poor compatibility, the use of FKM (Viton™) is not recommended for solutions containing acetonitrile.

Elastomers, such as FFKM and PTFE encapsulated gaskets or FEP encapsulated O-rings, are selected for their broad chemical resistance. FFKM is more expensive but can usually be reused. FEP encapsulated options tend to take a compression set, do not return to their original shape, and are usually single-use. FEP encapsulated O-rings also run the risk of splitting, exposing non-compatible materials to the inner elastomer in applications with routine manipulation or movement. Both materials must be used cautiously with trifluoroacetic acid due to potential degradation over time.

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Depending on application and qualification requirements, certifications for meeting USP Class VI, <88>, <87>, or <665> and Animal Derived Component Free (ADCF) may be required.

Single-Use Components

Single-use components offer both distinct advantages and specific challenges. Their selection over multi use components depends on various factors, including material compatibility, cost, operational flexibility, contamination risk, and environmental impact.

These components are typically made from elastomeric or polymeric materials and are disposed of after use. Compared to other aqueous and inorganic chemistry-based processes, where single-use components can be readily used, they have reduced applications in oligonucleotide and peptide manufacturing. This is primarily due to the reduced compatibility of common biopharmaceutical single use materials, such as High Density Polyethylene (HDPE), Low Density Polyethylene (LDPE), and silicone, with the organic solvents and reagents used.

For smaller operations, single-use PTFE-lined containers and PFA tubing may be suitable for delivering reagents or solutions. However, when PFA tubing is used, flow velocities and the risk of static charge build up should be addressed to ensure safe operation. Traditional biopharmaceutical industry single-use components are typically only used in powder transfer and downstream operations. Prior to use, unique material compatibility and leachable/extractable studies may be required for specific reagents and solvent solutions due to lack of industry information.

5.2.2 Application of Hygienic Design

Hygienic design principles were adopted by the biotechnology industry due to the need to control microbial growth within largely aqueous, growth-promoting processes. The application of hygienic design in oligonucleotide and peptide process equipment varies depending on the stage or unit operation of the process, scale, and commercial availability of suitable components.

Solvent and reagent feed systems do not typically require hygienic design. These systems are treated as chemical delivery systems and are not subject to cleaning. Hygienic design practices related to minimizing dead legs and hold-up volumes are only applicable when piping systems are not dedicated to a particular solvent or reagent.

Elements of the oligonucleotide synthesis process may require hygienic design features, not for microbial control, but to ensure clearance of amidites between batches. For these upstream solvent-heavy processes, biological control is typically not required (see Chapter 4, for additional information). However, equipment and process design for amidite vessels, amidite delivery systems, and synthesizers must be designed to support effective solvent flushing to ensure the integrity of and to prevent cross-contamination of the individual amidite solutions.

As the oligonucleotide manufacturing process transitions from solvent-based to aqueous-based, so does the need to incorporate hygienic design elements. In general, hygienic equipment and piping design principles are used as a standard for all aqueous process/product-contact applications post synthesis. See ASME BPE 2026 [68] for further details on hygienic design principles.

The application of hygienic design principles in peptide manufacturing processes is analogous to the approach for oligonucleotides. Amino acid preparation systems, amino acid delivery systems, and synthesis reaction systems may optionally employ hygienic design principles to maintain process integrity and to prevent cross-contamination of amino acid solutions. Hygienic equipment and piping design principles are used as the standard for aqueous process and product contact applications following synthesis.

5.2.3 Pressure Rating and Relief Devices

The pressure rating of vessels and equipment is an important factor to consider in the design of manufacturing processes. While pressure-rated equipment is not unique to oligonucleotide and peptide manufacturing, it remains a key safety element in overall process design and impacts other utilities such as sizing of emergency relief systems.

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Applicable codes such as International Fire Code (IFC) [69], Occupational Safety and Health Administration (OSHA) [52], American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code (BPVC) Section VIII Division 1 – Rules for Construction of Pressure Vessels [70] and PED [71] must be followed for discharge and hazard management. In addition, performing a thorough analysis of all relevant overpressure scenarios is required to appropriately specify relief devices in accordance to relevant code requirements such as American Petroleum Institute (API) 520/521 [72, 73].

5.2.4 Electrical Classification of Equipment

Due to the usage of flammable solvents throughout the process, certain process equipment may require electrical classification to meet safety and code requirements. Typical electrical classifications may include Class I, Division I or II, or ATEX Zones 0, 1, and 2 certifications for electrical equipment depending on the hazard level. The application of purged control panels or barrier cards for controls instrumentation may also be used depending on the application. See Chapters 9 and 11 for additional information.

5.3 Oligonucleotide Unit Operations and Equipment

Unit operations common in oligonucleotide manufacturing include:

- Synthesis and Reagent Delivery Systems
- Cleavage and Deprotection
- Tangential Flow Filtration (TFF)
- Chromatography and Buffer Preparation
- Annealing
- Concentration
- Conjugation
- Lyophilization

Specific equipment requirements depend on the unit operation and its associated process needs. Key design aspects of individual unit operations are discussed in the following sections.

Automation requirements for equipment also vary between unit operations. Highly recipe-driven processes like synthesis, chromatography, and lyophilization are typically fully automated. The level of automation for remaining unit operations may range from semi-automated to manual, depending on scale, user requirements, and the complexity of the process.

5.3.1 Synthesis

Solid-Phase Oligonucleotide Synthesis (SPOS) is the most commonly used method across the industry for manufacturing oligonucleotides. SPOS is typically carried out at clinical and commercial scale using a static compression column paired with a commercially available synthesizer skid. The static compression column used for synthesis is also commonly referred to as a fixed bed column. The description ”fixed bed” follows general terminology commonly used in chemical reactor design and is not intended to describe the state of the associated SPOS support, since the column may not be fully loaded to accommodate swelling of the support.

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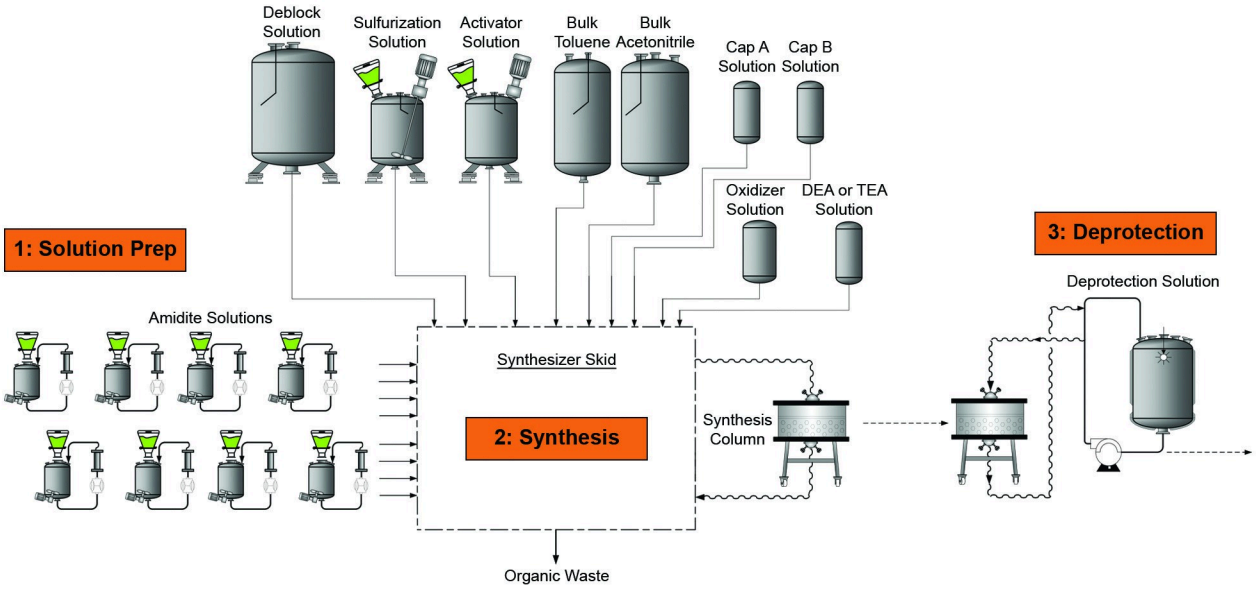
At a high level, a synthesizer skid shares certain features and instrumentation common to a chromatography skid. However, a synthesizer has the added capability to automatically switch between a large number of feed/supply additions, and the final product is accumulated on the column instead of being collected as fractions of the eluent stream. The number of unique amidite solutions and reagents feeding the synthesizer may require 12–24 inlet/feed ports.

Optional system features may also include recirculation through the column to improve coupling, sulfurization, or capping efficiency and a heat exchanger to control synthesis temperature.

In addition to the standard amidite inlet/feed ports needed, other reactant and rinse solution feeds described in Chapter 5 may include:

- Deblock solution
- Activator solution
- Sulfurization solution
- Oxidizer solution
- CAP A capping solution
- CAP B capping solution
- Acetonitrile
- Toluene
- DEA or TEA solutions (optional)
- Ethanol or other alcohol (optional)
- Cleavage solution (optional)

Figure 5.2: Synthesis Process Overview



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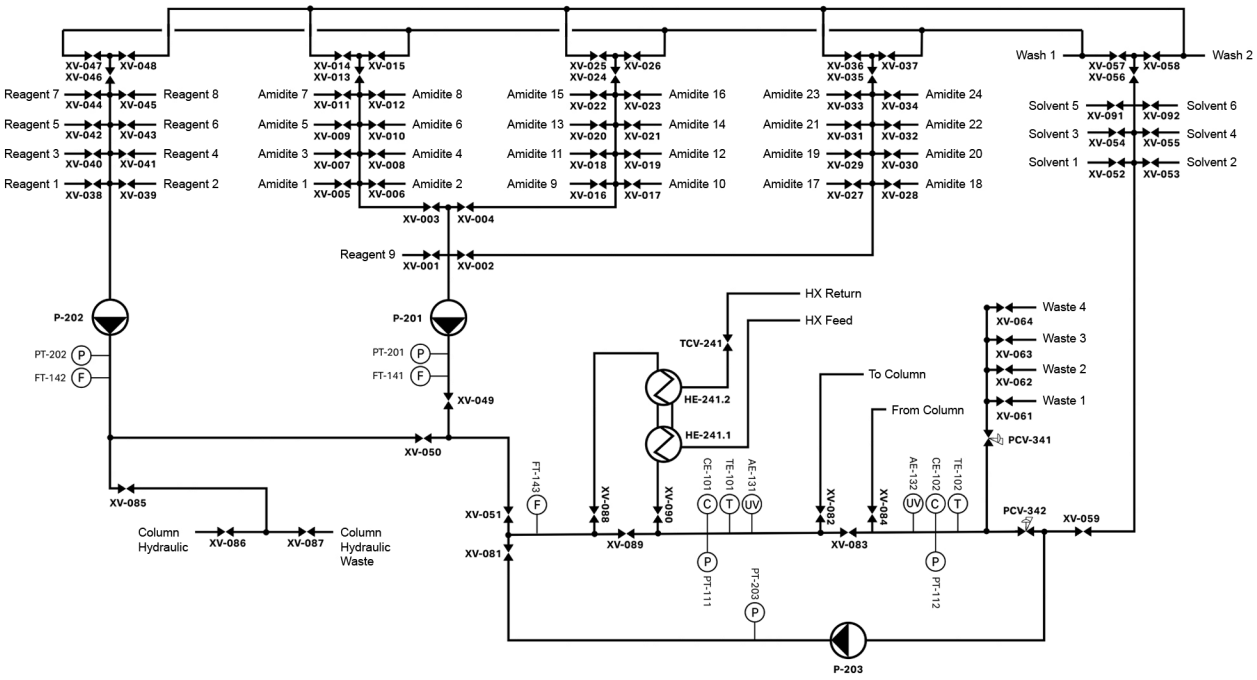
Amidite and reagent solutions along with solvent washes flow through the synthesizer skid, through the synthesis column where the solid-phase synthesis reactions occur. Post synthesis column, solutions and wash solvents return back through the skid for process monitoring and controls.

Instrumentation common to most synthesizer skids includes:

- UV-Vis; Located post synthesis column, and optionally pre column
- Conductivity/Temperature: Located both pre/post synthesis column
- Flow: Located on main feeds/pumps
- Pressure: Located both pre/post synthesis column

In general, the synthesis method or recipe is programmed to deliver specific volumes of amidite solutions or reagents at specific flow rates and/or pressures. Flow and pressure are used to control pumps and valves, while UV-Vis, conductivity, and flow (mass) totalization are used to monitor and control delivery end points.

Figure 5.3: Synthesizer Skid HMI P&ID View Schematic Example
Image courtesy of Cytiva, www.cytivalifesciences.com/en/us.



Spent reagents and solvent streams are typically routed from the skid to designated waste accumulation systems for disposal. Due to local waste treatment requirements, waste streams containing organic solvents used in the synthesis operation are typically collected for off-site processing. See Chapter 6 for additional details. Depending on scale of operation, cost savings evaluation, and sustainability goals, facilities may choose to segregate waste streams. Separate waste streams may include a non-halogenated mixed organic stream, halogenated organic stream, and an optional acetonitrile recovery stream.

The synthesis column is typically designed as a static compression column; however, a Dynamic Axial Compression (DAC) column may also be used. Loading and unloading of solid support is performed manually for a static compression column. If a DAC column is used, the dynamic function of the column can be used to aid in loading and unloading the column support.

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As an alternative to static compression columns, slurry phase stirred tanks or fluidized stirred-bed reactors, similar to those used for commercial peptide production, can be used for oligonucleotide synthesis. While less common, there are a few oligonucleotide products being produced for clinical or commercial use that employ this alternative reaction technology. For the purposes of this Guide, those alternative processes can be treated as functionally equivalent to static compression column technology.

As discussed in Chapter 3, the two main classes of solid-phase synthesis support commonly used in both static compression columns and fluidized stirred-bed reactors are Controlled Pore Glass (CPG) and polymeric (polystyrene-based) supports. Polymeric supports swell or contract when switching between solvents of varying hydrophobicity and gradually swell over the course of synthesis. These properties can result in a two- to three-fold difference between the minimum and maximum bed heights over the course of synthesis and must be factored into the column design and operation.

Synthesizer equipment, instrumentation, and associated preparation and delivery systems need to be electrically classified due to the presence of flammable solvents in the process.

Amidite Preparation and Delivery

Depending on scale, phosphoramidites may be prepped and delivered using glass Schott bottles, portable stainless steel vessels, or fixed stainless steel process vessels. Solid amidite powders are typically pre-weighed and dispensed in a hood or booth with procedural controls to prevent cross contamination.

Amidite preparation requires balances or load cells to measure amidite quantities, capability for manual powder additions, mass or flow totalization for acetonitrile delivery, and the capability to use co-solvents when required. Additional design considerations may include the capability for amidite drying via molecular sieves. Due to the high raw material cost of amidites, minimization of delivery piping line volumes to the synthesizer is also an important design consideration.

Solvent and Reagent Delivery Systems

Solvent and reagent delivery systems are key systems required by the synthesis process. A number of factors go into the design of these systems. See Chapter 6 for further information.

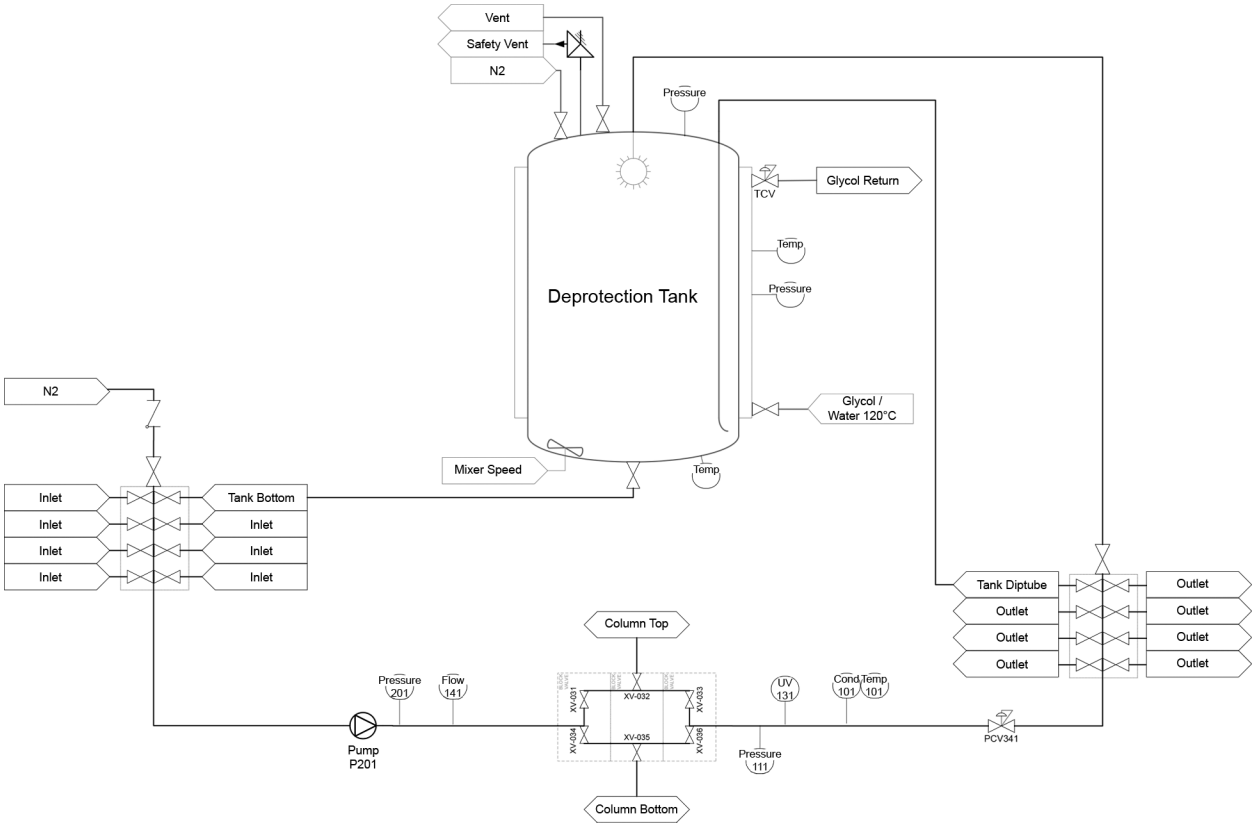
5.3.2 Cleavage and Deprotection

Cleavage and Deprotection (C&D) is a multistep process that can be conducted using various methods, depending on the oligonucleotide, the scale, and the reagents required.

When an on-column C&D strategy is employed, a temperature-controlled, jacketed reactor and custom pump system is connected to the synthesis column containing the support-bound oligonucleotide.

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Figure 5.4: Example of Cleavage and Deprotection System



C&D process equipment is available from the same commercial skid suppliers as oligonucleotide synthesizers. C&D process equipment can also be custom designed and constructed by the end user. These pump systems deliver the C&D reagents to the column under supervisory control using custom software to manage the sequence of flows. A custom C&D system offers 4–8 inlets for reagents and solvents, typically

- A secondary or tertiary amine, e.g., diethylamine or triethylamine (if backbone deprotection is not carried out as part of the synthesis)
- Acetonitrile
- Ammonium hydroxide or a mixture of ammonium hydroxide and methylamine in water
- Water
- DMSO
- Triethylammonium trihydrofluoride (TEA-3HF)

Prior to cleavage from the solid support, the cyanoethyl groups protecting the backbone of the oligonucleotide are removed. After the oligonucleotide is cleaved from the solid support, the resulting solution containing the protected oligonucleotide is then collected in the temperature-controlled reactor and further processed. The exact procedure depends on the nature of the protecting groups to be removed. In its simplest form, the solution is heated for a predefined duration of several hours until all protecting groups are removed, then cooled down and transferred to downstream processing.

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As an alternative to the on-column method, a manual approach may be employed. In this method, the solid support is first unloaded from the synthesis column and transferred to a filter reactor. The oligonucleotide-loaded support is then incubated with a suitable C&D solution for a predefined time and temperature. Subsequently, the solution containing the crude oligonucleotide is filtered off and transferred for downstream processing.

Reagents such as ammonium hydroxide and methylamine used for deprotection can build up pressure in excess of 30 psi upon heating in a closed vessel; thus, the temperature-controlled reactor needs to be pressure and temperature rated. The potential of operator exposure to ammonia or methylamine gas presents a safety hazard. Measures should be taken to protect operators via process containment, ventilation, gas detection systems, and Personnel Protective Equipment (PPE).

The process steps include critical heating, cooling and mixing requirements. Efficient heat transfer and mixing capability are required for scalable and reproducible operations. The C&D solution is first recirculated through the column at a defined temperature until the cleavage reaction is complete. Pumps are controlled by mass flow meters to ensure that the flow rate and the total volume delivery align with set points. Reaction end conditions can be controlled using in-line instrumentation such as conductivity or UV or simply by total volume of reagents and reaction time.

The use of TEA-3HF requires the flow path and reactor to be constructed of corrosion-resistant alloys and perfluorinated polymers and sealing materials to ensure chemical resistance.

C&D equipment and instrumentation need to be electrically classified due to the presence of flammable solvents in the process.

5.3.3 Tangential Flow Filtration

Tangential Flow Filtration (TFF) is a physical separation process that includes washing or using a buffer exchange to remove solvents, unreacted reagents, and chemical by-products from the in-process solution while retaining and/or concentrating the API product. Ultra-Filtration (UF) and Ultra-Filtration Diafiltration (UF/DF) unit operations are carried out using a TFF skid. Commercially available TFF skids and membranes suitable for processing oligonucleotide solutions are available in both semi-standardized and customized skid configurations.

Flat sheet cassettes are the most commonly employed UF filter style for oligonucleotide solutions. Hollow fiber or spiral wound modules are less common. Depending on the process and compatibility requirements with acetonitrile or other solvents, filter membrane materials of construction used are typically regenerated cellulose or polyethersulfone. UF filter pore size varies depending on the length of the oligonucleotide and cut-off values usually range between 1–10 kDa.

Additional design parameters usually include retentate flow and pressure, permeate flow and pressure, cross flow rate, and Trans-Membrane Pressure (TMP). Instrumentation includes flow rate, pressure, conductivity and pH for monitoring of wash end point conditions, and UV for monitoring/avoiding product break through.

TFF unit operations may be employed at various stages of the overall process.

TFF equipment and instrumentation may need to be electrically classified due to the presence of flammable solvents in the process.

5.3.4 Chromatography

Liquid chromatography is a physical separation process that relies on chemical interactions between the product, the stationary phase, and the mobile phase. It is widely used for product purification. Various chromatography methods and equipment platforms have been used to purify oligonucleotides successfully at both clinical and commercial scales. The two primary methods used include Anion Exchange Medium Pressure Liquid Chromatography (AEX-MPLC) and Reverse-Phase High Pressure Liquid Chromatography (RP-HPLC).

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Commercially available chromatography skids are available in both semi-standardized and fully customized skid configurations. Design considerations for a chromatography skid typically include pressure rating, flow capacity, temperature control, and instrumentation required to meet the purification method requirements. These may include instrumentation for flow measurement (both instantaneous and totalized), conductivity, NIR, and pH sensors.

Most commercial skids incorporate onboard method control with the ability to control product elution via step or linear gradients. Product fractions may be collected manually or using various automated control strategies including UV-Vis detection, conductivity, and flowrate triggers. Single-use fraction bags or product collection vessels are typically used for collecting product fractions.

While it is possible to conduct chromatography using non-flammable mobile phase solutions, many clinical and commercial scale oligonucleotide purification processes use buffers containing flammable concentrations of acetonitrile or alcohols. Skid design considerations may need to include electrical classification and other safety requirements associated with flammable solvents.

Buffer Preparation

Buffer solutions used in chromatography operations are typically prepared and supplied using dedicated vessels and delivery systems. Buffers may be batched then directly fed to the chromatography skid or buffer concentrates may be prepared upstream and then fed into an inline dilution skid that supplies the chromatography skid.

Design considerations for buffer preparation systems include:

- Salt or solids material handling
- Vessel temperature control
- Solution agitation and mixing capability
- Sample ports for pH and conductivity adjustment supply
- Adequate pressure and flow capacity to supply the chromatography skid

Due to the high halide concentrations present when using salt buffers, materials of construction are an important design consideration for buffer preparation vessels, piping, and associated equipment and instrumentation. Halide and corrosion-resistant alloys such as AL-6XN and Hastelloy C-22/276 should be evaluated in this application based on buffer composition and solution residence time. Single-use or polymer-based vessels may also be used depending on material compatibility and safety requirements.

Buffer preparation equipment and instrumentation may need to be electrically classified due to the presence of flammable solvents in the process.

5.3.5 Annealing

Annealing is the process of combining two single-stranded oligonucleotides to form a double-stranded molecule or duplex. The strands are mixed in a predefined ratio which is first determined experimentally during small-scale development. The annealing vessel should allow sampling of its contents to confirm that both strands are consumed during the reaction. If either strand is found to be unreacted and in excess, adjustments can be made to ensure a complete reaction before proceeding to the concentration or isolation step.

The annealing process is mediated by heat and requires precise temperature control to ensure reproducibility. Adherence to specified heat-up rates, hold times, and ramp-down rates need to be established during scale-up.

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The reaction is typically carried out in a jacketed stainless steel vessel connected to a temperature control unit. Annealing processes that have a lower maximum temperature requirement may be compatible with single-use equipment.

Design considerations for the annealing process include:

- The ability to weigh or measure single strand additions
- Accurate temperature measurement and control
- Optional measurement and adjustment of pH
- Sampling capability for off-line testing and verification of the single strand ratios

5.3.6 Concentration

Product concentration may be required at various steps throughout the process. Most commonly, these include post deprotection, post TFF/pre-annealing, and pre-lyophilization, depending on the overall process.

The concentration process and equipment may be performed using any one of these approaches:

- Tangential Flow Filtration (TFF)
- Bench top or cart-mounted rotary evaporators (rotovap)
- Simple stirred tank under vacuum evaporation conditions
- Thin-film or wiped-film evaporators

TFF, as previously described, can be used for concentration of the product in addition to performing tangential flow filtration or UF/DF.

Concentration pre-lyophilization assists in maximizing lyophilizer fill capacity. In order to address pre-lyophilization concentration requirements needed to achieve favorable properties for the final product post lyophilization, concentration of the product solution may be required prior to final drying.

General design considerations for vacuum concentration processes include:

- Vacuum pressure and flow rate
- Temperature control and heat transfer rate
- Instrumentation for pressure, temperature, and mass or level measurement

When using a thin-film or wiped-film evaporator, particular attention should be given to cleanability and minimizing thermal exposure to prevent product degradation.

5.3.7 Conjugation

Conjugation is the process of linking an oligonucleotide to a molecule that has targeted bioactive properties. The conjugation reaction can be executed using a variety of equipment configurations. Two common methods include on-column conjugation during SPOS and post-synthesis conjugation in a temperature-controlled reactor.

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Design considerations for conjugation typically include:

- Reactant delivery flow rates or totalization
- Temperature control and agitation
- Instrumentation for flow rate, level, pH, and temperature monitoring

5.3.8 Lyophilization

Lyophilization, also called freeze-drying, is the process used to remove water from the final oligonucleotide product solution and results in a final product in dry powder form. While it is feasible to employ other means to package bulk oligonucleotide containing solutions or use other drying techniques, the most common final step in the production of bulk oligonucleotides is bulk tray lyophilization. This process step is conducted using single-use trays with a water vapor permeable top membrane in a shelf lyophilizer.

Depending on end user requirements, other features that may be considered in Lyophilization equipment specification include:

- Capability for vial freeze-drying with auto stoppering
- Automated loading and offloading capability
- Clean-in-Place (CIP) capability
- Explosion-rated equipment capable of lyophilizing product solutions containing solvents

Isolation of the final dried powder, and the necessary transfer of that powder from the drying trays into final bulk containers, represents a significant contamination risk. See Chapter 4 for additional details on contamination control surrounding dried product handling.

5.4 Peptide Unit Operations and Equipment

Peptide manufacturing processes and equipment can vary throughout the industry. As discussed in Chapter 3, this equipment discussion will primarily focus on the commercial equipment train based on SPPS synthesis while attempting to capture key elements of the LPPS process that impact equipment choices.

Specific equipment requirements are dependent on the unit operation and associated process needs. Typical Critical Process Parameters (CPPs) and Critical Design Elements (CDEs) of individual unit operations are discussed further within this chapter. More details related to CPPs and their correlation to CDEs can be found in *ISPE Baseline® Guide: Commissioning and Qualification (Second Edition)* [74].

Automation considerations for equipment vary between unit operations. Highly recipe dependent processes like synthesis and chromatography are typically fully automated. The level of automation can vary from semi-automated to manual depending on production scale and user requirements. SPPS in particular, is more amenable to highly automated sequences because the elongation sequence can be done in a single reactor.

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5.4.1 Synthesis

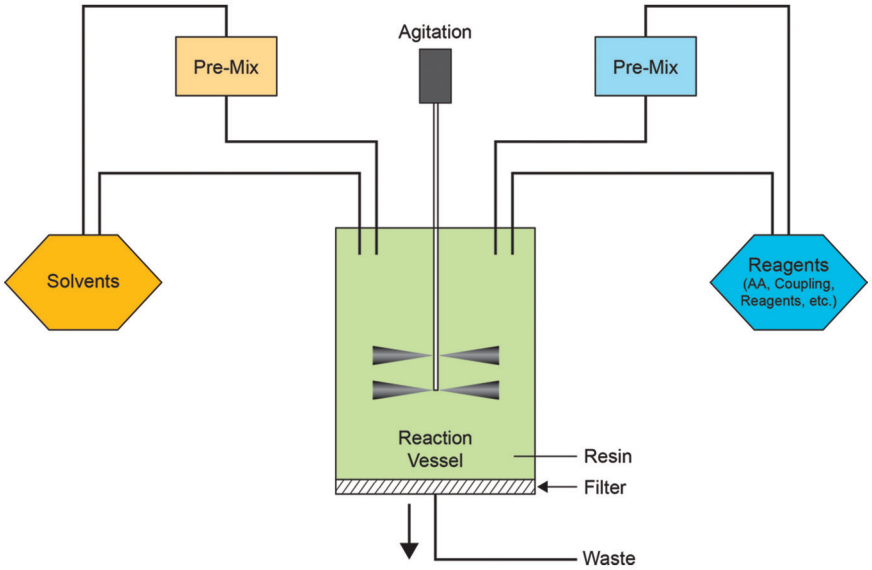
5.4.1.1 Solid Phase Peptide Synthesis

SPPS is traditionally carried out at clinical and commercial scales in a stirred tank fritted/filter reactor/vessel using a polymeric solid support to facilitate stepwise addition of amino acids. Solvents and reagents are pre-mixed in dedicated vessels and transferred to the synthesis reactor/vessel. Reaction feeds to the synthesis reactor/vessel include deprotection solutions, wash solutions to remove deprotection reagents, and amino acids and coupling solutions.

Real-time monitoring of reaction conditions (e.g., deprotection or coupling efficiency) may be achieved through Process Analytical Technology (PAT) integrated into a reactor/vessel recirculation line. Common PAT tools used to monitor the reactions include UV, UV-VIS, Refractive Index (RI), and Raman Spectroscopy. See further discussion in Chapter 12.

A typical schematic configuration of a SPPS reactor/vessel is shown in Figure 5.5.

Figure 5.5: SPPS Reactor System Schematic



5.4.1.2 Liquid Phase Synthesis (LPPS) and Hybrid Synthesis

LPPS and hybrid processes are amenable to traditional, agitated batch reactors. The peptide coupling reaction vessel must be equipped with solvent, reagent and solids charging capabilities, a robust agitation system to ensure solid dissolution, a means of temperature control for the vessel contents, and a sampling system. Because sequential elongation via LPPS typically involves purification and isolation between each addition, additional stirred tanks for a deprotection unit operation and a fritted reactor for precipitation and drying will also be included within the synthesis operation.

Hybrid approaches will typically combine SPPS equipment for elongation of peptide fragments with an LPPS stirred coupling reactor to couple them. The addition of a TFF module to the stirred tank reactor is a recent advance that allows multiple fragment couplings to be telescoped within the same LPPS reactor, rather than requiring purification and isolation as for sequential LPPS elongation.

Forward looking approaches, such as using soluble “tags” to eliminate peptide precipitation and isolation, even for sequential elongation of peptides show promise in further condensing the equipment train for LPPS. This is one example of the technologies discussed further in Chapter 12.

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5.4.1.3 Typical CPPs and CDEs – Synthesis

The following are typical critical process parameters during the synthesis step:

- Temperature: Temperature has a significant effect on reaction kinetics and therefore impacts the time required to complete a synthesis reaction, as well as crude purity
- Reaction times: Reaction times for deprotection, coupling of amino acids, mixing of the washes, solvent/reagent delivery times and drain times can affect the impurity formation and overall peptide quality
- Agitation speed: Adequate agitation is needed to maintain a homogeneous resin mix within the reactor. This also applies to LPPS reactors
- Reagent delivery volumes: Correct ratios depending on the synthesis approach to appropriately complete reactions
- Wash volumes: To properly clear reagents for subsequent reaction steps

The following are typical critical design elements that should be taken into consideration for the equipment to be used for the synthesis step:

- Temperature control: Temperature can be controlled by jacketing the reaction vessel, heat exchangers, heating/cooling mantles, or microwave heating. Microwave heating is typically limited to small scale peptide synthesis (< 20 liters) due to its limitation in uniformly heating large scale peptide synthesis reactors.
- Process times: Process times may be programmed in the automated sequence or handled manually by operators, depending on the degree of automation of the process equipment. Contact time and addition rates are often critical parameters. At larger scales the impact of transfer times should be considered as pumps speeds and transfer line sizing might have an impact on the addition rates.
- Volumes: The amount of solvents and reagents used for each operation must be controlled and monitored to ensure that the correct ratios are obtained for the chemical process. This can be done in multiple ways, e.g., level sensors, flow meters, metering pumps or by using pre-dispensed solids. Pre-dispensed solids can be charged into the synthesis reaction vessel via split-butterfly technology, O-ring canister technology or via the vessel handhole/manway using flexible barrier technology. Solids flow poorly and tend to be light and fluffy; therefore, it is recommended to keep the solid charging apparatus simple.
- Material of construction: Based on reagents used, typical materials of construction for synthesis reactors and pre-mix vessels are 316 L stainless steel and glass- or polymer-lined steel. Specific care should be taken for gasket material in solvent delivery systems used for peptide synthesis, to ensure chemical compatibility with commonly used solvent (for instance DMF).
- Filter screen sizing: Screen must be sized so that the solid in the SPPS operation does not bleed through, but not too small that washing and draining is blinded.
- Reaction vessel geometry: A balance is struck between sufficient mixing while ensuring efficient washing, and will also include considerations for resin swelling characteristics.
- Agitator design: stir blade design, stir speed, stir direction, and sparging using inert gas. If sparging is used then the flow of the gas should be controlled. Presence of the peptide-resin generates a high-viscosity slurry. Grinding of the resin from moving parts inside the reaction vessel must be avoided.

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- Inerting: Inertion of equipment with nitrogen or other inert gas may be used to minimize the risk of vapor explosion by ensuring the oxygen concentration is maintained below the Lower Oxygen Concentration (LOC) of the solvent. If the mixture is oxygen sensitive higher quality nitrogen can be used to achieve lower concentrations and in addition to ensure no ingress of moisture, etc.

5.4.2 Intermediate Processing

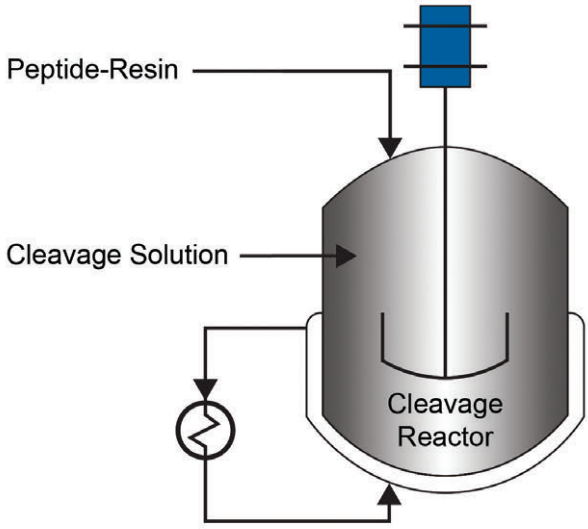
Post-SPPS elongation, the peptide is cleaved from the solid support via a cleavage reactor. For LPPS, the elongation process itself contains unit operations such as precipitation and stirred tank reactors that overlap with SPPS intermediate unit operations described in section 5.4.2.1.

5.4.2.1 Cleavage and Deprotection – SPPS

Following a successful SPPS the intermediate resin will be attached to a fully protected peptide molecule; further downstream processing requires the cleavage of the peptide-resin linker. As discussed in Chapter 3, the cleavage and deprotection steps can be performed together (hard cleavage) or separately (soft cleavage).

Either approach is typically conducted by transferring the peptide containing resin slurry to a dedicated cleavage reactor. A typical reactor schematic is shown in Figure 5.6.

Figure 5.6: Typical Cleavage Reactor Schematic



Typical CPPs and CDEs – Cleavage and Deprotection

The following are typical critical process parameters during the cleavage and deprotection step:

- Temperature: Cleavage and deprotection is an exothermic process and the temperature must be controlled to minimize impurity formation.
- Reaction times: Exposure, particularly to hard cleavage solutions, can lead to formation of unwanted impurities.
- Agitation speed: Adequate bed mixing of the peptide-cocktail slurry is required for reaction completion.
- pH: While hard cleavage often goes through uniform pH reactions, soft cleavage may utilize a variety of acid and base label groups requiring tighter pH controls.

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The following are typical critical design elements that should be taken into consideration for the equipment to be used for the cleavage and deprotection step:

- Temperature controls: For hard cleavage, it is more difficult to control exothermic reactions in larger scale equipment. When scaling cleavage and deprotection operations consider design impacts to ensure appropriate capabilities. Efficacy of jacket control for exothermic reactions decreases with reactor size; pre-cooling of the cleavage solution is a viable thermal control strategy for hard cleavage. There is also consideration of over-cooling which will prevent successful reaction kinetics.

- Process times: Process times can be programmed in the automated sequence or handled manually by operators, depending on the degree of automation of the process equipment. Contact time and addition rates are often critical parameters. At larger scales the impact of transfer times should be considered as pumps speeds and transfer line sizing might have an impact on the addition rates.

- Material of construction: Cleavage and deprotection reagents are often corrosive, toxic, and pose EH&S risks. Special care is needed to ensure chemical compatibility with the chemicals used. Refer to Section 5.2 for more details on the choices of materials for both equipment and elastomers. Typical materials of construction for cleavage reactors are Hastelloy C22 and/or polymer-lined steel.

- Reagents: Both sufficient quantities and composition are required to cleave, solubilize and scavenge the reaction. Sensitivity of reagent composition is more rigorous in soft cleavage as protecting groups need to be preserved.

- Agitator design: Presence of the peptide-resin generates a high viscosity slurry. The agitator should be designed to ensure a homogeneous solution.

5.4.2.2 Functionalization

For peptides that utilize oxidation/cyclization operations, the equipment should be designed in accordance with the stage of the process and the chemicals in use. In solution oxidation reactions use an additional stirred, temperature controlled vessel.

On-resin oxidation/cyclization can use the associated synthesis or cleavage equipment. The selection of the equipment will mostly be based on the materials of construction of the equipment used and its compatibility with the oxidizing agents in use.

Typical CPPs – Functionalization

The following are typical critical process parameters during the functionalization step:

- Temperature: Impacts the oxidation reaction rate and impurity formations

- Reaction time: Ensures that the desired oxidation has taken place

- Mixing speed: A controlled stirring speed ensures a uniform oxidation step

- pH: Potential impact on the oxidation efficiency and therefore product quality

Typical CDEs – Functionalization

The following are typical critical design elements that should be taken into consideration for equipment to be used for the functionalization step:

- Temperature and pH control: The vessel may need temperature control and/or pH probes to maintain the necessary conditions for the oxidation.

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- Process times: Process times can be programmed in the automated sequence or handled manually by operators, depending on the degree of automation of the process equipment.

- Material of construction: Usage will depend on the conditions needed for the oxidation/cyclization. The presence of stronger oxidizing agents (e.g., Iodine), may require the use of more resistant materials such as plastic or glass-lined vessels.

5.4.2.3 Precipitation and Drying

In the purification process stage, peptides are typically precipitated and dried. As discussed in Chapter 3, filter dryers, centrifuges, or cone dryers may be used. In addition, lyophilization can be used at this stage of the process, though it is more prevalent during the final isolation step.

Precipitation can be conducted in a separate mixed, temperature controlled vessel or in a filter dryer that combines the solid-liquid separation with the filtration and drying operation.

In both cases, product solution and anti-solvent are transferred into the vessel under agitation and cooled to facilitate the precipitation.

Typical CPPs and CDEs – Precipitation

The following are typical critical process parameters during the precipitation step:

- Temperature: Solubility is highly impacted by temperature as well as morphology.

- Addition time: Addition rates drive both morphology and have an exothermic impact.

- Agitation speed: Agitation conditions have a significant impact on morphology.

- pH: This is product dependent, but many peptides have their lowest solubility at their iso-electric point.

The following are typical critical design elements that should be taken into consideration for the equipment to be used for the precipitation step:

- Temperature control: Temperature is in general controlled by equipping the reaction vessel with a jacket. It should be taken into consideration that the efficacy of jacket control for exothermic reactions decreases with reactor size.

- Process times: Process times can be programmed in the automated sequence or handled manually by operators, depending on the degree of automation of the process equipment. Contact time and addition rates are often critical parameters. At larger scales, the impact of transfer times should be considered as pumps speeds and transfer line sizing might have an impact on the addition rates.

- Volumes: Incorrect anti-solvent volumes may lead to incomplete precipitation. Both sufficient quantities and composition of reagents are required to precipitate the peptide. Flow meters and metering pumps are commonly used.

- Monitoring instrumentation: PAT tools such as Raman Spectroscopy, FTIR, PVM, and FBRM can track particle formation; simpler tools such as refractive index, turbidity or temperature can also provide valuable data.

- Material of construction: Cleavage reagents are often corrosive, toxic, and pose EH&S risks. Special care is needed to ensure chemical compatibility with the chemicals used. Typical materials of construction for reactors used for the precipitation step are Hastelloy C-22 and/or polymer-lined steel. 316L SS may be more appropriate during isolation phase precipitation reactions where MTBE is a typical solvent.

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A filter dryer is similar to a filter reactor used in SPPS. It consists of a domed pressure vessel with a flat bottom fitted with a sintered mesh (typically 20 µm). The slurry is transferred into the vessel, solids settle, and pressure (0.5–3 bar, usually nitrogen) forces the supernatant through the filter. The agitator smooths the cake surface to prevent cracks and short-circuiting. After washing, the vessel is evacuated and heated for drying. Towards the end of the process, the agitator breaks the cake to accelerate drying. Once complete, the side discharge hatch is opened, and the agitator transfers the dried material for collection.

A horizontal centrifuge processes the batch in multiple sections, reducing cake height to 10–25 mm, which allows faster filtration and washing. For poorly filtering materials, this may require splitting the batch into more subsections, but overall processing time is typically shorter than using a single filter dryer and results in more consistent material. The subsections are then typically combined for subsequent drying via vacuum dryer or an equivalent.

Typical CPPs and CDEs – Intermediate Drying

The following are typical critical process parameters during the intermediate drying step:

- Temperature: Heating is required for efficient and effective drying while being balanced with product temperature stability.
- Pressure: Medium to high vacuum conditions are required for effective drying and to balance the thermal exposure of the peptide with operational time.
- Cake depth: Equipment design ranges for the product influence cake depth: too thin can lead to ineffective washing, while too thick produces a high pressure drop.
- Cake integrity: Improper formation and preservation of the cake bed can result in cracking and channeling, which results in ineffective washing.
- Rotational speed during filling and final rotational speed (centrifuge): For optimizing separation and removal of residual liquid.
- Sample filling weight (centrifuge)

The following are typical critical design elements that should be taken into consideration for the equipment to be used for the drying step:

- Temperature control: Temperature is in general controlled by equipping the reaction vessel with a jacket.
- Filter screen sizing: Screen must be sized so that the solid does not bleed through, but not too small that washing and draining operations blind the filter material.
- Material of construction: Reagents may be corrosive; therefore, use of higher alloys should be considered.
- Agitator design and motor power: High torque requirements for solid filters and stroke lengths, agitator considerations for cake smoothing and discharges.
- Discharging: Solid powder will need to be removed, consider discharge methodology from a yield, ergonomic and containment perspective. Active crude API is typically present at this point, making operator exposure a consideration, especially for potent compounds.

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5.4.3 Purification

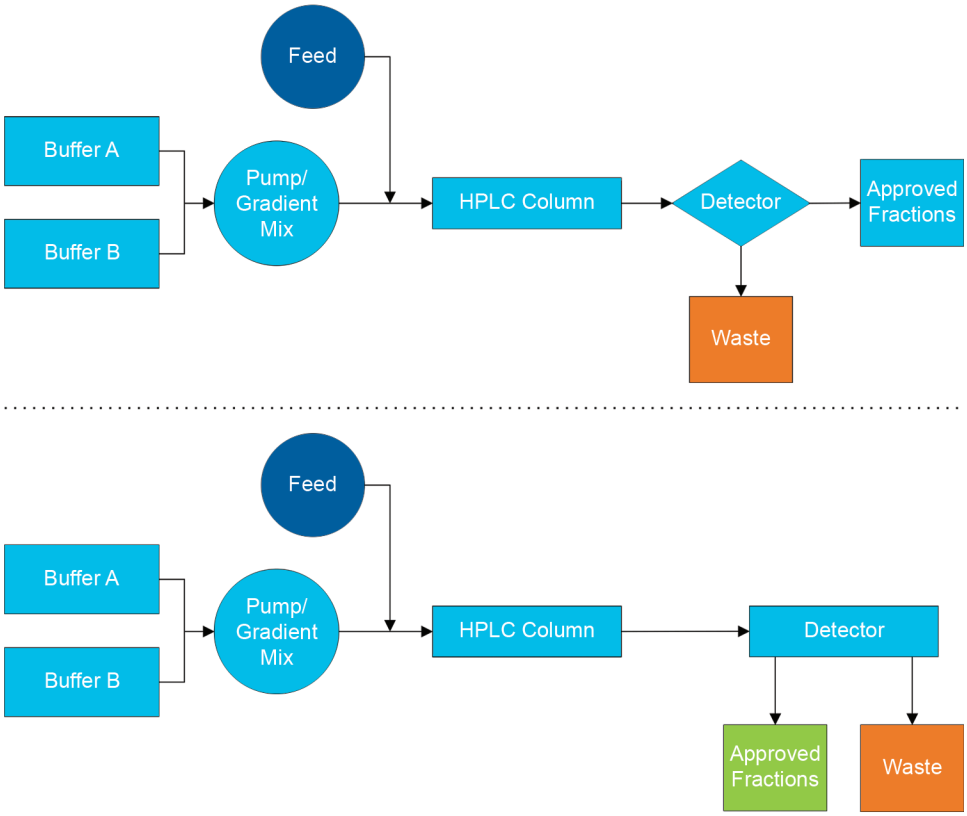
5.4.3.1 Chromatography

In chromatography, the crude peptide product is typically resolubilized or transferred to pool tanks feeding the chromatography system. As discussed in Chapter 3, the unit operations required will vary, but will typically include RP-HPLC. These vessels feed the chromatography flow skids connected to the columns associated with the step and the output flow is collected and segregated into waste and product portions. The method of segregation can vary from collection into smaller portable vessels and use of offline analytics (HPLC/UPLC analysis), to use of at-line PAT, to fully in-line PAT enabling outlet diversion to waste and product streams. See Figure 5.7. Concentration operations can be performed using chromatography or Tangential Flow Filtration (TFF).

With multiple recycles potentially occurring within chromatography operations, reloading the material to the column with a minimal number of loads results in a more concentrated solution.

Buffers are required to feed into all purification and salt exchange systems. At smaller scale, buffer may be prepared manually, but at larger scale, automated buffer tank preparation systems with appropriate supply tanks should be considered.

Figure 5.7: Typical HPLC Feed and Fraction Analysis Schematics



In-line dilution is possible for chromatography operations to reduce buffer hold tank sizes. It is more easily utilized for IEX than HPLC as the salt dilution has fewer complicating factors than in-line-mixing of water with organics.

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Typical CPPs and CDEs – Chromatography

The following are typical critical process parameters during the chromatography step:

- Process temperature for chromatography
- Process flow rates for chromatography
- Gradient design
- UV-Vis wavelength accuracy and response for chromatography
- pH of solutions

The following are typical critical design elements that should be taken into consideration for the chromatography step:

- Material of construction: In general chromatography systems are constructed in stainless steel. The impact of halogenated salts (chloride concentration) should be considered for stainless steel; a higher nickel alloy might in some cases be required.
- Temperature control: Temperature is in general controlled by equipping the chromatography system with heat exchangers.
- Flowrates for chromatography: Flow rate must be measured and regulated for in line dilution and gradients.
- Gradient design: The system shall be able to mix a gradient without gas bubbles reaching the column. Density or conductivity monitoring can be used to monitor the gradients.
- Fraction collection: Fractionation may be automated or handled manually by operators, depending on the degree of automation of the process equipment. The equipment used could vary from fixed pipes and jacketed tanks (for cooling the fractions) to simple mobile vessels.

5.4.3.2 Concentration and Size Exclusion Operations

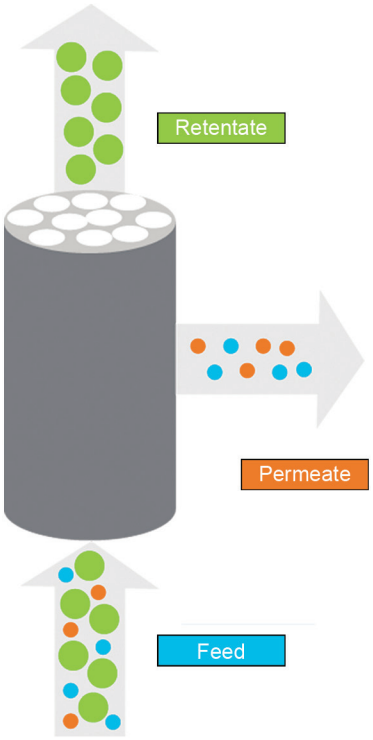
Tangential Flow Filtration (TFF)

The TFF unit operation has the benefit of allowing the process to concentrate the liquid stream using a coarse size-exclusion filter (UF) and remove organic solvents (from HPLC) or excess non-volatile salts (from IEX) during a buffer exchange process (DF) facilitated in the same equipment. These capabilities are not typically achievable using chromatography equipment alone.

The different types of membrane filtration are classified according to their membrane pore size, for example ultrafiltration (0.01 to 0.1 µm) and nanofiltration (0.001 to 0.01 µm). A typical operation recirculates the contents of a tank past the membrane filter at a relatively high pressure and returns the flow back to the tank, with the stream that has passed through the filter being referred to as the permeate and the stream returning to the tank referred to as the retentate. One of the critical parameters is the pressure difference between the retentate and permeate sides of the filter – the TMP.

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Figure 5.8: Tangential Flow Filtration Diagram



Ultrafiltration (UF), Diafiltration (DF), and Nanofiltration (NF) Considerations

A critical factor in UF performance is membrane selection. While pore size is the primary consideration, membrane structure and composition also influence behavior due to interactions with the product. During operation, a gel layer of solids forms on the membrane surface, affecting sensitivity to TMP, temperature, and solids concentration. If solids concentration becomes too high, this layer can severely restrict permeate flow. The maximum achievable concentration depends on peptide solubility and the solvent system, which ultimately determines the operational limits for UF.

For NF, selection of the pore size of the membrane with regard to the molecular weight of the desired product and those of the non-desired impurities is critical both for purification performance and to prevent against low yields due to loss of product to the permeate stream. Previously the material of construction of the membranes used would not have been suitable for prolonged contact with the solvents present, but materials compatible with the likes of DMSO, ACN, DMF, ethanol, methanol and DEA are now becoming available.

TFF equipment for peptide use should use temperature control to prevent excess heating through friction of the recirculating solution. For any peptide that has risks of aggregation, TFF equipment should be designed with lower shear pumps.

Typical CPPs and CDEs – Concentration and Size Exclusion

The following are typical critical process parameters during the concentration step:

- Process flow rates for TFF
- Process pressures (Transmembrane Pressure) for UF/DF
- Conductivity (or alternative end point detection) for UF/DF

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- Solution temperature (for both viscosity and product stability)

The following are typical critical design elements that should be taken into consideration for the equipment to be used for the concentration step:

- For UF/DF: In-line conductivity in the retentate and/or permeate lines may be used to monitor the process of desalting or concentration operations.
- Pressure for UF/DF: Measurements from the feed, retentate, and permeate lines from the filters are needed to optimize the filtration process.
- Flowrates for UF/DF: Flow into the filter system is also critical to ensure proper cross-membrane flow rates are maintained.
- Optical density in permeate helps detect potential product breakthrough.

5.4.4 API Isolation

5.4.4.1 Precipitation

Similar to the approach discussed for drying of the peptide intermediate before purification, precipitation is often required for final isolation and filter dryers can be used as the final isolation drying step. Similar parameters and equipment considerations apply as for the intermediate isolation discussed earlier in this chapter. With the exception of unique GMP considerations that can be understood through discussions in Chapter 4, users should reference the intermediate precipitation requirements in this chapter.

5.4.4.2 Final Drying

As detailed in Chapter 4, the final API isolation is often demarcated by a bioburden-controlling process closure operation such as a 0.22 µm filter.

This filtration should occur as close as possible to the final isolation unit operation with consideration for the cleanliness and process exposure environment. The equipment used for this should be single use and/or hygienic by design to ensure additional bioburden is not introduced in the filtration process.

Organic solvents are typically used to drive peptide precipitation in the final steps of drug substance isolation. As a result, residual low levels of organic solvents must be controlled to their ICH limits in the isolated product. Given that precipitation processes have the propensity to occlude residual liquid matrix components, the drying step is critical for this residual solvent control. Humidified nitrogen drying is effective in facilitating the removal of residual organic solvents at the final stages of drying. Nitrogen, having passed through a humidification system, can be adjusted to a sub-saturated condition by blending with dry nitrogen prior to introduction to the drying equipment. Residual moisture in the drying gas can act to modestly plasticize the amorphous product, providing a means of facilitating the diffusion of residual organic solvent to the surface of the drying peptide particles.

This method has been demonstrated to effectively drive down the levels of residual organic solvents to acceptable limits and can precede a dry nitrogen drying step to reach acceptable limits for both organic compounds as well as residual water in the final isolated peptide.

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Lyophilization

As discussed in Chapter 3, lyophilization has long been one of the most common ways to isolate final API peptides, and as discussed in this chapter, can also be used for the isolation of intermediates. The process requires loading trays (bulk), single use trays with permeable membranes or vials with the product solution and then running a pre-defined cycle to freeze and sublimate the solvents from the product.

Typical CPPs and CDEs – Isolation, Lyophilization

The following are typical critical process parameters during the lyophilization step:

- Chamber and shelf temperature: Critical to maintain the product temperature below the eutectic point and must be shown to be uniform both across and between shelves
- Chamber pressure: Maintained accurately to ensure it is sufficiently low during the lyophilization process to drive the drying process
- Drying Time and Temperature: For both primary and secondary drying cycles
- Rate of temperature change during the cycles
- Cycle length: Minimum drying cycle length with a determined maximum residual moisture must also be maintained throughout batches

The following are typical critical design elements that should be taken into consideration for equipment to be used in the lyophilization step:

- Chamber material of construction: Must withstand extreme temperatures and vacuum condition, stainless steel is often used.
- Seal material of construction: Silicone and EPDM are common for lyophilizer seals as a vacuum tight seal will outweigh needs for chemical compatibility since there is only vapor contact with the seals.
- Single use components: May be utilized and are common for lyophilization trays. For lyophilization trays, the plastics should be confirmed to be compatible with the used solvents.
- Heat distribution within shelves: Temperature distribution within the chamber is important to ensure a uniform heating process.
- Condenser capacity and temperature: Must be dimensioned for the intended load as temperature gradient and accumulation of ice are important factors in the maintenance of the appropriate vapor pressure gradient central to the operation.
- Vacuum performance: Vacuum level attainable affects sublimation rate and product temperature; insufficient vacuum can cause melt-back or collapse.

Filter Dryers, Centrifuges, and Cone Dryers

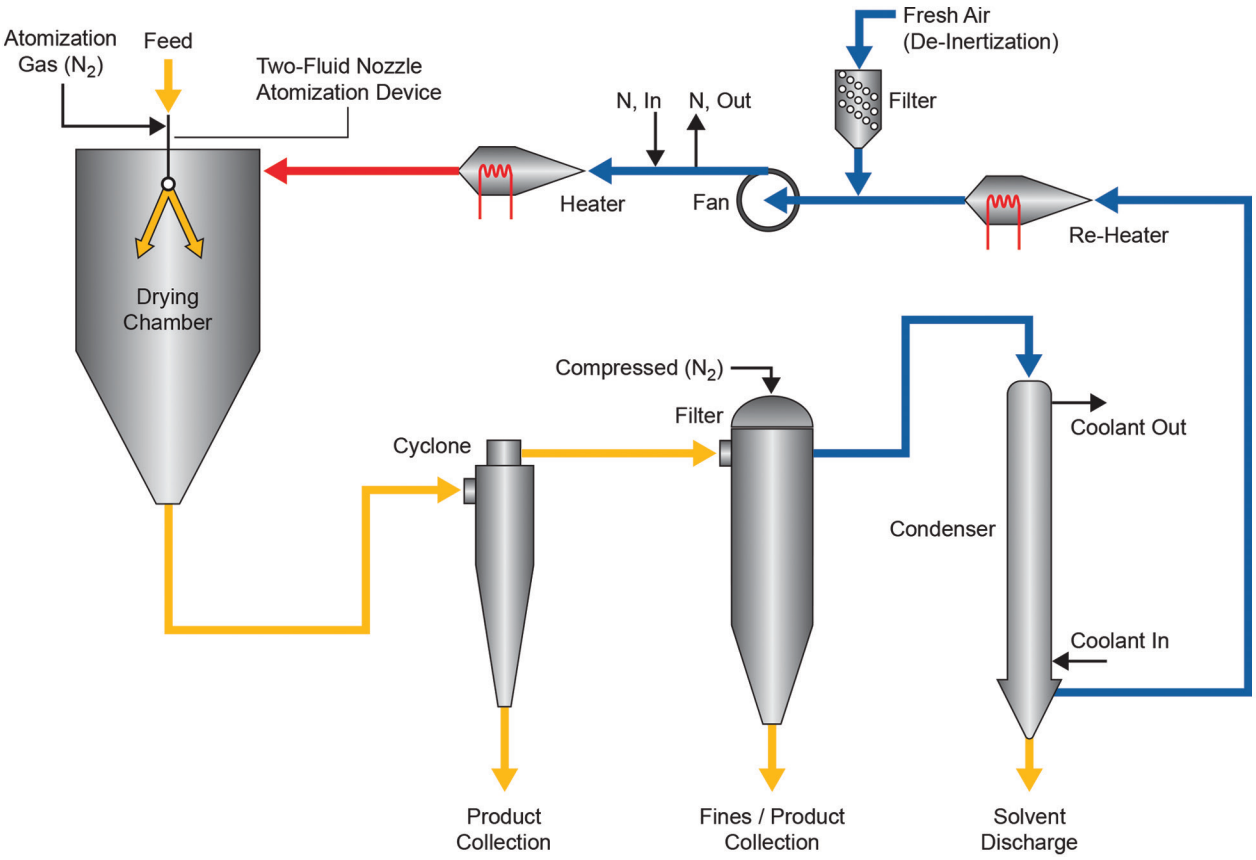
Isolation phase drying equipment of this type should have similar considerations as those outlined in the intermediate drying section, with the exception of GMP considerations, which are discussed in Chapter 4.

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Spray Drying

Spray drying takes solution, emulsion, or slurry containing the peptide API, atomizes it into fine droplets and feeds it into a drying chamber with a temperature-controlled gas stream. The droplets are rapidly evaporated with the gas as they move through the drying chamber forming solids which are collected in a cyclone separator or bag filter. By balancing the gas stream, residence time, and atomization, the effective temperature the solid API experiences is minimized, making spray drying a suitable high-throughput and continuous process for thermally sensitive products such as peptide APIs.

Figure 5.9: Flow Diagram of General Spray Dryer



Typical CPPs and CDEs Isolation – Spray Drying

Critical process parameters in spray drying will focus on defining and controlling the evaporation rate in the drying chamber to achieve a balance of throughput and thermal stability in steady state conditions. This is achieved by balancing the inlet gas mass flow rate and temperature against the atomized mass flow rate. The following are typical critical process parameters during the spray drying step:

- Feed gas mass flow rate (kg/hr): The higher the mass flow rate the higher the evaporative capacity.
- Inlet gas temperature (°C): The higher the gas inlet temperature the higher the evaporative capacity.
- Outlet gas temperature (°C): The principle of spray drying assumes that the drying chamber is at this temperature; this is therefore a useful controlling parameter of internal conditions.
- Droplet size (µm): The smaller the droplet sizes the higher the surface area to volume ratio which results in higher evaporative capacity.

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- Evaporation temperature and pressure of the bulk solvent: Although not a direct parameter, the physical properties of the bulk solution solvent will have major impacts on the dryer capacity as the different heats of vaporization will drive the minimum inlet temperatures required for evaporation.

- Residence time: A critical scaling parameter, it is calculated as the volume of the chamber divided by the gas flow rate.

In spray drying it will be critical to ensure the scalability of critical parameters, and this will be the main driver behind equipment conditions. The following are typical critical design elements that should be taken into consideration for equipment to be used for the spray drying step:

- Nozzle design: A variety of nozzles are available for spray drying which primarily change droplet size and injection rates. Smaller droplets produce spherical particles, while larger droplets produce toroidal particles. Nozzle type and resulting spray pattern can drive minimum diameter requirements to ensure wall encrustation does not occur.

- Chamber diameter (m): Chamber diameter has a large impact on residence times and certain minimum diameters are required based upon the desired particle size.

- Surface finishes: Ideally high surface finishes are used to help prevent encrustation and build up on drying chamber or collection surfaces.

- Volumetric: Surface Scaling: Control of drying chamber parameters is monitored by outlet gas temperature, consider the impact of surface area, materials of construction, and insulation in maintaining this control.

- Explosive and flammable gases: High volatility organics have noted uses in their lower evaporative temperature but come with flammable and explosive risks; this is controlled by using a N_2 gas stream.

  - Entrained flammables in N_2 must be condensed out

  - System must be kept oxygen free

- Collection system: Consider size and ability to swap in-process collection reservoirs based upon desired batch sizing and run times.

5.5 Final Product Packaging

Powder packaging into final containers can employ several different equipment options. Higher classification environments or closed environments (such as isolators and continuous liners) will typically be used based on the product risk assessment as discussed in Chapter 4, with the latter also providing a way to address higher-potency drug substance and general operator exposure considerations. For non-potent API, downflow booths may be employed from a product environment perspective (e.g., lyophilizer tray unloading).

Isolators may be equipped with humidity control and/or inert gas purging of the packaging space if required by the product. Suitable for larger bulk packaging operations, continuous liner systems may be more versatile than isolators, with humidity control aspects managed via a connected filter dryer or spray dryer.

Choosing between disposable flexible isolators and fixed isolators should take into account the size and features required for the operation as well as the potency of the peptide API. Flexible isolators allow for single use and easy disposal without cleaning, but may not offer sufficient containment for particularly highly potent compounds.

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5.6 Equipment Commissioning and Qualification

Process equipment used for GMP manufacturing of oligonucelotides and peptides requires documented Commissioning and Qualification (C&Q) activities. See *ISPE Baseline Guide: Commissioning and Qualification (Second Edition)* [74] for additional information on process equipment commissioning, qualification, and validation.

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6 Process Utilities and Support Systems

6.1 Introduction

A variety of process utilities and support systems are required for oligonucleotide and peptide manufacturing. These include utilities common to API manufacturing such as reagent and solvent delivery systems, process vent systems, emergency relief, and process waste systems, as well as utilities that are typical in biopharmaceutical manufacturing, such as pharmaceutical grade water, clean compressed gases, and Clean-In-Place (CIP) systems.

6.2 Reagent and Solvent Delivery Systems

Reagent and solvent delivery systems are a key component of oligonucleotide and peptide manufacturing facilities. Design factors related to these systems, including pressure/delivery and control systems, often impact the overall design and layout of the facility due to applicable code requirements. See further discussion in Chapter 8.

6.2.1 Delivery Systems and Solution Preparation

For larger scale production facilities, it may be preferred to supply synthesis-related solutions and reagents from bulk storage vessels. These vessels may be located remotely in a tank farm or in intermediate utility areas. Reagents may also be supplied from Intermediate Bulk Containers (IBCs) to an oligonucleotide or peptide synthesizer through hard-piped distribution systems. Automated delivery systems with integrated level and pressure monitoring mechanisms are often used to ensure a consistent supply of solvents and reagents. Switching systems for intermediate bulk containers are also frequently employed.

Reagent solutions can be sourced pre-formulated from specialty chemical suppliers or prepared on-site. Pre-made solutions are typically supplied in stainless steel pressure-rated intermediate bulk containers or in single-use totes connected to a fixed distribution system. For commercial scale production requiring large volumes of reagents, preparation of reagent solutions on-site from pure starting materials can be advantageous from a cost of goods perspective. The solutions used in synthesis are generally miscible liquid-liquid mixtures or easy to prepare dissolved solids in organic solvents, all of which can be prepared in agitated tanks or even be mixed in-line within the synthesizer.

6.2.2 Storage Conditions

Proper storage conditions are necessary to maintain solvent and reagent purity. Vessel inerting (e.g., dry N_2 blanketing) is typically used to both prevent moisture ingress, which may impact product quality, and to prevent ingress of oxygen as a protection against flammability hazards. Inerting systems are also used as part of solvent and reagent transfer, either by providing motive pressure for the transfer or by replacing the head space volume as solvents and reagents are consumed.

Consistent pressure control of the reagent solutions feeding the synthesizer is important. Depending on scale and pressure transfer requirements, bubble traps or membrane/vacuum-based degassers may be incorporated into the reagent delivery system to prevent entrained gases from entering the synthesizer or static compression column.

In general, if reagent delivery systems are housed in temperature-controlled facilities, additional temperature controls are not required. However, if reagent delivery or storage systems are located in non-temperature-controlled environments, additional temperature controls for reagents may be required.

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6.2.3 Piping and Equipment Considerations

Piping and valving for solvent delivery systems may have additional design considerations depending on code requirements such as ASME B31.3 – Process Piping [75]. Piping systems are typically fully welded in non-process interstitial spaces that do not have secondary containment or the ability for leak detection. As applicable, reagent piping systems may be designed to be drainable and to avoid dead legs/hold-up to support safe maintenance activities and facilitate change-over if the piping distribution is not dedicated to a single reagent. Valving within solvent piping systems may require fire-safe ball valves with chemically compatible elastomers. Fire-safe ball valves use ceramic or other non-flammable materials to seal off flow of flammable fluids in the event of a fire.

Magnetically coupled centrifugal pumps are commonly used in solvent distribution systems. Close coupled magnetic drive pumps offer leak-free operation by eliminating the need for mechanical seals, significantly reducing the risk of leaks and making them ideal for handling hazardous, corrosive, or toxic fluids. A key limitation of magnetically coupled centrifugal pumps is that they must remain liquid-filled at all times during operation or are prone to catastrophic failure.

6.3 Process Vent Systems

Depending on the scale of oligonucleotide and peptide manufacturing operations, a process vent system is utilized to manage flammability hazards, personnel exposure, and environmental emission control requirements associated with the use of volatile organic solvents.

Process vent systems divert inert gases and chemical vapors from the process equipment to a safe discharge location. Depending on local regulatory requirements and corporate environmental policies, process vent streams may also be tied to an emission control system to remove Volatile Organic Compounds (VOCs) before releasing emission streams to the atmosphere. Examples of emission control technologies include heat exchangers that condense chemical vapors into a liquid waste stream, adsorption systems such as activated carbon beds, and thermal oxidizers for incineration of VOCs.

To prevent a flammable environment within the process vent system, additional considerations for the process vent design may include the use of a low flow inert gas purge or sweep. Exhaust fans may also be used to maintain a slightly negative pressure environment encouraging flow away from process equipment.

6.4 Emergency Relief Systems

An important safety related system in oligonucleotide and peptide manufacturing is the emergency relief system. The emergency relief system connects overpressure relief devices throughout the process and routes chemical vapors and liquids to a safe location during an overpressure event. The emergency relief system prevents exposure of personnel to harmful chemicals and reduces the severity of a fire by diverting flammable solvents away from a fire zone. Individual relief devices are typically introduced into a central relief header using a top-feed “goose neck” connection, and the central relief header is sloped through the facility towards an emergency relief catch tank.

Designing these systems requires detailed calculations pertaining to the relief devices, piping system, and collection tanks and it is recommended that pressure relief specialists perform the design due to the several pressure relief scenarios that may be present within the process.

When two-phase flow relief scenarios are identified, additional design considerations may be required to either eliminate two-phase flow or to include a collection vessel to separate the vapor and liquid streams during a relief event. Additional resources for emergency relief design include: DIERS [76], API 520/521 [72, 73], NFPA [77], and ISO 4126 [78].

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6.5 Process Waste Systems

In oligonucleotide and peptide facilities, the use of organic solvents generally dictates that no process waste streams route to standard underfloor sanitary sewer systems and are instead collected. In peptide manufacturing, potency and toxicity concerns may also be a driver for waste collection and require strict containment boundary definitions around these compounds within a facility. As processes move towards aqueous conditions, traditional underfloor waste routing to pH balancing or neutralization systems may be used.

Process waste is typically segregated and collected for off-site processing. Based on local off-site waste processing options, segregation strategies may include separate organic waste and aqueous waste systems. Organic waste streams contain high organic solvent concentrations, while aqueous waste streams contain higher water content. In some cases, depending on the cost or disposal capability of local disposal contractors, it may be advantageous to segregate the halogenated waste as well.

Waste collection methodologies may consider different strategies to balance cost and facility implications beyond the facility arrangement considerations covered in Chapter 7. Use of small surge equipment and lift stations can manage space and cost effectively as operational scale increases, effectively pushing larger storage tanks, either above ground or underground, away from the process spaces to the exterior of the facility where waste handling can be more easily managed.

Opportunities for solvent collection and reuse within the synthesis process may be considered. Additional information related to direct solvent reuse and external solvent recycling is provided in Chapter 12.

Waste streams free of organic solvents are typically collected for on-site processing. This mainly includes waste generated from CIP systems. A waste neutralization system adjusts the pH of the waste stream and sends the neutralized waste stream to the sanitary sewer, which is further treated by local wastewater treatment facilities. Initial rinses of CIP cycles may need to be diverted to the aqueous waste collection for off-site processing to prevent the accidental release of non-compliant chemicals/concentrations.

Depending on code requirements such as ASME B31.3 [75], piping for waste systems may include requirements such as fully welded construction, secondary containment, and leak monitoring/detection capabilities.

6.6 Pharmaceutical Grade Water

Pharmaceutical grade water is a critical utility in the oligonucleotide and peptide manufacturing process. The specific grade or classification of water used depends on the application and the associated regulatory requirements.

Purified Water (PW) is typically used in the manufacture of non-sterile oligonucleotide and peptide APIs or drug substance intermediates. Water-for-Injection (WFI) is typically used for drug substance intermediates where water remains as part of the drug substance (e.g., frozen solution), drug product intermediates, final drug product, and sterile applications. Ultimately, end users must determine the required water quality based on a documented risk assessment, to define quality attributes to be monitored, sample requirements and testing, and demonstrate overall system control and adherence to the established specification.

Refer to following resources for water system design relevant to the industry:

- *ISPE Baseline® Guide: Water and Steam Systems(Third Edition)* [79]
- *ISPE Good Practice Guide: Membrane-Based Water for Injection System*s [80]
- *ISPE Good Practice Guide: Ozone Sanitization of Pharmaceutical Water Storage and Distribution Systems (Second Edition)* [81]

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- *ISPE Good Practice Guide: Critical Utilities GMP Compliance* [82]
- *ISPE Good Practice Guide: Sampling for Pharmaceutical Water, Steam, and Process Gases (Second Edition)* [83]
- *ISPE Good Practice Guide: Approaches to Commissioning and Qualification of Pharmaceutical Water and Steam Systems (Third Edition)* [84]

6.7 Clean Compressed Gases

Clean compressed gases are widely used in oligonucleotide and peptide manufacturing processes. Clean compressed gases include argon, nitrogen, other noble gases, and compressed air.

Depending on the application requirements, a point of use sterilizing-grade filter may be used to supply clean compressed gas to the process or process vessel as part of a contamination control strategy.

For additional information, refer to the following regarding gas system design relevant to the industry:

- *ISPE Good Practice Guide: Process Gases (Second Edition)* [85]
- *ISPE Good Practice Guide: Sampling for Pharmaceutical Water, Steam, and Process Gases (Second Edition)* [83]
- *ISPE Good Practice Guide: Critical Utilities GMP Compliance* [82]

Inertion and Use of Nitrogen and Argon

Inerting of process vessels and equipment is routinely required due to the extensive use of flammable solvents throughout the manufacturing process. Depending on the scale and consumption rates, the facility can be supplied by dewars or bottles, moving to exterior liquid nitrogen systems or on-site generation as usage rates increase and economic considerations of switch-over become more relevant.

Prior to the introduction of flammable solvents, vessels are purged with an inert gas to displace oxygen and reduce the potential for a flammable environment. Purging is typically performed at a higher pressure and may require multiple purge cycles, depending on vessel volume and the target residual oxygen concentration. Following initial vessel purging, a low-pressure continuous sweep or low-pressure static blanketing is used to maintain the inert process environment.

Nitrogen and argon are commonly used for inerting. While both nitrogen and argon are effective in inerting applications, argon is sometimes preferred for applications sensitive to moisture/water content. In oligonucleotide and peptide processes, these include solvent, reagent and amidite/amino acid feeds to the synthesis process. The higher density of argon allows it to be more effective in maintaining an inert state for longer durations during open operations; however, it is typically more expensive than nitrogen and the overall utility costs should be taken into account when considering the usage of argon versus nitrogen for specific applications. Some users may consider using argon due to its lower solubility in the synthesis solvents, while others may choose to use nitrogen as the primary inert gas throughout the process.

Prior to the use of equipment, it is standard practice to perform a pressure hold test using the inert gas to confirm system integrity. This may be integrated into the inerting procedure itself since the same gas supply is used.

Safety considerations associated with the use of nitrogen and argon include the incorporation of oxygen monitoring systems. Because nitrogen and argon displace oxygen and are asphyxiants, continuous oxygen monitoring is essential in process areas where releases could pose a safety risk.

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Clean Compressed Air

Clean compressed air can be used in downstream operations where the hazard of flammable solvents does not exist and inertion is not required.

6.8 Cleaning Systems

Repeatable, reliable and effective cleaning of oligonucleotide and peptide synthesis equipment depends on multiple interrelated factors. These include:

- The design and selection of processing equipment and interconnecting piping such that it can be effectively cleaned
- Understanding of the chemical and physical properties of the residues to be removed
- Knowledge and understanding of the cleaning mechanisms and the cleaning agents that facilitate these mechanisms
- Clear understanding of the factors affecting cleaning efficiency along with their application to specific equipment and cleaning cycles

The cleaning systems described in this section are intended to address the factors noted above and to review criteria for selection of cleaning strategies appropriate for both the equipment to be cleaned and the residues to be removed.

In general, the cleaning methodologies appropriate for a particular part of the process will tend to follow one of two approaches: solvent-based chemistry for upstream operations or aqueous cleaning chemistry for downstream operations (see Figure 6.2).

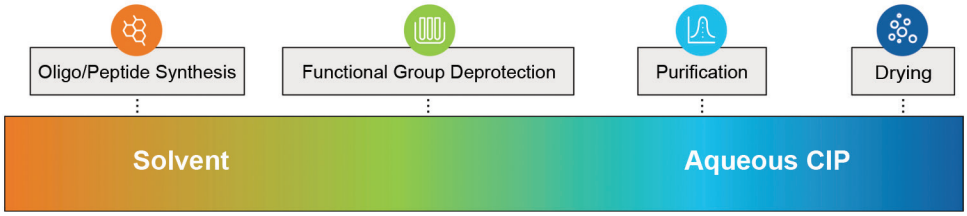
The solvents used for cleaning are often the same as those used in solution preparation, minimizing the introduction of new solvents specifically for cleaning. There are also applications where a combination of solvent-based and aqueous-based chemistries is required to address solubility and cleaning requirements. Regardless of whether solvents or aqueous detergents are utilized, the choice between single pass (once-through) and recirculated cleaning solutions should be assessed. Ultimately, it is the responsibility of the end user to conduct cleaning studies that verify the effectiveness of selected cleaning solutions and define the appropriate cleaning cycle requirements.

Refer to following regarding CIP system design relevant to the industry:

- *ISPE Guide: Cleaning Validation Lifecycle – Applications, Methods, and Controls* [86]
- CIP for Biopharmaceutical Processes, edited by Dale Seiberling, Informa Healthcare, 2008 [87]

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Figure 6.1: Solvent-Based Cleaning Methodology Transition to Aqueous-Based Cleaning



Clean-In-Place (CIP) Solvent-Based Systems

In-process stages where reactants and products are highly sensitive to water, such as amidite preparation and synthesis, cleaning is typically performed using solvent-based methods instead of aqueous solutions. In these cases, solvents are supplied from a centralized distribution system and delivered via spray devices to clean by solubilizing and removing residual soils from equipment surfaces.

Clean-In-Place (CIP) Traditional Aqueous-Based Systems

Traditional CIP systems can be used in the aqueous processing areas of oligonucleotide and peptide processes to facilitate removal of chemicals and residual soils from process equipment surfaces. These systems are available as fully automated skid-mounted systems. CIP systems include recipe-controlled cleaning cycles designed to achieve repeatable performance.

Clean-out-of-Place (COP) Systems

Equipment components that cannot be effectively cleaned in place may be removed and cleaned out of place using equipment such as COP cabinet washers or automated baths. These systems allow for controlled, reproducible cleaning of parts with complex geometries or limited cleanability when assembled.

6.9 Highly Potent Active Pharmaceutical Ingredients and Containment

In general, most oligonucleotide and peptide products are not classified as highly potent APIs. However, depending on occupational safety limit exposure data, enhanced isolation and containment measures may be required when manufacturing certain APIs. These measures may include the use of isolators, glove boxes or other engineered containment systems for handling API intermediates or the final drug substance. See ISPE Good Practice Guide: Containment for Potent Compounds [88] for additional design information.

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7 GMP Facility Approaches

7.1 Introduction

This chapter focuses on the GMP aspects of a facility designed for manufacturing oligonucleotides and peptides. The production and equipment layout, spatial segregation, adjacency requirements, operational GMP flows, environmental quality, and zoning requirements are all considerations discussed in this chapter.

Facility layouts can be varied and challenging, as the manufacturing processes for oligonucleotides and peptides use both bacteriostatic and bactericidal organic-solvent-based operations as well as aqueous-based processes. A well-designed oligonucleotide or peptide facility will offer a program that provides an efficient manufacturing train and seamless transition from the solvent-based environment to the aqueous-based environment. From a GMP perspective, the facility and program must appropriately protect the process and ultimately the product and end patient. However, because of the chemical reagents and solvents used, the GMP elements incorporated into the facility design must also be balanced with other critical architectural elements, such as hazardous material handling, environmental protection, and operator safety.

In addition to manufacturing operations, including GMP flows (i.e., personnel, materials, equipment, and waste), the facility layout needs to consider cleaning, maintenance, and quality control. The goal is to provide consistent, reproducible manufacturing of safe products while ensuring operational safety.

During facility program development, the following process drivers are directly related to, or can influence, the GMP facility design approach and should be considered:

- Program Contamination Control Strategy (CCS) (see Chapter 4), including:
  - Use of closed systems
  - Controlled-condition open operations
  - Requirements for process segregation
  - Process-adjacency requirements
  - Single-use technology and cleaning requirements
  - Environmental classification requirements for open operations
- Process scale, capacity, and business supply needs
- Flexibility for future process changes, new technologies, and expansion
- Code considerations for supporting systems. The facility implications of these topics are further addressed in specific discipline Chapters 8 Architectural, 10 Mechanical, and 11 Electrical.
- Automation approaches and considerations. See discussion in Chapter 9 Instrumentation and Controls.

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7.2 Room Classification Grades and Hygienic Design Elements

7.2.1 Facility Cleanroom Classification Grades

Classified cleanrooms may be considered for oligonucleotide and peptide production facilities as outlined in Chapter 4. The following are the cleanroom classifications discussed in this Guide:

- Controlled Not Classified (CNC)
- Grade D
- Grade C
- Grade B
- Grade A

ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition) [5] refers to Annex 1 [89]: *"Four (4) grades of cleanroom/zone are defined for the manufacture of sterile products: A, B, C, and D."* It indicates that these grades *"may be used to support the manufacture of other products that are not intended to be sterile."* The equivalence of these classifications to the ISO 14644-1, 2 [90] scheme are further defined within the *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5]. The Guide clarifies the following:

> *"specific process requirements dictate the appropriate environment for drug substance manufacturing operations…and once determined, the manufacturing area environments must conform to the required classification specifications."*

In this chapter, the grade system is the basis for discussion of applicable cleanroom classifications and facility design elements for oligonucleotide and peptide facilities. However, this is not intended to rule out using equivalent ISO classifications as mentioned previously. Additionally, we refer to CNC, which is often a factor in facility design, maintaining access and user-defined environmental controls without meeting cleanroom classification definitions set forth by industry standards and regulations.

7.2.2 Production Environment and Hygienic Facility Design Elements

The production environment in oligonucleotide and peptide facilities must combine hygienic design elements of Active Pharmaceutical Ingredient (API) and biologics facilities with safety features of facilities that process hazardous materials. As presented in Chapter 4, and as further discussed in this chapter's GMP layout section, the CCS shall include housing the manufacturing operations in appropriate environments. Where classified cleanrooms are selected (e.g., open aqueous operations), hygienic design features of cleanrooms should be applied. These features are discussed at length in the *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5]. The essential elements include the following:

- General cleanroom design features such as coving and finishing, and material of construction of surfaces that allow for appropriate cleaning and sanitization
- Appropriate layout features, such as air lock strategies (ingress and egress, personnel and material, pressurization hierarchy), and material pass-through designs (passive or active)
- Operational considerations such as gowning (choice of gown materials, level of gowning), occupant limits, equipment design, and local areas of protection

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- Heating, Ventilation and Air Conditioning (HVAC) design features such as air changes per hour, use of High-Efficiency Particulate Air (HEPA) filters, and room pressurization. These features are also further discussed in Chapter 10, Mechanical

- Plumbing and drain design features. These are further discussed in Chapters 6 and 10

7.3 GMP Facility Layout

7.3.1 Specific Oligonucleotide and Peptide Principles

As discussed in the process and equipment Chapters 3 and 5, oligonucleotide and peptide manufacturing consists of high-volume organic-solvent-based upstream processing followed by mostly aqueous-based downstream processing, all in GMP-compliant environments. The high volumes of organic solvents required to manufacture oligonucleotides and peptides often lead to bulk storage and distribution systems in which the quantities of hazardous material may significantly affect facility design. This is further discussed in Chapter 8 Architecture.

7.3.2 Layout Development Elements

An organized and structured layout development process is critical for designing an operationally efficient and GMP-compliant oligonucleotide or peptide facility. The following are some general product and process consideration examples with specific context for oligonucleotide and peptide manufacturing facility design:

- Future potential process changes and/or growth in scale: This includes bulk solvent storage areas and appropriate transfer capabilities

- Requirements for flexibility of manufacturing operations: Novel strategies for the synthesis of oligonucleotides are being developed that could significantly change the landscape, including the volumes of organic solvents required in current operations

- Segregating unit operations where required: Examples include dedicated solvent storage and transfer areas to concentrate areas of H-occupancy

- Providing necessary adjacencies: For example, downstream processes require large volumes of buffers. Proximity to buffer preparation and storage is advantageous

- Locating process support utilities near the manufacturing process but outside of the GMP manufacturing space: Examples include giving particular attention to waste lift stations and solvent delivery infrastructure, especially at larger scales

- Planning for the extent of required process transfer piping: This can be a significant consideration for these types of facilities, affecting cleanroom cleanliness, architectural construction type, and required spacing of suites

- Identifying open operations: For oligonucleotide and peptide facilities, this presents personnel and product protection implications, which can affect footprint and GMP facility flows

- Considering opportunities to increase process closure and containment: This can provide a pathway to reducing or eliminating classified cleanrooms, while also improving personnel protection and facility life safety profiles

- Providing adequate space for required process equipment. One example is column packing for solvent containing processes and implications for connectivity to other spaces

- Providing adequate space for storage: Examples include cold storage space within processing areas for intermediate chromatography fractions and storage requirements for often significant volumes of hazardous organic solvents required in these operations

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- Providing space and flows for required samples: This includes chromatography fraction analysis

- Providing design attention and adequate space for general (janitorial/housekeeping) functions as well as specific support functions: Examples include placement and protection of gas monitoring safety systems in hazardous occupancies

7.3.3 Cleanroom Considerations and Decision Tree

Although Chapter 4 outlines CCS considerations and risks based on typical process conditions expected in oligonucleotide and peptide facilities, the discussion is expanded here to include cleanroom approaches within the context of oligonucleotide and peptide GMP facility design. These principles reference substantial existing industry approaches on the topic as discussed earlier in this chapter and are meant to provide a framework for assessing GMP considerations for these modalities.

CNC and Grade D environments are typically appropriate for closed operations, including closed storage operations. Grade D is also an appropriate environment for critical or sensitive operations performed in a closed isolator. Grade D and CNC are appropriate environments for open operations performed under bactericidal/bacteriostatic conditions. Grade C is an appropriate environment for aqueous, open, low-bioburden operations. Grade B is used as the background for open aseptic operations such as operations performed in a Laminar Flow Hood (LFH) or Biological Safety Cabinet (BSC). Any requirements for aseptic operations would be based on the stated quality specifications of the drug substance.

Bacteriostatic operations, in particular, are defined as process steps for which there is no appreciable risk of increasing bioburden for the duration of the operation. This condition is achieved through the implementation of appropriate risk-based controls that effectively prevent microbial proliferation.

Whether for drug product manufacturing, aqueous-based drug substance manufacturing, or synthesis of oligonucleotides or peptides in bactericidal solvents, the rules for controlling, maintaining, and monitoring the cleanroom apply when classified cleanrooms are specified.

This includes the following:

- Cleanrooms shall be protected from adjacent areas with appropriate airlocks.

- Personnel working in Grade C and B cleanrooms shall wear dedicated plant uniforms with appropriate overgowning to access these spaces. This requirement does not apply to personnel limited to working in Grade D or CNC.

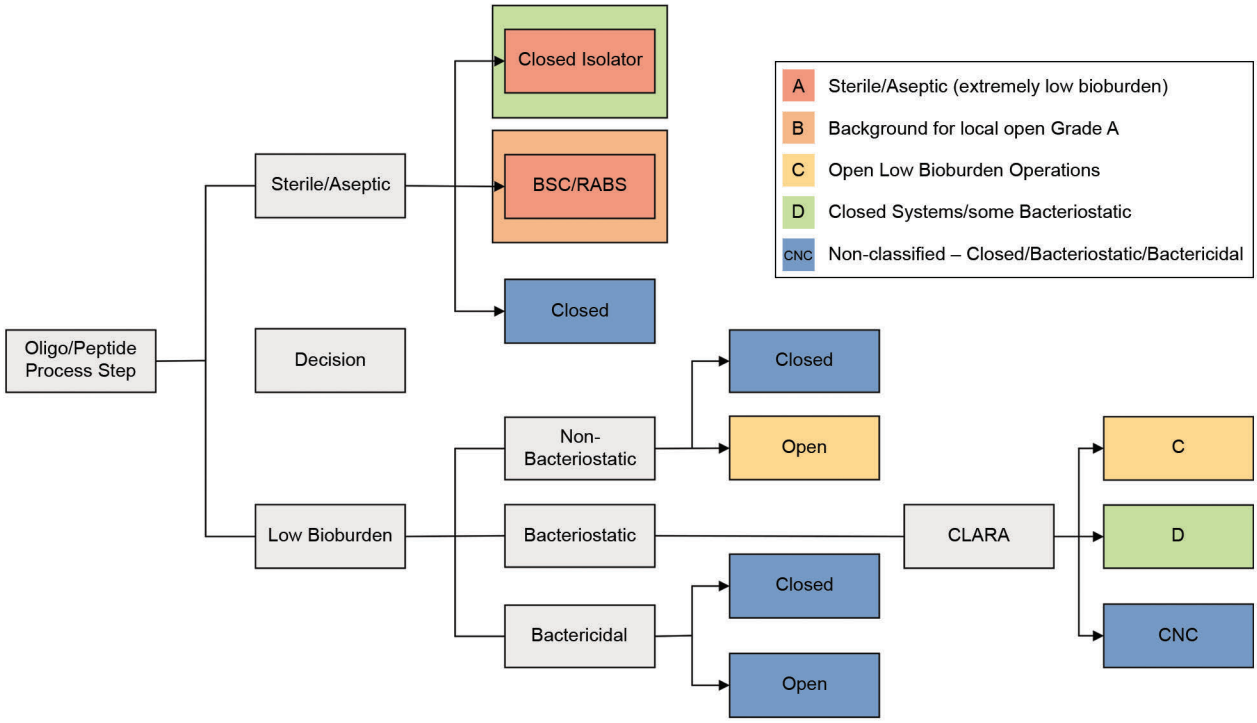
- The environmental monitoring of Grade C and Grade B cleanrooms will include dynamic testing (i.e., “in operation”) for viable and nonviable contaminants. Grade D requires static testing but leaves dynamic testing limits and applicability up to the manufacturer based on risk assessment and routine data.

- The classified cleanroom shall always meet the stated Grade B or C criteria during GMP operations.

Figure 7.1 provides a simplified strategy for determination of room classifications for the functions in an oligonucleotide or peptide manufacturing facility. A formal risk assessment (e.g., CLARA approach as defined by BioPhorum: Closure Playbook – closure analysis risk assessment [91]) should always be performed to determine appropriate room classifications for all functional areas. This process could include assessment of more nuanced operational situations, such as duration of open, controlled-condition processes and resulting environmental classification recommendations. In Figure 7.1, the determination has been simplified to “closed” and “open.” Detailed approaches and methodologies are included in reference materials such as the Biophorum Closure Playbook.

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Figure 7.1: Room Classification Decision Tree



7.3.4 Oligonucleotide and Peptide Facility Layout Models

Figures 7.2, 7.3 and 7.4 depict three layout models that could be applied in the design of an oligonucleotide or peptide facility: train, stack, and ballroom. These are not prescriptive rules for facility design; they provide a framework for thinking about GMP layouts for oligonucleotide and peptide processes and allow designers to consider a mixture of the concepts for a particular design. Layout selection should account for the process and equipment design to ensure that appropriate segregation and adjacencies (product, environment, or hazard classification) are maintained. Unique support system elements and segregation considerations are discussed later in this chapter.

Train

In a train layout configuration, discrete manufacturing suites are arranged according to the natural process flows through adjacent suites. As an example for an oligonucleotide manufacturing facility, the functional suites could be aligned according to this process scheme:

Synthesis → Cleavage/Deprotection → Chromatography → Downstream → Freeze-Drying

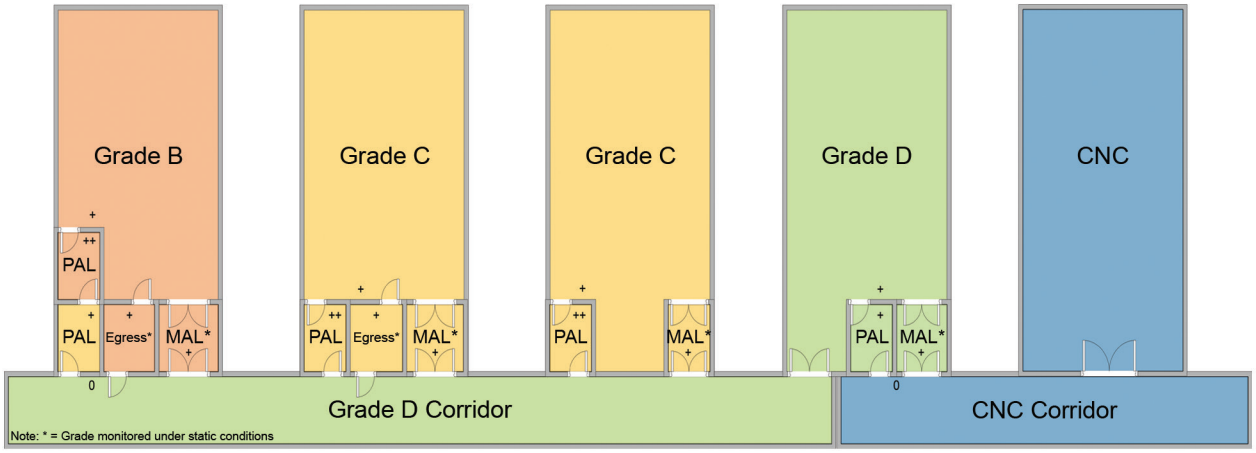
Both single and dual-corridor strategies can be used in train configurations. There are advantages and disadvantages to both.

Single-Corridor

A single-corridor train design consists of a common corridor for access to and egress from all functional areas. This model is well-suited for both single-product and multiproduct manufacturing when process containment is not the priority (e.g., non high-potent facility). The approach simplifies personnel, material, and waste flows and requires less real estate than a two-corridor strategy. Figure 7.2 encompasses multiple potential scenarios within the single-corridor model.

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Figure 7.2: Single-Corridor Manufacturing Layout

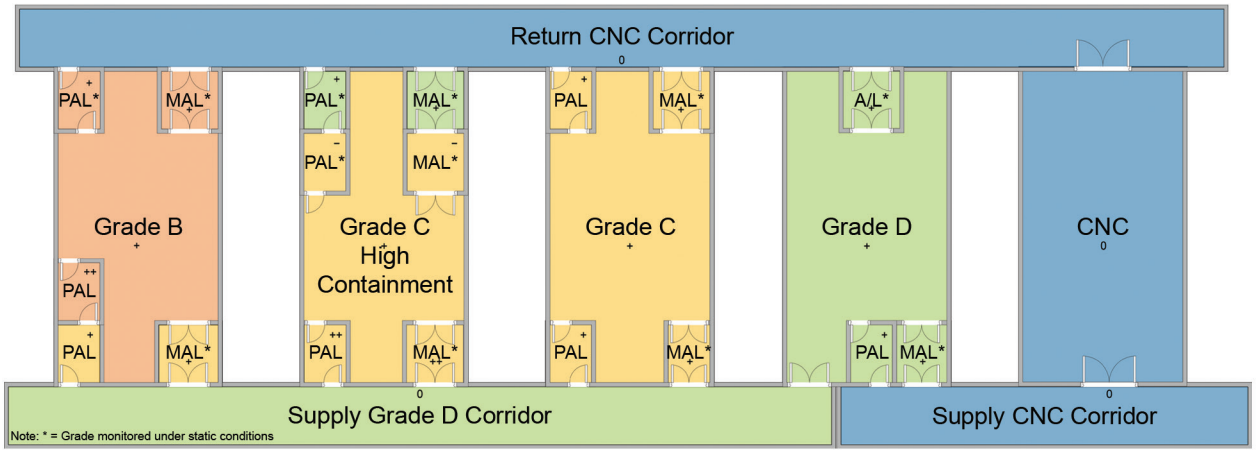


Ingress and egress air locks may or may not be shared. If shared, appropriate temporal segregation ensures that personnel and materials (raw materials and components) entering the cleanroom are not compromised by “soiled” personnel and materials (product and waste). The use of additional egress air locks further mitigates this risk.

Dual-Corridor

A dual-corridor design consists of a supply corridor for access to the functional areas described previously and a dedicated return corridor for egress from these functional areas. Similarly, the unit operations are often arranged sequentially, according to the manufacturing process flow. See Figure 7.3 for a representation of the dual-corridor model.

Figure 7.3: Dual-Corridor Manufacturing Layout



The dual-corridor similarly protects the manufacturing suites with ingress air lock transition points from the supply corridor. It also more robustly contains the individual functional areas by transitional egress air locks into the return corridor. This ensures soiled items and waste are never present in the supply corridor. Mechanical and support areas are accessed through the CNC areas. This can be advantageous for lyophilizer mechanical spaces or organic waste lift stations for example.

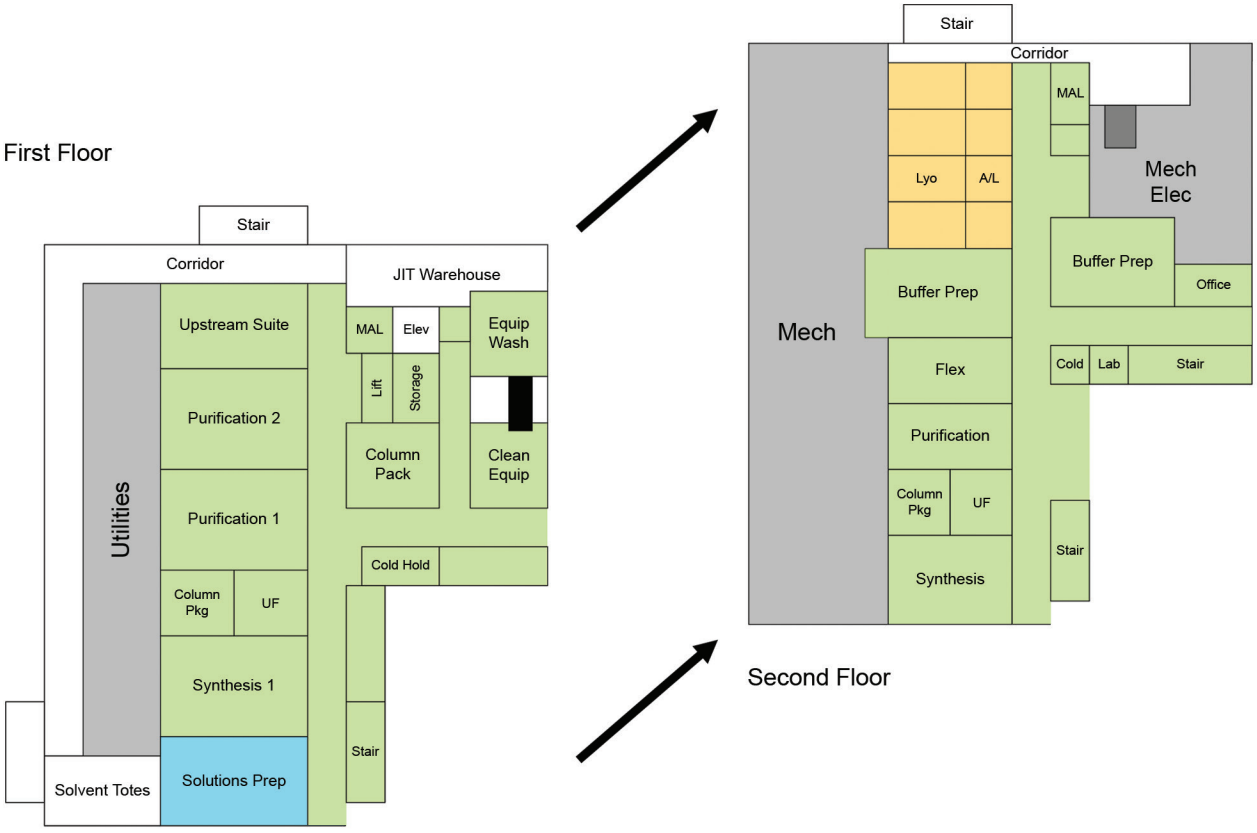
Shared personnel and material egress air locks are feasible, except in the high-containment scenario depicted that often requires special handling of waste materials. This scenario can apply in some peptide manufacturing product lines and is increasingly being discussed for siRNA oligonucleotides.

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Stack

The stack design segregates functional areas by vertically stacking the operations and services. A stack design typically contains isolated, multipurpose suites. The stack configuration is well-suited for facilities requiring a smaller footprint. The stacking arrangement also provides a natural segregation of environments and building occupancies. However, design complexity and cost can increase when choosing to locate flammable material quantities above grade level and transitions to and from functional areas may require elevators or stairs. For larger-scale operations, the stack approach can minimize piping runs, reducing cost of goods for high-value raw materials (e.g., feed of amidite solutions into synthesis) and enable gravity transfers of process solutions. See Figure 7.4 for an example stack configuration.

Figure 7.4: Stack Configuration



Ballroom

In a ballroom design, multiple functions are housed in a common space. Where no other hazards exist, ballrooms are especially efficient for housing closed systems that are effectively isolated from the environment and neighboring operations. The ballroom configuration can be well-suited for clinical and R&D production, as personnel can simultaneously monitor multiple operations.

For commercial oligonucleotide and peptide manufacturing, such a configuration offers space savings and operability advantages that must be carefully balanced against distinct requirements for different process areas. Compared with smaller scales, the solvent quantities in the upstream process areas often present step changes to the construction costs required to meet code requirements for building, electrical, fire protection, and HVAC. Conversely, as the process transitions to more aqueous operations, the use of cleanrooms may be favored depending on the process closure strategy and risk assessment.

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Strategically segregating these areas of different requirements can provide facility cost benefits over a single ballroom. Most facilities see another natural segregation break point as they transition toward bulk product (e.g., at the lyophilization step) where handling of the final drug substance can increase environmental control requirements.

7.3.5 Oligonucleotide and Peptide Support Systems

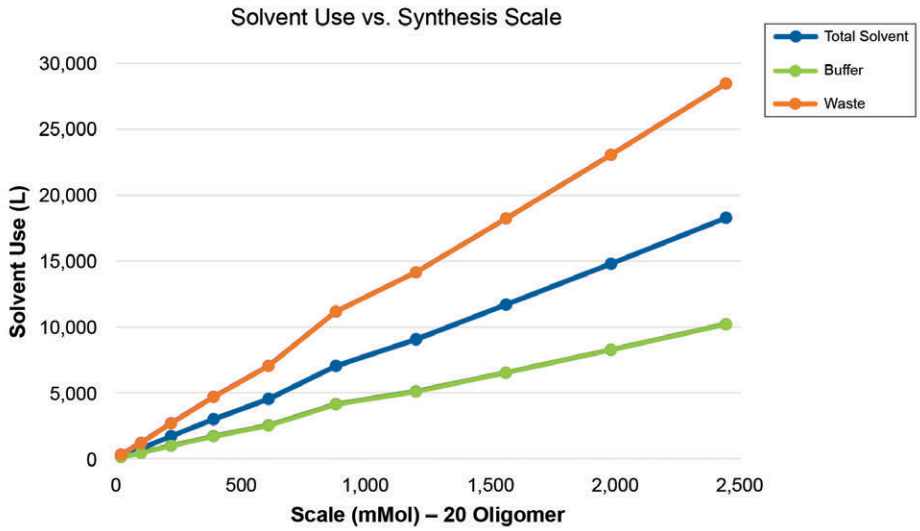
The distribution of organic solvents/raw materials and removal of solvent waste has significant implications for the design of oligonucleotide and peptide manufacturing facilities. Small-scale manufacturing may allow for bottle to drum quantities of raw materials that can be transported throughout the facility with minimal impact on facility design. However, larger-scale facilities will need to incorporate piping networks to supply and remove materials. This can also be seen with required safety systems around the process equipment. These can influence the desired GMP aspects of the facility design. Some items that should be addressed include the following:

- Location of piping runs (above ceiling, underground, laterally in lower grade spaces)
- Design philosophy of piping (coaxial pipe, leak detection, implications of concealed runs)
- Additional equipment (break tanks, lift stations, storage tanks, treatment skids)
- Cleaning considerations (aqueous Clean-In-Place (CIP)), solvent cleaning, recirculation of cleaning fluid)
- Buffer handling (separate space/grade of space, in-line dilution, location compared to point of use)
- General use of pressurized systems and impact on hazardous occupancy classification (e.g., H-2 versus H-3 for systems above 15 PSIG)
- Inerting systems and detection equipment
- Process venting and emergency relief systems

Consideration of these elements, along with other more typical service distribution systems such as process/instrument gases (e.g., air, nitrogen, and argon), process and potable water, heating/cooling water, emergency relief, and firewater, should be holistically considered as GMP layouts are developed. Batch size will affect these utilities and services. Figure 7.5 illustrates the impact of scale on typical organic solvent and waste usage.

Figure 7.5: Organic Solvent, Buffer, and Waste Volumes versus Scale for Typical 20 Oligomer Manufacturing Process

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7.4 Segregation Elements

As discussed previously, an oligonucleotide or peptide manufacturing facility needs to be designed with an understanding of adjacencies for the manufacturing program. Segregation considerations within the facility layout should strive to maintain these adjacencies where possible. Facility segregation elements for oligonucleotide and peptide facilities are discussed herein.

7.4.1 Product Protection Segregation Concepts

Segregation concepts are part of the overall CCS. These concepts should address the sources of potential product contamination described in Chapter 4, Section 4. Facility segregation concepts consist of three basic applications:

1. **Physical (Spatial):** Physical segregation can manifest as a separate area, separate room, closed system, or even separate HVAC systems. Closed systems provide physical segregation by isolating the process zone from the environment and can be an effective tool to mitigate cross-contamination risks.

   As described in *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5], closed systems can typically be housed in unclassified environments. Open systems are subject to contamination from the environment and from neighboring operations. Open systems may require separate classified cleanrooms, areas, protected environments (e.g., LFH, isolators, BSCs, Restricted Area Barrier System (RABS)), to adequately protect the process and product.

   Critical open processes should normally be performed in separate areas or rooms. Understanding the nature and risk of the open operation will help determine the level of physical segregation required.

2. **Chronological (Temporal or Time-based):** Chronological segregation relies on time-based separation, where activities are separated by time, using the same physical area for more than one function. An example is campaigned production or bidirectional Material Airlock (MAL) with the ingress and egress operations separated by time. Temporal segregation may also be supplemented with room/area clearance controls such as proceduralized cleaning between products. This approach aims to ensure that the previous material has been removed from the area and operational recipes, so that the equipment and room are in the correct state to receive the next product.

3. **Procedural (Standard Operating Procedure (SOP)-Based):** Procedural segregation involves using defined methods as outlined in SOPs to complete a task. An example is an incoming MAL material wipe-down procedure and concurrent gowning activities in the same area controlled by SOPs.

7.4.2 Hazardous Material Segregation Concepts

Oligonucleotide and peptide manufacturing have unique challenges compared with many other drug substance manufacturing schemes given the balance that must be struck between product protection and personnel and facility protection from the hazardous materials and operations associated with the manufacturing process.

Facility design and separation of unit operations, equipment, and areas are driven by multiple factors, including design requirements for the process, process scale, and local building and fire codes. These will inform decisions around managing volumes of ignitable organic solvents, including examples such as the location of ignitable/flammable buffers and the number of reagent vessels for synthesis operations. Logistical considerations for drug substance manufacturing will vary depending on the scale of the process and volume of organic solvent required, but includes many unique aspects:

- Storage location of hazardous materials such as flammable solvents, corrosives, toxics, and waste: A tank farm or dedicated solvent and waste distribution areas may be required, depending on the process scale, which influences volumes and must be aligned with local code requirements.

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- Buffer preparation and distribution: Buffer preparation is typically centralized, but buffer distribution may be remote local, using totes for manufacturing suites or functional areas.

- Single versus multilevel manufacturing: This may allow reagents/buffers to be fed by gravity into synthesizer/chromatography skids, reducing pumping needs. This must be balanced with code requirements that become more stringent for some materials (e.g., flammable solvents) on upper building floors.

- Minimized cleanroom manufacturing requirements: Closed operations can be performed in an unclassified space (see Chapter 4 for specific control strategies). However, open aseptic or low-bioburden aqueous operations will require classified cleanrooms.

- Local code implications: International Building Code (IBC), International Fire Code (IFC), and the National Fire Protection Association (NFPA), which is adopted by many jurisdictions, have specific requirements regarding construction, egress, and location of hazardous materials and operations, which will have an impact on facility design (see Chapter 8 for additional considerations and context).

- Requirements for electrical classification: These can affect facility design because of allowed materials/equipment, room penetration requirements, and how the runs are designed. Some of these requirements are discussed in more detail in Chapter 11 of this Guide.

7.5 Facility and Process Flows

The facility design should help mitigate the risk of product and raw material mix-ups, cross-contamination, and contamination from the environment, while mitigating the risk of operator exposure to hazardous conditions. The design should also provide for the quarantine of materials not released for production. An appropriate layout is an integral part of the approach to balancing these considerations. GMP flow diagrams (i.e., personnel, materials, equipment, waste, samples) can be used to visualize these risks within a proposed manufacturing layout. This can determine layout modifications necessary to meet requirements, or where infeasible, highlight areas where other methods, such as operational procedures, may need to be employed.

7.5.1 Program Development

Establishing proper flow of personnel, equipment, raw materials, product, samples, and waste is an important component in the design of GMP manufacturing facilities. The strategy should be established early in facility design and development. The GMP facility design can typically be achieved by considering the following roadmap elements:

- Determine requirements for manufacturing areas: What are the required functional areas?

- Determine required adjacencies and required segregations: What elements of the program should be close to support operations? Which elements should be segregated?

- Capture environmental classification requirements: Which process operations are aseptic, low bioburden, bacteriostatic, bactericidal, aqueous-based, organic-solvent-based, open, or closed and what does the CCS strategy imply for environmental classification requirements?

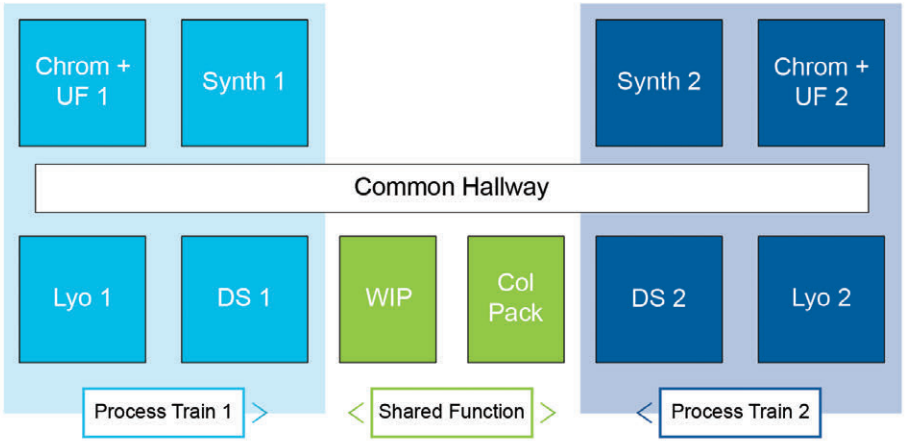
- Evaluate the GMP flows required to run the facility: Where is unidirectional flow required (for personnel, materials, equipment, waste, and samples)? Bidirectional flow is more generally acceptable in oligonucleotide and peptide manufacturing based on the typical risk profile as discussed in Chapter 4

- Iterate throughout the layout development process: How are hazardous chemical aspects balanced with the other roadmap requirements?

An example of functional adjacencies in a typical oligonucleotide manufacturing program is depicted in Figure 7.6.

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Figure 7.6: Example Manufacturing Adjacencies for an Oligonucleotide Manufacturing Program



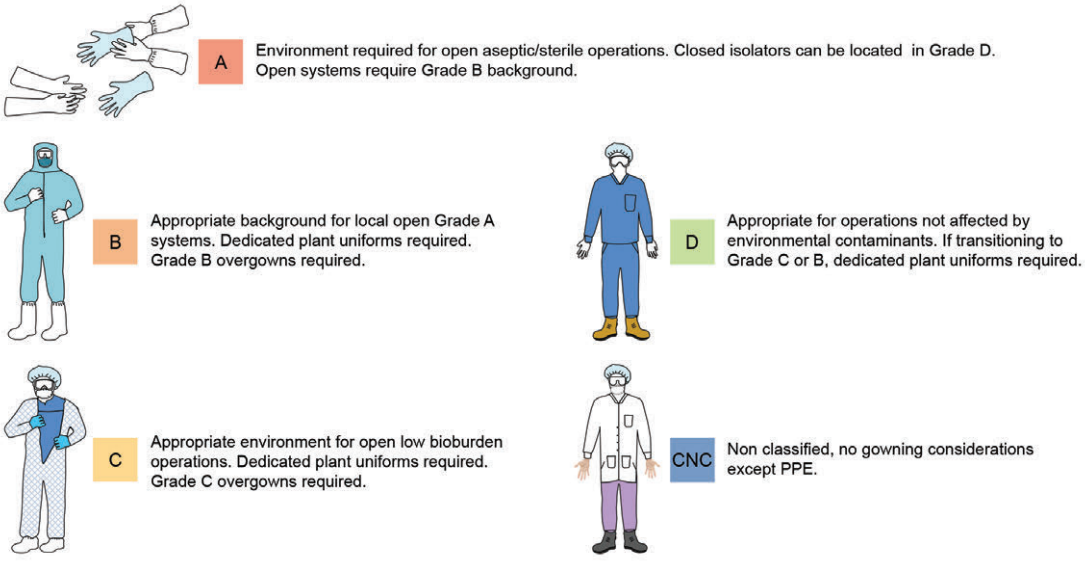
In regulated facilities, GMP flows must address potential opportunities for mix-ups and sources of contamination, including environmental, crossover, carryover, and raw materials. The proper flow and transitions of people, materials, equipment, and waste can effectively mitigate these risks.

7.5.2 Personnel Flow

In a GMP facility, the presence and circulation of personnel should not contribute to the contamination of the process or product. The personnel flow diagram will show how personnel enter, circulate through, and exit critical manufacturing areas including donning clean gowns and removing soiled gowns in appropriately designed transition zones. The goal is to provide appropriate protection and containment so that environmental contaminants and generated waste do not contaminate the facility or create crossover problems.

Just as cleanroom classification should be based on the stage of the process and process conditions, gowning should correlate to the specific functional area of the manufacturing facility. Gowning in areas where the processes are bactericidal will typically be focused on protection of the operator. Areas performing non bacteriostatic operations that are housed exclusively in closed systems can also be unclassified and therefore cleanroom gowning should not be necessary. In Figure 7.7, typical gowning approaches aligned with cleanroom classification decisions for oligonucleotide and peptide facilities are shown.

Figure 7.7: Typical Gowning by Facility Cleanroom Type



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For open low-bioburden or aseptic operations that are sensitive to environmental contaminants (e.g., open lyophilization and filling), cleanroom gowns commensurate with the bioburden specification of the product will be required.

Personnel flow diagrams should show personnel movement through the facility, gowning levels, and transitions according to environmental grade levels and should be harmonized with proposed gowning protocols. Potential environmental and cross-contamination issues can also be identified and minimized with these diagrams. Personnel occupying multiple environments may need to default to higher levels of gowning, as they may represent a higher risk of cross-contamination. The GMP personnel flow diagram can expose movements and transitions that are of high risk to process and product quality.

7.5.3 Material and Product Flow

As with personnel, the flow of raw materials, consumables (e.g., filter elements, elastomers, single-use bags), testing materials (e.g., Quality Control (QC)), can represent a risk for contamination of the process or the product. Most materials used in a GMP manufacturing facility are received from suppliers in an unclassified warehouse, where they are sampled, weighed, and dispensed. This process, and subsequent distribution to the manufacturing areas, should be conducted in a way that does not compromise the process or the environments housing the unit operations. This means using appropriate materials of construction, design, and procedural elements to guard against cross-contamination or environmental contamination.

Grade C or Grade B manufacturing areas are especially sensitive to the movement of personnel and materials, so the transition of materials into these spaces is critical to a successful program.

Material flow diagrams should be generated to provide an illustrative depiction of the CCS, which will detail the flow of these materials to and through these areas. The material flow diagrams can also identify points where potential cross-contamination may occur. Additionally, these diagrams may also detail areas where specific materials should be prohibited.

Of particular interest is the flow of the product as it transitions from manufacturing area to manufacturing area until it is finally delivered to the warehouse. Appropriate controls must be in place to mitigate the risk of mix-ups and potential cross-contamination.

7.5.4 Sample Flow

Sample flow diagrams will illustrate the locations for sample collection and the movement of these samples, including transitions from the manufacturing areas to QC. Sample control stations may be included on the diagram to detail where sample identification and labeling occur before exiting production. Areas of potential mix-ups can be identified with these diagrams.

7.5.5 Waste Flow

Regulatory authorities expect appropriate segregation of waste from process and product flows. Waste can take multiple forms, including solid, liquid, gas, toxic, hazardous, and flammable substances. Some waste types may also need to be segregated to prevent additional hazards. Waste flow diagrams can be generated to identify the locations and properties of different categories of waste. These waste flow diagrams will illustrate waste travel through the facility to designated waste staging areas, where they are stored safely until removal from the manufacturing site. From a GMP perspective, the waste flow diagram will depict potential intersections of waste and process/product flows. This can help elucidate appropriate engineering solutions for segregation of these streams. Hazardous waste generated in oligonucleotide and peptide manufacturing may require additional handling and packaging prior to exiting specified containment areas.

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7.5.6 Equipment Flow

Although prevalent in small-scale modular manufacturing environments, even large-scale manufacturing facilities will include some movement of equipment. Examples can include Environmental Monitoring (EM) test equipment by QC, calibration equipment by metrology, maintenance tools and equipment, and portable process equipment, including totes, carts, and components (e.g., filter housings) that will be used in the manufacturing areas and will often require cleaning, sanitization, maintenance, and storage. As with personnel and material flows, the flow of equipment could contaminate the process or environment housing the operation. Equipment flow diagrams can be generated to detail how equipment moves into and through GMP manufacturing areas, and which areas may restrict specific equipment based on area or equipment requirements. Points of potential cross-contamination can be identified with these diagrams.

7.6 GMP Facility Survey

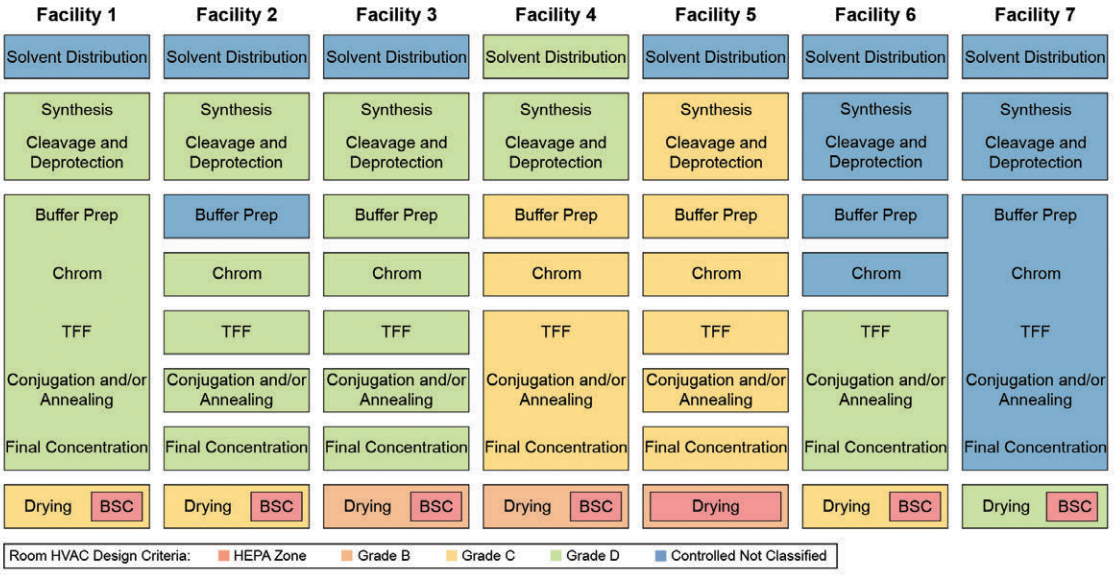
As outlined in this chapter, many elements figure into the successful design of a GMP facility for oligonucleotides and peptides synthesis. As discussed in Chapter 4, oligonucleotide and peptide drug substance facilities that produce products for parenteral route administration often have similar process considerations.

Although classified cleanrooms are indicated for open low-bioburden and sterile manufacturing, they may also be appropriate for a given functional area based on assessed risk, nature of the operation(s), preferred personnel and material GMP flows, desired adjacencies, or other strategic reasons, such as core business model drivers (e.g., Contract Manufacturing Organization (CMO) versus innovator).

Different levels of suite segregation are employed based on throughput and flexibility needs for a facility. This can influence the cleanroom classification of neighboring suites based upon common material flows.

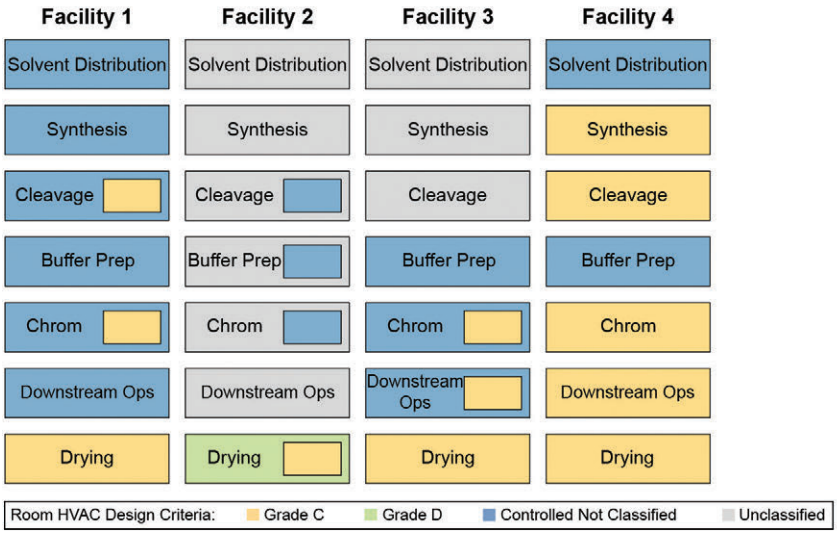
In this section, a review and interpretation of different approaches that have been taken in existing facilities for unit operation segregation and cleanroom classifications is performed, in the context of applicable chapter guidance. The intent is to provide a framework for facility designers, though the strategy for an individual facility must ultimately be supported by a risk assessment and data that support the CCS once the facility is operational. See Figures 7.8 and 7.9 and subsequent discussion for oligonucleotide and peptide facility segregation and classification surveys by unit operation. A baseline facility is also presented for discussion.

Figure 7.8: Survey of Current Oligonucleotide Manufacturing Facilities—Cleanroom Classifications and Segregation of Functional Areas



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Figure 7.9: Survey of Current Peptide Manufacturing Facilities—Cleanroom Classifications and Segregation of Functional Areas



7.6.1 Solvent Distribution, Synthesis, Cleavage, and Deprotection

Oligonucleotide and peptide synthesis up to cleavage operations are typically performed in bactericidal solvents. Isolation of solvent storage and distribution systems from the core process functions in unclassified environments is a well-established philosophy that aligns with the guidance in *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* [5]. Most of the peptide facilities in our survey have embraced this approach. These operations are located in technical/mechanical unclassified areas as the process conditions adequately mitigate the risk of contamination from the environment. Facility enhancements focus solely on operational safety considerations. For multiproduct facilities, segregation of unit operations is commonly used. Facilities dedicated to a single product often adopt a ballroom approach based on formal risk assessment. Local protection in an otherwise unclassified room (e.g., booths or soft wall containment) is a common approach for handling crude API when operations cannot be closed. This approach typically provides adequate protection for product and personnel.

The use of unclassified spaces is consistent with Closed Processing Environments (CPEs) or otherwise protected operations. In the case of the oligonucleotide examples, a possible driver for locating bactericidal solvent processes in a Grade D environment could include a desire for better connectivity with other functional areas that are classified cleanrooms. In general, flammable solvent code requirements for airflow may provide sufficient air handling capability to meet cleanroom airflow rates for Grade D. The implementation of a Grade C environment in some oligonucleotide and peptide facilities (e.g., Oligonucleotide Facility 5 and Peptide Facility 4) suggests a process with perceived risk from environmental contaminants. This would be atypical based on the expected process conditions at these steps. However, in the case of future technologies that may include alternative synthesis operations with aqueous or non-bacteriostatic process conditions, this facility approach would provide flexibility to adapt to the new conditions.

7.6.2 Buffer Preparation

The survey of oligonucleotide and peptide manufacturing facilities underscores multiple strategies for buffer preparation. Both ballroom and segregated buffer preparation strategies can be inferred, prioritizing connectivity and construction simplicity over declassification of individual operations. If both non-bacteriostatic and bacteriostatic/bactericidal buffers are prepared open for different products, segregation may be a consideration. Grade D may also be employed as part of a strategy to protect more stringent adjacent cleanrooms.

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Bactericidal or bacteriostatic buffers, for example, containing acetonitrile, are not likely affected by viable contaminants so performing buffer preparation in an unclassified environment can be justified. Aqueous buffer preparation that is performed under controlled conditions (e.g., duration, temperature, pH) with a subsequent process closure operation (e.g., 0.22 µm filtration) can typically be housed in a CNC environment with little risk to the process. However, endotoxin levels appropriate for intrathecally administered oligonucleotides, for example, should be considered in the risk assessment.

A Grade C environment is rarely indicated for buffer preparation, except for open operations with growth promoting buffers that will not be sterile filtered before use. Grade C environments on their own (without downstream filtration) do not generally provide adequate long-term hold conditions for aqueous buffers with regard to endotoxin burden. Ease of facility movement approaches that seek to minimize air locks may lead to Grade C buffer area classification based on the requirements for other adjacent unit operations.

7.6.3 Chromatography, Tangential Flow Filtration (TFF), Conjugation/Annealing, Downstream Operations, Final Concentration

Closed chromatography operations can be performed in unclassified environments as suggested by several of the oligonucleotide and peptide facilities. Open chromatography operations in which non-bactericidal or non-bacteriostatic conditions exist may justify Grade C. When systems are closed, they can be located in a ballroom with other unit operations, such as TFF. In Oligonucleotide Facility 1, for example, a Grade D ballroom approach leverages closed process downstream manufacturing operations and single-use equipment.

Oligonucleotide Facilities 4 and 5, and Peptide Facility 4 have segregated Grade C operations, suggesting potentially open chromatography operations where the process environment is non-bactericidal/bacteriostatic or shared. Facility renovation scenarios can also be a factor, where limited available square footage enables segregation but not additional airlocks to change classification between unit operations. In this way, segregation of closed operations may also otherwise facilitate logistics or respond to building or scale-up planning constraints and considerations. Oligonucleotide Facility 4 includes Grade C for TFF, annealing/conjugation, and final concentration, suggesting open operations in non-bacteriostatic conditions. Given that these operations are also collocated in a common ballroom, this implies a limitation to a single batch occupancy of the room from a cross-contamination risk perspective.

For peptide facilities, the final isolation unit operation will often be a filter dryer or a lyophilizer. In Facility 3, filter drying as the final isolation step is captured within “downstream operations” (with an alternate product/process pathway to lyophilization: “drying”) and acknowledges handling of the final API within a localized Grade C air supply with a CNC background. This could include upstream process closure via (CPE), such as 0.22µ filtration. In Peptide Facility 2, a similar approach is captured within “drying”, where the use of Grade D implies a strategy to protect the localized Grade C air supply environment (such as a booth or cabinet).

7.6.4 Drying/Isolation

The disposition of the product from an oligonucleotide or peptide API plant should be the driver for the design of the manufacturing suite for the final product. Typically, the final API is lyophilized and packaged. Lyophilized powders are generally bacteriostatic. Lyophilized oligonucleotides and peptides are typically packaged as “low-bioburden drug substance.”

Lyophilization operations are open operations. Oligonucleotide Facilities 3 and 4 depict a Grade B background for loading the lyophilizer within a BSC or LFH. This provides a local environment that would meet the Annex 1 recommendations for sterile products. The US Food and Drug Administration (FDA) allows for a Grade A filling environment of sterile products in a background of Grade C.

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Oligonucleotide Facility 5 was designed with the entire unloading room as a HEPA-protected zone with Grade A airflow conditions and no local enclosure. Given the difficulty of maintaining a zero-bioburden (Annex 1 Grade A) specification in a human occupied space, the room is designated as Grade B, implying that the lyophilized product specification is not intended to meet Annex 1 Grade A. Furthermore, this approach may create limitations and risks with regard to operator exposure that are potentially better served by isolators.

Oligonucleotide Facilities 1 and 2 depict local HEPA protection enclosures in a Grade C background. This design would be consistent with open low-bioburden operations as long as the desired bioburden specification can be met consistently and reliably. This logic would also apply to the peptide facility approaches that use open Grade C or local Grade C enclosures for dryer operations. Other measures such as the use of protective Lyoguard trays help further mitigate the risk of contamination from the environment.

Facility 7 has been included within the oligonucleotide Figure 7.8, as a proposed ISPE manufacturing baseline facility that could be considered for both modalities based on the following appropriate risk mitigation measures and approaches:

- The solvents used in oligonucleotide and peptide manufacturing facilities are typically bactericidal. These can be stored in appropriate containers that protect the solvents from mix-ups or cross-contamination. There is no concern for microbial contamination and therefore an unclassified space is appropriate for the housing of these containers.

- The synthesis and cleavage and deprotection steps are performed under bactericidal conditions and therefore the systems used for these operations can be housed in unclassified environments. Any intermediate peptide isolation could be handled in appropriate local protective environments.

- The preparation of aqueous buffers can be performed under “controlled conditions” and then closed by appropriate CPEs (e.g., sterile filtration). Closed systems can be placed in unclassified environments as stated throughout the regulatory guidelines.

- Technologies and operations exist that can successfully close chromatography, filtration (e.g., TFF, concentration), conjugation, and annealing operations. Per Annex 1 and other guidelines, these can also be housed in unclassified environments. If appropriately closed, these operations can also be co-located in a ballroom.

- Most sterile or aseptic operations performed exploit the use of closed isolation technologies (i.e. closed isolators). As stated in Annex 1, Part IV, and other regulatory guidelines, closed isolators may be housed in a Grade D environment. Because oligonucleotide and peptide drug substances do not typically need to meet Annex 1 conditions, the use of an isolator is more typically driven by personnel protection than product protection. However, it also provides a robust compliance strategy that is cost-efficient for facilities. Note the approach in Peptide Facility 3 to protect personnel and product for final API pack-out of a filter dryer (“downstream ops”) with a localized Grade C air supply within a CNC background.

To consider this ISPE facility design approach, appropriate system closure and controls must be demonstrated. A formal risk assessment (e.g., CLARA approach as defined by BioPhorum: Closure Playbook – Closure Analysis Risk Assessment [91]) should be performed to confirm the suitability of the environments selected for all unit operations and the extent to which any open operations meaningfully affect the risk profile. This approach has been successful in other modalities such as cell culture based monoclonal antibody API manufacturing facilities.

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8 Architecture

8.1 Introduction

This chapter outlines the architectural considerations for designing oligonucleotide and peptide manufacturing facilities, which present distinct challenges compared with other types of manufacturing environments. These facilities operate at the intersection of biologics GMP principles and the higher-safety, higher-hazard conditions typical of API manufacturing. Accordingly, this chapter focuses specifically on this intersection, covering architectural design guidance not previously addressed within the ISPE Baseline Guide Series. It also focuses on the importance of an integrated facility design approach in which process requirements, hazardous materials usage and handling, and utility infrastructure are woven into the architectural framework to support safe, efficient, and compliant operations.

Continuing from the foundation laid in Chapter 7, this chapter discusses the layout development and conceptual design of the production and support spaces. It also examines other functional areas including warehouse, laboratories, offices, and utilities. Planning and adjacencies for the areas that fall outside the production core are just as critical to achieving a holistic and integrated facility design that addresses GMP requirements, hazardous occupancy considerations, and ensures compliance with applicable building and fire codes.

8.2 Key High-Level Considerations

This section outlines the categorical elements that need to be addressed when designing an oligonucleotide or peptide manufacturing facility.

8.2.1 Code Considerations

From site and development regulations to specific considerations of the building and hazard classes within it, regulations are promulgated nationally and locally. The full range of applicable standards must be carefully considered in the design process. Largely because of the quantity of hazardous materials they store and use, oligonucleotide and peptide facilities will have unique considerations that influence architectural facility design. These considerations include more stringent and robust construction types, fire protection, secondary containment systems, and life safety systems. For example H-class occupancy may be required in processing suites. When codes and jurisdictional requirements differ, the more stringent regulations should be followed.

Insurance underwriting requirements also need to be considered (e.g., FM Global). Appropriate stakeholders need to review these considerations during preliminary design to address fire hazards and associated approaches. In this Guide, the US codes are used to illustrate considerations that may apply, with specific jurisdiction regulations ultimately governing the design process.

8.2.2 Site Selection

Site selection should align with the company's business strategy and should consider immediate and anticipated future operational demands. Building and fire codes, environmental regulations, and local zoning and land development regulations will specify restrictions on land use, setbacks, and safety measures that require thorough regulatory review for these facility types. Consideration should also include site access and the site circulation required for emergency and delivery vehicles, including firefighting equipment access and chemical tanker trucks. Depending on scale, additional access to railways for bulk solvent supply and waste management may be a possible consideration.

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8.2.3 Process Flows

Oligonucleotide and peptide manufacturing facility efficiency and safety relies on careful planning, from overall functional group adjacencies to core production and production support space configurations. These elements influence how personnel, materials, products, and waste move through the facility, setting the basis for operational effectiveness and regulatory compliance. Refer to Chapter 7 for more in-depth discussion.

8.2.4 Utility Infrastructure

Unique facility infrastructure includes the location of solvent tank farm areas and routing of chemical delivery and waste piping through appropriately sized racks to and from the external tank farm areas and the processing areas inside the building. The placement of these areas on the site must also consider floor sloping and trench drains required for containment in high-hazard-occupancy spaces, and adjacency of required technical spaces to production areas. These elements all influence oligonucleotide and peptide production facility building size and shape and have an important influence on operational continuity and facility operational efficiency.

8.2.5 Production Scale

As with other manufacturing facilities, production scale significantly affects the overall facility design. The scale directly influences production equipment size, volume of materials handled, and utility capacity. Understanding the range of scales that the facility must accommodate is fundamental for ensuring that the design meets current and future operational demands.

8.3 Oligonucleotide and Peptide Facility Design Considerations

8.3.1 Land Use Regulations

Land use regulations meaningfully impact the design of all facilities, and this is heightened for oligonucleotide and peptide facilities. Legally binding land covenants and fire safety regulations are two influential categories in this regard. Facility insurance carriers may have their own additional requirements.

These needs must be identified early in a project:

- Allowable uses: Permissible uses for a property, such as residential, office, or manufacturing, must be correctly defined in terms of the unique oligonucleotide and peptide facility hazards.
- Lot coverage: The amount of built and open space allowed preserves the natural environment and maintains permeable surfaces that can be used to control stormwater runoff.
- Building setbacks: Building setbacks maintain a minimum distance from property lines to site buildings and decrease the potential for fire to spread. They also increase road visibility, safety, access to utilities, and aesthetic site appeal.
- Building heights: Height and the number of stories help preserve the character of the area around a site and control development density.
- Parking requirements: Parking must meet facility operational needs and comply with requirements for the facility type.
- Stormwater management: The site must independently manage generated stormwater while ensuring hazardous materials are not released accidentally.

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- Environmental considerations: Specific regulations can include noise and emission generation limits and environmental impact studies (e.g., wind/wake analysis, etc.).

8.3.2 Fire Safety Regulations

Fire safety regulations, often referred to as building codes, are construction regulations that specify minimum requirements for a facility and provide a reasonable level of safety for building occupants, firefighters, and emergency responders. The following are key architectural areas covered by fire safety regulations:

- Use and occupancy: The quantities of hazardous materials can, depending on scale and other factors, result in the facility being classified as a higher-risk hazardous use (e.g., IBC High Hazard Group H Occupancy) that incurs more restrictive requirements meant to mitigate these hazards.

- Building area and height: Three important factors: occupancy, construction type, and allowable area, are interactive and must be determined early to establish the initial parameters of the building regulations.

- Construction type: Facility materials of construction and combustibility must be identified, including for primary structural components (i.e. columns, floors, and roofs) and interior and exterior walls.

- Fire protection systems and features: In addition to additive materials that increase the fire resistivity of a particular material or assembly, such as structural members, active fire protection and life safety systems (e.g., sprinklers, detection systems, alarm systems, and smoke control) contribute to the building and fire compliance strategy.

- Occupant egress: Life safety is the primary goal of building regulations. Occupant travel distance to safe exit will be commensurate with risk and affects facility room design and overall aspect ratios for the facility through quantity and location of exit pathways. For example, common path of travel distances, the point at which occupants have two distinct paths to an exit, are specifically regulated.

- Accessibility: Ability to use the space for individuals with physical disabilities, as per jurisdiction-specific regulations.

8.3.3 High Hazard Considerations

In this section, specific fire safety regulation considerations for hazardous occupancies are highlighted and discussed.

- Hazardous material inventory: To evaluate and properly design a hazardous process, it is necessary to create an inventory and categorize aggregate quantities of all the materials expected to be used and stored in the facility, including hazardous waste. Open use considerations lead to more stringent facility implications than closed or storage uses.

- Fire separation distance: In addition to distances from hazardous buildings to property lines discussed previously, minimum distances between hazardous buildings must be considered along with possible consideration for fire rating of exterior building walls as a mitigation approach for tighter sites.

- Explosion control: Solvents used in manufacturing oligonucleotides and peptides can heighten the risk of a potential explosion. Mitigation strategies and requirements to address this hazard are codified and include mechanical ventilation and deflagration venting to relieve explosions while avoiding structural collapse. Deflagration venting significantly affects the design of processing rooms, building structure, and explosion relief panels enclosing the building exterior, so mechanical ventilation is generally preferred as a preventive method. However, some jurisdictions and insurers may require both. See discussion in Chapter 10 Mechanical, Chapter 11 Electrical, and Chapter 9 Instrumentation and Controls; for more detail on facility explosion control considerations.

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- Secondary containment: Secondary containment is typically a regulated requirement for hazardous materials such as flammable liquids. A typical sizing approach would be to accommodate the larger maximum volume of two scenarios: the volume of the single largest vessel or container within the affected space or the volume of water discharged from the sprinkler system over a specified duration. Depending on the building location and the required volume, containment may warrant depressed slabs, drains to holding tanks, lift stations, and/or pits with floor-level grating.

- Health hazards: Besides the flammable hazards typically associated with these facility types, health hazard chemicals, including corrosives, toxics, and highly toxics, must be addressed for unique implications.

- Hazardous storage: Hazardous storage is extensively regulated. Small-volume storage can be indoors, with larger volumes driving the need for large exterior bulk tanks. Exterior storage and flammable liquid dispensing station locations should accommodate property line and occupied building setback distances. All hazardous liquid storage, dispensing, and use must consider spill containment.

8.4 Conceptual Design Layout Development

8.4.1 Facility Functions and Adjacencies

Identification of key facility functions is essential to define initial space allocation requirements. The typical functional groups in oligonucleotide and peptides manufacturing facilities include the following:

- Administration and amenities: Offices, collaboration areas, and personnel support areas, including locker rooms and break rooms

- Manufacturing: Core process areas (e.g., upstream process suite, downstream process suite, lyophilization) and process support spaces (e.g., in-process cold room, equipment wash and storage)

- Laboratories: Quality control, analytical development, and process development

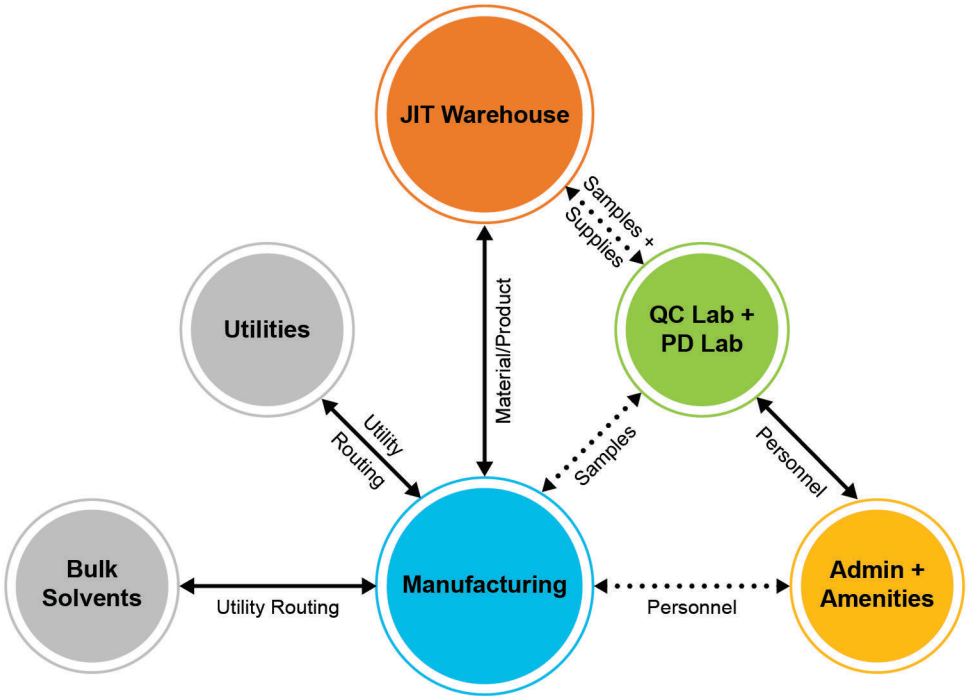
- Warehouse: Storage for raw materials, consumables, Work-In-Progress (WIP), finished bulk (including temperature-controlled storage where necessary), equipment storage, and maintenance supplies

- Utilities: Infrastructure for HVAC, purified water, facility support systems and process gases, process solvents and waste, and emission recovery systems

Adjacencies of these major functional areas within a manufacturing facility are important to support efficient material, personnel, and product movement. In oligonucleotide and peptide manufacturing facilities, high hazard considerations add a layer of complexity. Certain functional areas, such as manufacturing, warehouse, and laboratories, may need to be positioned along an exterior wall to maintain compliance with applicable codes. Figure 8.1 illustrates typical primary (solid lines) and secondary (dashed lines) adjacency relationships in oligonucleotide and peptide facilities.

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Figure 8.1: Example Facility Functional Adjacencies



8.4.2 Site Design

Site planning must consider high hazard and land use elements. These aspects will influence the location of these elements on the site: tank farm, loading docks, logistics and service vehicular access, circulation, employee and visitor access, site security, and parking areas. Relative building locations must follow regulations and consider potential future building expansions. Deflagration venting, if required, must be coordinated to minimize exposure risks to pedestrian pathways, roadways, and other site elements. HVAC intake locations should also be sited away from parking and loading areas to mitigate potential air quality and safety risks.

8.4.3 Building Core and Shell

Building construction is fundamental to these facility types and a higher structural importance factor is often required to minimize facility risk to the public, reinforce business continuity, and protect critical assets.

Column bay spacing must minimize obstructions while meeting structural requirements. Wide column bays provide flexibility and reduce interference with equipment layouts and circulation. However, long spans can create deep beams that reduce usable vertical clearance. Open-web-truss systems can address this by allowing utility ducts and pipes to pass through. Vertical clearances are important for process flexibility and functionality. Adequate vertical height allows program stacking, such as placing amidite preparation above synthesis to leverage gravity feed, which reduces transfer complexity and improves process flow. Adequate floor-to-floor and floor-to-roof heights are necessary to allow for above-ceiling mechanical systems, utility routing, and maintenance access.

Structural systems and materials must be selected to support both current and future needs. Moment frames can allow for open circulation and flexible utility routing but are more expensive and require larger structural members. Shear walls and brace frames are less expensive but can limit circulation paths and utility routing. Concrete structures have benefits such as durability and inherent fire resistance, but can complicate future modifications due to embedded reinforcements.

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8.4.4 Air Locks

Air locks are transitional spaces designed with controlled pressure and interlocked doors to facilitate movement between areas with different room classifications and/or pressures. They primarily safeguard clean areas from contamination from adjacent spaces with lower classifications. In some cases, air locks also serve to maintain containment boundaries.

Anticipated capacity of required personnel gowning and material movement must be considered to maintain efficient operations. See Chapter 10 Mechanical for an outline of the different air lock pressurization strategies and their application and Chapter 7 GMP Facilities for specific facility examples.

8.4.5 Ventilated Enclosures

Ventilated enclosures, such as fume hoods and glove boxes, play a critical role in controlling exposure to hazardous vapors and ensuring a safe working environment. At a small scale, standard fume hoods provide containment for processes involving smaller setups and material handling. For intermediate-scale operations, walk-in hoods accommodate larger equipment that exceeds benchtop capacity while maintaining controlled airflow for operator protection. However, at larger production scales and higher throughputs, equipment size and process requirements often make using standard ventilated enclosures impractical. Instead, dedicated processing areas with localized exhaust, specialized ventilation controls, and closed process systems are used to ensure safe operation.

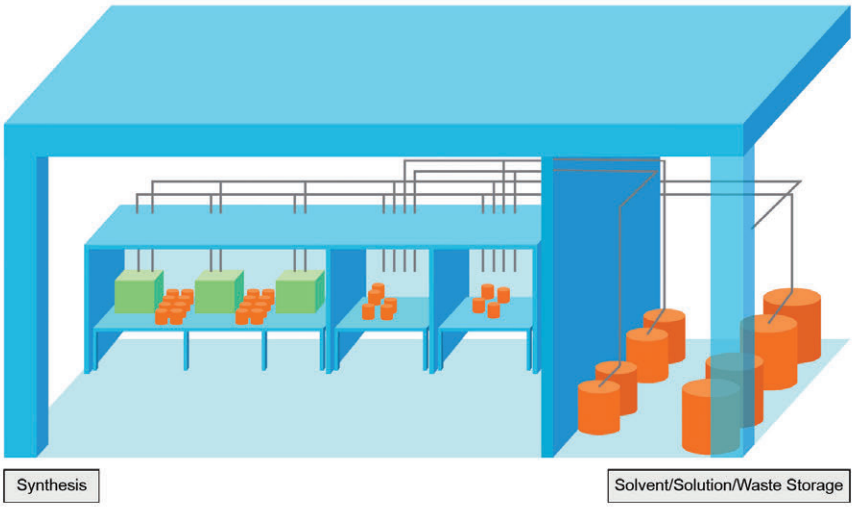
8.4.6 Production Scale and Facility Configuration

Oligonucleotide and peptide manufacturing facilities increase in complexity as production scales up, requiring different configurations to accommodate changes in equipment size, material handling, and process integration. The following three examples represent facility configuration realms and considerations, progressing from small-scale benchtop operations to large-scale commercial manufacturing.

8.4.6.1 Small Scale

For early stage development or low volume production, configurations rely on standard benchtop fume hoods and smaller walk-in hoods for handling reagents, some solution preparation, and process operations (see Figure 8.2). Solution storage consists of smaller portable vessels that can be moved by hand, and, in some cases, moved directly into the process suite for use.

Figure 8.2: Small-Scale Oligonucleotide Synthesis

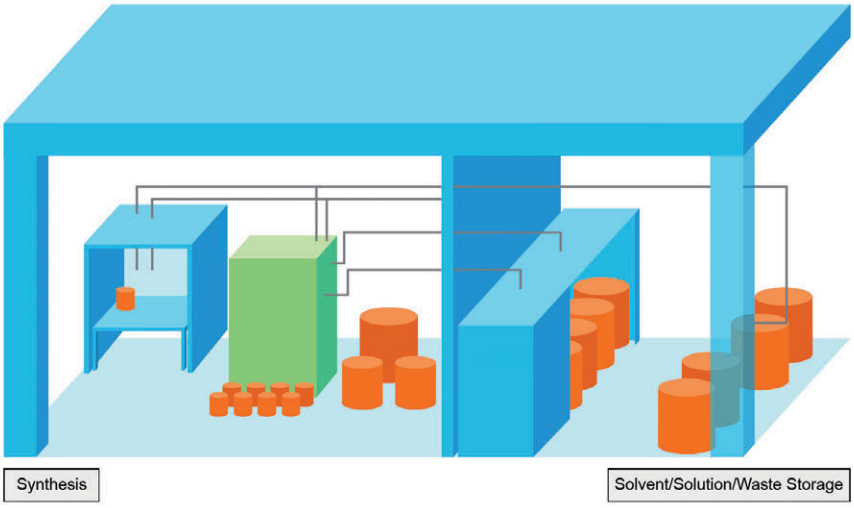


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8.4.6.2 *Pilot Scale*

For clinical-scale production, fume hoods may be replaced with entire rooms designed for flammable liquid usage to support larger-volume vessels and processing equipment (see Figure 8.3). Depending on volumetric usage, solution preparation may be co-located in the suite, using a walk-in fume hood, or it may be in an adjacent area. Solvent storage and distribution systems are more structured, often with dedicated transfer lines.

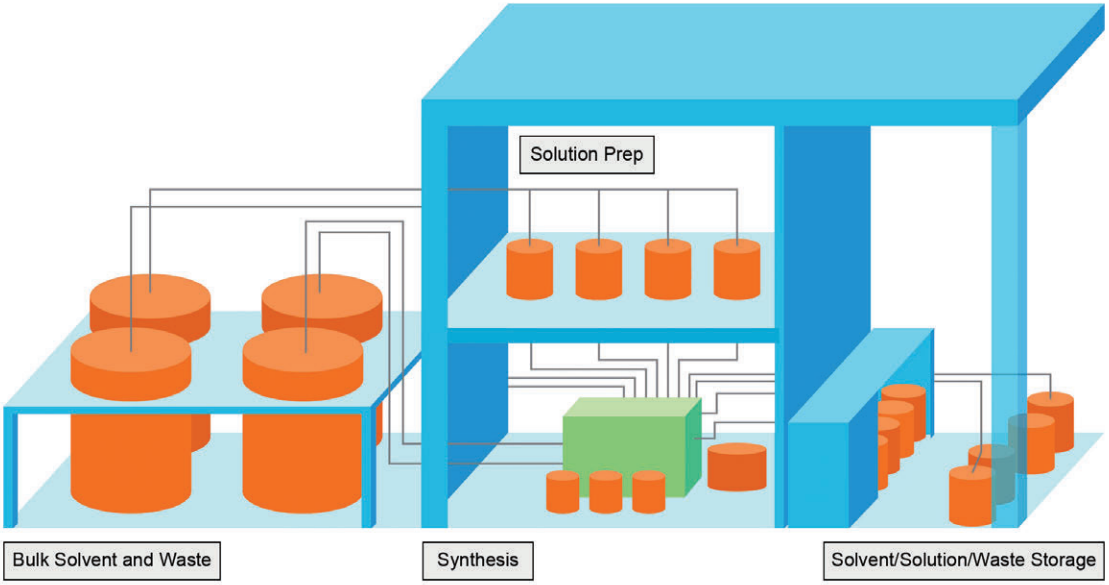
Figure 8.3: Pilot Scale Oligonucleotide Synthesis



8.4.6.3 *Commercial Scale*

Configurations designed for higher throughput production incorporate fully integrated solvent storage and distribution systems (see Figure 8.4). A tank farm is used for bulk solvent and waste storage and is connected to mixing vessels in a solution preparation area. The solution preparation vessels feed directly into the process unit operation through a distribution system.

Figure 8.4: Commercial Scale Oligonucleotide Synthesis



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8.5 Architectural Design Opportunities

Design of manufacturing facility exteriors and interiors presents a unique opportunity to showcase advanced technology and processes. Thoughtfully designed spaces not only can reflect innovation, but can enhance the experience for employees and visitors. With the right balance, the design can create an inclusive, supportive environment for manufacturing operators, laboratory users, warehouse staff, and administrative personnel.

8.5.1 Exterior Building Design

8.5.1.1 Facade

The building exterior provides a platform to convey company identity, image, and vision (see Figure 8.5). Intentional building forms and appropriate building materials can balance functionality with visual appeal. Building presence can be elevated by moving beyond a bland box toward a visually engaging facade.

Figure 8.5: Building Facade Example
Image courtesy of CRB Group, www.crbgroup.com.



8.5.1.2 Tank Farms and Pipe Racks

Tank farms are a functional necessity for oligonucleotide and peptide manufacturing facilities, particularly at larger production scales (see Figure 8.6). Integrating the tank farm into the site plan is critical for building and zoning code compliance and operational efficiency. Tank farms and pipe bridges should be placed strategically to align with overall site vehicular circulation for solvents and waste supply vehicle access, and to enable landscaping or screening structures to minimize visual impact from public roads and/or adjacent properties.

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Figure 8.6: Typical Tank Farm Configuration
Images courtesy of Nitto Avecia, www.avecia.com.



8.5.2 Interior Building Design

Interior spaces require thoughtful planning to ensure functionality and an enhanced experience for employees and visitors.

8.5.2.1 Entry Lobby

The lobby serves as the facility's first impression and is a critical branding tool. Design elements are expected to communicate cleanliness, precision, and technological sophistication and highlight the organization's process, science, and mission.

8.5.2.2 Break Rooms

These spaces cater to employees and guests, offering a comfortable environment for relaxation and informal meetings. Placing break rooms on the exterior walls will help maximize natural lighting into the spaces. Diverse and flexible seating arrangements and amenities can accommodate occupants and prompt efficient use of space.

8.5.2.3 Process Spaces

Thoughtful and strategic planning of the manufacturing areas and adjacent spaces, such as control rooms and corridors, can provide opportunities for glazing along shared walls and provide views into the manufacturing operations from outside the controlled spaces (see Figure 8.7). Borrowed light from nearby exterior windows can enhance visibility and ambiance in the manufacturing space. Providing views into process spaces is also beneficial from a touring and oversight perspective. Facilities designed with visibility into manufacturing areas can enable external personnel, such as visitors or auditors, to observe operations without needing to wear gowns, thereby minimizing risks and operational disruptions (see Figure 8.8).

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Figure 8.7: Process Space Viewing Corridor
Image courtesy of CRB Group, www.crbgroup.com.



Figure 8.8: Process Space Viewing to Corridor
Image courtesy of CRB Group, www.crbgroup.com.



Even with stringent safety considerations, such as explosion-controlled environments in large scale solvent handling areas, designs can incorporate architectural glazing features. Large-scale chemical processing facilities have used glass walls on exterior facades to balance safety requirements with enhanced visibility and architectural aesthetics.

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9 Instrumentation and Controls

9.1 Introduction

This chapter outlines general considerations for instrumentation and controls, along with oligonucleotide and peptide manufacturing facility-specific considerations. This chapter assumes the reader is familiar with standard instrumentation and controls used in modern pharmaceutical facilities and associated GMP requirements. Instrumentation and control aspects that may be unique to oligonucleotides and peptides involve combining practices typical of either API or biopharmaceutical manufacturing and determining which practices to apply. Instrumentation and controls used in oligonucleotide and peptide manufacturing also need to be designed to accommodate flammable solvents used throughout the process. This includes associated code and safety requirements.

9.2 Instrumentation Considerations

9.2.1 General

In general, instrumentation used in oligonucleotide and peptide manufacturing and associated specification requirements are the same as those used in the pharmaceutical or chemical industries.

Instrument specifications include but are not limited to the following:

- Measurement range
- Accuracy
- Tolerance
- Materials of construction
- Electrical requirements
- Control requirements
- Cleanroom and manufacturing environment requirements

Additional specifications for instrumentation used in oligonucleotide and peptide manufacturing may include the following:

- Hygienic or nonhygienic process application
- Hazardous area classification or nonhazardous area classification

Considerations related to hygienic or nonhygienic process applications are described in Chapter 6.

Instrumentation used in hazardous or electrically classified areas must meet applicable code requirements, including but not limited to National Electrical Code (NEC) [48] or Atmosphères Explosibles (ATEX) [50]. Approaches may include selecting an instrument that meets all electrical classification requirements, including intrinsically safe installation methods, or a combination of these. See Chapter 11, Electrical, for additional information.

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Besides technical specifications, instrumentation used in GMP manufacturing is evaluated using a criticality assessment. Examples of criticality classifications for GMP instruments may include the following:

- GMP critical
- GMP noncritical
- Non-GMP
- For Information Only (FIO)

Criticality classifications are used to identify the potential level of quality impact that individual process control instruments may have, and to define activities such as calibration frequencies and out-of-tolerance nonconformance response procedures. Methodology and definitions utilized in GMP criticality assessments can vary including direct and indirect impact.

9.2.2 Instrumentation Installation and Wiring Methods

Installation of instruments should follow all requirements of the applicable electrical area classification and methods to limit exposure of non-resistant materials. Considerations for these requirements may include the following:

- Isolators and intrinsically safe
- Sealed conduits
- Flex conduit
- Wire ways
- Purge panels

Field instrumentation should be wired to control panels, remote Input/Output (I/O) panels, and Motor Control Centers (MCC) that should preferably be located in nonrated spaces. Placement of these panels should be considered for optimally reducing distance, quantity, and ease of access for maintenance and calibration activities.

When wiring devices to control devices or junction boxes with multiple sources and/or destinations the design and methods should follow standards laid out by the Institute of Electrical Electronics Engineers (IEEE), NEC, and the International Electrotechnical Commission (IEC), and other applicable national or international organizations. Some of these topics include the following:

- Separation of power and signal wiring
- Cable sizing
- Wire coloring
- Materials
- Ferrules
- Proper wire and device labeling
- Easily available and complete drawing sets

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- Fuses and circuit breakers
- Separating different voltage ratings
- Finger safe coverings on power sources
- Grounding

9.2.3 Calibration and Preventive Maintenance

Instrumentation used in the GxP manufacture of oligonucleotides and peptides should include routine calibration and preventive maintenance pending end-user requirements. Calibration process and adjustment tolerances, and calibration and PM frequencies should be established before use, with activities scheduled and recorded following approved standard operating procedures, often using a Computerized Maintenance Management System (CMMS) or similar. See *ISPE GAMP® Good Practice Guide: A Risk-Based Approach to Calibration Management (Second Edition)* [122] for additional information.

9.3 Control System Considerations

9.3.1 System Architecture

The network architecture and integration of associated equipment should be considered during facility design. A decentralized architecture is the most basic configuration. Process and utility equipment are installed with local control systems, and all recipe creation and control is performed at the equipment level. Hybrid architecture enables integration of process equipment with the plant network. Control networks should be segregated from enterprise and corporate networks to improve security and network efficiency. Considerations for a hybrid system architecture include the following:

- Data integrity
- Automated backups and disaster recovery
- Data visualization
- Remote troubleshooting support
- Remote alarming/interlocking with supporting utilities
- Digitized production (i.e., Manufacturing Execution Systems (MES), Laboratory Information Management Systems (LIMS), Enterprise Resource Planning (ERP) per ISA 95)
- Redundancy of critical components
- Domain authentication
- Cybersecurity
- Network performance
- Change management and validation

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9.3.2 Control System Hardware

In general, both Programmable Logic Controllers (PLCs) and Distributed Control Systems (DCSs) can be used for process controls.

Selection of the control system PLCs depends on factors such as complexity, network integration, redundancy, speed requirement, and future expansion. For simple control systems, an all-in-one solution may be used that integrates the Central Processing Unit (CPU), I/O, and/or Human–Machine Interface (HMI). This can help reduce costs and shorten development time, but may be limited in processing power, functionality, and total I/O count. For more complex systems, the following should be considered when selecting PLC hardware:

- Processor speed and memory
- Quantity of network ports for network segregation and redundancy
- Spare I/O capacity for future expansion
- Programming language (ladder logic, Sequential Function Chart [SFC])
- Communication protocols available for instrumentation
- Ability for replacement

The control system HMI may be chosen based on the PLC selected with some additional hardware considerations.

For DCS, a Supervisory Control and Data Acquisition (SCADA) server may be used with a combination of thin clients and tablets. Thin client(s) may be installed at the remote I/O panel and multiple tablets can be provided for flexibility as production staff performs operations throughout the plant.

For process equipment with local control systems and dedicated PC, separation of the PC and operator interface should be considered to mitigate the risk of component failure. A panel-mounted monitor may be used with a din rail-mounted industrial PC as opposed to a panel-mounted PC.

Development or validation environments within PLC and server hardware should be considered. This may allow testing of security patching, testing new or modified code, or adding new equipment to existing production environments.

In some high-risk scenarios, a Safety Instrumented System (SIS) may be implemented. See Section 9.5 for additional information.

Control system hardware should be selected with appropriate environmental protection. Electrical panels located within process areas are recommended to have a NEMA4X/IP66 rating for liquid protection and corrosion resistance. Electrical panels, conduit, and instrumentation should be compatible with cleaning agents used in the facility.

9.3.3 Panel Purge Controllers

When possible, control panels should be located remotely, outside of electrically classified areas. For control panels located within the electrically classified boundary, several protection methods are available based on area classification, panel size, power requirements, and operational needs. Common options include explosion-proof enclosures, panels designed with Intrinsically Safe (IS) components, usage of IS circuit extensions (IS barriers), or pressurization with integrated purge controllers. A purge controller supplies low-volume, low-pressure compressed air to an electrical or controls cabinet and maintains positive pressure with respect to the surrounding environment. This prevents flammable vapors from being present in the electrical cabinet.

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Three types of purge controllers include:

- Type X (Ex pxb): Type X purge systems create a pressurized and continuously purged environment in the enclosure, removing any hazardous atmosphere within and maintaining positive pressure to prevent hazardous gases from entering. It provides the highest level of protection among the purge controller types and allows electrical equipment rated for safe (nonhazardous) areas to operate in Class I Division 1/ATEX Zone 1. Once purged, the enclosure is deemed safe. Type X systems must include shutdown or alarm mechanisms if the internal pressure is lost. They are often interlocked to prevent power from going to internal equipment until purging is complete.

- Type Y (Ex pyb): Type Y purge controllers also use pressurization to maintain a safe environment inside the enclosure, but it is designed to reduce the internal classification of the enclosure from Class I, Division 1/ATEX Zone 1 to Class I, Division 2/ATEX Zone 2. Type Y controllers require continuous monitoring to maintain positive pressure and provide alerts or shutdowns if pressure is lost. However, the degree of monitoring and interlock requirements are generally less stringent than for Type X.

- Type-Z (Ex pzc): Type-Z purge controllers provide the lowest level of purge protection, reducing the enclosure's classification from Class I, Division 2/ATEX Zone 2 to a nonhazardous environment. Type-Z controllers do not require a fail-safe or interlock mechanism. If the panel loses purge pressure, the operator is notified via an alarm. In a Class I Division 2/ATEX Zone 2 environment, electrical panels should be supplied with Type-Z purge for explosion protection when local installation to the process is required. Manual or automatic purge controllers may be used; the latter removes human error by automatically performing a high-flow-rate purge.

See Figure 9.1 for examples of integrated purge controllers.

Figure 9.1: Examples of Integrated Purge Controllers
Images used with permission from Pepperl+Fuchs, www.pepperl-fuchs.com/en-us.



9.3.4 Control System Software

Control system designs must address several key considerations to meet regulatory standards, maintain process control, and ensure product quality. Control systems are critical to managing, monitoring, and recording parameters that directly affect the manufacturing process.

The control system software typically includes multiple layers such as firmware, operating systems, and application software. The latter can be standard, user-configurable, or custom-built. User-configurable and custom-built software is configured or developed specifically for the process and should be designed and tested following GAMP® guidance.

Oligonucleotide synthesis consists of a repetitive series of instructions with minor variations among the repetitions. The oligonucleotide synthesis control system should have suitable graphical aids to enable generation and review of these repetitive sequences of operations. All software should provide real-time feedback on critical process control parameters.

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Operators need to interact with the system quickly and efficiently, especially during alarms or interlocks. Critical safety and process alarms should be clearly defined and managed. The control system should distinguish between warnings and critical alarms to avoid alarm fatigue and ensure rapid operator response when necessary. Critical and noncritical process parameters should be recorded in the data historian.

The control system should be able to interface with higher-level systems such as LIMS, MES, and ERP systems to streamline data management, reporting, and scheduling. Integration with LIMS allows seamless data transfer for product quality analysis. MES can manage production schedules, equipment status, and track lot histories. ERP systems can be used to manage resources, materials, and inventory.

9.4 Control System GMP Related Considerations

The GMP-related requirements and best practices for controls systems for oligonucleotide and peptide manufacturing are the same as those for the rest of the pharmaceutical industry. These include but are not limited to 21 CFR Part 11 [92]/EudraLex Annex 11 [93] requirements, GAMP best practices [94], and associated qualification requirements.

9.4.1 21 CFR Part 11/EudraLex Annex 11 Requirements

An overview of 21 CFR Part 11/EudraLex Annex 11 related design requirements include the following:

- Electronic records and historical data
- User access security and related permission levels
- Electronic signatures
- Audit trails
- Backup and restore capability
- System validation
- Training
- Data integrity

9.4.2 GAMP Considerations

GAMP is guidance to ensure automated systems used in pharmaceutical manufacturing are properly validated and comply with regulatory requirements.

GAMP guidance and considerations should be used to design and evaluate all areas related to facility control to promote patient safety, product quality, and data integrity. All control systems should be evaluated to ensure that they can be maintained throughout the expected lifecycle of the process.

Key concepts of GAMP include the following:

- Product and process knowledge
- Lifecycle approach
- Scalability

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- Quality risk management
- Leverage supplier activity

See *ISPE GAMP® 5 (Second Edition)* [96] and associated GAMP subtopic guides for additional information.

9.4.3 Control System Qualification Considerations

Commissioning and qualification considerations for oligonucleotide and peptide manufacturing control systems are again similar to those for the general pharmaceutical industry.

Overall, control system qualification should address 21 CFR Part 11 [92]/EudraLex Annex 11 [93] requirements as applicable, follow GAMP guidance, and ensure that process requirements are met. In alignment with International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) Q8 (Pharmaceutical Development) [27], Q9 (Quality Risk Management) [28], and Q10 (Pharmaceutical Quality System) [29], qualification activities should verify that the control system supports consistent operation within the defined control strategy, with particular focus on Critical Process Parameters (CPPs) that impact Critical Quality Attributes (CQAs).

The GAMP lifecycle for commissioning and qualification of control systems includes planning, specification, configuration and coding, verification, and reporting.

From an ICH Q8/Q9/Q10 [27, 28, 29] perspective, qualification verifies three layers of control:

1. The control system, in interaction with physical components and instruments, is fit for purpose and capable of executing the defined control strategy.
2. Potential failures in CPP control are identified through risk assessment; unacceptable risks are mitigated to an acceptable level (e.g., via additional controls, redundancy, or enhanced testing).
3. The system supports an ongoing state of control through robust data management (archiving and trending) and immediate notification of loss of control via critical alarms.

Commissioning and qualification activities are developed to ensure all GMP user requirement functional specification and configuration requirements are met.

Additional considerations for control system qualification may include leveraging of vendor commissioning testing and documentation, such as Factory Acceptance Tests (FATs) and Site Acceptance Tests (SATs).

9.5 Safety Related Process Controls

When designing the process controls or building an automation system for an oligonucleotide or peptide manufacturing facility, additional safety system-related considerations may exist. These include both safety-related code requirements and industry best practices for solvent handling.

9.5.1 Safety Instrumented System (SIS)

A SIS rated control system is an engineered hardware and software solution that detects hazardous conditions and initiates a safe shutdown of all or part of a process or system. SIS rated control systems can provide a separate layer of protection to reduce the likelihood or impact of a safety event. SIS systems typically operate in parallel with process control systems. However they are distinct, segregated systems designed based on Safety Integrity Level (SIL) requirements. It is the responsibility of the end user to determine SIS applicability and to perform SIL evaluations for SIS design requirements and implementation.

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9.5.2 Hazardous Atmosphere Monitoring

A significant number of the unit operations involved with oligonucleotide or peptide manufacturing include flammable solvents. Besides meeting code requirements for electrically classified areas, it is common to include monitoring devices for Lower Explosive Limit (LEL) for the detection of flammable liquid or vapor leaks and oxygen (O_2) for the detection of inert gas leaks in-process areas. If a solvent leak or oxygen deficient environment is detected, monitoring devices trigger local visual and audible alarms to alert personnel of the potential hazard. This could also apply to leak detection systems within piping systems or spill containment areas.

In general, the hazardous atmosphere monitoring system may be considered as a safety-related process monitoring and notification system and different from a life-safety system. Besides notifying personnel of a possible process hazard, LEL and O_2 devices may be interlocked with process controls to shut off solvent supply or inert gas supply to an affected manufacturing area or to safely relieve pressure from pressurized systems.

Application and local code requirements may guide whether the hazardous atmosphere monitoring system and associated components require formal SIS and SIL evaluation and certification. See IEC 61508 [96] and IEC 61511 [58] for additional information and requirements.

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10 Mechanical

10.1 Introduction

Chapter 10 focuses on aspects of mechanical design and cleanroom environmental controls unique to the oligonucleotide and peptide industry. Where applicable, design topics common to other GMP facilities may be mentioned, but are not discussed in detail.

See the mechanical design chapters of the following ISPE Baseline Guides for additional information:

- *ISPE Baseline Guide: Sterile Product Manufacturing Facilities (Third Edition)* (2018) [4]
- *ISPE Baseline Guide: Biopharmaceutical Manufacturing Facilities (Third Edition)* (2023) [5]
- *ISPE Baseline® Guide: Oral Solid Dosage Forms (Third Edition)* (2016) [97]
- *ISPE Baseline Guide: Active Pharmaceutical Ingredient (Second Edition)* (2007) [3]

Applicable industry references for HVAC design may vary by region. Examples include:

- *ISPE Good Practice Guide: Heating, Ventilation, and Air Conditioning (Second Edition)* [98]
- American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE) [99]
- Federation of European Heating, Ventilation and Air Conditioning Associations (REHVA) [100]
- Chartered Institution of Building Services Engineers (CIBSE) [101]

Additional systems and utilities included in the mechanical design of GMP facilities but not discussed in detail in Chapter 10 include:

- HVAC cooling/heating water and glycol systems
- Process cooling/heating water and glycol systems
- Steam for HVAC, process, or humidification
- Energy recovery utilities
- Fire suppression systems

There are several unique considerations that apply to oligonucleotide and peptide manufacturing facilities. A central design challenge is the incorporation of safety elements for handling flammable solvents while meeting GMP design requirements for cleanroom environmental controls.

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10.2 HVAC Design for Environmental Controls

The basis of HVAC cleanroom environmental controls is well established in industry and involves supply of conditioned, filtered, or HEPA-filtered air (depending on area classification) at predetermined flow/air exchange rates while maintaining a pressure cascade scheme between rooms. The highest pressures are typically assigned to process areas with more stringent classification and the highest risk of product contamination. In addition, room temperature is typically controlled within specified ranges. Humidity is typically monitored and may be controlled in accordance with process and environment requirements. An environmental monitoring program is used to demonstrate operation within cleanroom classification specifications. Environmental control framework is reviewed with particular emphasis oligonucleotide and peptide facility considerations that apply.

10.2.1 Temperature

HVAC temperature is typically controlled for personnel comfort, where materials have storage temperature requirements, and where process solutions may be affected (e.g., handling Dimethyl Sulfoxide (DMSO), that has a freezing point near ambient temperature). Safety considerations related to temperature in processing and storage areas are noted in Section 10.3.7.

10.2.2 Relative Humidity

Humidity, temperature, and their combined influence on the manufacturing environment are related to bacterial and fungal growth. The potential for personnel shedding at low Relative Humidity (RH) levels has been discussed in other ISPE Baseline Guides, and the same considerations apply in the oligonucleotide and peptide facility.

Reagents and amidites used in synthesis areas and freeze- or spray-dried finished drug substances are often hygroscopic. Absorbed moisture could adversely affect product or process. Material handling and packaging practices should be considered when establishing humidity control requirements for these areas. Furthermore, the high and low humidity control ranges and corresponding absorption potentials should be considered when establishing specifications and methods for calculating drug substance concentration.

Humidity level maintenance and its effect on static-charge distribution is a relevant consideration in areas in which flammable solvents are handled.

10.2.3 Air Exchange Rates

Although there is no regulatory requirement for air exchange rates in most oligonucleotide and peptide manufacturing facilities, design often aligns with industry standards. For example, though not directly applicable for these facilities, the US FDA 2004 Guidance for Industry on Sterile Drug Products Produced by Aseptic Processing – Current Good Manufacturing Practice [102] contains a suggestion that 20 AC/hour should be adequate for ISO 8 and suggests that cleaner spaces should have higher ventilation rates. *ISPE Good Practice Guide: Heating, Ventilation, and Air Conditioning (Second Edition)* Section 6.1.2.1 [98] provides additional generalized design parameters. In some cases, lower air exchange rates may be sufficient to achieve the desired room classification.

Overall air exchange rate requirements may not be driven solely by the Contamination Control Strategy (CCS). In areas where high volumes of solvents are used, higher exchange rates may still be required for safety. Suites containing high volumes of solvents often require increased room exhaust and local exhaust points which results in higher overall air exchange.

10.2.4 Pressurization and Air Locks

Pressure cascades, directional airflow, and air locks have been extensively covered in other ISPE Baseline® and Good Practice Guides. General design considerations from those sources apply, with the potential for specific considerations for oligonucleotide and peptide facilities.

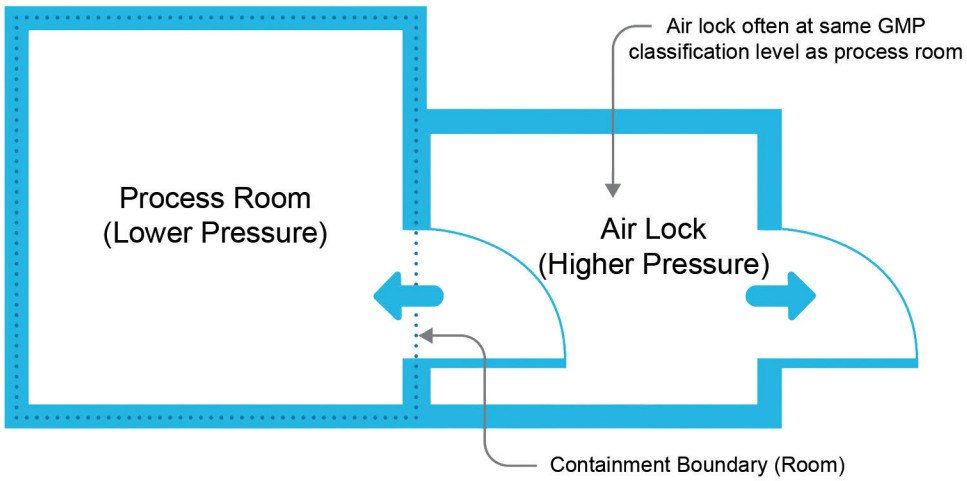
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The hazardousness and/or flammability of many of the constituents used in the manufacture of oligonucleotides and peptides pose unique code-mandated directional airflow requirements. These can clash with generally accepted cleanroom pressurization practices. Standard cleanroom pressure cascade design typically involves designating the most critical production suite(s) as the most pressurized within a pressure cascade schema, with decreasing pressurization levels of adjacent suites.

Cleanrooms in oligonucleotide and peptide manufacturing that serve as high-volume flammable chemical use areas where open processing may be performed require negative or inward airflow for chemical containment. For example, rooms within peptide and oligonucleotide manufacturing areas are often classified from both a cleanroom environmental classification (e.g., Grade D) and hazardous electrical classification (e.g., Class I Division 2) perspective. This may necessitate cleanroom production suites to be negatively pressurized with respect to adjacent air locks or corridors, depending on the entry strategy and overall corridor design. See Chapter 7 for additional content on corridor and classification discussions of different approaches to facility and equipment layouts.

In an example of this design approach, air locks or corridors adjacent to production suites are the most positively pressurized in the overall pressure cascade, with pressurization decreasing to the remaining adjacent areas. Similar approaches may apply to high potent containment zones for peptides. This type of design strategy is used to ensure that both safety and cleanroom environmental control requirements based on risk assessment are met. An example configuration is shown in Figure 10.1.

Figure 10.1: Example of Cleanroom Suite Pressurization used for Containment of Flammables



Alternatively, condition-based ventilation can be considered in areas with potentially hazardous environments. Airflow and pressurization set points can be inverted when local Lower Explosive Limit/Oxygen (LEL/O2) sensors detect an unsafe condition requiring containment. With this approach, the cleanroom pressure cascades can resemble a standard design during normal processing operations but can shift operating modes quickly for life-safety hazard events. The design approach to room pressurization and airflow direction should be reviewed across multiple expertise areas to ensure that regulatory and Authority Having Jurisdiction (AHJ) code requirements are met. Further details and code considerations are provided in Table 10.1.

Regulatory guidance that quantifies differential pressure requirements exists for sterile processing (FDA Sterile Guidance [102] and EU Annex 1 [89]), and though these requirements are typically not applicable in oligonucleotide and peptide drug substance manufacturing facilities, they are often part of good engineering practice. Table 10.1 lists typical pressurization design values by application.

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Table 10.1: Typical Pressurization Design Values by Application [5]

Condition	Differential Pressure (Inches of Water Gauge/Pascals)
Areas of differing classification	≥ 0.04 in. w.g. (10 Pa)
Areas of the same classification, when separation is desired (optional)	≥ 0.02 in. w.g. (5 Pa)
Bubble and sink air locks	≥ 0.03–0.06 in. w.g. (7.5–15 Pa)
Bubble and sink air locks (each door)	≥ 0.02–0.06 in. w.g. (5–15 Pa)
UC and natural ventilation	Per the design

10.3 HVAC Design for Containment and Safety

Design of HVAC systems for oligonucleotide and peptide manufacturing facilities requires special focus on the hazardous elements, which is based on an understanding of the manufacturing process, the quantity and types of chemical solutions used at each step, process containment, and the flammable and hazardous conditions present. Besides applicable codes and regulations, considerations should address operational requirements along with business and insurance policies.

Codes and regulations include the following:

- Local building codes (often refer to IBC, particularly when special flammable occupancy classification is required)
- Local fire codes applicable to flammable and combustible liquid storage (often refer to or align with NFPA1, NFPA 30, NFPA 45, or IFC etc.)
- Local fire codes applicable to flammable and combustible liquid processing
- Local electrical codes (which typically reference NEC, ATEX, etc.)

HVAC design for areas utilizing hazardous or flammable materials often requires monitoring the concentration of the hazard(s), dilution of the hazard(s) within a space using fresh supply air, and containment of volatile chemical hazards to prevent fugitive emissions. This is as required by code or safety design philosophy and as aligned with architectural design and electrical classification strategies. Design strategies for addressing the presence of flammable vapors or hazardous materials are outlined further in Sections 10.3.1 through 10.3.7.

10.3.1 Once-Through/100% Outside Air

A 100% Outside Air (OA) design strategy may be used to avoid creating a flammable environment and/or reduce the risk of exposure of personnel to toxic materials. This may be dictated by the process or may be a design consideration that includes factors such as risk, codes, capital cost, operating cost, and sustainability factors. Because 100% OA systems are not efficient, sustainability considerations may include use of energy recovery systems. Once-through air systems may also be used in localized areas in which hazardous compounds or solvents are handled in the open.

10.3.2 Recirculated Air Systems and Emergency Ventilation

Designs that recirculate air locally within specific process areas may be used. The risk of solvent or hazardous material release should be evaluated, and appropriate monitoring, interlocking, alarming, and notification strategies should be applied.

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If recirculation of airflow is considered for an area in which flammable or hazardous materials (e.g., ammonia hydroxide release from cleavage and deprotection reaction) may be present, design considerations may include an emergency purge mode of operation. When local sensors (such as LEL and ammonia) signal that the concentration of flammables or hazardous chemicals exceeds the safe design limit, the Air Handling Unit (AHU) and exhaust fan systems switch to a once-through (100% OA) airflow mode. If proposed, this design must be thoroughly reviewed to ensure all AHJ and code requirements are properly met or if it should instead be considered as an alternative means-and-methods design approach. International Mechanical Code (IMC) Section 502, Required Systems [103], applies in many jurisdictions, and further refers to specific instances in which recirculation is not permitted, including those detailed in IMC Section 510, Hazardous Exhaust Systems.

10.3.3 Monitoring for Hazardous Conditions

As noted previously, where applicable, monitoring of LEL and hazardous chemical concentration and interlocking associated HVAC systems may be employed to detect and respond to the presence of unsafe conditions. Regulations typically prohibit recirculation of air when solvent concentrations exceed 25% of LEL. If recirculated air systems are used, an interlock may be required to switch the system to 100% OA when LEL sensors within the room or inside return ducts reach a value of 25% of LEL. Selection and configuration of sensors used to detect hazardous chemicals (e.g., ammonia hydroxide) will depend on the expected worst-case concentration, location, and industrial hygiene data, including occupational exposure limits. As with other manufacturing facilities, there may also be areas with potential to create an O_2-deficient environment because of inert gas release, fire protection systems, or refrigerant. The potential for these hazards should be considered, and the appropriate monitoring strategy (e.g., O_2 monitoring) should be selected.

10.3.4 Hazardous Material Removal

Fume hoods, ventilated enclosures, and point exhaust (e.g., movable “trunks”) are intended to reduce the risk of hazardous materials exposure and prevent dangerous conditions. These are typically directly connected to 100% exhaust. When used in cleanroom environments in which a differential pressure balance is maintained, design often favors a Constant Air Volume (CAV) approach where exhaust at each point, enclosure, or fume hood does not vary with use or sash position. However, a Variable Air Volume (VAV) solution may be implemented in a laboratory setting for energy saving. The *ISPE Good Practice Guide: Heating, Ventilation, and Air Conditioning (Second Edition)* provides further recommendations for the removal of hazardous materials [98].

10.3.5 Fire Dampers

Depending on ductwork size and sheet metal gauge, fire dampers at fire control area boundaries may be required to maintain one-hour fire-rated wall ratings. Fire dampers are required by code for cases in which ductwork penetrates walls with two-hour or three-hour ratings. Additional detailed design considerations will vary by application and applicable code.

10.3.6 Fire Protection

Fire protection systems used in oligonucleotide and peptide manufacturing facilities are not unique, and design requirements noted in other Baseline® Guides are applicable.

Attributes of fire protection systems specific to oligonucleotide and peptide facilities, as permitted by local code and authorities, include the following:

- Containment of sprinkler discharge/fire suppression water may be required in hazardous occupancy flammable storage and processing areas. The collected water may require testing and abatement prior to discharge. See Chapter 8 for additional details.

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- Clean agent extinguishing systems (gaseous suppression) or high-pressure mist fogging systems may be considered, pursuant to approval from the AHJ, in high-risk/high-value areas, to protect areas in which water extinguishing systems are undesirable (e.g., because of integration with materials/chemicals). HVAC design must include evaluation areas in which these systems are applied.

- Alarms from a fire alarm system may, through a Building Management System (BMS) or process utility control system, interlock with the distribution of flammable solvents and reagents.

- Special or unique process hazards may require special agent chemical extinguishing systems. If the extinguishing agent presents the risk of creating an oxygen-depleted environment, ambient oxygen monitoring and alarming may be required.

10.3.7 Additional Design Considerations

Additional safety related design considerations for HVAC include:

- The manufacturing area layout and pressurization plan should incorporate isolation airlocks (pressure bubble or sink) to keep flammable vapors from migrating through the building

- Application may require higher exhaust velocities and/or wind wake study to prevent reentrainment and/or to meet code requirements

- Consideration should be given to the flammable properties of reagents and buffers containing acetonitrile, alcohols, or other solvents stored at ambient conditions and the potential for a spilled pool of flammable liquid to create a flammable atmosphere

- Humidity level maintenance is a relevant consideration in areas in which flammable solvents are handled. In dry environments, humidification can significantly reduce the electrostatic voltage produced when personnel interact with common materials such as floors, work surfaces, and containers (NFPA 77 A3.3 [104]). However, in moderate ranges of RH, an increase in RH may not substantially reduce the risk of igniting materials because of static discharge

- The strategy for HVAC system calibration, maintenance, and periodic interlock testing should be considered during the design process to ensure that equipment and the BMS/Building Automation System (BAS) can be serviced safely in the applicable environment

- Explosion-Proof (XP), spark-proof, or intrinsically safe electrical components, instruments, and equipment should be provided for potentially flammable exhaust air streams. Non-XP fan motors may be used if located outside of the exhaust air stream or electrical classification area. Refer to Chapter 9 Instrumentation and Controls and Chapter 11 Electrical for additional information

- Where applicable, exhaust fans should be Air Movement and Control Association (AMCA) Type B or A spark resistant, nonferrous construction as required by code

- For chemical and/or flammables storage and usage rooms, it may, depending on vapor density, be necessary to have both high- and low-wall exhaust-duct extraction points to serve these areas (per IMC code)

- When open processing is performed, consideration should be given to engineering controls (e.g., ventilated enclosures, discussed further in Chapter 8 Architectural), potential for personnel exposure, material handling, and assessment of hazardous exhaust code applicability

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10.4 AHU and Exhaust Zoning

As with other pharmaceutical manufacturing facilities, an oligonucleotide or peptide facility that handles multiple products is often divided into distinct AHU zones. The number of AHUs required depends on several factors such as process segregation (for multiproduct facilities), room classification, hazardous chemical use, and occupancy type. It is crucial to consider the risks to both the product and personnel, and the preferred air filtration and monitoring technologies. Separate zones should be used to prevent cross-contamination by returning air from different processes or classifications. In addition, separate HVAC zones should be used when serving process rooms that handle flammable and hazardous chemicals.

For oligonucleotide and peptide manufacturing facilities where 100% OA is not used, it may be desirable to consider separating HVAC distribution for the following areas:

- Upstream processes:
  - Rooms may require once-through airflow depending on process operations
  - Areas with flammable materials or with open processing
  - Typically including synthesis, solution preparation, and cleavage and deprotection areas
- Downstream processes:
  - Rooms may require once-through airflow depending on process operations
  - Areas that may require higher-grade cleanroom classification
  - Final bulk drug substance filling operations
  - Typically including buffer preparation, chromatography, TFF, annealing, concentration, and lyophilization areas
- Potent or otherwise hazardous compound handling areas
- Chemical storage areas:
  - Storage areas for flammables or other hazardous materials
  - Areas that require H-2 or H-3 hazard classifications per IBC

10.5 Resource Conservation and Sustainability Targets

Corporate sustainability policies should be reviewed for feasibility relative to GMP from the CCS along with AHJ-approved flammable storage and processing requirements. Sustainability and energy use considerations are provided in the *ISPE Good Practice Guide: Heating, Ventilation, and Air Conditioning (Second Edition)* [98].

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10.6 Monitoring and Control Systems

As with other pharmaceutical manufacturing facilities, a BMS/BAS is typically used to control and monitor HVAC systems and utilities, historical trend data, and communicate alarm conditions. The BMS/BAS system may control or interface with subsystems including HVAC, utilities, lighting, security, life safety, and backup power infrastructure. Typically, a separate system is used for alarming and managing critical cleanroom environmental data. This system may be stand-alone or incorporated into a common control system architecture. Additional monitoring of critical environmental and safety parameters, including LEL and O_2 concentration, are often incorporated into the BMS/BAS or process control system.

10.7 Commissioning and Qualification (C&Q) of HVAC and Mechanical Systems

The Commissioning and Qualification (C&Q) approach is considered in parallel with specification and design. Clearly defined user requirements and risk assessment are frequently used to determine C&Q for each system within a manufacturing facility as would be done for other types of facilities. Extensive C&Q guidance is provided in *ISPE Baseline Guide: Commissioning and Qualification (Second Edition)* [74].

10.8 Cost Considerations for HVAC Systems

The trade-off between capital expenditure and operating cost needs to be considered, along with design for sustainability and with consideration of applicable codes and in close cooperation with the local AHJ. Selection of a central station, primary/secondary AHU, or 100% OA topology is a major cost consideration, and requires careful evaluation of the aforementioned requirements with an understanding of the short- and long-term objectives for the facility.

Additional considerations that impact long-term operational costs and sustainability targets may include the following:

- Exhaust air heat recovery, dew-point-based discharge air temperature reset schedules
- Low temperature (≤ 130°F or < 54°C) heating hot water systems to allow for use of heat recovery chillers
- Water-to-water heat pump units or other low-grade heat sources
- Enterprise Content Management (ECM) or Variable Speed Drive (VSD)/Variable Frequency Drive (VFD)-driven fan motors for capacity control
- Application of variable volume unoccupied setback of HVAC systems

Cost-saving strategies should include consideration of the area electrical classification material/chemical storage requirements and personnel safety relative to airflow/air turnover and LEL monitoring where applicable.

10.9 Materials of Construction

Because sporicidal agents are frequently used to sanitize oligonucleotide and peptide facilities, stainless steel is often specified for exposed surfaces. Galvanized steel is typically suitable for ductwork construction, although in some cases the portion of the duct connecting to low-wall exhaust or return registers may be specified as stainless steel, as there is potential to contact sporicidal sanitizing agents. Stainless steel duct systems may be required for certain chemical exhaust systems if they are exposed to open chemicals and/or fumes or vapors.

Protective coatings may also be used on heat transfer coils that have potential to contact sporicidal sanitizing agents. This is often seen on cold room evaporators.

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As noted in Section 10.3.7, code may dictate materials of construction in cases where nonferrous or nonsparking requirements apply.

10.10 Maintenance of HVAC Systems

The development of a comprehensive maintenance program is applicable to oligonucleotide and peptide facilities and defined in the *ISPE Good Practice Guide: Maintenance (Second Edition)* [105]. More detailed recommendations specific to mechanical and HVAC systems are provided in *ISPE Good Practice Guide: Heating Ventilation, and Air Conditioning (Second Edition)* [98]. Besides the recommendations in these guides, there are several maintenance considerations that apply to oligonucleotide and peptide facilities (along with some API manufacturing sites).

Periodic safety interlock testing may be advisable as part of a comprehensive maintenance plan. This may include:

- Routine calibration and/or testing of applicable LEL/O_2 sensors
- Verification of applicable LEL/O_2 HVAC interlock functionality
- Because HVAC systems, including airflow/air turnover rates and humidity, are important engineering controls linked to maintaining a nonflammable environment and static-charge mitigation, the level of monitoring and maintenance of associated equipment and systems may, depending on the level of redundancy applied, be approached with increased rigor

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11 Electrical

11.1 Introduction

Electrical design for oligonucleotide and peptide facilities encompasses several critical aspects. Foremost among these are compliance with local and national codes, compliance with electrical classification requirements for hazardous areas, and alignment with GMP cleanroom standards. Additional design considerations include ensuring system reliability, appropriate redundancy, and maintenance of electrical systems.

11.2 Electrical Area Classification

Oligonucleotide and peptide processing facilities routinely handle flammable or combustible chemicals while also being subject to GMP facility requirements. Design approaches in these facilities must therefore consider applicable codes, flammable chemical inventories, and operational details as a basis for classification, as well as the extent of hazardous area locations within the facility. These elements are essential for the proper design, installation, inspection, maintenance, and operation of electrical systems and equipment within electrically classified areas.

Early development of hazardous area classification documentation across multiple disciplines is critical to establishing a foundation that can be refined throughout the design cycle to achieve an appropriate finalized state prior to construction and commissioning.

11.2.1 Applicable Codes

Electrical area classification definition must comply with all applicable local and national codes, as well as requirements from AHJs and other stakeholders such as FM Global or similar loss prevention entities.

Regardless of variation between these requirements by region or particular company and site, the objective remains the same: to identify electrical classification appropriate to the hazardous nature of areas and ensure the safe selection and installation of code rated electrical equipment. Applicable design codes may include, but are not limited to:

United States

- NFPA 1 Fire Code [106]: Overall governing document that provides guidance for defining electrical area classifications and incorporates NFPA 30
- NFPA 30 Flammable and Combustible Liquids Code [107]: Provides specific requirements for the development of the extent of electrical area classification
- NFPA 70 National Electrical Code [48]: Covers the requirements for electrical installations in hazardous classified locations. Defines permitted equipment, wiring methods, purged/pressurized enclosures, and installation requirements in classified locations
- NFPA 497 Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas [108]: Provides information on specific flammable gases and vapors, flammable liquids, and combustible liquids for proper selection of electrical equipment in hazardous locations
- NFPA 499 Recommended Practice for the Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas [109]: Provides information on combustible dusts as it relates to the proper selection of electrical equipment

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Europe

- EU ATEX Directive 99/92/EC (ATEX 153): Minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres [110]
- EU ATEX Directive 2014/34/EU (ATEX 114): Equipment and protective systems intended for use in potentially explosive atmospheres [50]
- IEC 60079-10-1/10-2 Classification of Areas, Explosive gas atmospheres/ Explosive dust atmospheres [111]: Supports design, construction, operation, and maintenance of equipment used in hazardous locations including assessment of ignition sources

11.2.2 Identification of Flammable Chemical Hazards

A thorough evaluation of hazardous flammable chemicals used within the oligonucleotide or peptide facility needs to be conducted, detailing key properties used for classification. These include:

- Flammability group
- Flash point
- Auto-Ignition Temperature (AIT) with corresponding temperature class (T-Code)
- Lower explosive limit (%LEL)
- Vapor density (relative to air = 1)

11.2.3 Basis for Classification

Establishing a basis for classification requires evaluation of how each flammable chemical is used throughout the facility. This assessment includes:

- Quantities of flammable chemicals
- Use cases of flammable chemicals
- Open or closed processing operations including waste disposal activities and potential leak sources
- Identification of chemical storage locations

11.2.4 Boundaries for Hazardous Electrically Classified Areas

Boundaries for electrically classified areas define hazard requirements and equipment design considerations from an electrical perspective. Documentation typically includes:

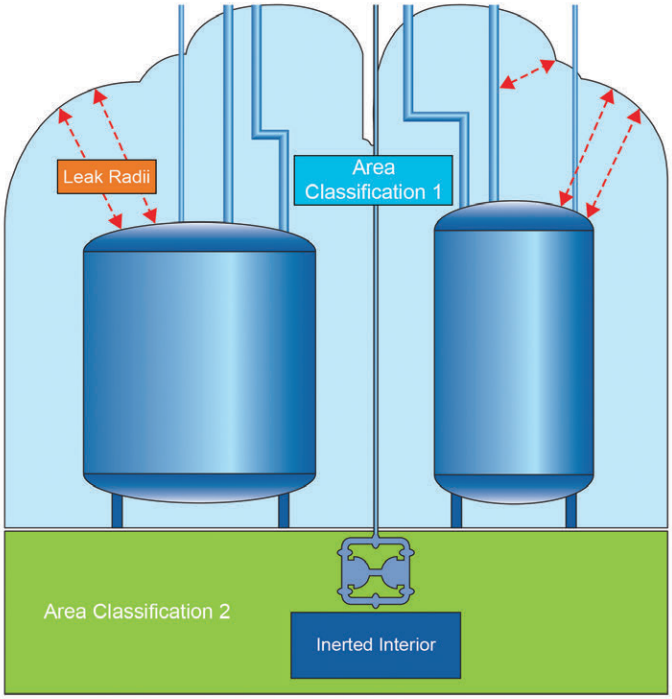
- Defined area classification extent or boundary
- Identification of leak or point sources with corresponding distances and elevations
- Differentiation between Division 1 versus Division 2 – reflecting normal operations versus abnormal operations (normal open operations versus point source leaks caused by abnormal operations such as accidental rupture)
- Clearly defined basis for the electrical area classification

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Integral to the basis for evaluation and documentation is coordination with other disciplines: process, process utilities (solvent delivery and waste systems and container design specifications), architectural (occupancy, fire zoning, containment requirements) and mechanical disciplines (air pressurization schemes) to ensure these elements support compliance with associated electrical classification requirements and hazardous area boundaries.

Figure 11.1 provides a general visual representation of electrical area classification delineating different hazardous zones. Further requirements and details are captured in applicable code sections and must be evaluated thoroughly and ultimately approved by the AHJ.

Figure 11.1: Example Electrical Area Classification Scheme for Tank Farm Equipment



11.2.5 Device Selection and Location Considerations for Electrically Classified Areas

Based on the facility electrical area classification scheme, careful selection and placement of electrical devices and fixtures, including lighting, receptacles, power panels, and control panels, is required to ensure safe access and operability. The amount of equipment installed within electrically classified areas should be minimized, and requirements should be aligned across all design disciplines. All electrical equipment, instrumentation, and electrical devices within those boundaries must be identified and addressed using one or more of the following approaches:

- Specified to meet the corresponding classification requirements
- Relocated to an appropriate location outside of the electrically classified area
- Intrinsically safe barriers/installation methods, allow for the location of certain low voltage devices within certain classified areas
- Purging of electrical panels, can be used to maintain an inert environment around devices that would otherwise be unable to be located within the boundary

Electrically rated equipment carries a significant cost premium when applied across an entire facility design strategy. For example, relocation of unrated lighting above hazardous classified spaces can significantly reduce overall program cost.

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Emergency power sources for both emergency lighting and exit signage should be located outside cleanrooms and hazardous areas whenever possible. When equipment must be installed within electrically classified areas, usage of a properly rated enclosure suitable for both the electrical classification and environmental conditions is required.

Hazardous area documentation in combination with applicable local and national electrical codes should guide the selection of electrical devices and wiring methods. There are several protection methods recognized under these codes. Evaluation of these methods is required to determine the best solution based on lifecycle cost and impact on facility operations.

11.3 Electrical Power Supply

Electrical power supply design must consider the operational needs of the facility, including tolerance of power outages and the potential impact to the product.

11.3.1 Power Types and Considerations

To meet facility power requirements, three primary power types are applied:

- Normal Power: Normal power provided by electric utilities is generally reliable but remains susceptible to disturbances such voltage sags, swells, switching interruptions and occasional long-term outages.

- Auxiliary Power: Auxiliary or back-up power augments the normal power supply and can be provided by multiple source types, including generator sets.

- Uninterruptible Power Supply (UPS): UPS systems provide power to critical continuous operation systems, augmenting the normal power and auxiliary power supply, using energy storage technologies such as batteries, flywheels and fuel cells.

Power requirements for all systems including building HVAC and utilities, process equipment, QC laboratory instruments, and computer systems should be assessed for continuous power requirements and classified as normal power, auxiliary power, or uninterruptible power.

For systems that must remain operational but can sustain brief power interruptions, a typical approach for auxiliary or back-up power involves an automatic start diesel driven generator paired with an automatic transfer switch. Loads are typically restored to the generator within 60 seconds or less (optional standby).

For critical systems that cannot tolerate interruptions, a UPS system or batteries provide seamless continuous conditioned power that are typically then supported by auxiliary power once online. Design considerations for a UPS system include usage of a central UPS versus local devices. Location considerations for a central UPS include load ratings for floors along with accessibility and logistics to support battery maintenance.

Additional design considerations include:

- Control systems, instrumentation, and data collection systems: Static UPS is typically necessary to maintain system operation and data integrity, including processes and cleanroom areas with continuous monitoring requirements.

- Process equipment systems: Some process equipment systems can tolerate power interruptions if the control system and instrumentation are fed via UPS while the sub-equipment (e.g., pumps, agitators, compressors) is on auxiliary or normal power. In the event of power loss, the control system holds the process equipment at a safe state while also maintaining process instrumentation and data collection. If interruption to the operation of a process equipment system is determined to be unacceptable, UPS should be used for the entire system.

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- Cleanroom environmental controls and monitoring: Maintaining airflow operation and pressurization schemes within the cleanroom areas is the primary objective. Some equipment may tolerate brief power interruptions depending on the active monitoring requirements for the classified space. Controllers generally require continuous power, static UPS to ensure continuity.

- Code required back-up power: Backup power may be required where ventilation considerations are part of the hazardous occupancy strategy. See Chapter 10 for further details.

- General power and lighting: Convenience power, general lighting, and non-process critical electrical functions typically do not require special backup considerations.

11.3.2 Electrical Receptacles

As with power classification, a similar evaluation applies to electrical receptacle types.

- Dedicated receptacles: Powered by their own circuit breaker and can be used for critical loads that need to be isolated from other circuits and loads with larger power requirements.

- Convenience receptacles: Intended for general purposes, less sensitive usages.

11.4 Lighting

The lighting system design within cleanroom areas should take into consideration personnel tasks and operational requirements as well as support process equipment cleaning programs and visual inspection needs. If cleanroom lighting design cannot meet all process equipment cleaning program visual inspection needs, supplemental instrumentation (i.e., internal vessel lighting) and procedural solutions (i.e., standardized external lighting sources and visual inspection procedures) must be implemented.

General requirements for cleanroom lighting design are discussed in Sections 11.4.1 through 11.4.4.

11.4.1 Design Light Levels and Color Temperature

As for other pharmaceutical manufacturing facilities, illumination levels should be matched with the tasks performed in each space. Core processing operations require the highest illumination levels, with subsequent reduced levels, typically applied at air locks and at supply and return corridors. Adjacent spaces should transition gradually to allow visual adaptation.

Typical design lux values at 76.2-centimeter (foot-candle values at 30 in) work plane for suites include:

- Supply and return corridors: 430–540 lux (40–50 foot-candles)

- Personnel and material air locks: 540–650 lux (50–60 foot-candles)

- Core processing suites: 750–970 lux (70–90 foot-candles)

Refer to national and local standards, such as the Illumination Engineering Society (IES) [112], or the International Commission on Illumination (CIE) [113].

A consistent color temperature throughout the facility is recommended. Color temperatures of 3500 or 4000 kelvin provide bright, appropriate lighting for process suites/environments.

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11.4.2 Light Level Distribution and Available Ceiling Space

Achieving uniform light level distribution within each space is generally dependent on ceiling heights and available ceiling space, affecting the number of light fixtures needed to meet lux design requirements.

The cleanroom ceilings must also accommodate air diffusers, sprinklers, process utility panels, and instrumentation requiring close multidiscipline coordination. Recessed light fixtures are preferred over surface-mounted fixtures to minimize airflow disruptions and are easier to clean.

11.4.3 Lighting Control

Lighting control selections shall comply with applicable national and local codes such as the International Energy Conservation Code (IECC) [114].

Preferred control strategies following IECC include:

- Corridors: Supply and return corridors automatically turn on the lighting to not more than 50% power and manual switch(es) bring light levels to 100% with manual control to allow occupants to turn off lights. Occupancy sensors automatically turn off lights within 20 minutes after all occupants have left the space, 5 minute timeout recommended.

- Air Locks: Manual on with manual control to allow occupants to turn off lights and occupancy sensors automatically turn off lights within 20 minutes after all occupants have left the space, 5 minute timeout recommended.

- Processing Suites: Dual level manual control that allows the occupant to reduce the connected lighting load in a reasonably uniform illumination pattern by not less than 50% with manual control to allow occupants to turn off lights.

11.4.4 Emergency Lighting and Illuminated Exit Signs

Emergency egress lighting is required within processing suites by applicable national and local codes (e.g., IBC [115], NFPA 101 [116]).

Self-illuminating exit signs with a 10–20year life span can be utilized within process suites to minimize operational disruption. However, these devices usually contain radioactive isotopes and require registration with agencies such as the US Nuclear Regulatory Commission (NRC) [117] along with controlled disposal.

11.5 Fire Detection and Alarm

The fire detection and alarm system, including the placement of initiation and notification devices shall comply with the applicable local and national codes and standards. Design considerations include:

- Heat detector versus smoke detector

- Interaction with fire suppression systems (e.g., two points of detection)

Placement of fire alarm devices within cleanroom areas should consider and balance area classification requirements around wipe-down and general fit-and-finish as appropriate to the spaces while fulfilling requirements for the number and types of devices to serve the safety purpose.

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Remote location for audible notification may be possible for some spaces if adequate coverage is provided from adjacent rooms or spaces. Consideration must be given to the location, sound projection characteristics, and size of the adjacent space.

Interaction with LEL detection/reporting in process, lights and horns, evaluated with the area classification.

See Chapter 9 Instrumentation and Controls, for handling of interlocks with process equipment around spill events and LEL detection.

11.6 Grounding (Earthing)

Equipment grounding must be carried out in accordance with local and national code requirements. One typical example is the requirement for individual ground conductors for each power circuit, thereby disallowing certain wiring methods.

Supplemental grounding should be provided to ensure that all metallic parts are at an equal potential to minimize static discharges and instrumentation errors.

Fixed mounted equipment, metallic platforms, stairs, etc., should be bonded directly to steel enclosures or frames from local ground bus. Provisions for bonding portable equipment shall also be included, especially considering different operational scales and unit operations (e.g., amidite solution vessels). Common methods include:

- Spring-clamp grounding leads
- Coiled cable or retractable grounding/earthing reels bonded to a local ground bus

Solvent storage and delivery systems, especially those associated with toluene or similar non-polar solvents, require particular focus for ensuring grounding requirements. Practices may include grounding systems and grounding verification procedures to support bulk tanker truck offloads, design of bonding/grounding cables between gasketed flanges, hoses, or other non-conductive piping components, and procedural bonding/grounding of portable process vessels along with procedures for periodic verification of grounding system integrity.

11.7 Low Voltage Systems

Communication systems are a requirement and an important backbone of the facility, including voice and data network systems, wireless access points, security systems, and environmental monitoring systems.

A comprehensive system architecture should be developed to address:

- Hardware requirements
- Software, I/O and PLC quantities
- Grounding and power types
- Network capacities and configurations
- Communication protocols appropriate for OEM skids and facility process needs

Facility space must consider hazardous classification requirements and cleanroom classification requirements discussed in Sections 11.2 and Section 11.9, respectively.

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In addition to considerations for the area with regard to devices, locations and protection for servers and switches must also take into account appropriate areas while balancing cable length limitations.

11.8 Cleanroom Access Control and Air Lock Door Interlocking

Electronic access control will often be required for processing areas or other restricted areas of the facility. The access control system shall be coordinated closely with door hardware. Interlocks for air lock doors must be integrated with air pressurization schemes and visual, go/no-go, status indicator lights provided.

In cleanroom locations where door crash bars are not permitted, emergency override switches must enable emergency egress and be interfaced to local audible alarms and the fire alarm system.

11.9 Electrical Apparatus and Wiring Methods in Clean Areas

The cleanroom areas supporting oligonucleotide and peptide operations will be selected based on perceived risk to the product and will have attendant facility and equipment design requirements for microbial and particulate control and cleaning considerations that apply to electrical devices and wiring methods. Whenever feasible, electrical equipment should be located outside of these areas.

Where equipment must be installed within cleanrooms, design and equipment selection considerations include:

- Use of properly rated and sized enclosures and sealants appropriate for the cleanliness conditions is indicated.
- Electrical apparatus, weatherproof devices/covers, closed cell gasketing, sealed membrane speaker systems and wiring methods that are fully compatible with cleanroom requirements.
- Maximize the use of recessed boxes and wiring methods to minimize exposed surfaces.
- Provision of airtight seals for recessed outlets, plenum-rated back boxes and application of an approved silicone sealant at conduit or cable at entry into the box from behind, inside the box, where the conductors enter, and at the wall opening to the perimeter of the box.
- Minimize horizontal surfaces, and, where required, design supports with spacers to allow approximately sufficient space for manual cleaning of surfaces and walls.
- Use compression terminals and bolted bonding connections for reliability and cleanability.
- Provide a heat shrink type cover where wire strands are exposed to improve cleanability.
- Devices used in rated clean spaces shall be suitable for use with jet spray washdown procedures.

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11.10 Maintenance Considerations

Electrical system maintenance should be considered during the design of a facility and included as part of an overall maintenance program (see *ISPE Good Practice Guide: Maintenance (Second Edition)* [105]).

Electrical preventive maintenance typically includes routine inspections, testing, cleaning, and planned replacement or repair. Inspections may include technology such as thermal imaging to identify loose or degraded connections. Electrical maintenance requirements have historically been determined at the discretion of the owner/operator of a facility with consideration given to insurer requirements and regional/local codes and standards. Increasingly, standards such as NFPA 70B [118] are defining prescriptive maintenance expectations for new or extensively renovated facilities.

Where system criticality and redundancy are high, electrical design may require redundant or a separate power supply to allow future electrical system maintenance without impacting business continuity. Further, maintenance activities in sensitive electrical classification zones must broadly consider process operational status and impact of any required downtime to achieve safe isolation. The type of access, room side or above ceiling, must be considered based on the ceiling construction type and available space.

The preferred approach is to minimize maintenance activities within electrically classified spaces, and this must be coordinated with other design disciplines for equipment placement, and the construction of walkable cleanroom ceilings where appropriate, to enable maintenance and unobstructed access.

Emergency power sources should be arranged in a way that eliminates or minimizes the impact to cleanroom operations during routine maintenance and repair activities.

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12 Future Developments Impacting Oligonucleotide and Peptide Production Facilities

12.1 Oligonucleotide Production Facilities

12.1.1 Introduction

The first DNA synthesizers in the 1980s used vessels for agitated bed technology. In the following decades however, oligonucleotide manufacturers settled on synthesis flow-through static compression column. This became the industry standard for Solid-Phase Oligonucleotide Synthesis (SPOS) using the phosphoramidite method. These flow-through oligonucleotide synthesizers allowed better automation, fast chemistry, and low cycle times.

Currently, almost all commercial oligonucleotide products also undergo freeze-drying or lyophilization as the final step in the API facility. Lyophilization has advantages, such as minimum risk during operation, nondynamic shelf temperature control, and vacuum chamber pressures. But it also has long cycle times of three to five days, and commercial unit scale-up capability is limited.

With a growing oligonucleotide pipeline poised to apply to larger patient populations, these standard operations and others present inherent limitations that will hamper existing production facilities' ability to support the expected industry growth in cost, sustainability, and scalability. These challenges foretell significant changes to the baseline facilities of the future.

- Cost: Oligonucleotides are one to two orders of magnitude more expensive than small-molecule drugs. This is because of their size, the complexity of the phosphoramidite monomers, and the copious reagents and solvents required to drive the reactions to completion.

- Sustainability: Oligonucleotide manufacturing generates large quantities of waste, resulting in an average of 4300 kg of waste per 1 kg of drug substance produced [120]. This mainly consists of organic waste from the synthesis and aqueous waste from the purification of the oligonucleotide. The typically small doses required and infrequent dosing help reduce overall waste relative to therapeutic effect. Nonetheless, this remains an area for further improvement.

- Scalability: The current maximum scale of commercially available oligonucleotide synthesis equipment is ~1.8 moles per batch, which corresponds to ~7 kg of an antisense oligonucleotide 20–mer per synthesis run. This clearly constrains the ability to provide capital-efficient production capabilities for metric-ton quantities of oligonucleotides per year.

Many improvements and optimizations to the existing SPOS process are underway to address these issues. In this chapter, the focus is placed on new technologies and approaches that could significantly change oligonucleotide facilities and their corresponding manufacturing facility requirements.

12.1.2 Optimization Opportunities for Existing Processes

12.1.2.1 P5 Chemistry

In 2021, chemists at Bristol Myers Squibb and The Scripps Research Institute in California introduced an innovative Phosphorus (V)-based platform for oligonucleotide synthesis. These new P(V) reagents have comparable reaction rates to the P(III) reagents typically used to synthesize oligonucleotides and can be used in established oligonucleotide synthesis equipment.

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This P(V) platform offers some advantages over the traditional phosphoramidite method:

- Versatility: The P(V) platform allows the design of oligonucleotides with potentially superior pharmacokinetics and higher efficacy.

- Efficiency: The P(V) method is more direct, requiring fewer synthetic steps. By eliminating the oxidation step, it shortens the elongation cycle and potentially reduces overall reagent consumption and the potential for side reactions.

- Stability: Compared to P(III) reagents, P(V) reagents are less susceptible to moisture. They are stable crystalline solids that are simple to handle and store.

  Despite these benefits, the P(V) method has not yet achieved widespread adoption in large-scale industrial settings. Some challenges limit its integration into commercial oligonucleotide manufacturing:

- Adoption and characterization: The phosphoramidite method has been extensively optimized and integrated into manufacturing workflows over decades. A transition to the P(V) platform requires changes in established protocols and the change in chemistry may introduce new side products or impurities that have not been fully characterized.

- Supply chain and cost: The P(V) reagents are so far not commercially available in large quantities and at a competitive cost when compared to P(III) reagents.

12.1.2.2 Azide-Based Reagents

Mesyl (methanesulfonyl) phosphoramidite oligonucleotides are emerging as a new class of molecules backed by recent industry data and widely considered for commercial adoption. The data shows increased bioavailability, duration of action, and broadened therapeutic index for antisense oligonucleotides. This leads to key potential advantages for manufacturing:

- Reduced batch size requirements for future therapeutics

- Redress for anticipated scale challenges

Despite these potential benefits, the synthesis of mesyl phosphoramidate oligonucleotides involves using mesyl azide solutions, which are a highly dangerous class of molecules. Mesyl azide in particular is explosive, and the azide ion is highly toxic. This compound and similar others can complicate process design and add complexity to facilities, including considerations for:

- Safe transport of reagents to site, storage and sampling at the site

- Safe use in processes, minimizing explosion, and exposure hazards

- Waste handling and treatment considering typical excess stoichiometric presence of the azide, resulting in unreacted material that requires special consideration; crucially, typical oligonucleotide synthesis waste is acidic, which can trigger formation of hydrazoic acid (HN_3), a highly volatile, energetic, and toxic byproduct, if mixed with mesyl azide solution

- Quench reaction generates less hazardous waste but presents unique kinetic and energetic hazards

- Evolution of N_2 gas during reactions requiring equipment and process considerations around synthesis columns

- Best practice to conduct detailed Process Hazards Analysis (PHA) to document and manage risks involving such reagents

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12.1.2.3 Solvent Recycling

Several technologies are under evaluation for processing spent solvent and mixed reagent effluent from oligonucleotide facilities. This section focuses primarily on recovery and purification of acetonitrile used in synthesis, and its removal from downstream operations. The European Pharma Oligonucleotide Consortium (EPOC) recently published a paper providing a comprehensive risk assessment of the potential carryover of synthesis reagents and byproducts into recycled acetonitrile [125]. It also recommends stringent segregation of synthesis waste streams to maximize the purity of the acetonitrile prior to recycling. Based on this information, the primary compounds of concern, considering both their potential impact and their likelihood of survival through the recovery process, are water and triethylamine.

The following is a survey of technologies providing potential solutions for the focus areas mentioned, whether in combination or paired with other technologies. Section 12.2.2 focuses on specific peptide modality solvents of interest and delves into unique regulatory considerations.

Distillation

Distillation is an energy-intensive but highly effective unit operation for separating materials with different boiling points. However, the presence of azeotropes in common synthesis solvents may necessitate multistage distillation to achieve the desired purity. Efficient use of this type of equipment would likely require significantly increased process scales.

Suspension Melt Crystallization

Suspension melt crystallization is a continuous process that uses a crystallizer to chill a feed stream to the point of crystal formation for the target molecule. This is an excellent separation option for liquid streams with close boiling points and azeotropes, is less energy-intensive than traditional distillation, and can achieve very good purity results. This technology may be paired with a preceding distillation step to remove dissolved solids.

Pervaporation

Pervaporation is a unit operation involving a highly selective membrane. Select materials in a feed stream adhere to a membrane. Permeate, is then evaporated on the permeate side of the membrane because of a cross-membrane pressure or temperature difference.

Pervaporation appears to be an energy-efficient alternative to distillation and may be particularly useful for removing acetonitrile from the mostly aqueous chromatography waste streams. It may also enable traditional on-site neutralization; thereby alleviating an often outsized subset of the typical oligonucleotide facility hazardous waste generation.

Adsorption

Adsorber columns could be designed to remove contaminants from acetonitrile wash streams. Adsorption matrices can be selected to capture key solvents, such as water and Dichloroacetic Acid (DCA), that would hamper synthesis in an acetonitrile wash. Adsorption columns have a finite capacity and need to be regenerated or disposed of after each use. This technology requires pairing with other separation techniques such as distillation to recover material sound enough to reuse in synthesis.

12.1.2.4 Process Analytical Technologies (PAT)

As process analytical technologies (PAT) have become more prevalent as characterization tools during process development, more commercial applications are becoming available to enable PAT as a long-term component of the control strategy. A survey of different approaches specific to oligonucleotide unit operations is presented here. There is additional discussion about barriers to adoption and anticipated modification needs for existing facilities to incorporate such technologies in Section 12.2.2.1.

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Synthesis

Oligonucleotide chain elongation reactions occur on a solid support and it is impractical to take solid support samples during synthesis. PAT can provide indirect insight into the column flow and reaction profiles by monitoring the pre- and post- column liquid properties. Fourier Transform Infrared (FTIR) spectroscopy and Raman are two PAT technologies that show promise [119] for reducing manufacturing failures through early detection and increasing throughput via leaner operations. There is also the potential to reduce the amount of product release testing [120].

Some potential areas of PAT application are:

- Phosphoramidite identification
- Understanding detritylation reactions
- Optimization of synthesis washes
- Transition analysis

Chromatography

Ultraviolet-Visible (UV-Vis) spectroscopy, which is generally installed in chromatography systems as a PAT to guide fraction collection, only provides concentration information, not purity. Thus, fractions need to be collected and analyzed off-line. Real-time purity monitoring systems, typically relying on in-line mass spectrophotometry, could increase facility throughput and minimize cold storage locations for fractions within downstream process areas.

Freeze-Drying

Partly owing to its long cycle time, parameters for lyophilization are typically selected conservatively to minimize the possibility of product collapse or meltback during processing. Product cannot be easily directly monitored because monitoring would require thermocouple probes placed directly in the product. However, there are PAT approaches for monitoring the process noninvasively and thus a potential to improve cycle times with optimized batch parameters:

- Capacitance manometer versus pirani gauge pressure for real-time primary drying end-point determination. The transition point between the primary drying (sublimation) phase secondary drying (desorption) phase is otherwise prone to being set conservatively as a fixed time duration to avoid incomplete sublimation, accounting for gas property variability with the Pirani method.
- Tunable diode laser absorption spectroscopy for water vapor concentration and gas velocity for real-time product temperature can provide highly sensitive measurements (parts per billion) of sublimated water vapor gas concentration to consistently determine the average product temperature and transition the equipment to the secondary drying phase.
- Manometric temperature measurement for chamber versus condenser pressure rise to determine real-time product temperature during primary and secondary drying.

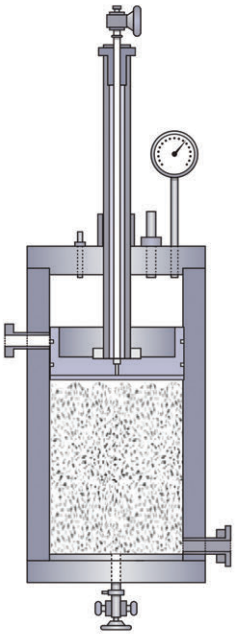
12.1.2.5 Dynamic Axial Compression Column

Many commonly used synthesis solid supports are swellable polystyrene beads that are designed to expand to accommodate the growing oligonucleotide chain. When preparing a standard static synthesis column with these solid supports, it is common practice to intentionally leave an empty space, called a headspace, above the filled bed to accommodate swelling during synthesis. This provides necessary space for chain elongation but decreases synthesis solvent wash efficiency and increases solvent wash volumes.

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Dynamic Axial Compression (DAC) columns can allow better control of both elements (see Figure 12.1). These columns could be used to dynamically adjust the column bed height during SPOS.

Figure 12.1: Typical Dynamic Axial Compression Column



This concept has been successfully executed at laboratory scale [126], showing no product quality impacts and demonstrating large wash volume reductions (50% reduction).

Scaling DAC operation up to large scale has been harder. Relying on column pressure feedback to the piston height controller, as is common for chromatography column DAC, appears to damage the beads and degrade flow characteristics through the swelling and contracting packed bed. Current practice, in use at the multi-kilo scale, has achieved more modest wash volume reductions of 20%, by using fixed, unidirectional piston height adjustments over the course of synthesis in lieu of live pressure feedback [127]

12.1.2.6 Alternative Hydrophobicity-Based Chromatography Process

Hydrophobic Interaction Chromatography (HIC) works through the same chromatographic separation strategy discussed for reversed-phase Chromatography (RP-HPLC) in Chapter 3. It separates product-related impurities based on the presence or non-presence of the final Dimethoxytrityl (DMT) protecting group (or other terminal hydrophobic functional group).

Unlike the RP-HPLC method, HIC is performed in purely aqueous buffers. This reduces the total solvent consumption required for oligonucleotide manufacturing and eliminates flammability risks from solvent-based chromatography, reducing facility costs. However, the bactericidal/bacteriostatic nature of the buffers declines and the necessary environmental control for the process increases. The product elutes as a relatively dilute fraction under low-salt conditions and requires further downstream processing to remove the final DMT protecting group, concentrate the product, and remove the salts.

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12.1.3 SPOS Process Alternatives

12.1.3.1 Stirred-Bed Synthesis

In contrast to the static column format with loosely packed solid support standard used for SPOS, stirred-reactor synthesis remains the workhorse for peptide synthesis via SPPS. Although this process also requires specialized equipment, the reaction dynamics are inherently more scalable, and SPPS production with this technology has achieved production outputs of several metric tons per year. Consequently, stirred-bed technology has therefore generated interest for its potential to bring large-scale production of oligonucleotides to similar scales to support future demand.

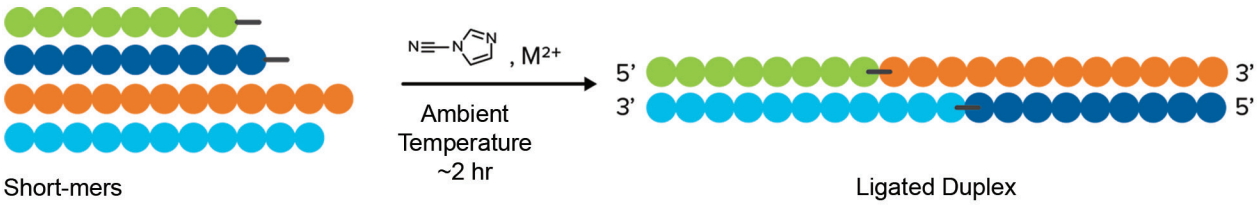
There are several advantages to stirred-bed synthesis in addition to scalability. However, one drawback is the sustainability profile, where wash solvent usage is increased relative to the plug-flow characteristics of a static compression column. Additionally, the elongation cycles of oligonucleotide synthesis are fast. This means that the contact time of reagents and the solid support in a single step can be very short and reaction conditions must be adapted. Stirred tank equipment may also be leveraged in several other novel oligonucleotide synthesis approaches being developed at present.

12.1.3.2 Chemical Ligation

Chemical ligation involves joining several shorter oligonucleotide “shortmers” together by a chemically mediated ligation reaction. This could leverage stirred tank reactors for the ligation and perhaps boost scalability. This technology could enable alternative purification strategies, as the relative difference between uncoupled sequences from a short desired sequence is much greater compared to uncoupled sequences and a typical full-length desired sequence. Capital expenditures of future facilities per kilogram could be reduced. Ligation methodologies also allow existing column-based synthesis equipment to be reused to synthesize the shortmers.

Figure 12.2 illustrates a duplex approach, though modified approaches are also being explored to address the yield challenges inherent in the future potential scale increases for longer molecules (e.g., sgRNA).

Figure 12.2: Schematic Illustration of the Chemical Ligation of Duplex Oligonucleotide
Image courtesy of Nitto Avecia, www.avecia.com.



12.1.3.3 Enzymatic Synthesis

Many organizations are pursuing using bioengineered enzymes to assemble oligonucleotide therapeutics as an alternative to phosphoramidite-based SPOS. These methods generally rely on using enzymes in an aqueous chemistry regime to couple monomers or blockmers, including the following:

- Template-independent Enzymatic Oligonucleotide Synthesis (TiEOS)
- Enzymatic ligation
- Template-dependent approaches

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There are many compelling reasons to pursue this technology:

- The potential reduction in Process Mass Intensity (PMI) by substituting large volumes of organic solvents and reagents for smaller volumes of aqueous solutions
- The reduction or substitution of flammable liquids from processes can simplify facility design and potentially reduce capital investments and facility operating costs
- Waste for some enzymatic processes may be dischargeable to municipal wastewater after limited treatment, alleviating reprocessing and incineration costs
- The potential for a less complex impurity profile and ideally enabling removal of the standard chromatographic purification step
- Potential to overcome the coupling efficiency limitations of phosphoramidite chemistry, enabling synthesis of longer sequences

Adopting enzymatic processes does shift facility focus on bioburden control in the upstream operations akin to traditional biotechnology processes as compared with solvent regime systems of SPOS outlined in Chapter 5 of this Guide.

Enzymatic processes are still in development and numerous questions remain, including how to address the much longer cycle times, final purity profiles, extensive Ultra-Filtration (UF) requirements, and increased quantity of sequence-related reaction kinetic effects. Additionally, there will be a need for the development of a unique raw materials supply chain that will initially push facility costs up and countervail the potential benefits.

12.1.3.4 Liquid Phase Synthesis

Liquid Phase Oligonucleotide Synthesis (LPOS) involves the same chemistry as SPOS but in solution rather than on a solid support. Chemistry is performed by sequentially charging reactants to a stirred vessel during each synthesis cycle. Excess reagents are removed at the end of each cycle by precipitation followed by filtration and washing, or by diafiltration of the solution using a membrane.

In terms of facility considerations, standard reaction and filtration vessels are used for the precipitation approach, whereas an UF skid with retentate and permeate vessels are used for the membrane filtration approach. LPOS thus captures the advantage of alleviating the scale limitations associated with the specialized SPOS equipment and enables production in conventional pharmaceutical chemical facilities. This would be expected to lower capital expenditures, particularly as scale increases.

However, LPOS reaction times are generally significantly longer than SPOS reaction times, which complicates the scale up benefits. Currently, most LPOS processes struggle to achieve equivalent final purities and yields, relative to SPOS, but progress is continuing to be made.

Similar organic solvents to SPOS are used for LPOS so the same facility hazardous occupancy classification would generally be appropriate.

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12.1.4 Downstream and Isolation Process Alternatives

12.1.4.1 Multicolumn Countercurrent Solvent Gradient Purification (MCSGP)

MCSGP is relevant to both oligonucleotide and peptide modalities and discussed in Section 12.3.

12.1.4.2 Continuous Ultra-Filtration (UF)

With batch processing, the size of Ultrafiltration (UF) systems, vessels, and piping becomes notably large. Therefore, the equipment surface areas and internal piping fluid holdup preclude the use of late-stage equipment for earlier clinical development, and much smaller skids are necessary. Three scales of equipment are often necessary to cover all phases of product development.

Continuous processing for Ultrafiltration/Diafiltration (UF/DF) and concentration have been demonstrated using similar TFF cassette technology and/or spiral membranes. Cassette manufacturers have already modified internal flow channels to enable single-pass operation. Adopting the continuous approach would offer the following advantages:

- Reduction of typical UF processing times of eight to 24 hours
- Reduction of cold storage footprint and movement logistics associated with large in-process pools driven by batch processing of multiple synthesis runs
- Potential to align with parallel processing for chromatography, spray-drying, and continuous annealing

Reducing the single-pass nature of continuous UF reduces the flow range and vessel/piping sizes given continuous UF's single-pass nature. This enables equipment designs with much wider scale capability and reduces capital investment associated with changing production scales.

12.1.4.3 Spray-Drying

Spray Drying is relevant to both oligonucleotide and peptide modalities and is discussed in Section 12.3.

12.1.4.4 Liquid/Frozen API

Liquid/Frozen API as an alternative to dried API/DS is relevant to both oligonucleotide and peptides modalities and is discussed further in Section 12.3.

12.2 Peptide Production Facilities

12.2.1 Introduction

While recombinant synthesis of peptides historically represented the primary synthetic approach for large-volume therapeutic peptides, Solid Phase Peptide Synthesis (SPPS), first exemplified by Merrifield in the 1960s, has more recently become the favored technology. Owing to benefits including rapid development timelines, route flexibility, and the ability to introduce non-canonical amino acids and main-chain chemical modifications, SPPS has enabled the synthesis of increasingly complex peptides at commercially relevant scales that were previously inaccessible.

This fundamental SPPS methodology has remained relatively unchanged for nearly a half century, with the ability to markedly improve intermediate purity of the growing peptide chain and eliminate isolation of challenging amorphous intermediate compounds remaining key differentiators in its continued success.

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Nevertheless, obstacles remain for commercial peptide production. These challenges to the current state of the large-scale process are currently being magnified by the success of Incretin drugs that are driving the need for multimetric ton production:

- Cost: Though they are still uncommon in commercial peptides, expensive and complex non-canonical amino acids have already prompted efforts to reduce stoichiometric excesses while still attempting to achieve full reaction completion. As these components become more prevalent in the coming years, and, as volume of products increases, these synthesis considerations will continue to drive the need to improve reagent consumption, solvent use, and cycle time. High volume and high value complex peptides pose a significant potential cost risk if synthesis failures occur late in the process. This has led to increased scrutiny of conventional SPPS equipment and methodologies with a focus on improving robustness and yield.

- Sustainability: Incremental improvements in solvents, cleavage reagents and solid supports have broadened SPPS flexibility and reduced the environmental footprint of SPPS processes. However, increasing environmental regulatory concerns continue to challenge reliance on traditional polar aprotic solvents, accelerating the search for safer and greener alternatives that yield acceptable solubility and similar kinetic profiles. Furthermore, process optimization is playing an increasingly important role in reducing the environmental footprint of established solvent systems.

- Available capacity: Though an influx in investment being driven by market demands is addressing this issue in some respects, these same dynamics have the potential to further entrench the process limitations associated with SPPS. This underscores a unique opportunity to rethink process, equipment, and facility paradigms for forward-looking organizations that can adopt more flexible, future-ready approaches evolve with technology and manufacturing innovations.

Several opportunities exist to optimize conventional SPPS processes and to explore more significant modifications to the peptide process for synthesis, purification, and isolation. This chapter will examine both areas, with the former providing context for future adjustments to baseline facility considerations, while the latter envisions possibly more significant changes to the face of peptide manufacturing facilities.

12.2.2 Optimization Opportunities for Existing Processes

12.2.2.1 Process Analytical Technologies (PAT)

As discussed briefly in Section 12.1.2.4, PAT technology is increasingly becoming available for commercial applications. Incorporation of PAT faces challenges such as capital cost and enabling modifications potentially needed for existing facilities. For new facilities, the bar for implementing these technologies should be lower and the industry is uniquely positioned in a growth phase to pivot the baseline approach. Some of the barriers to adoption in existing facilities include the ability of equipment to support online monitoring and integration of automation into facility controls infrastructure and recipes. Conversely, a reduction of analytical support from an in-process laboratory can potentially simplify facility logistics and alleviate required footprint and manufacturing proximity requirements.

Synthesis

Both SPPS and LPPS are good candidates for efficiency gains due to their highly repetitive unit operations and PAT tools can magnify these gains via application across cycles.

Significant volumes of solvent are used to remove excess reagents and byproducts in SPPS reactors, with usage that can vary widely with scale and equipment design. Implementation of online monitoring and quantitative impurity analysis of wash effluent enables in situ optimization tailored to specific equipment. This approach can reduce solvent consumption and enable solvent reuse and recycling strategies. Applicable tools include automated sample and chromatographic analysis, online spectroscopic measurement, and bulk fluid property measurements.

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However, a key limitation is that fully realizing these benefits may require a change in the traditional wash regime from stirred tank to plug-flow to take full advantage of the approach. This implies meaningful changes to the process and the equipment. Short of these investments, some existing stirred tanks systems can utilize plug-flow-like conditions depending on geometry and potentially gain some of the benefit.

Monitoring for end of reaction following coupling is routinely performed using colorimetric tests such as the Kaiser or Chloranil test for assessing bead color and extent of reaction. This method of assessing end of reaction is manually performed and qualitative in nature only.

Cleavage of peptide from a sample of bed resin is possible, with the subsequent cleaved solution being analyzed following the derivatization. This approach would lend itself to automation and quantification but still requires the complex automated sample preparation systems for cleavage and isolation as well as associated analytical tools capable of quantifying extent of reaction. Liquid phase peptide synthesis, by virtue of its solution-based reaction, is a more suitable analytical substrate as it can be readily sampled and analyzed in situ using a variety of tools.

Different wash volume optimization approaches where an initial step towards PAT can be made by simply monitoring reactor outlet conditions, reaction end-point monitoring by PAT would initially require equipment modification to facilitate monitoring within the traditional stirred tank batch reactor.

Chromatography

For peptide purification, the traditional approach requires chromatographic separation unit operations, which represents a significant analytical workload. Similar to oligonucleotides, the complexity of the crude impurity profile, typically requires RP-HPLC analysis. This has the same chromatographic fraction pooling and individual testing requirements as discussed for oligonucleotides prior to combining into a single product pool.

Online monitoring of product purity has the potential to provide selective pooling and can remove the need for fractionation. RP-HPLC analysis is commonly used, but the time required for analysis must be considered, as it directly impacts the allowable flow rate for column operation and overall cycle time.

Online analysis of the chromatographic effluent by mass spectrometry is rapid, with multiple masses informing collection activities, thus representing a more scalable approach. Online mass spectrometry can be initiated with pre-determined UV-Vis thresholds to ensure data acquisition efficiency and determine start/end points of the product peak.

One current limitation is the availability of technology that would enable a mass spectroscopic approach at commercial scale. In addition to more efficient processes, facility needs for proximate fraction storage and pooling areas can potentially be reduced by this approach.

Crystallization, Precipitation, Isolation

Process Analytical Technology (PAT) tools can play a crucial role in the control of crystallization and isolation processes. Tools, such as FTIR and Raman spectroscopy (in-solution), FBRM, and PVM (crystalline phase), allow for continuous monitoring of key parameters such as solute concentration, particle size, and crystal morphology. This real-time data supports the optimization of crystallization conditions, including temperature, cooling rates, and antisolvent addition, to control nucleation and crystal growth effectively.

Furthermore, by integrating PAT data into feedback and feedforward control systems, manufacturers can maintain consistent product quality while reducing process variability. Adoption of these approaches can improve wash efficiency and filtration time parameters required to reach required impurity levels.

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While crystallization has been studied extensively, isolation by precipitation of amorphous solids can similarly benefit from PAT. Real-time analysis of particle size, including the formation of larger aggregates of primary particles can benefit process cycle time and particle drying behavior. Additionally, early detection of undesirable phase behavior, such as oiling out of high concentration peptide phases, can help prevent challenging isolation conditions.

Equipment modification is required to support PAT for isolation and crystallization to detect conditions within the vessel, which can present a barrier in existing facilities. Moreover, certain techniques, such as Raman spectroscopy, can involve a more expensive approach and there is currently no clear consensus on the best technology for large scale applications. A successful commercial methodology will likely require a combination of tools capable of monitoring both solution- and solid-phase processes attributes. This comes with challenges around the solid-phase process in particular, where variability in products may drive different approaches to technology used.

Drying

As has been noted earlier for oligonucleotides, lyophilization has been the standard for drug substance drying during the commercial era of peptide production. The process is time consuming, with batch processing on the order of days, and energy intensive.

Future advances in PAT include those non-invasive approaches mentioned in the oligonucleotide section: capacitance manometers, tunable diode laser adsorption spectroscopy, and manometric temperature measurement. Though these methods are improvements over older approaches, the correlation between product temperature and off-gas water content has yet to be established sufficiently for acceptance as a product end point for validation purposes. The recent introduction of single-use and autoclavable and remote temperature sensors may allow companies to bridge this perceived gap in the future.

12.2.2.2 Green Chemistry

In traditional Solid-Phase Peptide Synthesis (SPPS), coupling efficiency declines with peptide length, increasing impurity formation, lowering yields, and necessitating more purification steps, all of which raise Process Mass Intensity (PMI). The main SPPS waste stream is N, N-Dimethylformamide (DMF), accounting for up to 90% of waste. As discussed in the Regulatory chapter, DMF regulations have tightened in the European Union due to aspects such as toxicity and carcinogenicity.

Semaglutide represents a large scale commercial peptide produced using a yeast-based expression system that has reduced PMI and organic solvent usage, in particular. Notwithstanding, this method introduces several additional challenges that have limited broader uptake as a true alternative to chemical synthesis methods: limitations of flexibility of the platform to produce non-natural amino acids, limiting the breadth of addressable therapeutic targets, extended development timelines, and ability to control impurities, which simplifies regulatory disposition.

Given the commercial advantages of maintaining a synthetic approach, most green chemistry work is focused on the development of alternative organic solvents with better sustainability profiles. Though literature documents progress on this front, it is clear that multiple different approaches will likely need to be employed for various process considerations to match the versatility and effectiveness of DMF. For example, (N-Butylpyrrolidinone (NBP)) reduces racemization and aspartimide formation while limiting resin swelling and solvent recovery, γ-valerolactone (GVL) and other green solvents also often limit resin swelling and face additional challenges such as amino acid reactivity. Further still, binary solvent mixtures, which can tune polarity for different SPPS steps, face a balancing act between less polar environments (e.g., DOL with NFM or NBP) that favor coupling, and polar environments (e.g., DMSO with DOL, MeTHF, or EtOAc) that favor Fmoc removal and correlate with peptide purity.

12.2.2.3 Solvent Recycling

Traditional chemical process approaches to solvent recovery have included distillation, with mechanical vapor recompression as a lever to reduce energy usage. Entraining solvents can address azeotropes, but generally require additional distillation columns, increasing capital expense and energy expenditure. In addition to these considerations, the pharmaceutical industry has unique barriers to adoption of solvent recycling.

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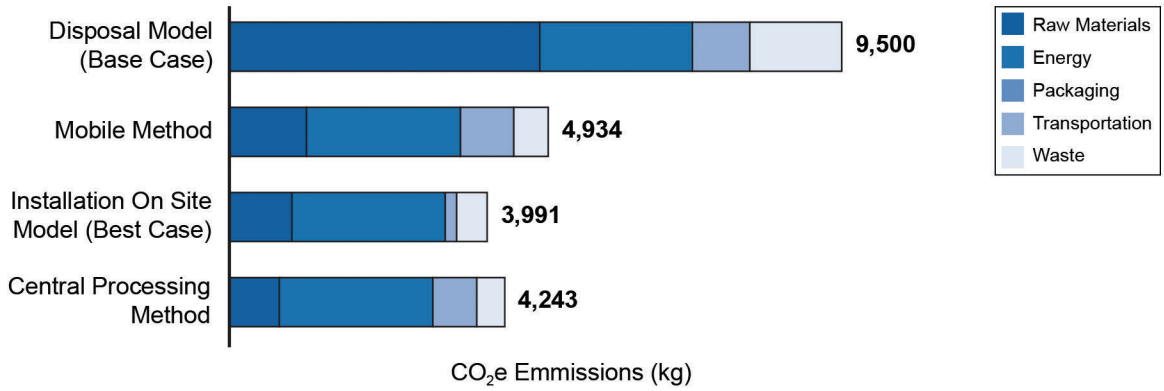
One challenge in implementing solvent recovery strategies is the evolving regulatory landscape. Historically, it was acceptable to include specifications for recovered solvent in initial submissions, allowing flexibility for its future introduction into the manufacturing process. However, regulatory expectations have shifted to expect the filing to provide product comparability data using both virgin and recovered solvents. This shift places a greater responsibility on manufacturers to establish and validate solvent recovery technologies early in development. Furthermore, regulatory scrutiny may be more stringent within a product portfolio as compared to a single-product reuse case, limiting the potential overall benefit for some organizations.

In addition, one of the cornerstone solvents of traditional SPPS, DMF, poses particular difficulty for recycling due to the potential for decomposition and history of nitrosamines in finished APIs, setting a difficult regulatory environment for its reuse.

To mitigate these regulatory challenges, some organizations have adopted alternative strategies, such as partnering with external recycling operators that then sell recovered material for non-GMP uses to maintain a holistically more sustainable approach that reduces reliance on incineration.

As discussed within Section 12.1.2.3, pervaporation or membrane distillation is one approach currently being used in tandem with flash distillation, especially for acetonitrile mixed with water in downstream chromatography. This method can effectively remove any remaining peptide residues and uses a strategy that isolates the recovered solvent to return to the same unit operation that generated the waste. Such strategies can provide a positive impact on supply chain resilience and reduce emissions compared to an incineration model.

Figure 12.3: Emissions Summary by Operating Model



As discussed in Chapter 2, changes to CMR materials in the REACH classifications (human health impact) are driving greater consideration of changes to the process as emissions from facilities using these chemicals are being more tightly controlled in some jurisdictions, potentially requiring capital investment in vent abatement technologies. Some alternative funding strategies such as tax incentives are being considered to reduce the impact of capital expenditure.

12.2.2.4 Deprotection Washing Efficiency

A significant percentage of the raw material used in SPPS process is consumed during the washing after deprotection, which traditionally has been a 20% piperidine/80% DMF solution.

Improvements to this approach include flushing the deprotection reactor headspace with nitrogen to drive off basic compounds and significantly reduce the required wash volumes prior to the next step. This implies facility modifications such as modification of N_2 gas piping distribution infrastructure, bulk tank sizing and footprint on a site, and pressure control automation and equipment.

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Additionally, the reuse of solvent washes can be mediated through holding tanks and piping arrangements that can achieve more than one use of a given solvent tank volume. This additional facility tank capacity, piping connectivity, and automation presents significant renovation challenges for modifying an existing facility and is thus more likely to be employed in build-out of new production trains or facilities.

12.2.3 SPPS Process Alternatives

Liquid phase peptide synthesis methods initially gave way to SPPS as the approach of choice because of the difficulty isolating complex amorphous intermediates and the difficulty rejecting impurities in these isolates demonstrated difficult challenges to LPPS feasibility at commercial scale.

However, liquid phase chemistry retains many significant advantages over solid phase chemistry:

- Reaction kinetics
- No resin swelling limitations
- Overall reagent usage reduction and overall usage of solvents in the process (PMI)
- In-situ reaction sampling versus offline process requiring cleavage, deprotection, precipitation, and testing

As methods to resolve LPPS manufacturing challenges gain traction, the manufacturing benefits of newer chemistries are also potentially significant:

- Facility throughput and equipment occupancy
- Simpler and broader equipment train options
- Greener solvent alternatives, especially for DMF and NMP
- Overall cost of materials
- Wash solvent usage reduction, especially at larger scales, impacting solvent lifecycle costs from raw material to waste incineration
- Simplified and reduced solvent and waste infrastructure
- Reduced facility hazardous occupancy construction costs

The continuing momentum of these techniques has the potential to further reduce facility footprint dedicated to SPPS as the main engine of peptide elongation, allowing an increasingly specialized role for the technology in generating difficult fragments until they can be made fully obsolete.

12.2.3.1 Hybrid SPPS/LPPS Processing

Notwithstanding isolation challenges for LPPS, it has been established that LPPS can produce short peptide chains with significantly lower levels of related substances, and markedly higher purities than those of longer chains of amino acids produced through SPPS.

This has led to the establishment and commercial success (Fuzeon, Exenatide, Tirzepatide) of a hybrid liquid-phase/solid-phase approach to combine SPPS techniques with LPPS fragments, synthesized without the use of resin sequestration. This hybrid approach has enabled meaningful reductions in the large solvent volumes inherent with SPPS-only processes.

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Continued advances in fragment selection and design, including advancing modeling tools capable of predicting epimerization risk as well as fragment crystallinity, are creating additional opportunities to identify crystalline fragments that can expand the role of LPPS synthesis in commercial peptide production.

In parallel, additional innovations in process equipment, such as solvent resistant membranes with decreasing molecular weight cut-offs for retention, have made tangential flow filtration a viable unit operation to conduct LPPS synthesis without requiring intermediate isolations.

12.2.3.2 Chemo-Enzymatic Peptide Synthesis (CEPS)

The use of enzymes in peptide synthesis (CEPS) offers a better environmental footprint for peptide synthesis and provides valuable tools for long-chain peptide, cyclized peptide, and peptide conjugate processes. Transpeptidases that ligate peptide fragments in a controlled, site-selective manner are well-known in scientific literature (*Nuijens et al. Front. Chem. 7:829,* (2019)) [121]. The enzymatic nature of this approach provides an epimerization-free alternative to fragment condensation using traditional coupling reagents.

Key disadvantages are the current need for stoichiometric excess of peptide fragments to achieve commercially-viable yields and the need for early selection of a lead compound prior to process development to account for sequence motifs that can be recognized by the enzymes. This presents a barrier to the adoption of this technology for full amino acid chain elongation but employing CEPS for certain sections and combining with traditional SPPS methods is nearer at hand.

Facility simplification benefits can be expected to include reduction of solvent storage and handling equipment, including a reduction of capital cost compared to the hazardous occupancy facilities required for traditional SPPS approaches.

Despite these advantages, the development of robust supply chains for enzymes suitable for large-scale manufacturing remains a key challenge. The need to produce, characterize, and ensure consistent availability of specialized enzymes introduces additional cost and scalability considerations that must be addressed for widespread commercial implementation.

12.2.3.3 Tag-Assisted Liquid Phase Peptide Synthesis

Tag-Assisted Liquid Phase Peptide Synthesis (TAPS) attaches the elongating peptide to a tag selected for solubility properties in certain solvents. Thus, the tag-bound peptide can be made insoluble, allowing isolation from impurities and side products via precipitation or phase separation. Equipment options for the isolation include precipitation reactors or separation via TFF.

More importantly, the TAPS approach allows the peptide synthesis to use traditional small-molecule solution-phase reactors instead of specialized SPPS reactors. This can potentially have a dramatic effect in expanding the worldwide installed capacity for synthetic peptide production to include large-volume small-molecule reactors.

While TAPS holds significant promise for the manufacturing of synthetic peptides, it remains under development, and no commercial implementations have yet been established. Isolation of tag-bound peptides requires further optimization and is influenced by the peptide sequence and the specific synthesis cycle. Furthermore, cleavage and subsequent removal of the tag from the peptide present challenges, particularly for soft cleavage operations. Additionally, the tags supply chain will represent a significant COGs cost until broad capabilities are developed at a commercial scale.

From a speed to market perspective, since both SPPS and TAPS rely on the same underlying chemistry and building blocks (Fmoc-AA), the expected impurity profile of the peptide API should remain consistent. This compatibility enables the sponsor to take advantage of SPPS strengths in earlier clinical phases and switch to TAPS without jeopardizing their regulatory strategy.

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12.2.4 Downstream Process Alternatives

12.2.4.1 Multicolumn Countercurrent Solvent Gradient Purification (MCSGP)

MCSGP is relevant to both oligonucleotide and peptide modalities and is discussed in Section 12.3.

12.2.4.2 Periodic Countercurrent Chromatography (PCC)

PCC utilizes a rotating scheme of serial columns, overloading the first column until breakthrough onto the second column. In this approach, the first column can then be processed in traditional fashion (washing, elution, and prep for next injection). A third column subsequently catches breakthrough from loading of the second, and the cycle continues, creating a continuous loading and elution process.

One limitation of this approach is the requirement for use of isocratic (instead of linear gradient) elution, which limits its resolution power. This often means the most effective application is to use PCC primarily in an initial capture role where the peptide is separated from weaker binding impurities with large differences in molecular properties.

Despite this limitation, PCC greatly increases the utilization of resin capacity compared to batch operations and the capture approach enables effective removal of impurities without requiring a separate traditional wash cycle. While PCC systems require more operating equipment, this is often offset by reduced solvent consumption and improved process throughput.

12.2.5 Isolation Process Alternatives

As discussed in Section 12.1.1, the traditional freeze drying approach to isolating the final product is a burden for large scale peptide production. Different process alternatives are being considered to enable the industry to move away from freeze drying as an essential unit operation.

12.2.5.1 Precipitation as a Final Isolation Step

Recent publications have highlighted amorphous precipitation as an effective method for peptide isolation. This technique typically removes solvent and inorganic byproducts through soluble filtration techniques, such as TFF using appropriately sized membranes. After removal of these residual inorganic compounds, solvents can be exchanged to provide system that lends itself to amorphous solid isolation of the peptide.

One challenge for peptide products is that they can be prone, at critical solvent compositions, to transforming into tacky or viscous morphologies. To mitigate this, various strategies for precipitation, including reverse addition and co-addition of solvent/antisolvent systems to enable rapid compositional changes. In addition, careful control of composition, and the use of thermal cycling can facilitate particle aggregation that can improve filtration, washing, and solvent rejection during the isolation process.

Another challenge with this approach is the potential for residual solvents to become trapped within the isolated peptide product at levels that are unacceptable for drug specifications without significant further processing and/or risk to the product. Humidified drying using water with nitrogen is one solution that can provide for moderate plasticization of the surface of the drying solid, thus reducing residual solvent mass transfer resistance and thereby improving cycle time and overall drying performance.

Implementation of these processes would require custom equipment to facilitate solvent system additions and nitrogen sparging under vacuum drying conditions and pack-out capabilities. Such requirements would introduce equipment not typically used at this final drug substance stage. This has implications for both the equipment design and the level of cleanliness of the packaging room to support the endotoxin and bioburden control strategy developed.

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12.2.5.2 Spray Drying

Spray Drying is relevant to both oligonucleotide and peptide modalities and is discussed in Section 12.3.

12.2.5.3 Liquid/Frozen API

Liquid/Frozen API approaches for drug substance is relevant to both oligonucleotide and peptide modalities and is discussed in Section 12.3.

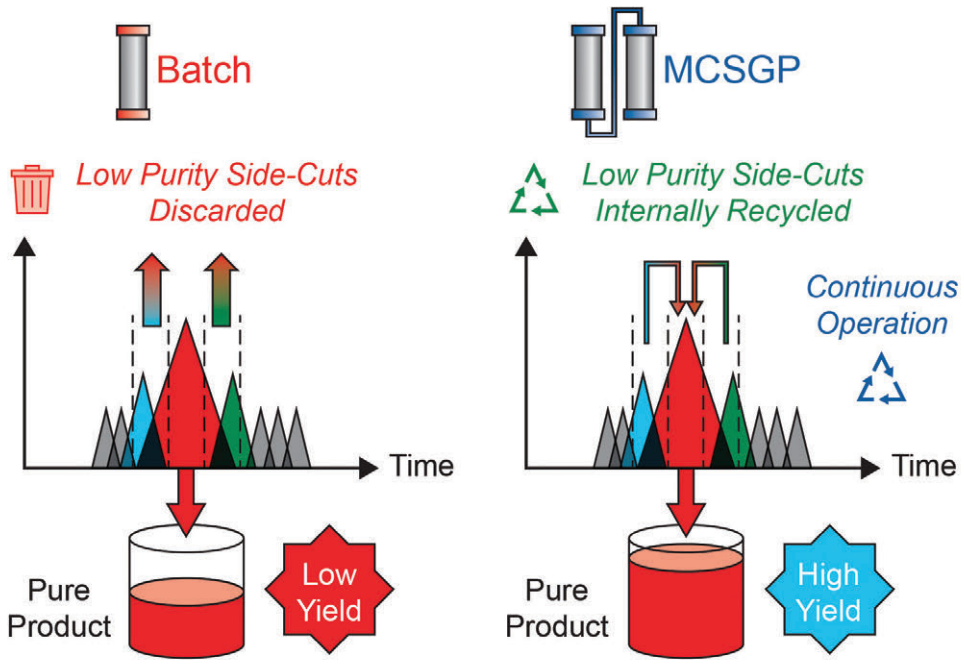
12.3 Future Development Considerations Relevant to Both Oligonucleotide and Peptide Modalities

12.3.1 Multicolumn Countercurrent Solvent Gradient Purification (MCSGP) as an Advance to Conventional Purification Methods

MCSGP is an advanced chromatographic technique that replaces traditional single column batch chromatography with a two-column approach, operating the columns in a countercurrent configuration (see Figure 12.4). The ability to internally recycle partially purified fractions (side cuts) offers a yield advantage over batch chromatography when side cuts are not routinely recovered in batch mode.

MCSGP also offers a clear advantage over traditional batch chromatography when considering the analytical workload associated with the two approaches to purification. Furthermore, the elimination of fraction pooling and corresponding analysis increases potential throughput and reduces facility in-process fraction storage requirements.

Figure 12.4 MCSGCP Visualization
Image Courtesy of YMC America, www.ymcamerica.com.



MCSGP is generally considered cost-effective for large-scale pharmaceutical manufacturing, where higher material throughput and equipment utilization provide better return on investment than for smaller-scale manufacturing. However, the initial capital investment represents one of the main challenges to adoption because specialized skids and control software cannot be retrofitted to existing chromatography equipment. Its adoption also requires an additional investment in specialized operational expertise.

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The potential for increased yield and decreased solvent use with MCSGP should be evaluated on a case-by-case basis as the applicability will vary based on batch mode operations. For example, highly productive batch processes that utilize high loading density and/or do not require recycling of side cuts may not realize significant productivity or solvent use benefits from MCSGP.

For peptides, the frequent use of high-pressure RP chromatography may lead to counter-productive flow rate reductions or reduced bed heights when multiple columns are run in series. MCSGP may be best utilized as the second polishing step of the typical chromatographic process where high purities are required and a significant quantity of material requires side cut recycling. This approach could allow the first chromatography step to run at high loading densities, with or without a gradient. If designed appropriately, this may enable additional advanced technologies such as periodic countercurrent chromatography (PCC) as discussed in the peptide section earlier in this chapter.

12.3.2 Spray Drying as an Alternate Final Drugs Substance Drying Unit Operation

Commercial Potential

Spray drying is a common method of isolating products across many industries at a wide range of scales, but it has not traditionally been used outside of oral solid dose and specialty chemical applications. Key potential advantages of spray drying include high throughput, continuous or semi-continuous operation capable of much larger scales. Comparative energy usage and production time benefits are significant. For both oligonucleotides and peptides, the technology represents a compelling potential alternative to lyophilization for final drug substance drying.

Spray-drying is a closed and continuous process with a residence time of seconds. Aqueous solution processing rates using conventional large-scale equipment are practical up to 5 kg/hr. This means an equivalent 20-kg-lyophilization batch could be processed in a few hours compared with three to five days. This provides the opportunity to support larger scales much more efficiently than lyophilization from a capital-investment-per-kg-throughput standpoint.

The ability to fill product directly into the final container represents a significant improvement from a labor and operational perspective when compared to bulk powder handling and filling operations of multiple lyophilization product trays.

With the increasingly common expectation that facility designs accommodate potent peptides, high containment considerations are already established at commercial scales for spray drying, including vacuum operation, contained discharge, safe change filters, and clean-in-place. In addition, the flexibility of the unit operation provides several other development advantages:

- Scaling up a spray drying process is a relatively straightforward process once the critical quality parameters have been identified, which is possible to do at a small or laboratory scale
- Capability to dry either aqueous or solvent solutions, especially considering lower heat of evaporation for solvents, this can help alleviate degradation risks of spray drying
- Ability to control particle size and shape
- Subsequent storage and handling of the spray dried powder can provide favorable product matrices that discourage degradation as compared to lyophilized product, which could make ambient storage conditions viable

Barriers to Adoption

Spray drying relies on higher operating temperatures than freeze-drying and temperature sensitivity of the product can be a key barrier, potentially causing unacceptable degradation during processing. In particular, hold-up within the chamber can increase degradation risk and raise challenges around yield. Product testing for oligonucleotides to determine molecule resilience is relatively nascent.

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Some approaches to address temperature sensitivity include selection of excipient mixtures and recent developments in electrostatic drying, which can force volatiles to the surface of droplets to maximize solvent evaporation, providing the opportunity to achieve comparable drying characteristics as traditional spray drying at reduced processing temperatures.

Additionally, other aspects must be successfully established through the commercial scale-up process of this technology, such as final moisture content to enable bacteriostatic conditions, bulk density and powder handling characteristics, and acceptable product hold-up/loss comparability to lyophilization.

Line losses can be significant when using spray-drying equipment for smaller lots, making this technology most appropriate for large-scale facilities or near-continuous operation. Similarly, cleaning requirements also lead to line loss considerations for multiproduct facilities.

Facility Considerations

Spray-drying equipment will have unique facility considerations compared to lyophilization and may take up a substantial amount of three-dimensional space in a facility, often involving a stack-up with access to filters, spray balls, and pack-out areas required at multiple platform or floor levels within classified space.

12.3.3 Liquid/Frozen API as an Alternate Approach to Dried Drug Substance

The manufacturing process for oligonucleotide and peptide drug products typically involves re-dissolution of isolated solids prior to compounding the liquid drug product formulation. The inherent variability in powder characteristics, such as bulk density, surface area, crystallization level, flowability, and residual moisture can impact the consistent dispersion and dissolution rate of the API potentially impacting the robustness of drug product manufacturing.

As an alternative, a solution API stored in a liquid or frozen state can simplify the transition to drug product without the complexities of dissolving powdered drug substance. The successful history of transitioning monoclonal antibody solution APIs into sterile drug products provides a framework for an appropriate facility design and microbial contamination control strategies that may be applied to oligonucleotide and peptide solution APIs.

This approach could simplify the overall manufacture of peptide drug substances by eliminating the final isolation step and any associated upstream steps driven by this step. This approach is contingent upon individual solubility characteristics, concentration needs for drug product processing, and stability of the liquid matrix under routine storage conditions.

Facility modifications may include the addition of controlled rate freezing equipment and different storage requirements and capabilities in warehouses, depending on individual product characteristics and variability of requirements for different products or product families within a given production facility portfolio. This can include considerations around contamination risk and implementation of product handling risk mitigation involving equipment, handling, and cleanroom classification for the unit operation [35].

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13 Appendix 1 – Phosphoramidite Chemistry

Phosphoramidite chemistry enables highly efficient and selective reactions charactherized by high yield and minimal by-product formation via the strategic use of various protecting groups. Phosphoramidites are comprised of the following key components:

1. A ribose or deoxyribose group, analogous to those in RNA or DNA.

2. A nitrogenous base attached to the 1' carbon of the ribose sugar, corresponding to Adenine, Cytosine, Guanine, or Thymine bases in DNA, or Uracil in RNA. Adenine, cytosine, and guanine have variable protecting groups to block outlying amino groups, which are reactive during synthesis. Thymine and uracil do not have any outlying amino groups, so these do not require protection.

3. A variable functional group bound to the 2'-carbon position or occasionally bridging the 2' to 4' positions confers nuclease resistance on the final therapeutic molecule. Common examples include 2'-H and 2'-F for DNA oligonucleotides and 2'-O-Me and 2'-O-MOE for chemically modified RNAs. In RNA oligonucleotides, specialized protecting groups protect the 2'-hydroxyl group of the ribose sugar. TBDMS and 2'-O-triisopropylsilyloxymethyl are the most commonly employed protecting groups.

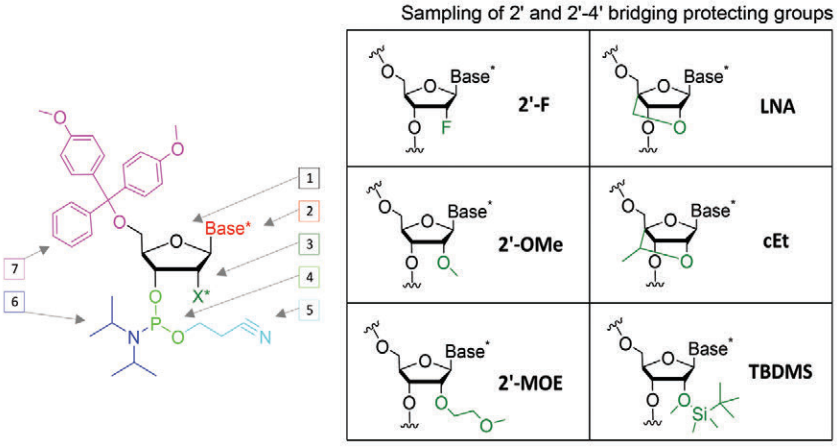
4. A phosphite group, the starting structure for a stable phosphate or phosphorothioate backbone in the final oligonucleotide.

5. A cyanoethyl group attached to the phosphite group blocks reactive sites along the oligonucleotide backbone.

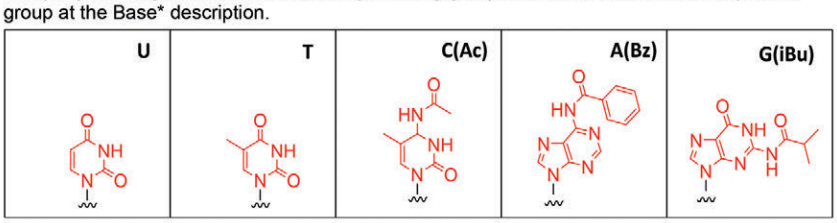
6. A diisopropylamino group stabilizes the phosphoramidite and protects the reactive coupling site until activation is required.

7. A Dimethoxytrityl (DMT) group bound to the 5'-hydroxylposition prevents undesired chain elongation of the oligonucleotide during the synthesis cycle.

Figure 13.1: Key Elements of Phosphoramidite Chemistry



Sampling of nitrogenous bases and their protecting groups. Bound to the ribose/deoxyribose group at the Base* description.



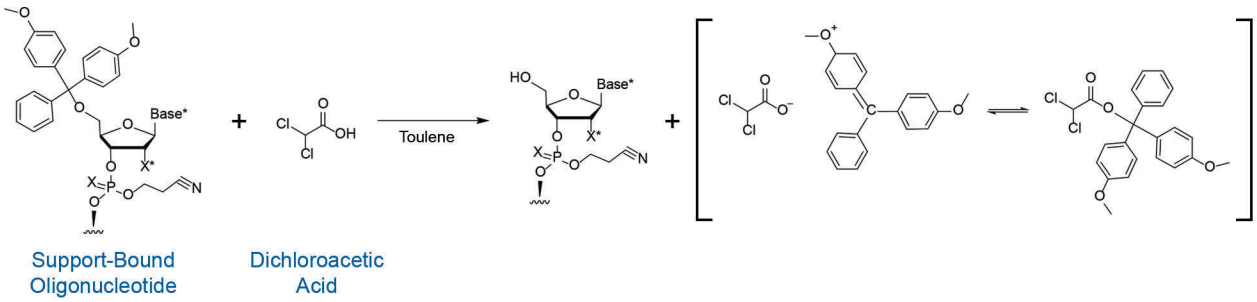
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14 Appendix 2 – SPOS Reactions

14.1 Detritylation

Figure 14.1: Deblock Reaction



Support-Bound Oligonucleotide | Dichloroacetic Acid

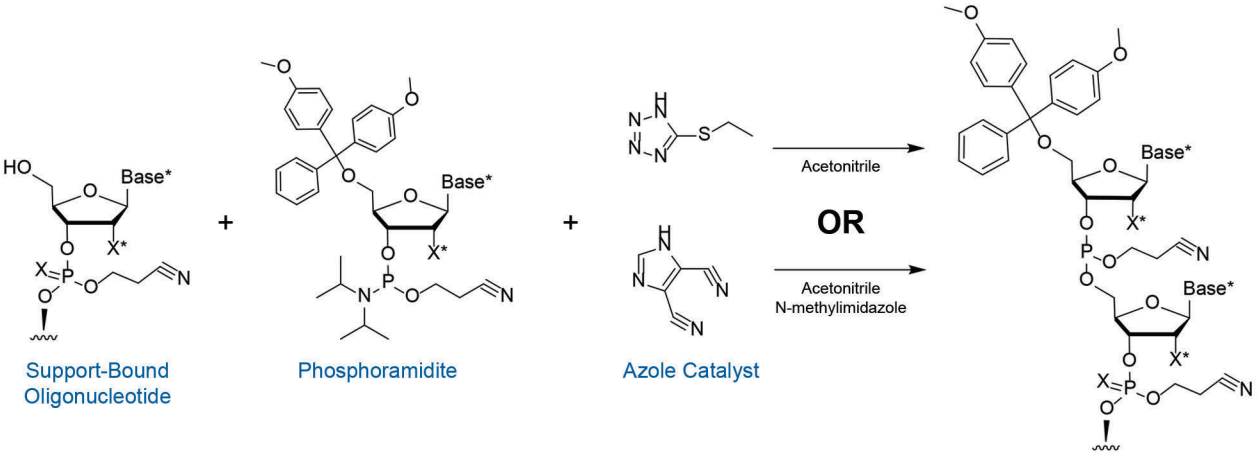
The DMT cation released in the reaction under these conditions absorbs light in the visible spectrum, enabling reliable, real time monitoring of reaction progress to completion using UV-Vis detection with sensors.

Excessive contact of deoxy (2'-H) oligonucleotides to strong acids can lead to unintended depurination, resulting in the removal of purine groups along the oligonucleotide. While insufficient acid exposure can produce truncated products, as the affected site remains protected throughout the remainder of the current cycle, to elongate in subsequent cycles, yielding an oligonucleotide sequence that is missing one nucleotide at that position.

Both of these classes of impurities can be challenging to remove during purification. Selecting an appropriate strength acid can help control impurities, as can optimization of the contact time, equivalents, and temperature. The volume of deblock solution used may also be adjusted in a base-dependent manner and over the course of synthesis to optimize the detritylation reaction. In practice, gradually increasing the quantity of deblock reagent volume as the synthesis progresses is more commonly employed than varying the volume based on base chemistry alone.

14.2 Coupling

Figure 14.2: Example Coupling Reactions



Support-Bound Oligonucleotide | Phosphoramidite | Azole Catalyst

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The addition of an azole-based catalyst in the activator solution drives off the diisopropylamino group which "activates" the amidite for coupling to the 5' hydroxyl group exposed by detritylation. The mixture of amidite solution and activator solution is typically recirculated through the synthesis column to ensure appropriate activation of the amidite and sufficient contact with the coupling sites on the oligonucleotide chain. The result is a highly efficient reaction which typically exceeds 98% yield.

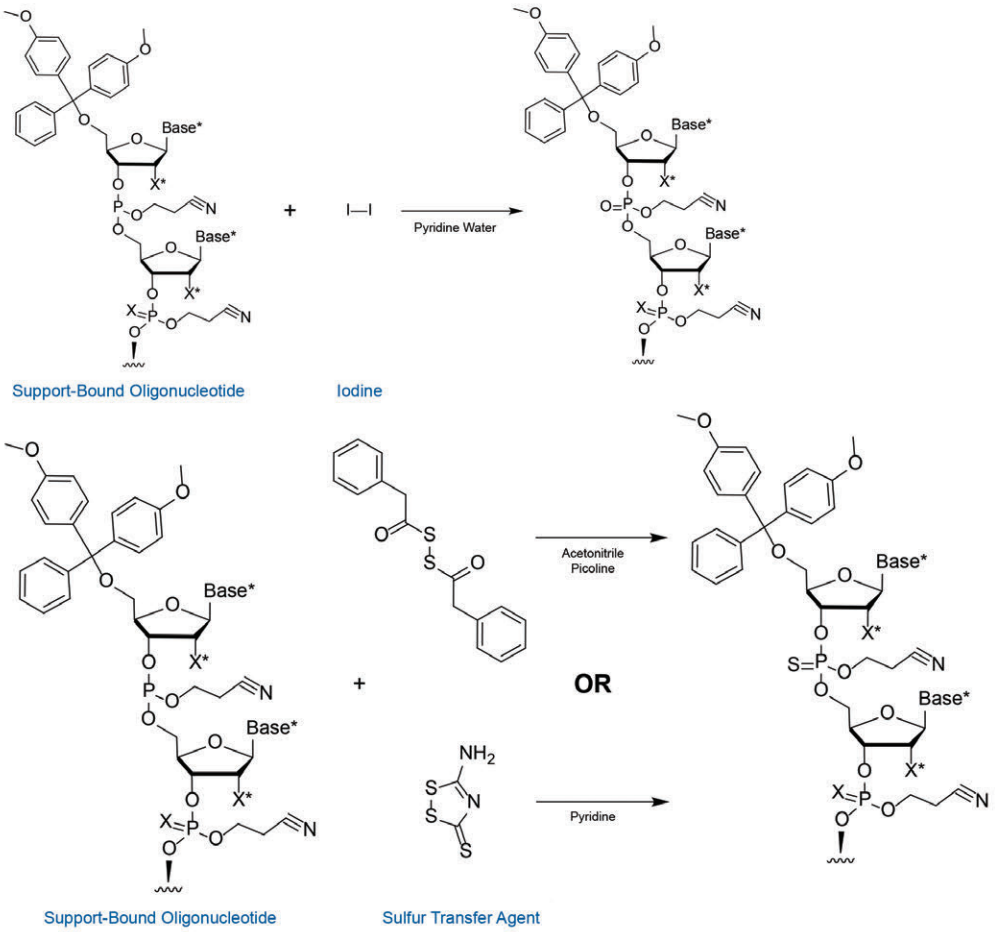
Phosphoramidites are readily hydrolyzed to H-phosphonates in the presence of water, which renders them unreactive. As such, strict measures should be taken to control the water content in the amidite solutions and activator solution. RNA phosphoramidites have greater sensitivity to moisture than DNA phosphoramidites due to hydrolysis resulting in chain cleavage.

Azole-activated amidites may be delivered in molar excess to the number of reactive sites to maximize reaction yield. Different phosphoramidite chemistries and activator solutions may require differing ratios of the solutions and contact times to achieve complete activation of the phosphoramidite.

Furthermore, insufficient contact time with the coupling mixture can impact reaction yield, while excessive contact time can result in undesired double-coupling (n+1 impurity) via detritylation. Optimal conditions for coupling vary for amidites of different bases and with differing functional groups. Equivalents of activator and amidite solution may be modulated based on the specific chemistry per coupling and even the amidite's specific location in the sequence. The coupling reaction is typically performed at ambient temperature.

14.3 Backbone Oxidation

Figure 14.3: Oxidation and Thiolation Reactions



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These reaction steps stabilize the backbone of the oligonucleotide. Oxidation may be performed before or after capping reactions, whereas thiolation is commonly carried out prior to capping, as the capping reagents have been shown to inhibit the sulfurization reaction. Some have even demonstrated that thiolation reaction by-products cap unreacted coupling sites, which may allow for elimination of capping steps in reaction cycles involving thiolation.

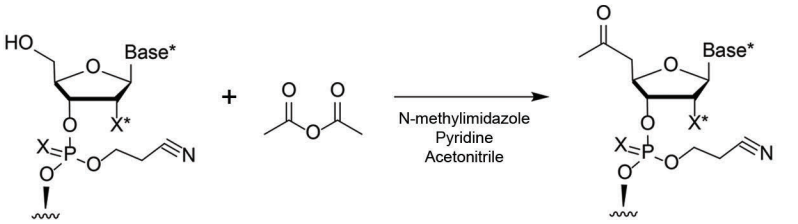
When both phosphate and phosphorothioate backbone chemistries (commonly called mixed backbone) are used in a product, the oxidation solution must be of an appropriate strength to avoid oxidizing the phosphorothioate linkages. This is accomplished by aging the solution before use, which consists of holding the solution at a controlled temperature for several days to weeks. Elevating the temperature appears to reduce the aging time required.

Thiolation and oxidation reactions are commonly carried out at ambient temperature. The equivalents of either thiolation or oxidation solution and their contact time with the elongating oligonucleotide chain are the primary optimization parameters. However, neither excessive contact time nor excess reagent beyond the minimum required for a successful reaction has adverse impacts on the product.

Once thiolation and oxidation reaction parameters have been developed for a synthesis platform process, they tend to be equally effective regardless of the recently coupled amidite's chemistry, nucleobase identity, and location in the overall sequence.

14.4 Capping

Figure 14.4: Capping Reaction



The coupling reaction yield for any given amidite addition is less than 100%. As a result, a small fraction of sites remains unreacted following the coupling step. These sites, if left unprotected, could be subject to later coupling with a different amidite, resulting in truncated and incorrect oligonucleotide sequences which are difficult to separate from the desired product. The capping step in the synthesis cycle exists to mitigate these risks.

Due to the small number of sites that need to be capped, capping is typically performed in great molar excess of the ~2% of uncoupled sites. Excessive capping equivalents and prolonged capping time have been shown to reduce yield and increase impurities. The capping reaction is typically performed at room temperature to balance reaction efficiency with product stability.

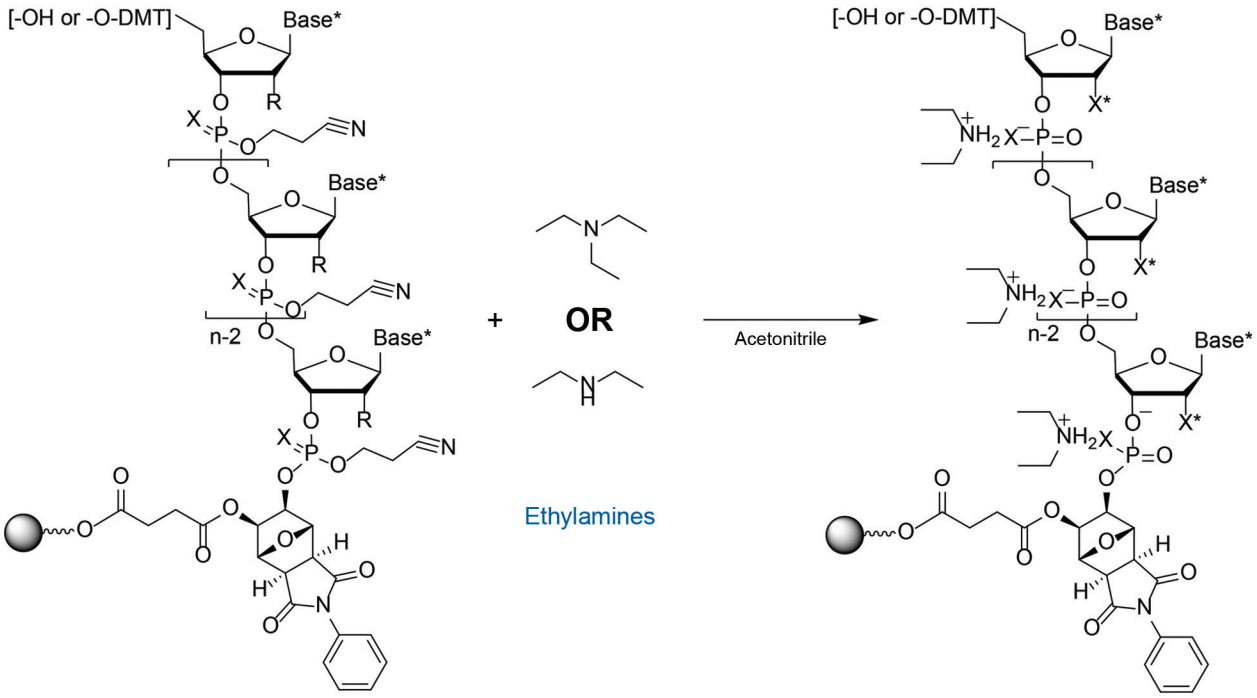
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15 Appendix 3 – Deprotection Reactions

15.1 Backbone Deprotection Reaction

Figure 15.1: P-Deprotection Reaction

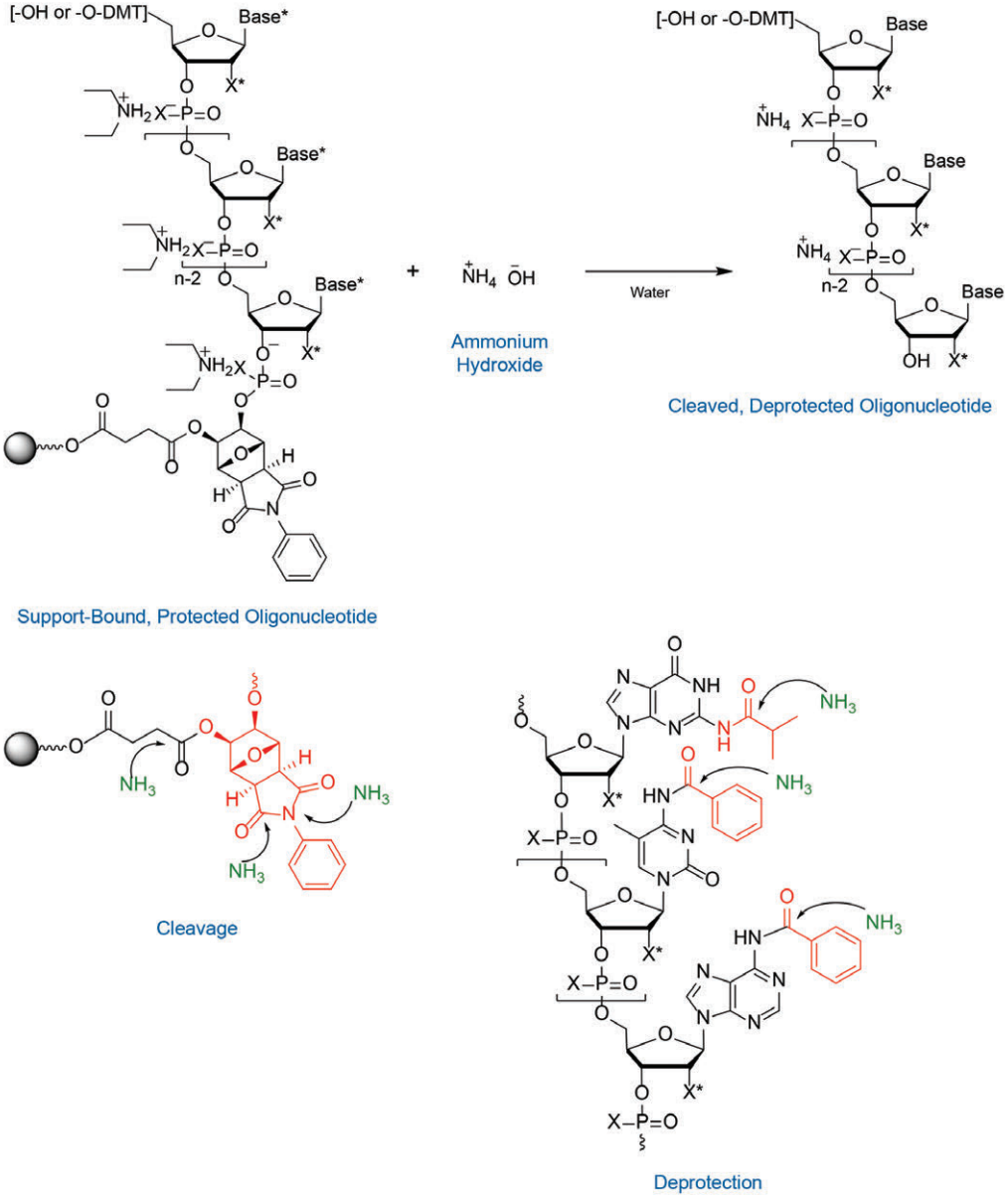


The P-deprotection reaction is typically performed with significant molar excess of either DEA or TEA. The DEA reaction is much faster and quite exothermic, resulting in fluids exiting the column at elevated temperatures (potentially above 50°C). The reagent is typically delivered at room temperature. Delivering excess reagent and/or excessive contact time between the reagent and the oligonucleotide appears to have no negative impacts.

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15.2 Base Deprotection Reaction

Figure 15.2: Base Deprotection, Cleavage, and Deprotection Reactions



The deprotection solution cleaves the oligonucleotide and failure sequences from the solid support. This reaction is rapid and occurs at room temperature. The result is a crude oligonucleotide dissolved in the deprotection reagent. Heating the oligonucleotide crude in the deprotection reagent to temperatures up to 55°C over a period of several hours results in base deprotection, which removes the remaining phosphoramidite protecting groups described in Appendix 1. Optimization of this reaction is primarily driven by equivalents, temperature, and time. Excessive reaction conditions can result in degradation of the oligonucleotide product. The specific reaction by-products are dependent on the reagents used, the chemistry of the cleaved linkage on the solid support, and the phosphoramidite protecting groups.

In the case of RNA oligonucleotides, a following desilylation step is needed to remove the specialized TBDMS protecting groups blocking 2'-hydroxyl sites. This step can be carried out with triethylamine trihydrofluoric acid (TEA-3HF) after the oligonucleotide has been cleaved from the solid support. Treatment with TEA-3HF is relatively fast and is not sensitive to moisture.

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17 Appendix 5 – Glossary

17.1 Acronyms and Abbreviations

ACN	Acetonitrile
AET	Antimicrobial Effectiveness Testing
AEX	Anion Exchange Chromatography
AEX-MPLC	Anion Exchange Medium Pressure Liquid Chromatography
AHJ	Authority Having Jurisdiction
AHU	Air Handling Unit
AIT	Auto-Ignition Temperature
AMCA	Air Movement and Control Association
AOC	Antibody-Oligonucleotide Conjugate
API	Active Pharmaceutical Ingredient
ASEAN	Association of South-East Asian Nations
ASHRAE	American Society of Heating, Refrigeration and Air Conditioning Engineers (US)
ASME	American Society of Mechanical Engineers (US)
ASO	Antisense Oligonucleotide
ATEX	Atmosphères Explosibles (EU Directives on explosive atmospheres)
ATMP	Advance Therapy Medicinal Product
BAS	Building Automation System
BMS	Building Management System
BPCI	Biologics Price Competition and Innovation
BPE	Biopharmaceutical Engineering
BPVC	Boiler and Pressure Vessel Code
BSC	Biological Safety Cabinet
C&D	Cleavage and Deprotection
C&Q	Commissioning and Qualification
CAV	Constant Air Volume
CBER	Center for Biologics Evaluation and Research
CCS	Contamination Control Strategy
CDC	Centers for Disease Control and Prevention (US)
CDE	Critical Design Element
CDER	Center for Drug Evaluation and Research (US)
CDSCO	Central Drugs Standard Control Organization (India)
CEPS	Chemo-enzymatic Peptide Synthesis
CFU	Colony-Forming Unit
CIBSE	Chartered Institution of Building Services Engineers (UK)
CIP	Clean-In-Place
CLARA	Closure Analysis Risk Assessment
CMA	Critical Material Attribute

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CMMS	Computerized Maintenance Management System
CMO	Contract Manufacturing Organization
CMR	Carcinogenic, Mutagenic, Reprotoxic
CNC	Controlled Not Classified
COP	Clean-out-of-Place
CPE	Closed Processing Environment
CPG	Controlled Pore Glass
CPP	Critical Process Parameter
CPU	Central Processing Unit
CQA	Critical Quality Attribute
CTQ	Critical to Quality
CUPA	Certified Unified Program Authority (California, US)
DAC	Dynamic Axial Compression
DCA	Dichloroacetic Acid
DCS	Distributed Control System
DDS	Detailed Design Specification
DEA	Diethylamine
DF	Diafiltration
DMF	Dimethylformamide
DMSO	Dimethyl Sulfoxide
DODT	3,6-Dioxa-1,8-Octanedithiol
DS	Drug Substance
EC	European Commission (EU)
ECM	Enterprise Content Management
EDT	1,2-Ethanedithiol
EM	Environmental Monitoring
EMA	European Medicines Agency (EU)
EPOC	European Pharma Oligonucleotide Consortium (EU)
ERP	Enterprise Resource Planning
FAT	Factory Acceptance Test
FDA	Food and Drug Administration (US)
FEP	Fluorinated Ethylene Propylene
FIO	For Information Only
FRS	Functional Requirements Specification
GAMP®	Good Automated Manufacturing Practice
GMP	Good Manufacturing Practice
GVL	γ-Valerolactone
HAM	Hazardous Atmosphere Monitoring
HDPE	High Density Polyethylene
HEPA	High-Efficiency Particulate Air (also sometimes called High-Efficiency Particulate Arresting)
HIC	Hydrophobic Interaction Chromatography

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HMI	Human–Machine Interface
HVAC	Heating, Ventilation, and Air Conditioning
IBC	International Building Code
ICH	International Conference on Harmonization
IEC	International Electrotechnical Commission
IECC	International Energy Conservation Code
IEEE	Institute of Electrical Electronics Engineers (US)
IESNA	Illumination Engineering Society of North America (US)
IEX	Ion Exchange Chromatography
IFBA	International Federation of Biosafety Associations
IFC	International Fire Code
IMC	International Mechanical Code
I/O	Input/Output
IP	Internet Protocol
IP-RP-HPLC	Ion-Paired Reversed-Phase High Performance Liquid Chromatography
IS	Intrinsically Safe
LDPE	Low Density Polyethylene
LEL	Lower Explosive Limit
LFH	Laminar Flow Hood
LIMS	Laboratory Information Management System
LOC	Lower Oxygen Concentration
LPOS	Liquid Phase Oligonucleotide Synthesis
LPPS	Liquid Phase Peptide Synthesis
mAb	monoclonal Antibody
MAC	Maximum Allowable Carryover
MAL	Material Air Lock
MAQ	Maximum Allowable Quantity
MCC	Motor Control Center
MCSGP	Multicolumn Countercurrent Solvent Gradient Purification
MES	Manufacturing Execution System
MHRA	Medicines and Healthcare products Regulatory Agency (UK)
mRNA	messenger Ribonucleic Acid
MTBE	Methyl Tertiary-Butyl Ether
MTR	Material Test Record
MWCO	Molecular Weight Cut-Off
NBP	N-Butylpyrrolidinone
NC	Not Classified/Controlled
NEC	National Electrical Code (US)
NF	Nanofiltration
NFPA	National Fire Protection Association (US)
NIH	National Institutes of Health (US)

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NIR	Near Infrared Reflectance
NMPA	National Medical Products Administration (China)
NRC	Nuclear Regulatory Commission (US)
OA	Outside Air
OSHA	Occupational Safety and Health Administration (US)
OSWG	Oligonucleotide Safety Working Group
PADS	Phenylacetyldisulfide
PAT	Process Analytical Technology
PCC	Periodic Countercurrent Chromatography
PES	Polyethersulfone
PFA	Perfluoroalkoxy Alkane
PHA	Process Hazard Analysis
PIC/S	Pharmaceutical Inspection Cooperation Scheme
PLC	Programmable Logic Controller
PMDA	Pharmaceuticals and Medical Devices Agency (Japan)
PMI	Process Mass Intensity
PPE	Personal Protective Equipment
PRE	Pitting Resistance Equivalent
PSM	Process Safety Management
P(V)	Phosphorus(V)
PW	Purified water
QbD	Quality by Design
QC	Quality Control
RA	Risk Assessment
RABS	Restricted Area Barrier System
REACH	Registration, Evaluation, Authorization, and Restriction of Chemicals (EU)
REHVA	Federation of European Heating, Ventilation and Air Conditioning Associations (EU)
RH	Relative Humidity
RI	Refractive Index
RP-HPLC	Reversed-Phase High Performance Liquid Chromatography
SAL	Sterility Assurance Level
SAT	Site Acceptance Test
SAX	Strong Anion Exchange Chromatography
SCADA	Supervisory Control and Data Acquisition
SDS	Software Design Specification
SFC	Sequential Function Chart
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SOP	Standard Operating Procedure
SPOS	Solid-Phase Oligonucleotide Synthesis
SPPS	Solid-Phase Peptide Synthesis

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TAPS	Tag-assisted Liquid Phase Peptide Synthesis
TDLAS	Tunable Diode Laser Absorption Spectrometry
TEA	Triethylamine
TEA-3HF	Triethylamine Trihydrofluoric Acid
TFA	Trifluoroacetic Acid
TFE	Thin Film Evaporation
TFF	Tangential Flow Filtration
TiEOS	Template-independent Enzymatic Oligonucleotide Synthesis
TIPS	Triisopropylsilane
TMP	Trans-Membrane Pressure
TOM	2'-O-triisopropylsilyloxymethyl
UF	Ultrafiltration
UPS	Uninterruptible Power Supply
URS	User Requirements Specification
USP	Unites States Pharmacopeia (US)
VAV	Variable Air Volume
VFD	Variable Frequency Drive
VOC	Volatile Organic Compound
VSD	Variable Speed Drive
WFI	Water-for-Injection
WHO	World Health Organization
WIP	Work-In-Progress
XP	Explosion-Proof

17.2 Definitions

Activator

A reagent—typically a weak acid or nucleophilic catalyst—that converts the phosphoramidite monomer into a reactive intermediate, enabling efficient coupling to the 5'-hydroxyl of the growing oligonucleotide during SPOS.

Aggregate Quantities

The total combined amount of hazardous materials stored or used in a facility or area of a facility. This metric determines regulatory compliance and influences facility design requirements.

Amidites (Phosphoramidites)

Nucleotide derivatives used as monomer units in SPOS to enable controlled, sequential coupling reactions. See Appendix 1 for more specific chemical information.

Anion Exchange Chromatography (AEX)/Strong Anion Exchange Chromatography (SAX)

A charge-based liquid chromatography technique in which negatively charged analytes (such as oligonucleotides) bind to a positively charged stationary phase and are eluted by increasing ionic strength or altering pH. SAX resins contain quaternary ammonium functional groups and are commonly used for oligonucleotide and peptide separations.

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Annealing (Duplexing or Hybridization)

The process by which two complementary single-stranded RNA molecules, the sense and antisense strands of an siRNA, are combined under controlled thermal conditions to form a double-stranded siRNA duplex via sequence-specific base pairing interactions (predominantly Watson-Crick).

Antisense/Antisense Oligonucleotide (ASO)

An oligonucleotide strand with a sequence that is complementary to the sense strand or target RNA sequence, allowing it to bind specifically through base-pairing interactions. In industry, ASO may be used more broadly and includes double-stranded oligonucleotides.

Aseptic

A state free of contaminating organisms. Distinguished from sterile where sterile is free of any live organisms, where aseptic is free from contaminating organisms. (e.g., a bioreactor is sterile and aseptic before inoculation, and aseptic only after inoculation).

Authority Having Jurisdiction (AHJ)

Local governmental authority charged with interpretation of governing codes and local requirements to render decisions on design approaches that affect permitting for projects, including allowable alternate means and methods to meet requirements.

Bactericidal

A solution or condition that reduces the level of viable contaminants, typically 3 log10 reduction within 24 hours (e.g., antiseptics, antibiotics, high temperature, pH extremes, radiation, organic solvents).

Bacteriostatic

A solution or condition that inhibits microbial growth/proliferation for the duration of the operation; requires some level of reduction, but not enough to meet the bactericidal definition (e.g., 20% ethanol, WFI storage and distribution).

Capping

The process step in SPOS that terminates unreacted 5'-hydroxyl groups on support-bound oligonucleotides after the coupling reaction.

Carryover Contamination

Residues remaining on process contact surfaces from a previous operation due to insufficient cleaning or sanitization, causing contamination of the subsequent process or product.

Classified Space

A cleanroom that is within the particulate and microbial contamination limits specified in ISO Standard 14644-1 classification units (“In Operation”), European Community (EC) Grades A, B, C, D (“At Rest” and “In Operation”), and in Annex 1 of the EudraLex.

Cleavage/Deprotection (Ammonolysis)

Chemical treatments used to release the fully assembled oligonucleotide/peptide from the solid support and to remove protecting groups that had been required during synthesis to prevent undesired side reactions. This operation transfers the immobilized oligonucleotide/peptide into the solution-phase.

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Closed Process

A scheme in which all manufacturing systems used in a process are closed. See Closed System.

Closed Processing Environment (CPE)

A manufacturing environment that only houses closed systems and processes. Appropriate environmental controls and specifications are user-defined and commensurate with the risk.

Closed System

A system that isolates the process zone from its manufacturing environment and prevents ingress of environmental contaminants during product contact. Process zone is limited by equipment. System closure requires two steps:

a. Removal of latent contaminants within the process zone to a level commensurate with the process requirements, whether aseptic or low bioburden, prior to process contact

b. Prevention of ingress of environmental contaminants during process contact

Conjugation

The chemical coupling of an oligonucleotide or peptide molecule to a non-nucleotide moiety through a covalent bond to impart specific physicochemical or biological properties.

Contamination

The unplanned introduction of a contaminant, whether by a viable or non-viable particulate, with a potential to adversely impact process or product quality. In drug substance contamination events, the contaminate is typically homogeneously distributed throughout the batch.

Control Area

A space within a building (or outdoors) designed to contain hazardous materials, limited to specific Maximum Allowable Quantities (MAQs) by fire-rated walls, floors, and roofs, essentially creating a compartmentalized “hazard zone” to prevent fire spread and ease code compliance, with rules differing for indoor and outdoor areas and height above grade.

Controlled Condition Open Processing

Open process where the process conditions mitigate the risk of contamination from the environment while the process solution or product solution is exposed to the environment prior to rendering it closed (e.g., by sterile filtration). Examples of controlled conditions include: all synthesis reagents and operations and bactericidal chromatography operations.

Controlled Not Classified (CNC)

An unclassified manufacturing environment where user-defined critical control parameters (e.g., temperature, humidity, access, pressurization, pest control, viable and non-viable particulates, etc.) are controlled as per process requirements. Unclassified environments are generally not appropriate for open processing except where “Controlled Condition Open Processing” applies.

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Corrosives

Substances that can destroy or irreversibly damage other materials or living tissue upon contact. They are commonly acids or bases and require special handling and storage to prevent harm.

Coupling

The chain-elongation reaction in SPOS, where an activated phosphoramidite monomer reacts with the 5'-hydroxyl group of the growing strand to form a phosphite triester linkage, adding one nucleotide to the sequence.

Critical Design Element

A specific feature, component, or functional requirement that is vital for the safety, structural integrity, or core performance of a system, product, or process.

Critical Process Parameter (CPP)

A measurable input (input material attribute or operating parameter) or output (process state variable or output material attribute) of a process step that must be controlled to achieve the desired product quality and process consistency.

Critical Quality Attribute (CQA)

Physical, chemical, biological or microbiological properties or characteristics that need to be controlled (directly or indirectly) to ensure product quality (ISPE PQLI [123]). Physical, chemical, biological or microbiological properties or characteristics that should be within an appropriate limit, range, or distribution to ensure the desired product quality (ICH Q8 [27]).

Cross-Contamination

Contamination of a system by a neighboring system. Open systems are especially prone to cross-contamination. Closed systems are less prone to cross-contamination because the process zone is isolated from the manufacturing environment.

Deblocking/Deblock

The acid-mediated removal of the 5'-Dimethoxytrityl (DMT) protecting group from the terminal hydroxyl group of the growing oligonucleotide chain on the solid support. Also known as detritylation. Deblock refers to the reagent solution used to perform this activity, commonly a strong, halogenated acid such as DCA, TCA, or TFA in DCM or toluene.

Decontamination

The neutralization and/or removal of contaminating or dangerous substances. Distinguished from sterilization.

Deflagration

A rapid combustion process that propagates through a gas or dust mixture at subsonic speeds. It can cause significant pressure buildup, requiring mitigation strategies in hazardous facilities.

Deflagration Venting

A safety feature that relieves pressure during an explosion by allowing gases to escape without causing structural collapse. It is commonly used in facilities handling flammable solvents or chemicals.

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Depurination

The acid-catalyzed loss of a purine base (adenine or guanine) from the deoxyribose or ribose backbone of an oligonucleotide (primarily occurs during detritylation reactions). Once formed, this site leads to chain cleavage under basic conditions.

Detectability

The ability to discover or determine the existence, presence, or fact of a hazard.

Detritylation

See Deblocking.

Duplex Oligonucleotide

The double-stranded nucleic acid product formed by annealing two complementary single-stranded oligonucleotides.

Dynamic Axial Compression (DAC) Columns

A chromatography column design that maintains a constant, controlled compression force on the packed stationary phase bed during operation via a movable piston and/or a hydraulic head assembly. When applied to SPOS, this design can be used to accommodate swelling and contraction of polymeric solid supports.

Flammable

Any material capable of ignition and combustion under standard operating conditions, typically defined for liquids by a flash point below 93°C, requiring specific handling and engineering controls in GMP manufacturing.

Flash Point

The lowest temperature at which a liquid gives off sufficient vapor to form an ignitable mixture with air near its surface.

Glazing

The installation of glass in windows, doors, or walls for aesthetic and functional purposes. In industrial settings, glazing can provide visibility into process areas while meeting safety requirements such as fire resistance or explosion control.

Harm

Where a hazard or risk could result in a negative impact to a patient due to lack of ability to detect or monitor the risk element.

Hazard

A real or potential condition, situation or agent that can cause harm to people, product, facility or process. A hazard can cause damage or loss of a system, equipment or to the environment. Any circumstance in the production, control and distribution of a pharmaceutical that can cause an adverse health effect.

Highly Toxics

Materials that pose an extreme health hazard even in very small quantities. These substances can cause severe injury or death and demand stringent containment and handling protocols.

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Hydrophobic Interaction Chromatography (HIC)

A hydrophobic-interaction-based, aqueous liquid chromatography method that uses a mildly hydrophobic stationary phase and a high-salt aqueous mobile phase to separate compounds by differences in hydrophobicity.

Hygienic

Pertaining to equipment, components, and systems that by design, materials of construction, and operation provide for the ability to clean and maintain a level of cleanliness.

IBC High Hazard Group H Occupancy

A classification under the International Building Code for facilities handling hazardous materials in quantities that pose significant risks. This designation triggers stringent requirements for construction, fire protection, and life safety systems.

Latent Bioburden Contaminants

Pre-existing contaminant in hidden or dormant form, found within process systems that need to be removed to effectively close the system before use. For example, contaminants within a TFF cassette that are eliminated by the introduction of a bactericidal preparation/solvents prior to processing.

Liquid Phase Oligonucleotide Synthesis (LPOS)

An alternative to SPOS but in solution rather than on a solid support. Chemistry is performed by sequentially charging reactants to a stirred vessel during each synthesis cycle. Excess reagents are removed at the end of each cycle by precipitation followed by filtration and washing, or by diafiltration of the solution using a membrane.

Low Bioburden Process

A process where the level of bioburden is limited to a user-defined criteria greater than zero, as determined by risk assessment (e.g., maintaining the bioburden in chromatography eluate to < 10 CFU/100 mL)

Lower Explosive Limit (LEL)

The minimum concentration of a flammable gas or vapor in air below which it will not ignite or explode when exposed to an ignition source

Lyophilization

A low-temperature, low-pressure dehydration process in which a frozen product is dried by sublimation of ice during primary drying and desorption of bound water during secondary drying, producing a stable, dry solid while preserving product structure and activity.

Manufacturing Environment

The space housing a process. It includes the conditions, equipment, personnel, materials, and waste, occupying that space. In open processing, the manufacturing environment constitutes part of the process zone. In closed processing, the manufacturing environment is outside the process zone.

Moment Frame

A structural system using rigid connections between beams and columns to resist lateral forces such as wind or earthquakes. Moment frames provide open floor layouts but require larger structural members.

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Multicolumn Countercurrent Solvent Gradient Purification (MCSGP)

A chromatographic technique that arranges multiple columns in a countercurrent configuration. This setup enables the direct process/internal recycling of partially purified fractions (side cuts), thus overcoming the yield-purity trade-off of batch chromatography.

Multiproduct Facility

A facility that supports production of two or more products, either in a campaigned or concurrent manner.

Multiuse Equipment

Equipment, components, assemblies, and systems that are designed for repeated use. Multiuse equipment is designed for cleaning and sanitization (or sterilization for sterile or aseptic operations) between uses. As opposed to single-use systems.

Nucleobase

The nitrogenous base (purine or pyrimidine) that provides the informational identity of a nucleotide.

Nucleoside

The core chemical precursor to nucleotides, consisting of the nitrogenous base and sugar, but lacking a phosphate group.

Nucleotide

The basic structural unit of nucleic acids and oligonucleotides, consisting of three covalently linked components: a nitrogenous base (purine or pyrimidine), a sugar (typically ribose or deoxyribose), and one or more phosphate groups.

Oligonucleotide

A short polymer of nucleotides linked by phosphodiester (or analogous) bonds. Oligonucleotides can exist as single-stranded or double-stranded (duplex) molecules.

Open Process

A scheme in which not all the manufacturing process systems or process zones are closed. See Open System.

Open System

A system that is subject to the contaminants in the manufacturing environment. Process zone is not limited by equipment. Distinguish from International Building Code definitions.

Oxidation

The process step in solid-phase oligonucleotide synthesis (SPOS) that converts the trivalent phosphorus (P(III)) linkage formed during the coupling reaction into the pentavalent phosphate (P(V)) linkage, producing a chemically stable phosphodiester backbone.

Peptide

A short chain of amino acids linked by peptide bonds, produced synthetically or biologically, typically containing fewer than 100 residues and serving as a biologically active or structural molecule in pharmaceutical applications.

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Process Analytical Technology (PAT)

A system for controlling manufacturing processes through real-time measurement of quality or performance attributes of raw or in-process materials with the goal of ensuring consistent product quality.

Process Closure (formerly known as "functional closure")

The act of closing a system (or multiple systems), by removing latent contaminants (viable and non-viable) prior to process contact, and preventing ingress of contaminants during process contact. See Closed Process.

Process Zone

The environment that includes all elements (e.g., raw materials, chemicals, products, contaminants, personnel, waste) that may contact the process or product. In a closed system, the process zone is limited by the "process zone boundary", which isolates the process zone from the manufacturing environment. In an open system, the process zone includes the "manufacturing environment".

Purified Water (PW)

A purified water that meets compendial standards for the jurisdiction in which is generated (e.g., USP, EP, JP). The TOC limit < 500 ppb, conductivity < 1.3 µS/cm, and bioburden limit < 100 CFU/ml.

Residual Risk

Risk remaining after protective measures or safeguards have been put into place.

Reverse-Phase Chromatography (RP-HPLC)

A hydrophobic-interaction-based liquid chromatography method that uses a nonpolar stationary phase and a polar mobile phase to separate compounds by differences in hydrophobicity.

Risk

The potential for an event or situation to cause harm, loss, damage, or other negative consequences. Risk is typically expressed as a combination of the consequences of an event, and likelihood of occurrence, and may include other measures such as detectability.

Risk Assessment

Systematic process of organizing information to support risk decisions. This includes identification of risk(s) and evaluation of the nature, severity and likelihood of those risk(s).

Risk Management

Systematic application of quality management procedures for assessing, controlling, communicating and reviewing risk.

Sanitization

Operation (e.g., thermal, filtration, chemical, or irradiation) that reduces the level of bioburden in a system, solution, or component to an established specification. Typically qualified as a minimum 3 log reduction (99.9%) in bioburden.

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Secondary Containment

An engineered system that captures leaks or spills from primary containers to prevent environmental contamination. It often includes features like depressed slabs, trench drains, or holding tanks.

Segregation

The separation of materials, operations, areas, personnel, equipment, or streams, to prevent mix-ups, carryover or cross-contamination, or overlap.

Sense

An oligonucleotide strand whose sequence matches the target mRNA sequence and complementary to the antisense strand. For a duplex oligonucleotide, it is the non-active strand.

Separation

The action of moving materials, operations, components, personnel, equipment or streams, apart by physical, procedural or temporal means.

Severity (Impact)

For the purpose of the risk assessment, the risk will be evaluated in terms of the severity of impact to process/product Quality; risk to patient and any regulatory impact.

Shear Wall

A vertical structural element designed to resist lateral forces by transferring them to the foundation. Shear walls are common in buildings requiring high stability but can limit interior flexibility.

Solid-Phase Oligonucleotide Synthesis (SPOS)

A stepwise chemical process in which an oligonucleotide is assembled one nucleotide at a time on an insoluble solid support, using protected phosphoramidite monomers and cyclic coupling, capping, oxidation, and deblocking reactions.

Solid-Phase Peptide Synthesis (SPPS)

A stepwise chemical process in which a peptide chain is assembled sequentially from amino acids on an insoluble solid support.

Spray Drying

A rapid, thermal dehydration process in which a liquid feed is atomized into fine droplets and contacted with a stream of heated drying gas, causing the solvent to evaporate and produce a dry powder.

Sterile

Free from living organisms. The Sterility Assurance Level (SAL) requirements is different for drug substances and drug products.

Sterilization

Operation (e.g., thermal sterilization, sterile filtration, chemical sterilization, or irradiation) that renders a system, solution, or component sterile. Typically qualified as a minimum 6 log reduction in living microorganisms including spores.

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Structural Importance Factor

A design parameter used in building codes to account for the critical nature of a structure during events such as earthquakes or high winds. Facilities with higher importance factors are designed to withstand greater forces to ensure safety and continuity.

Sulfurization

The process step in Solid-Phase Oligonucleotide Synthesis (SPOS) that converts the trivalent phosphorus (P(III)) linkage formed during coupling into a pentavalent phosphorothioate (P(V)) linkage by substituting a sulfur atom in place of one of the non-bridging oxygen atoms of the phosphate group. Also commonly referred to as thiolation.

Tangential Flow Filtration (TFF)

A membrane-based ultrafiltration process in which liquid flows tangentially across the membrane to separate, concentrate, or exchange buffers for biomolecules such as oligonucleotides and peptides. Commonly referred to as Ultrafiltration (UF) in oligonucleotide processing.

Thin Film Evaporation (TFE)

A low-pressure, low-temperature concentration process in which a liquid product is spread into a thin, turbulent film on a heated surface, driving evaporation of volatile components.

Toxics

Chemicals that can cause harm to living organisms through ingestion, inhalation, or skin contact. Toxic substances vary in severity and require controlled exposure limits to ensure safety.

Water-For-Injection (WFI)

A purified water that meets compendial standards for the jurisdiction in which is generated (e.g., USP, EP, JP, etc.). The TOC limit < 500ppb, conductivity < 1.3 µS/cm, bioburden limit < 10 CFU/100ml, and endotoxin < 0.25 EU/mL.

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